Clearly the greatest numerical value of $F$ is $g$ and this occurs when $\odot - h = 90°$ or 270°. Now by (72) and (73),

$$g^2 = \cos^2 \alpha \cos^2 \epsilon + \sin^2 \alpha,$$

or
$$g = (1 - \sin^2 \epsilon \cos^2 \alpha)^{\frac{1}{2}},$$

a formula which gives the maximum numerical value of $F$. Also $h$ is given by
$$\tan h = \tan \alpha \sec \epsilon,$$

so that, for the maximum value of $F$, the sun's longitude is given by
$$\odot = \tan^{-1}(\tan \alpha \sec \epsilon) + 90° \quad (\text{or } 270°).$$

Thus the dates can be calculated when the parallax factor attains its greatest numerical value. If the interval between the two dates is six months, then $F_1 = -F_2$ and the parallax is given by

$$\Pi = \frac{(m_2 - m_1)\cos \delta}{2F_2}.$$

The practical method for the measurement of stellar parallax consists in taking photographs, at or near the most favourable dates determined by consideration of the parallax factor, of the region of the sky in which the star whose distance is to be measured (called the "parallax star") is situated. Several faint stars are chosen as "comparison stars" and the measurement of a plate leads, in effect, to the determination of quantities $(\alpha_1 - \alpha_0)$, which we have denoted by $m_1$, for the several comparison stars concerned. In the same way, a plate taken about six months later will enable the quantities $(\alpha_2 - \alpha_0)$ or $m_2$ to be derived. The general formula (71) is then applied. We shall discuss the details of the photographic method more fully in Chapter XII. As a rule about a score of plates, each with three or four distinct exposures, taken at the appropriate epochs are necessary for a reliable determination of the parallax of a single star. The parallax of the nearest star, Proxima Centauri, is 0″·76 and with modern instruments parallaxes as small as 0″·005 can be measured with fair accuracy.

**130.** *The parsec and light-year.*

The distance corresponding to a parallax of 1″ is called a *parsec*. If $d$ is this distance, $\sin 1'' = a/d$, and as $a = 1{\cdot}496 \times 10^8$ km and $\sin 1'' = 1/206265$, the parsec is thus equivalent to 206265 astronomical units or $1{\cdot}496 \times 10^8 \times 206265$ km or $3{\cdot}086 \times 10^{13}$ km.

A star whose parallax is $0''{\cdot}010$ is at a distance of 100 parsecs and generally, if the parallax is $n/1000$ seconds of arc, the corresponding distance is $1000/n$ parsecs. In popular books on astronomy, the *light-year* is generally adopted as the unit of stellar distance—it is the distance traversed by light in one year. From the known value of the velocity of light, it is easily found that 1 light-year $= 9{\cdot}46 \times 10^{12}$ km. Thus 1 parsec $= 3{\cdot}26$ light-years.

## EXERCISES

**1.** If $a$ and $b$ are the equatorial and polar radii of the earth (assumed spheroidal), show that the greatest value of the angle of the vertical is

$$\tan^{-1}\frac{a^2 - b^2}{2ab}. \qquad [\textit{Ball.}]$$

**2.** Show that, if two bodies have declinations and hour angles $\delta$, $H$ and $-\delta$, $H$, respectively, and equal horizontal parallaxes, their parallaxes in right ascension are equal.

**3.** Show that the parallax in declination of a planet observed from a place in latitude $\phi$ vanishes if

$$\tan\phi = \tan\delta\cos H,$$

$\delta$ and $H$ being the planet's declination and hour angle, respectively, and the earth being assumed spherical.

**4.** Show that, if the horizontal parallax $P$ of a body is small so that $\sin^2 P$ may be neglected, the apparent daily path of the body as seen from a place in latitude $\phi$ is a small circle of radius $90° - \delta + P\sin\phi\cos\delta$, described about a point depressed $P\cos\phi\sin\delta$ below the pole.

**5.** Assuming the sun's horizontal parallax to be $8''{\cdot}80$, show that the time during which the sun is below the horizon at either pole is increased owing to parallax by an amount equal to 7 cosec $\epsilon$ minutes, $\epsilon$ being the obliquity of the ecliptic. [*M.T.* 1903.]

**6.** Assuming the earth to be spherical, show that parallax increases the apparent semi-diameter of the moon in the ratio $\sin z' : \sin(z' - \psi)$, where $z'$ is the apparent zenith distance of the moon's centre and $\psi$ is the angle subtended at the moon by the observer and the earth's centre. [*Ball.*]

**7.** The apparent position of a point $S$ $(\alpha, \delta)$ on the celestial sphere is displaced a small distance $SS'$ along the great circle towards a point $Q$ $(\alpha_0, \delta_0)$ so that $SS' = k\sin SQ$, $k$ being small. Show that the increments in R.A. and declination are

$$\Delta\alpha = k\sin(\alpha_0 - \alpha)\cos\delta_0\sec\delta,$$
$$\Delta\delta = k\{\cos\delta\sin\delta_0 - \sin\delta\cos\delta_0\cos(\alpha_0 - \alpha)\}.$$

Apply these to obtain formulae for the changes in the co-ordinates of a star due to parallax and aberration. [*M.T.* 1929.]

**8.** Prove that the cosine of the angle between the directions in which a star is displaced on the celestial sphere by annual aberration and by annual parallax is

$$\sin 2(\odot - \lambda)\cos^2\beta\,[4\sin^2\beta + \cos^4\beta\sin^2 2(\odot - \lambda)]^{-\frac{1}{2}},$$

where $\beta$, $\lambda$, $\odot$ are the latitude and longitude of the star and the longitude of the sun, respectively. [*Coll. Exam.*]

**9.** The satellite Phobos revolves in the plane of the equator of Mars in a period of $7^h\,40^m$, the radius of its orbit being 2·79 times the radius of Mars. The rotation period of Mars is $24^h\,40^m$. Show that allowing for parallax the time between rising and setting of Phobos at stations on the equator is $4^h\,16^m$, and that Phobos is never seen from latitudes above 69°. [*M.T.* 1925.]

**10.** Show that if a planet be observed from a place $P$ on the earth's surface, and from $O$, the centre of the earth, its angular radii $S$ and $S'$ will be related by the formula

$$\sin S' = \frac{\sin(\delta' - \gamma)}{\sin(\delta - \gamma)}\sin S,$$

where $\gamma$ is defined by the relation

$$\tan\gamma = \cos\tfrac{1}{2}(\alpha' - \alpha)\tan\phi'\sec\{\theta - \tfrac{1}{2}(\alpha + \alpha')\},$$

where $\phi'$ is the geocentric latitude of the place, $\theta$ the sidereal time of the observation, $\alpha'$ and $\delta'$ the planet's right ascension and declination as seen from $P$, and $\alpha$ and $\delta$ its corresponding co-ordinates as seen from $O$. [*Ball.*]

**11.** If $Z$ be the zenith, $C_1$ and $C_2$ two planets, $O$ the middle point of $C_1C_2$, $D$ the true distance $C_1C_2$, and $D'$ the apparent distance, show that $D' - D = Q\cos ZR$, where $R$ is a point on the arc $C_1C_2$, such that $OR = \gamma$, where $Q\cos\gamma = (P_1 + P_2)\sin\frac{1}{2}D$ and $Q\sin\gamma = (P_1 - P_2)\cos\frac{1}{2}D$, $P_1$ and $P_2$ being the horizontal parallaxes of the two planets. [*M.T.* 1928.]

**12.** A star is observed at longitude $\lambda$, latitude $\beta$. Owing to annual parallax its longitude is observed to change through $0''{\cdot}5$. What is the maximum change in its latitude, and at what times of the year do the maxima and minima of latitude and longitude occur? What is its distance? [*M.T.* 1920.]

**13.** If $\alpha$, $\delta$ be the right ascension and declination of a star with annual parallax $\Pi$; $p$ and $D$ the position angle and distance of an adjacent star with negligible parallax; and if $\rho$, $\theta$, $\lambda$, $\mu$ be auxiliary quantities defined by the equations

$$\rho\cos\theta = \sin\delta', \qquad \lambda\cos\mu = \cos\delta'\sin(\alpha' - \alpha),$$

$$\rho\sin\theta = \cos\delta'\cos(\alpha' - \alpha), \qquad \lambda\sin\mu = \rho\cos(\theta + \delta),$$

where $(\alpha', \delta')$ are the sun's co-ordinates, show that the parallax corrections necessary to be applied to the observed position angle and distance, in order to obtain their *heliocentric* values are $\Pi\lambda\cos(p + \mu)\operatorname{cosec} D$ and $\Pi\lambda\sin(p + \mu)$, respectively, if we assume the earth's orbit to be a circle. [*Ball.*]

## CHAPTER X

# PRECESSION AND NUTATION

**131.** *Introductory.*

The phenomenon of precession was discovered by Hipparchus in the second century B.C. By comparing contemporary observations with observations made about a century and a half earlier, he was led to the conclusion that the longitudes of the stars appeared to be increasing at the rate of 36″ per annum (the modern value is about 50″) while, as far as he could detect, their latitudes showed no definite changes. There are two possible explanations; either all the stars examined had real and identical motions in longitude—an improbable hypothesis—or the fundamental reference point, the vernal equinox $\Upsilon$ from which longitudes are measured along the ecliptic, could no longer be regarded as a fixed point on the ecliptic. Now $\Upsilon$ is defined to be one of the two points of intersection of the ecliptic and the equator on the celestial sphere; the observations showed no changes in the latitudes of the stars and therefore it was legitimate to conclude that the ecliptic was a fixed plane. According to the second hypothesis (which was adopted by Hipparchus), it was necessary to assume that the equator and, in consequence, the vernal equinox moved in such a way that the longitudes of the stars increased uniformly by an amount in accordance with the observations.

In Fig. 92 let $L\Upsilon M$ denote the fixed ecliptic, $T\Upsilon R$ the celestial equator at time $t$ and $T\Upsilon_1 R$ the celestial equator one year later. In one year the vernal equinox has moved from $\Upsilon$ to $\Upsilon_1$ and thus the longitude of a star $S$ has increased from $\Upsilon D$ to $\Upsilon_1 D$, that is, by about 50″. The uniform backward movement of $\Upsilon$ along the ecliptic is called the *precession of the equinox*. Now Hipparchus satisfied himself that the obliquity $\epsilon$ of the ecliptic had suffered no appreciable change and it therefore followed that the motion of the equator must be such that the pole $P$ moved from $P$ to $P_1$ around $K$ in a small circle, $KP$ or $KP_1$ being the obliquity $\epsilon$. As $KP$ is perpendicular to the great circle joining $K$ and $\Upsilon$, and

$KP_1$ is perpendicular to the great circle $K\Upsilon_1$, the angle $PKP_1$ is equal to the arc $\Upsilon\Upsilon_1$. From the known movement of $\Upsilon$ along the ecliptic it is easily seen that the pole $P$ describes the small circle of which $K$ is the pole in a period of about 26,000 years.

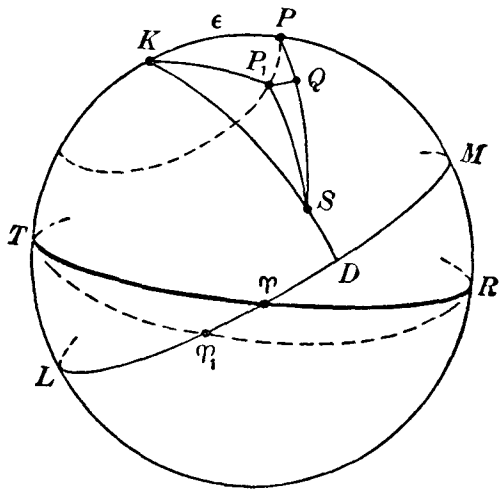



Fig. 92.

Newton first gave the correct dynamical explanation of precession. Consider the familiar phenomenon of an ordinary top (Fig. 93) spinning rapidly about its axis $OC$ inclined at an angle $i$ to the vertical $OD$, and with its point $O$ on a rough horizontal table. If $G$ is the centre of gravity of the top, its weight acting vertically downwards along $GE$ has a moment about $O$ which at first sight would appear to result in increasing the angle $COD$ (that is, $i$) and in bringing the surface of the top almost immediately into contact with the table. But owing to the rapid rotation about $OC$, the motion is very

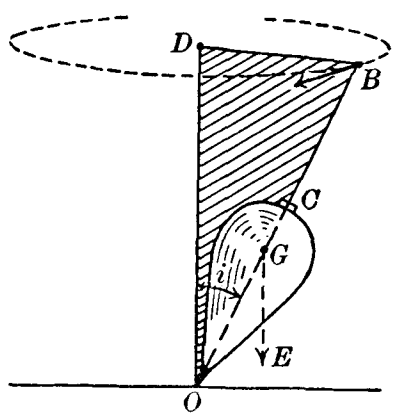



Fig. 93.

much different, for the axis $OC$ is seen to move uniformly around the vertical, so that any point $B$ on the axis describes a circle about a point $D$ on the vertical $OD$, and the axis $OC$ sweeps out a conical surface. Instead of falling towards the table, the axis $OC$ is at any moment moving at right angles to the plane containing $OC$ and $OD$. This motion of the axis is called *precession*. Consider now the earth spinning rapidly about its polar axis under the forces of attraction exerted by the sun and moon. If the earth were a sphere, the density of its matter at any point depending only on the distance from the centre, the direction of the solar and lunar attraction would pass through the centre and there would be no cause operating to change the direction of the rotational axis. But the earth is actually a spheroid, with the equator as principal plane, and as the sun and moon are not situated in the equatorial plane—except on two occasions during their respective orbital periods when their declinations are zero—the direction of the gravitational attractions of the sun and moon do not pass through the earth's centre and, accordingly, the attractive forces have moments about the centre. Consider the sun only. Then the moment of the solar attraction would appear to have the effect of making the earth's equatorial plane move towards the ecliptic, just as in the case of the top (Fig. 93), the moment of the top's weight about the point $O$ would appear to cause the axis $OC$ to approach the table. The actual dynamical result as regards the earth is similar to that in the example of the top; the earth's axis, at any instant, moves at right angles to the plane containing the axis and the sun and, accordingly, it has a conical motion about the direction given by the pole of the ecliptic; in other words, the pole of the equator will describe a small circle around the pole of the ecliptic $K$, as in Fig. 92, and the vernal equinox $\Upsilon$ moves backward along the ecliptic in the direction $\Upsilon\Upsilon_1$. The principal action of the moon on the direction of the earth's axis is of a similar nature. The combined retrograde movement of the equinox $\Upsilon$ along the ecliptic in the direction $\Upsilon\Upsilon_1$ is called the *luni-solar precession*, about two-thirds of the effect being due to the moon and the remainder to the sun. Certain effects of a periodic character are excluded in the definition of luni-solar precession; reference will be made to these later (section 134).

**132.** *The effect of precession on the right ascension and declination of a star.*

Let $\psi$ denote the luni-solar precession in one year. Then as we have seen, the latitude of a star is unaffected, while the longitude increases at the rate of $\psi$ or 50″ per annum. In Fig. 92 let $\Upsilon\Upsilon_1$ or $P\hat{K}P_1$ represent $\psi$. If $(\alpha, \delta)$ and $(\alpha_1, \delta_1)$ are the co-ordinates of the star $S$, referred to the equators $\Upsilon R$ and $\Upsilon_1 R$ respectively, we shall have: $PS = 90° - \delta$, $P_1S = 90° - \delta_1$, $K\hat{P}S = 90° + \alpha$, $K\hat{P}_1S = 90° + \alpha_1$, $KS = 90° - \beta$, $P\hat{K}S = 90° - \lambda$, $P_1\hat{K}S = 90° - \lambda_1$, where $\beta$ is the star's latitude and $\lambda$, $\lambda_1$ the longitudes referred to $\Upsilon$ and $\Upsilon_1$ respectively. Also $KP = KP_1 = \epsilon$.

We shall first investigate the value of $\delta_1 - \delta$. Through $P_1$ draw the small circle arc $P_1Q$ perpendicular to $PS$. We regard $\psi$ as a small quantity and so $QS$ is sensibly equal to $P_1S$; hence $QS = 90° - \delta_1$ and $PQ = \delta_1 - \delta$ or, as we shall write it, $\Delta\delta$. Now $P_1P$ is perpendicular to $KP$ and as $K\hat{P}Q = 90° + \alpha$ we have that $P_1\hat{P}Q = \alpha$. Hence

$$\Delta\delta \equiv PQ = PP_1 \cos \alpha \qquad \text{.....(1).}$$

Now $P\hat{K}P_1 \equiv \Upsilon\Upsilon_1 = \psi$ and

$$PP_1 = P\hat{K}P_1 \sin \epsilon = \psi \sin \epsilon \qquad \text{.....(2).}$$

Hence, from (1) and (2),

$$\delta_1 - \delta \equiv \Delta\delta = \psi \sin \epsilon \cos \alpha \qquad \text{.....(3).}$$

This equation gives the yearly change in declination due to the luni-solar precession $\psi$.

Consider now the change in right ascension. We shall denote $\alpha_1 - \alpha$ by $\Delta\alpha$. From the triangle $KPS$, by the cosine-formula **A**, we have

$$\sin \beta = \cos \epsilon \sin \delta - \sin \epsilon \cos \delta \sin \alpha \qquad \text{.....(4)}$$

and from the triangle $KP_1S$,

$$\sin \beta = \cos \epsilon \sin \delta_1 - \sin \epsilon \cos \delta_1 \sin \alpha_1 \qquad \text{.....(5).}$$

In (5) we write $\delta_1 = \delta + \Delta\delta$, $\alpha_1 = \alpha + \Delta\alpha$ and expand; keeping only first order terms in $\Delta\alpha$ and $\Delta\delta$, we shall have

$$\sin \delta_1 = \sin \delta + \Delta\delta \cos \delta,$$

$$\cos \delta_1 = \cos \delta - \Delta\delta \sin \delta,$$

$$\sin \alpha_1 = \sin \alpha + \Delta\alpha \cos \alpha,$$

so that (5) becomes

$$\sin\beta = \cos\epsilon\,(\sin\delta + \Delta\delta\cos\delta) - \sin\epsilon\,(\cos\delta - \Delta\delta\sin\delta)\,(\sin\alpha + \Delta\alpha\cos\alpha) \quad \ldots\ldots(6).$$

Subtracting (4) from (6) and neglecting the term in $\Delta\alpha.\Delta\delta$ we obtain

$$\sin\epsilon\cos\alpha\cos\delta\,\Delta\alpha = (\cos\epsilon\cos\delta + \sin\epsilon\sin\alpha\sin\delta)\,\Delta\delta.$$

But from (3), $\Delta\delta = \psi\sin\epsilon\cos\alpha.$

Hence $$\alpha_1 - \alpha \equiv \Delta\alpha = \psi\,(\cos\epsilon + \sin\epsilon\sin\alpha\tan\delta) \quad \ldots\ldots(7).$$

This last equation enables us to compute the yearly change in right ascension due to the luni-solar precession $\psi$.

133. *The effect of precession on right ascension and declination (alternative method).*

Owing to precession, the spherical triangle defined by the star and the poles of the equator and ecliptic, that is $KPS$, is changed after an interval of one year to the triangle $KP_1S$ (Fig. 92). We can thus suppose that the latter triangle is obtained by applying infinitesimal changes $\Delta\lambda$, $\Delta\alpha$ and $\Delta\delta$ to the values of $\lambda$, $\alpha$ and $\delta$, which occur in the elements of the triangle $KPS$. The elements $\epsilon$ and $90° - \beta$ remain constant. From the triangle $KPS$, by formula **A**,

$$\sin\delta = \cos\epsilon\sin\beta + \sin\epsilon\cos\beta\sin\lambda$$

whence, by differentiation,

$$\cos\delta\,\Delta\delta = \sin\epsilon\cos\beta\cos\lambda\,\Delta\lambda \quad \ldots\ldots(8).$$

But by formula **B**,

$$\cos\alpha\cos\delta = \cos\beta\cos\lambda \quad \ldots\ldots(9).$$

Hence, from (8) and (9)

$$\Delta\delta = \Delta\lambda\sin\epsilon\cos\alpha \quad \ldots\ldots(10).$$

If $\Delta\lambda$ is the change in $\lambda$ due to luni-solar precession in one year, $\Delta\lambda = \psi$ and we obtain the formula (3) which gives the annual change in declination.

From (9), by differentiation,

$$\sin\alpha\cos\delta\,\Delta\alpha + \cos\alpha\sin\delta\,\Delta\delta = \cos\beta\sin\lambda\,\Delta\lambda$$

and by means of (10) this becomes

$$\sin\alpha\cos\delta\,\Delta\alpha = \Delta\lambda\,[\cos\beta\sin\lambda - \sin\epsilon\cos^2\alpha\sin\delta] \ldots\ldots(11).$$

But, by formula **C**,

$$\cos\beta\sin\lambda = \sin\delta\sin\epsilon + \cos\delta\cos\epsilon\sin\alpha.$$

Hence (11) becomes, after a little simplification,

$$\Delta\alpha = \Delta\lambda\,[\cos\epsilon + \sin\epsilon\sin\alpha\tan\delta]$$

which, on writing $\psi$ for $\Delta\lambda$, gives formula (7) of the previous section.

**134.** *Nutation.*

In (11) of the previous section, $\Delta\lambda$ represents the change of the equinox in longitude in one year, and perhaps it is assumed that $\Delta\lambda$ is constant, to be identified with a value of about $50''$ per annum which leads to the revolution of the equinox around the ecliptic in the period of about 26,000 years. Now, $\Delta\lambda$ would be constant provided that the declinations of the sun and the moon remain fixed and had non-zero values for, then the gravitational attraction of each on the spheroidal earth would be unaltered and the direction of each attraction would not pass through the earth's centre; the result would be that each body would exert a constant couple on the earth.

First consider the changing position of the sun in the ecliptic. The couple now tending to tilt the earth's axis is of the variable form $b\,(1-\cos 2\odot)$, where $b$ is a constant partly depending on the obliquity. By integration, the change of longitude in $t$ years is of the form

$$a_1 t + l\sin 2\odot \qquad \ldots\ldots(12)$$

(small terms being omitted), in which $\odot$ is the sun's longitude and $a_1$, $l$ are constants whose values are obtained from dynamical theory. The term $a_1 t$ indicates a progressive and uniform motion of $\Upsilon$ along the ecliptic and terms of this character are generally referred to as *secular terms*, and in the particular problem under discussion as *precessional terms*. Thus $a_1 t$ represents the part of the precession in $t$ years due to the sun.

The term $l\sin 2\odot$ is periodic in character; it is zero when the sun's longitude is $0°, 90°, \ldots 360°$. If $\Upsilon_2$ represents the equinox as defined by the term $a_1 t$ and $\Upsilon_3$ the equinox as defined by (12), it is clear that the distance between $\Upsilon_2$ and $\Upsilon_3$ oscillates between the values $+\,l$ and $-\,l$, and that the period of the changes in the value of $\Upsilon_2\Upsilon_3$ is six months. The term $l\sin 2\odot$ is called a *periodic term.*

In a similar way the corresponding nature of the moon's orbit will lead to principal terms of the form

$$a_2 t + m \sin 2\mathbb{C} \qquad \ldots\ldots(13),$$

where $\mathbb{C}$ denotes the moon's longitude.

Thus combining the effects of the sun and moon the motion of the equinox can be described, so far as the preceding argument is concerned, by the formula (principal terms only)

$$at + l \sin 2\odot + m \sin 2\mathbb{C} \qquad \ldots\ldots(14),$$

where $a$ is written for $a_1 + a_2$.

We have assumed in the foregoing that the plane of the moon's orbit is coincident with the ecliptic. Actually the moon's orbital plane is inclined at an angle (denoted by $i$) of $5° 09'$ to the ecliptic and we have now to examine the effects of this inclination on the motion of the equinox $\Upsilon$.

In Fig. 94 let the dotted circle intersecting the ecliptic at $N_1$ and $N$ represent the intersection of the moon's orbital plane at

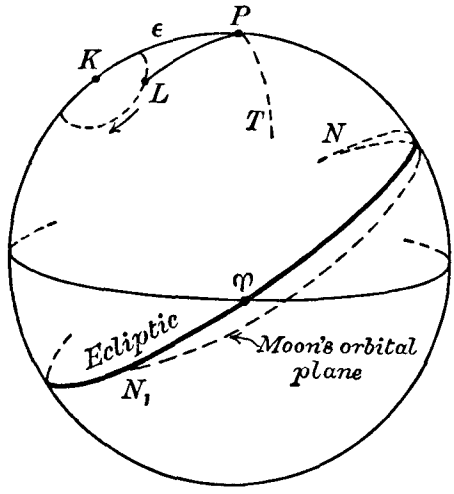



Fig. 94.

a given time with the celestial sphere. The pole of the moon's orbital plane will be at $L$ where the arc $KL$ is the inclination $i$ or $5° 09'$. If the pole $L$ is regarded for the moment as a fixed point on the celestial sphere, the effect of the moon's attraction on the spheroidal earth would be to cause the pole $P$ to move

around $L$ in a small circle $PT$ of radius $LP$. Firstly, this would lead to the backward movement of $\Upsilon$ along the ecliptic; secondly, the angular distance between $K$ and $P$ would not remain constant or, in other words, the obliquity of the ecliptic would be altered. But from considerations of dynamical theory, as well as from observation, it is known that the moon's orbital plane is not fixed with respect to the ecliptic, but moves in such a way that the pole $L$ describes a small circle around $K$ (in the direction of the arrow) in a period of 18·6 years. We infer that so far as the moon is concerned, the backward movement of the equinox $\Upsilon$ along the ecliptic will consist of two parts; one part due to the action of the moon on the supposition that the pole of its orbit coincides with $K$, and the other depending on the position of the moon's orbital plane with respect to the ecliptic, that is to say, on the position of the pole $L$ which can be defined in terms of the longitude of the node $N_1$ or $N$ and the inclination $i$. The second part will evidently be periodic in character and the principal terms of the mathematical expression are of the form

$$b \sin \Omega + c \sin 2\Omega \qquad \ldots\ldots(15),$$

where $\Omega$ is the longitude of the moon's ascending node $N$ and $b$ and $c$ are constants whose values are obtained from dynamical theory.

Combining (14) and (15) we have the following expression (principal terms only) giving the motion of the equinox on the ecliptic:

$$at + b \sin \Omega + c \sin 2\Omega + l \sin 2\odot + m \sin 2\leftmoon \quad \ldots\ldots(16).$$

The first term, $at$, gives the *luni-solar precession in longitude* and the remaining terms—all periodic in character—give the *nutation in longitude*. The nutation in longitude is denoted by $\Delta\psi$.

In the statements made in this section only the more important terms for the nutation in longitude have been mentioned; we shall refer more fully to these later.

### 135. *Nutation in the obliquity.*

We have already indicated in the preceding section that, inasmuch as the pole of the moon's orbital plane does not coincide with the pole of the ecliptic, the obliquity will not remain constant. As the motion of $L$ around $K$ is periodic, the changes

in the obliquity are also periodic. If $\Delta\epsilon$ denotes the change in the obliquity, the principal terms are found to be of the form (we include one term due to the sun):

$$\Delta\epsilon = b_1 \cos\Omega + c_1 \cos 2\Omega + l_1 \cos 2\odot + m_1 \cos 2\leftmoon \quad \ldots\ldots(17).$$

These terms, being periodic, give the *nutation in the obliquity.*

We now investigate the changes in the right ascension and declination of a star due only to the change $\Delta\epsilon$ in the obliquity, the ecliptic being regarded as fixed. In Fig. 95, let $P$ be the pole

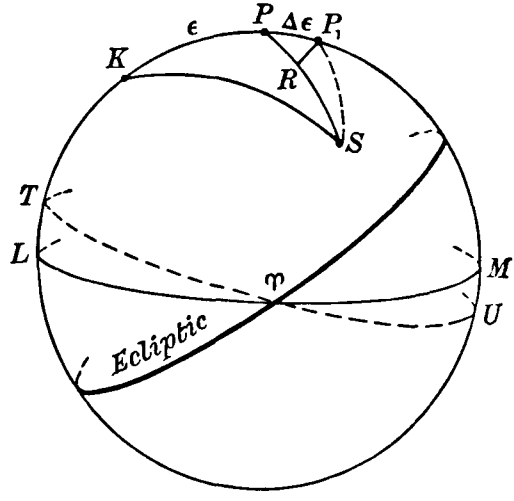



Fig. 95.

of the equator $L\Upsilon M$ which allows for luni-solar precession only. Let $P_1$ be the pole of the equator $T\Upsilon U$ which allows in addition for the nutation in the obliquity. Finally let the co-ordinates of a star $S$ referred to these two equators be $(\alpha, \delta)$ and $(\alpha_1, \delta_1)$. Then $PS = 90° - \delta$, $P_1S = 90° - \delta_1$, $KP = \epsilon$, $KP_1 = \epsilon + \Delta\epsilon$, $K\hat{P}S = 90° + \alpha$, $K\hat{P}_1S = 90° + \alpha_1$, $P\hat{K}S = 90° - \lambda$ and $KS = 90° - \beta$. From $P_1$ draw a small circle arc $P_1R$ perpendicular to $PS$. As $\Delta\epsilon$ is a small quantity, $P_1S$ and $RS$ are sensibly equal and therefore $PR = \delta_1 - \delta$ which we shall denote by $\Delta\delta$. Also $P_1\hat{P}R = 90° - \alpha$. Then

$$PR = PP_1 \cos P_1PR,$$

or

$$\delta_1 - \delta \equiv \Delta\delta = \Delta\epsilon \sin\alpha \quad \ldots\ldots(18).$$

From the triangle $KPS$, by the sine-formula **B**,

$$\cos\alpha\cos\delta = \cos\lambda\cos\beta \qquad \ldots\ldots(19)$$

and from the triangle $KP_1S$, similarly,

$$\cos\alpha_1\cos\delta_1 = \cos\lambda\cos\beta \qquad \ldots\ldots(20).$$

Write $\alpha_1 - \alpha = \Delta\alpha$. Then with sufficient accuracy, keeping only the first powers of $\Delta\alpha$ and $\Delta\delta$, we have by simple expansion

$$\cos\alpha_1 = \cos\alpha - \Delta\alpha\sin\alpha,$$
$$\cos\delta_1 = \cos\delta - \Delta\delta\sin\delta.$$

Hence from (20), neglecting the product term in $\Delta\alpha.\Delta\delta$,

$$\cos\lambda\cos\beta = \cos\alpha\cos\delta - \Delta\delta\cos\alpha\sin\delta - \Delta\alpha\sin\alpha\cos\delta,$$

and using (19) we have

$$\Delta\alpha\sin\alpha\cos\delta = -\Delta\delta\cos\alpha\sin\delta \qquad \ldots\ldots(21).$$

But $\Delta\delta$ is given by (18); hence

$$\Delta\alpha = -\Delta\epsilon\cos\alpha\tan\delta \qquad \ldots\ldots(22).$$

Formulae (18) and (22) give the changes in the declination and right ascension of a star due only to the nutation $\Delta\epsilon$ in the obliquity.

They may also be derived simply as follows. From the triangle $KPS$, we have by formula **A**,

$$\sin\delta = \sin\beta\cos\epsilon + \cos\beta\sin\epsilon\sin\lambda.$$

Regarding $\lambda$ and $\beta$ as constant, as the nutation in obliquity does not affect these co-ordinates, we obtain by differentiation

$$\cos\delta\Delta\delta = -\Delta\epsilon(\sin\beta\sin\epsilon - \cos\beta\cos\epsilon\sin\lambda).$$

But by **C**,

$$-\cos\delta\sin\alpha = \sin\beta\sin\epsilon - \cos\beta\cos\epsilon\sin\lambda.$$

Hence $\Delta\delta = \Delta\epsilon\sin\alpha,$

which is equation (18).

Also by differentiating (19) we obtain

$$\sin\alpha\cos\delta\Delta\alpha + \cos\alpha\sin\delta\Delta\delta = 0,$$

which is the same as (21) and the resulting formula (22) is derived as before.

### 136. *Planetary precession.*

Up to this point it has been assumed that the ecliptic is a fixed plane. Let us first examine what is meant by a "fixed

plane". Consider the celestial sphere with the sun as centre. Now if we are given any two points on the sphere, not diametrically opposite, one and only one great circle can be drawn through them and accordingly we may specify the great circle in terms of the two points. Let us suppose that the two points on the sphere are given by two stars which, for simplicity, will be assumed at an infinite distance. The assumptions made will dispose of any necessity to consider aberration, parallax and proper motion. Then we can say that the plane containing the sun and the two stars is a fixed plane relative to the sun. In this sense, the ecliptic is not quite a fixed plane.

Consider the ecliptic $A\Upsilon B$ (Fig. 96) at 1900·0, which we shall assume to be coincident with a fixed plane as just defined. We shall thus consider the ecliptic $A\Upsilon B$ to be a fixed plane of reference, and we shall refer to it as the "fixed ecliptic". We

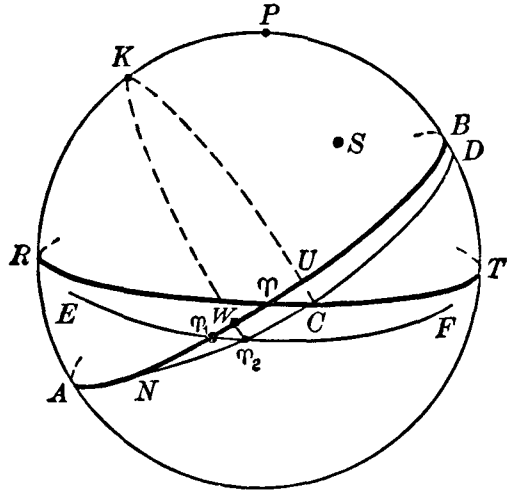



Fig. 96.

shall suppose that one year later the ecliptic is $NCD$, intersecting the fixed ecliptic in $N$. This change in the position of the ecliptic is produced by the mutual gravitational attraction of the planets, giving rise to secular variations in the inclination, and in the longitude of the node $N$ of the earth's orbital plane, with reference to the fixed plane of reference. Due to this cause alone,

the equinox moves in one year from $\Upsilon$ to $C$; this motion of the equinox is called *planetary precession*, which we shall denote by $\lambda'$. The value of $\lambda'$ is about $0''{\cdot}12$ per annum. It is evident from Fig. 96 that the effect of planetary precession is to decrease the right ascensions of all stars by the same yearly amount, namely $\lambda'$, without altering their declinations. It is also clear from Fig. 96 that a minute change in the obliquity results which, due to this cause, decreases at the rate of about $0''{\cdot}47$ per annum. Actually it is found that the displacement in $t$ years of the equinox $\Upsilon$—along the equator in the direction $\Upsilon C$—is given with sufficient accuracy by a formula of the type $ct + dt^2$ and that the change in the obliquity is given by a similar expression. The coefficients in each case are very small.

Consider now the combined effect of the luni-solar and planetary precessions. In Fig. 96 $A\Upsilon B$ and $R\Upsilon T$ are the ecliptic and equator for 1900·0. One year later the fundamental planes are $NCD$ and $ET_1\Upsilon_2$. $\Upsilon\Upsilon_1$ is the luni-solar precession in one year. As all the changes in the fundamental planes are small we shall have, with sufficient accuracy, that $\Upsilon_1\Upsilon_2 = \Upsilon C$. $\Upsilon_1\Upsilon_2$ is the planetary precession in one year. Join $K$ to $C$ and $K$ to $\Upsilon_2$ by great circle arcs cutting the fixed ecliptic in $U$ and $W$ respectively. Then $\Upsilon U$ or $\Upsilon_1 W$ is the planetary precession in longitude. Its value is clearly $\lambda' \cos\epsilon$ per annum.

We now define *general precession*. In one year (we are still excluding nutation), the equinox moves from $\Upsilon$ to $\Upsilon_2$. Referred to the fixed ecliptic $A\Upsilon B$ and the equinox $\Upsilon$, the longitude of $\Upsilon_2$ can be written $360° - p$, where $p$ is clearly a small quantity. The quantity $p$ is called the general precession (in longitude). When we are dealing with the positions of the fundamental planes separated by a short interval—a few years, say—the general precession can be represented geometrically with high accuracy. We shall consider the interval of one year for which Fig. 96 has already been used. Then the general precession in longitude is $\Upsilon W$, that is $\Upsilon\Upsilon_1 - \Upsilon_1 W$, or the luni-solar precession minus the planetary precession in longitude.

As we have seen, planetary precession produces a slight change in the obliquity.

We now give the numerical expressions for the general

precession $p$ and the obliquity at a time $t$ years after 1900; the first is $p = 50''{\cdot}2564 + 0''{\cdot}000222t$ per annum, and the second is

$$\epsilon = 23° 27' 8''{\cdot}26 - 0''{\cdot}4685t.$$

**137.** *The mean equator and the mean co-ordinates of a star.*

In Fig. 96, $ATB$ and $RTT$ are respectively the ecliptic and equator at the beginning of 1900 (this is usually written 1900·0); these are regarded as fixed planes of reference. After one year the luni-solar and planetary precessions produce changes in the ecliptic and equator so that for 1901·0 these planes are now $NT_2D$ and $ET_2F$. $NT_2D$ and $ET_2F$ are defined to be the *mean ecliptic* and *mean equator* for 1901·0. Also the equinox $T_2$ is the *mean equinox* for 1901·0.

In this chapter we shall omit the general consideration of the proper motion of a star and we shall now define provisionally the *mean position* of a star. The mean position of a star at any time is its position on the celestial sphere centred at the sun, and referred to the mean equator and equinox at that time. In practice we consider only the mean position with reference to the equinox at the beginning of a year. It is to be noted that in our definition no cognisance is taken of nutation, aberration, annual parallax and (provisionally) of the star's proper motion.

We now investigate the problem: to derive the mean position of a star for 1901·0, given the mean position for 1900·0. Let $(\alpha, \delta)$ be the co-ordinates referred to the mean equator and equinox for 1900·0, and $(\alpha_1, \delta_1)$ referred to the mean equator and equinox for 1901·0. Due to luni-solar precession, the star's longitude has increased in one year by $\psi$. Formulae (7) and (3) are immediately applicable to give the changes in right ascension and declination due to this cause. We have, rewriting these formulae,

$$\alpha_1 - \alpha = \psi(\cos\epsilon + \sin\epsilon \sin\alpha \tan\delta) \quad \text{.....(23),}$$
$$\delta_1 - \delta = \psi \sin\epsilon \cos\alpha \quad \text{.....(24).}$$

But planetary precession decreases the right ascension in one year by $\lambda'$ ($\Upsilon_1\Upsilon_2$ in Fig. 96); hence the complete expression for $\alpha_1 - \alpha$ is

$$\alpha_1 - \alpha = (\psi\cos\epsilon - \lambda') + \psi \sin\epsilon \sin\alpha \tan\delta \quad \text{...(25).}$$

The declination is unaltered by planetary precession, and hence (24) is the complete equation for the change in declination.

Put $m = \psi \cos\epsilon - \lambda'; \quad n = \psi \sin\epsilon$ .....(26).

Then $\alpha_1 - \alpha = m + n \sin\alpha \tan\delta$ .....(27),

$\delta_1 - \delta = n \cos\alpha$ .....(28).

The quantities $m$ and $n$ are functions of the precessional constants and of the obliquity, all of which vary slowly with the time; consequently $m$ and $n$ vary slowly with the time.

The following table* gives the values of $m$ and $n$ at several epochs (as $n$ occurs in both formulae (27) and (28), it is tabulated both in time and angular measures).

*Table: values of* m *and* n.

| | $m$ | $n$ | $n$ |
|---|---|---|---|
| 1800 | $3^s{\cdot}07048$ | $1^s{\cdot}33703$ | $20''{\cdot}0554$ |
| 1850 | $3^s{\cdot}07141$ | $1^s{\cdot}33674$ | $20''{\cdot}0511$ |
| 1900 | $3^s{\cdot}07234$ | $1^s{\cdot}33646$ | $20''{\cdot}0468$ |
| 1950 | $3^s{\cdot}07327$ | $1^s{\cdot}33617$ | $20''{\cdot}0426$ |

Taking the year as the unit and writing $\frac{d\alpha}{dt}$ and $\frac{d\delta}{dt}$ for the rates of change of the co-ordinates $\alpha$ and $\delta$ due to precession, we rewrite (27) and (28):

$$\frac{d\alpha}{dt} = m + n \sin\alpha \tan\delta \qquad \text{.....(29),}$$

$$\frac{d\delta}{dt} = n \cos\alpha \qquad \text{.....(30).}$$

*Example.* To calculate the mean right ascension of $\alpha$ Orionis for the epoch 1902·0, given that for 1900·0 the mean co-ordinates are: $\alpha = 5^h\ 49^m\ 45^s{\cdot}481, \quad \delta = +\ 7^\circ\ 23'\ 18''{\cdot}41.$

We first compute the annual precession given by formula (29). From the table above, for 1900·0,

$$m = 3^s{\cdot}0723, \quad n = 1^s{\cdot}3365.$$

$$\log n = 0{\cdot}12597$$

$$\log \sin\alpha = \bar{1}{\cdot}99956$$

$$\log \tan\delta = \bar{1}{\cdot}11284$$

$$\bar{1}{\cdot}23837$$

Hence $n \sin\alpha \tan\delta = 0^s{\cdot}1731$

Also $m = 3^s{\cdot}0723$

Therefore $\frac{d\alpha}{dt} = 3^s{\cdot}2454$

* Newcomb: *Spherical Astronomy*, p. 406.

The annual precession in right ascension for this star is thus + 3$^s$·2454, and in two years the change in the right ascension due to precession is + 6$^s$·491. Hence the mean right ascension referred to the mean equinox of 1902·0 is obtained by adding 6$^s$·491 to the mean right ascension for 1900·0, and the result is:

Mean right ascension (1902·0) = 5$^h$ 49$^m$ 51$^s$·972.

In a similar way, the mean declination for 1902·0 is found to be + 7° 23′ 20″·20.

138. *The secular variation.*

The procedure adopted in the example just given is correct only when the interval of time concerned is a small number of years. It is to be noticed that on the right-hand sides of (29) and (30) the values of $\alpha$ and $\delta$ are varying quantities as well as $m$ and $n$, and this fact must be taken into account when the interval of time exceeds, say, 5 to 10 years. The rate of change of $\frac{d\alpha}{dt}$ per century is defined to be the *secular variation* in right ascension. Now $\alpha$ is a function of $t$ (in years, measured say from 1900·0) and if $\alpha_0$, $\frac{d\alpha_0}{dt}$ and $\frac{d^2\alpha_0}{dt^2}$ denote the values of the right ascension, of $\frac{d\alpha}{dt}$ and of $\frac{d^2\alpha}{dt^2}$ at $t = 0$, that is for 1900·0, then $\alpha$ can be written

$$\alpha = \alpha_0 + t\frac{d\alpha_0}{dt} + \frac{t^2}{2}\frac{d^2\alpha_0}{dt^2} \qquad \text{.....(31).}$$

If $s$ denotes the secular variation in right ascension, $\frac{d^2\alpha_0}{dt^2}$ is the rate of change (per annum) of $\frac{d\alpha_0}{dt}$, that is, $\frac{d^2\alpha_0}{dt^2} = \frac{s}{100}$. Hence from (31),

$$\alpha - \alpha_0 = t\left(\frac{d\alpha_0}{dt} + \frac{st}{200}\right) \qquad \text{.....(32).}$$

Now, in general,

$$\begin{aligned}\frac{s}{100} \equiv \frac{d^2\alpha}{dt^2} &= \frac{d}{dt}(m + n\sin\alpha\tan\delta)\\ &= \frac{dm}{dt} + \sin\alpha\tan\delta\frac{dn}{dt} + n\cos\alpha\tan\delta\frac{d\alpha}{dt}\\ &\quad + n\sin\alpha\sec^2\delta\frac{d\delta}{dt} \qquad \text{.....(33).}\end{aligned}$$

Now from the values of $m$ it is found that, for 1900·0, the annual rate of change of $m$ is given by

$$\frac{dm}{dt} = + 0^s{\cdot}0000186,$$

and, similarly,

$$\frac{dn}{dt} = - 0''{\cdot}0000853.$$

Also $\frac{d\alpha}{dt}$ and $\frac{d\delta}{dt}$ in (33) are given by (29) and (30). Hence the value of $s$ can be computed.

The formula in declination corresponding to (32) is:

$$\delta - \delta_0 = t\left(\frac{d\delta_0}{dt} + \frac{s_1 t}{200}\right) \qquad \ldots\ldots(34),$$

where $s_1$ is the secular variation in declination.

In the principal catalogues, quantities called the *annual variation* in right ascension and in declination are tabulated for each star. The annual variation in right ascension, for example, is the value of $\frac{d\alpha}{dt}$ due to annual precession (for the appropriate epoch, say 1900·0) together with the effect of proper motion, which we have hitherto excluded from our discussion. The secular variations $s$ and $s_1$ are also given.

*Example.* To compute the mean co-ordinates of $\epsilon$ Octantis for 1950·0, given that the mean co-ordinates $(\alpha_0, \delta_0)$ for 1900·0 are

$$22^h\ 8^m\ 49^s{\cdot}30 \quad \text{and} \quad - 80^\circ\ 56'\ 14''{\cdot}7.$$

From Boss's *Preliminary General Catalogue*, the annual variations in right ascension and declination for 1900·0 are respectively

$$+ 6^s{\cdot}9931 \quad \text{and} \quad + 17''{\cdot}699,$$

and the secular variations $s$, $s_1$ are

$$s = - 0^s{\cdot}5926 \quad \text{and} \quad s_1 = + 0''{\cdot}468.$$

Then, by (32), $\alpha$ being the mean right ascension for 1950·0,

$$\begin{aligned} \alpha - \alpha_0 &= 50\,(6^s{\cdot}9931 - \tfrac{50}{200} \times 0^s{\cdot}5926) \\ &= 50\,(6^s{\cdot}9931 - 0^s{\cdot}1481) \\ &= 5^m\ 42^s{\cdot}25. \end{aligned}$$

Hence the mean right ascension for 1950·0 is

$$(22^h\ 8^m\ 49^s{\cdot}30 + 5^m\ 42^s{\cdot}25) \quad \text{or} \quad 22^h\ 14^m\ 31^s{\cdot}55.$$

Also by (34),

$$\delta - \delta_0 = 50\,(17''\cdot699 + \tfrac{50}{200} \times 0''\cdot468)$$
$$= +\ 14'\ 50''\cdot8.$$

Hence the mean declination for 1950·0 is

$$(-\ 80°\ 56'\ 14''\cdot7 + 14'\ 50''\cdot8) \text{ or } -\ 80°\ 41'\ 23''\cdot9.$$

**139.** *The true equator and the true co-ordinates of a star.*

To fix our ideas we shall consider a particular time and date, say, 1931 March 9, $15^h$ U.T. The *true equator*, true ecliptic and the *true equinox* are the actual equator, ecliptic and equinox respectively, at this instant, and their respective positions with reference to a fixed ecliptic, say, at 1900·0 are dependent on the amount of precession and nutation computed for the interval between 1900·0 and the instant in question. We define the *true place* of a star at this instant as its position on the celestial sphere, centred at the sun, referred to the true equinox and equator of the date.

Let us denote the mean co-ordinates of a star for 1900·0 by $(\alpha_0, \delta_0)$, the mean co-ordinates for 1931·0 by $(\alpha, \delta)$ and the true co-ordinates for 1931 March 9, $15^h$ U.T. by $(\alpha_1, \delta_1)$. Also let $\tau$ denote the fraction of the year between the beginning of the year and the date in question. Then, considering the change in the right ascension of the star due to precession and nutation, we have

$\alpha_1 - \alpha_0 =$ precession in R.A. for $(31 + \tau)$ years
$+$ the effect of nutation on the R.A.

We write $\alpha_1 - \alpha_0 = (\alpha - \alpha_0) + (\alpha_1 - \alpha)$.

Now $\alpha_1 - \alpha =$ the difference between the true R.A. of the star, referred to the true equinox of date, and the mean R.A. referred to the mean equinox of the beginning of the year (1931·0); it is made up of

(i) precession in R.A. for the interval $\tau$,

(ii) the effect of nutation in R.A. for the date in question.

Also $\alpha - \alpha_0$ is the difference of the mean co-ordinates for 1931·0 and 1900·0. We thus have the general rule:

To derive the true co-ordinates of the star at $(1931 + \tau)$, the mean co-ordinates for 1900·0 (say) being given, firstly compute

the mean co-ordinates for 1931·0 (by the procedure illustrated on p. 241) and, secondly, compute (i) the precession for the interval $\tau$, and (ii) the effect of nutation. Denote by $\Delta\alpha_1$, $\Delta\alpha_2$ the effects on the right ascension due to (i) and (ii) respectively.

Then by formula (27)—which gives the *annual* change in R.A. due to precession—we have for the interval $\tau$,

$$\Delta\alpha_1 = \tau\,(m + n \sin\alpha \tan\delta).$$

Consider now nutation. As we have seen, nutation changes the longitude of a star and also the obliquity of the ecliptic. For the date concerned, let $\Delta\psi$, $\Delta\epsilon$ denote the nutation in longitude and in the obliquity respectively. $\Delta\psi$ consists of the periodic terms given in (16) and $\Delta\epsilon$ is given by (17). The change in the right ascension due to $\Delta\psi$ is given from (7) by

$$\Delta\psi\,(\cos\epsilon + \sin\epsilon \sin\alpha \tan\delta),$$

and the change in R.A. due to $\Delta\epsilon$ is given from (22) by

$$-\Delta\epsilon \cos\alpha \tan\delta.$$

Hence the total change $\Delta\alpha_2$ in right ascension due to nutation is given by

$$\Delta\alpha_2 = \Delta\psi\,(\cos\epsilon + \sin\epsilon \sin\alpha \tan\delta) - \Delta\epsilon \cos\alpha \tan\delta.$$

But $\alpha_1 - \alpha$ (the difference between the true R.A. at date and the mean R.A. for the beginning of the year) is $\Delta\alpha_1 + \Delta\alpha_2$; thus we can write

$$\alpha_1 - \alpha = (m\tau + \Delta\psi\cos\epsilon) + \sin\alpha\tan\delta\,(n\tau + \Delta\psi\sin\epsilon) - \Delta\epsilon\cos\alpha\tan\delta \quad \ldots\ldots(35).$$

But from (26), $m$ and $n$ are defined in terms of the luni-solar precession $\psi$ and the planetary precession $\lambda'$ by

$$m = \psi\cos\epsilon - \lambda', \quad n = \psi\sin\epsilon \quad \ldots\ldots(36).$$

Substituting in (35) the expressions for $\cos\epsilon$ and $\sin\epsilon$ given by (36), we obtain

$$\alpha_1 - \alpha = \left(\tau + \frac{\Delta\psi}{\psi}\right)(m + n\sin\alpha\tan\delta) + \frac{\lambda'\Delta\psi}{\psi} - \Delta\epsilon\cos\alpha\tan\delta \quad \ldots\ldots(37).$$

Suppose now that all the precessional and nutational quantities are expressed in seconds of arc, and that we write

$$\left.\begin{aligned} A &= n\left(\tau + \frac{\Delta\psi}{\psi}\right); \quad B = -\Delta\epsilon; \quad E = \frac{\lambda'\Delta\psi}{15\psi} \\ a &= \tfrac{1}{15}(m/n + \sin\alpha\tan\delta); \quad b = \tfrac{1}{15}\cos\alpha\tan\delta \end{aligned}\right\} \quad \ldots(38).$$

Then $$\alpha_1 - \alpha = Aa + Bb + E \qquad \text{.....(39),}$$

in which the right-hand side is expressed uniformly in seconds of time.

In (39), the quantities $A$, $B$ and $E$ are computed from the known interval $\tau$, the nutations $\Delta\psi$, $\Delta\epsilon$ corresponding to the date concerned, and the luni-solar and planetary precessions $\psi$ and $\lambda'$; they are independent of the position of the star. They are known as the *Besselian Day Numbers* and are tabulated in the almanacs for every day in the year.

The quantities $a$ and $b$ depend on the values of $m$ and $n$ and on the co-ordinates of the star; they change slowly with the time, but over considerable intervals they may be regarded as constants associated with the particular star concerned.

In a similar way, we have the analogous formula in declination:

$$\delta_1 - \delta = \left(\tau + \frac{\Delta\psi}{\psi}\right) n \cos\alpha + \Delta\epsilon \sin\alpha \qquad \text{.....(40),}$$

in which $n$ and $\Delta\epsilon$ are supposed expressed in seconds of arc.

Put $$a' = \cos\alpha; \quad b' = -\sin\alpha \qquad \text{.....(41).}$$

Then, using the values of $A$ and $B$ in (38), we have

$$\delta_1 - \delta = Aa' + Bb' \qquad \text{.....(42).}$$

Equations (39) and (42) enable us to calculate the *true* co-ordinates $(\alpha_1, \delta_1)$ at any date when the mean co-ordinates $(\alpha, \delta)$ at the beginning of the year are given.

The same equations enable the complementary problem to be solved, namely, the calculation of the mean co-ordinates for the beginning of the year from the true co-ordinates of date.

It is to be remembered that, in this discussion, we have omitted the effects of the star's proper motion on the co-ordinates.

The calculations may also be carried out in another manner. From (38) and (39), we can write

$$\alpha_1 - \alpha = \tfrac{1}{15} mA/n + E + \tfrac{1}{15} A \sin\alpha \tan\delta + \tfrac{1}{15} B \cos\alpha \tan\delta \qquad \text{.....(43).}$$

In this equation the day numbers $A$ and $B$ are expressed in seconds of arc, while $E$ is in seconds of time. This corresponds to their tabulation in the almanacs. Also from (41) and (42)

$$\delta_1 - \delta = A\cos\alpha - B\sin\alpha \qquad \text{.....(44).}$$

Let $f$, $g$, $G$ be defined by

$$f = \tfrac{1}{15} mA/n + E; \quad g \sin G = B; \quad g \cos G = A \qquad \ldots\ldots(45).$$

Then

$$\alpha_1 - \alpha = f + \tfrac{1}{15} g \sin (G + \alpha) \tan \delta \qquad \ldots\ldots(46),$$

$$\delta_1 - \delta = g \cos (G + \alpha) \qquad \ldots\ldots(47).$$

In these formulae, $g$ is given in seconds of arc and $f$ in seconds of time.

The quantities $f$, $g$ and $G$ are known as *Independent Day Numbers* and, like the *Besselian Day Numbers*, they are tabulated for every day in the almanacs.

**140.** *The apparent place of a star.*

We have seen that the *true place* of a star at any instant is its position on the celestial sphere, centred at the *sun*, with reference to the true equinox and equator at that instant. The true place is thus independent of the effects of aberration and annual parallax. The *apparent place* at any instant of a star or other heavenly body is defined to be its position on the celestial sphere, whose centre is the *earth's centre*, with reference to the true equinox and true equator at that instant. We thus have:

Apparent place = true place plus the corrections due to aberration and annual parallax.

The formulae (25) and (26) of Chapter VIII give the corrections to the right ascension and declination of a star, due to aberration, in the form $Cc + Dd$ and $Cc' + Dd'$ respectively. The corrections due to annual parallax are given in equations (65$a$) and (66$a$) of Chapter IX. With a little manipulation these formulae can be written in a similar form. As a result we find:

$$\text{Apparent R.A.} = \text{True R.A.} + (C + \Pi Y)c + (D - \Pi X)d \qquad \ldots\ldots(48),$$

$$\text{Apparent Dec.} = \text{True Dec.} + (C + \Pi Y)c' + (D - \Pi X)d' \qquad \ldots\ldots(49).$$

**141.** *Reduction from mean place to apparent place* (*or* vice versa).

We shall suppose that $(\alpha_0, \delta_0)$ are the mean co-ordinates of a star for some such epoch as 1900·0. It is required to determine the apparent co-ordinates $(\alpha', \delta')$, for example, at 1931 March 9, $15^h$ U.T.

The first step is to compute the mean co-ordinates $(\alpha, \delta)$ for

the mean equinox of 1931·0 by the principles of section 138. Using (39), (42) and (48), (49) we shall have

$$\alpha' - \alpha = Aa + Bb + Cc + Dd + E \quad \ldots\ldots(50),$$

$$\delta' - \delta = Aa' + Bb' + Cc' + Dd' \quad \ldots\ldots(51).$$

These formulae give the differences between the apparent co-ordinates at the date concerned and the mean co-ordinates for the beginning of the year. The quantities in capital letters are the *Besselian Day Numbers* and the quantities in small letters are the *Besselian star constants*. For the highest accuracy, the latter should be calculated for the year in question, as strictly they are not constant, but slowly varying, quantities.

We may also carry out the computations by means of (46) and (47) of section 139 and (27) and (28) of Chapter VIII in terms of the *independent day numbers* as follows:

$$\alpha' - \alpha = f + \tfrac{1}{15} g \sin(G + \alpha) \tan\delta + \tfrac{1}{15} h \sin(H + \alpha) \sec\delta \quad \ldots\ldots(52),$$

$$\delta' - \delta = g \cos(G + \alpha) + h \cos(H + \alpha) \sin\delta + i \cos\delta \quad \ldots\ldots(53).$$

The formulae can also be used in deriving the mean co-ordinates for the beginning of the year when the apparent co-ordinates at the date are known.

**142.** *Cataloguing the stars.*

Let us suppose that a star is observed with the meridian circle on 1931 March 9 at $15^h$ U.T. After the instrumental and refraction corrections have been applied, we obtain the apparent right ascension $\alpha'$ and the apparent declination $\delta'$ corresponding to the true equinox of date. By applying the formulae (50) and (51)—or (52) and (53)—we obtain the star's mean co-ordinates $\alpha$ and $\delta$ referred to the beginning of the year 1931. For the purposes of a catalogue it is convenient to refer the positions to the mean equinox of a common epoch, say, 1900·0. The mean co-ordinates $(\alpha_0, \delta_0)$ referred to this common epoch are then obtained according to the procedure in section 138.

**143.** *The numerical values of the nutation in longitude and in the obliquity of the ecliptic.*

In (16) we have indicated the nature of the principal terms of the nutation in longitude, $\Delta\psi$—they are the periodic terms in

(16)—and in (17) the principal terms of the nutation in the obliquity, $\Delta\epsilon$. The coefficients $b$ and $b_1$ are not quite constant, varying slowly with the time. If $T$ denotes the time interval expressed in centuries and reckoned from 1900·0 the expressions (16)—nutation only—and (17) take the following numerical forms:

$$\Delta\psi = -(17''\cdot233 + 0''\cdot017T)\sin\Omega + 0''\cdot209\sin 2\Omega - 1''\cdot273\sin 2L - 0''\cdot204\sin 2\mathbb{C} \quad \ldots\ldots(54),$$

$$\Delta\epsilon = (9''\cdot210 + 0''\cdot001T)\cos\Omega - 0''\cdot090\cos 2\Omega + 0''\cdot552\cos 2L + 0''\cdot088\cos 2\mathbb{C} \quad \ldots\ldots(55),$$

in which we have written the sun's mean longitude $L$ in place of the sun's true longitude $\odot$.

In calculating the Besselian Day Numbers and the Independent Day Numbers, many other terms which are not given in (54) and (55) are used. The complete expressions for $\Delta\psi$, $\Delta\epsilon$ are given in the *Explanatory Supplement to the Astronomical Ephemeris.*

The coefficient of $\cos\Omega$ in (55) for any epoch is called the *constant of nutation.*

## EXERCISES

**1.** If $P$, $K$ are the poles of the equator and ecliptic, and $X$ is a star such that $P\hat{X}K$ is 90°, show that $X$ has no precession in right ascension.

**2.** Make a rough estimate of the present declination and right ascension of the point of the celestial sphere which was First Point of Aries in 120 B.C., the date when the precession was discovered. [*M.T.* 1921.]

**3.** The equatorial co-ordinates of a star on the ecliptic are $(\alpha, \delta)$ and its longitude is $\lambda$. If $\Delta\alpha$, $\Delta\delta$ and $\Delta\lambda$ are the annual precession in right ascension, declination and longitude respectively, prove that

$$\Delta\alpha \cot\alpha \cos^2\delta = \Delta\delta \cot\delta = \Delta\lambda \cot\lambda. \qquad [\textit{Coll. Exam.}]$$

**4.** At a certain date two stars came simultaneously to the horizon of a place in latitude $\cot^{-1}(\sqrt{3}\sin\epsilon)$ at $0^h$ sidereal time. At a later date, when the precession had amounted to 60°, show that the same stars came simultaneously to the horizon of another place, whose latitude was $\epsilon + \cot^{-1}(2\tan\epsilon)$, at sidereal time $6^h$, $\epsilon$ being the obliquity of the ecliptic. [*M.T.* 1929.]

**5.** Show that a southern star of declination $-\delta$ and right ascension $\alpha$ must at some epoch have been visible from a station in north latitude $\phi$, provided that

$$\sin\delta\cos\epsilon + \cos\delta\sin\epsilon\sin\alpha < \cos(\phi - \epsilon),$$

where $\epsilon$ is the obliquity of the ecliptic. Obtain formulae for calculating the date at which it ceased (or will cease) to be visible. [*M.T.* 1926.]

**6.** Show that, owing to precession, the position angle $\theta$ of a non-polar double star alters at a rate given by $\frac{d\theta}{dt} = +1{\cdot}4 \sin \alpha \sec \delta \sin \epsilon$ degrees per century.

The approximate position of the double star 22 Cygni is ($19^h\ 22^m$; $+ 27°\ 30'$). The observed position angles (increasing) for the mean epochs 1830·0 and 1905·0 are respectively 256°·0 and 311°·8. What is the real change in position angle? [*Lond.* 1926.]

**7.** Owing to precession the interval between two consecutive transits of a star across a given meridian differs from a mean sidereal day. If $\beta + \epsilon > 90°$, show that this difference vanishes if the star's longitude $\lambda$ is given by

$$\sin \lambda = \cot \beta \cot \epsilon,$$

where $\beta$ is the star's latitude and $\epsilon$ is the obliquity of the ecliptic.

**8.** The co-ordinates of a star for 1900·0 are: $16^h\ 56^m\ 12^s$; $\delta = + 82°\ 12'$. Show that the annual precession in R.A. for 1900·0 is $- 6^s{\cdot}30$.

**9.** Prove that the complete transformation from the true place to the apparent place of a star is as stated in equations (48) and (49).

**10.** Two stars rise simultaneously at a site of latitude $\phi$ at local sidereal time $T$. Assuming that the proper motions of the two stars are negligible, show that they will again rise simultaneously after a fraction

$$\frac{1}{\pi} \cot^{-1}(\sin \epsilon \tan \phi \sec T + \cos \epsilon \tan T)$$

of the precessional period. Show further that the local sidereal time of their rising will then be $\pi - T$. [*Glas.* 1972.]

CHAPTER XI

# THE PROPER MOTIONS OF THE STARS

**141.** *Definition of proper motion.*

In 1718 it was discovered by Halley that the positions of certain bright stars had altered appreciably, since the time of Hipparchus, in relation to the general stellar background. Suppose for the moment that all stars with the exception of Arcturus are at an infinite distance from the sun, so that they form a definite system of fixed reference points. The comparison of the observations of Arcturus, made in the days of Hipparchus and of Halley, showed conclusively that this star had moved in the interval of about twenty centuries through the considerable angle of about one degree, with reference to the stars in its immediate neighbourhood on the celestial sphere. This suggested that Arcturus had a definite space-velocity relative to the sun, and that its distance was finite.

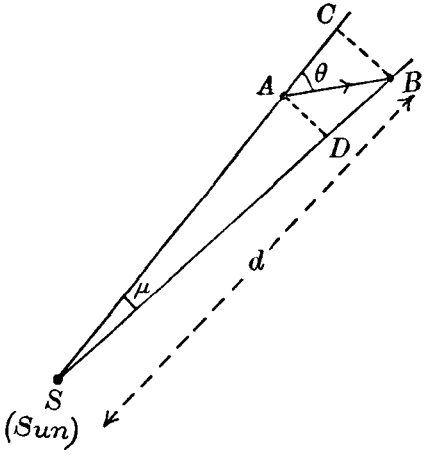


Fig. 97.

Consider Fig. 97. Let $S$ be the sun and suppose that a star with a space-velocity relative to the sun moves in the course of a year from $A$ to $B$. The star's track in space can be assumed to be a straight line; certainly no observations during so long an interval as two centuries have suggested that the path of a single star deviates by a measurable amount from a straight line. Let $\mu$ denote the angle $ASB$. Then $\mu$ is the angle through which the star is seen to move in one year; it is called the *proper motion* of the star, and is generally measured in seconds of arc per annum.

**145.** *Relation between proper motion, tangential velocity and parallax.*

In Fig. 97 let $BC$ and $DA$ be drawn perpendicularly to $SA$. Let $d$ denote the distance $SB$ (in kilometres) of the star from the sun and $\theta$ the angle between its direction of motion $AB$ and the direction $SA$. Let $V$ denote the star's linear velocity from $A$ to $B$, expressed in kilometres per second, so that, if $n$ is the number of seconds in one year, we have

$$AB = nV \quad \text{km} \qquad \ldots\ldots(1).$$

Let $v$ denote the component of the linear velocity at right angles to the direction $SA$ (the line of sight); $v$ is called the *tangential velocity* or *cross-velocity*, and it is expressed in kilometres per second. For the star with the largest proper motion the value of $\mu$ is about $10''$ per annum, and in every instance we can assume that $A\hat{S}B$ is a very small angle so that, in Fig. 97, $AD$ and $CB$ may be taken to be equal; also $AD$ or $CB$ is the distance described in one year perpendicularly to $SA$, so that

$$AD = CB = nv \quad \text{km} \qquad \ldots\ldots(2).$$

Now $CB = d\sin\mu$, so, if $\mu$ is in seconds of arc, we can write

$$CB = d\mu \sin 1'' \qquad \ldots\ldots(3).$$

Let $\Pi$ denote the star's annual parallax and $a$ the radius of the earth's orbit in kilometres; then the distance $d$ and the parallax $\Pi$ are related (Chapter IX) by the formula

$$\sin \Pi = a/d$$

or, since $\Pi$ is small, by

$$d = \frac{a}{\Pi \sin 1''} \qquad \ldots\ldots(4),$$

where $\Pi$ is expressed in seconds of arc. Hence from (2), (3) and (4) we have

$$v = \frac{\mu a}{n\Pi} \qquad \ldots\ldots(5).$$

We now insert the values of $a$ and $n$ in (5); $a = 149{\cdot}6 \times 10^6$ km and $n = 31{\cdot}56 \times 10^6$ (the number of seconds in a year). Hence we derive

$$v = 4{\cdot}74\,\frac{\mu}{\Pi} \qquad \ldots\ldots(6),$$

a relation which gives the cross-velocity $v$ in kilometres per second when the values of $\mu$ and $\Pi$ are known.

For example, the annual proper motion $\mu$ of Capella is 0″·439 (the methods of measuring proper motion will be described later) and its parallax $\Pi$ is 0″·075. Inserting these values in (6), we find that the tangential velocity $v$ is 27·7 km per second.

146. *Radial velocity.*

In one year (Fig. 97), the distance of the star has increased from $SA$ to $SB$ or, with sufficient accuracy, from $SA$ to $SC$, that is, by the distance $AC$. The rate at which the star's distance from the sun is changing owing to its linear velocity in space is called the *radial velocity* with respect to the sun. If $\rho$ denotes the radial velocity in kilometres per second, we have

$$AC = n\rho = AB \cos \theta.$$

But by (1), $AB = nV$; hence

$$\rho = V \cos \theta \qquad \text{.....(7).}$$

The values of the radial velocity $\rho$ can be determined for the brighter stars by spectroscopic methods, directly in terms of kilometres per second. As in section 124 (*d*), we accept such results here without explaining the spectroscopic principles involved.

When a star is increasing its distance from the sun (as in Fig. 97), the radial velocity is defined to be positive; when its distance is diminishing, the radial velocity is negative.

Now $BC = AB \sin \theta$; hence, from (1) and (2),

$$v = V \sin \theta \qquad \text{.....(8).}$$

If $v$ is determined from (6) and $\rho$ is known from spectroscopic observations, we then have the two equations (7) and (8), from which the star's linear space-velocity $V$ and the angle $\theta$ can be determined. For Capella, as we have seen, $v = 27{\cdot}7$ kms. per second and $\rho$ is found to be + 30·2 kms. per second. Hence

$$V \sin \theta = 27{\cdot}7 \quad \text{and} \quad V \cos \theta = 30{\cdot}2.$$

By an easy calculation, it is found that $V = 41{\cdot}0$ kms. per second (this is the space-velocity of Capella relative to the sun) and $\theta$ is $42\frac{1}{2}°$.

147. *The measurement of proper motion.*

Referring again to Fig. 97, we notice that the component of the star's linear space-velocity which gives rise to proper motion

is the tangential velocity in the direction $AD$ or $CB$; the radial velocity by itself does not affect the direction in which the star is seen. Now $A$, $D$ and $S$ are coplanar; hence on the celestial sphere with the sun as centre, the star will appear to move along a great circle. In Fig. 98 let $S$ be the position of the star, say, at 1930·0 and $T$ its position one year later. The great circle arc $ST$ is thus the annual proper motion $\mu$. Let $(\alpha, \delta)$ and $(\alpha_1, \delta_1)$ be the co-ordinates of $S$ and $T$ respectively, referred to the *same* equinox and equator—say the mean equinox and equator of 1930·0. Then the differences $(\alpha_1 - \alpha)$ and $(\delta_1 - \delta)$ are due to the annual proper motion. We shall write:

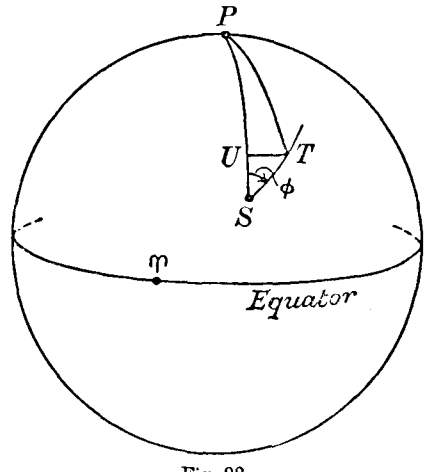


Fig. 98.

$$\alpha_1 - \alpha = \mu_\alpha; \quad \delta_1 - \delta = \mu_\delta \qquad \text{.....(9).}$$

Then $\mu_\alpha$ and $\mu_\delta$ are the components of the proper motion in right ascension and declination respectively. We shall call $\mu$ the *total proper motion*, which is always expressed in seconds of arc per annum.

Draw the small circle arc $UT$ parallel to the equator. Then $UT = U\hat{P}T \sin PT$; also $UT = ST \sin\phi$, where $\phi$ is the *position angle* $PST$. $\phi$ is measured from the meridian joining the star to the north pole $P$ from 0° to 360°, in the direction shown by the arrow in Fig. 98. But $U\hat{P}T = \mu_\alpha$, $PT = 90° - \delta_1$ and $ST = \mu$. Hence

$$\mu_\alpha \cos \delta_1 = \mu \sin \phi$$

or, writing $\delta$ for $\delta_1$ since $\mu$ is small, we have

$$\mu_\alpha = \mu \sin \phi \sec \delta \qquad \text{.....(10).}$$

As $\mu$ is supposed to be expressed in seconds of arc per annum, $\mu_\alpha$ is also given by (10) in terms of the same units. Similarly, $SU = \delta_1 - \delta = \mu_\delta$, so that

$$\mu_\delta = \mu \cos \phi \qquad \text{.....(11),}$$

$\mu_\delta$ being expressed in seconds of arc per annum. If $\mu_\alpha$, $\mu_\delta$ are known, the formulae (10) and (11) are sufficient to enable the values of $\mu$ and $\phi$ to be calculated.

The values of $\mu_\alpha$ and $\mu_\delta$ for the brighter stars (down to about the ninth magnitude) can be obtained by meridian observations separated by a long interval of time. We shall illustrate the procedure by means of an example. In the *A.G. Catalogue* the mean co-ordinates of Arcturus for 1875·0 are

$$14^h\ 9^m\ 57^s{\cdot}63;\quad +\ 19^\circ\ 50'\ 2''{\cdot}6.$$

These mean co-ordinates are derived from several meridian observations of the star, which, for simplicity, we shall suppose to have been made round about the beginning of the year 1875. After the instrumental and refraction corrections have been applied, each meridian observation yields the apparent co-ordinates referred to the true equinox of date; after the application of the formulae for precession, nutation and aberration, the mean co-ordinates for the epoch 1875·0 are obtained. This is done for each of the observations; when the average is taken for the several observations, the results, already quoted, are the mean co-ordinates of the star for the beginning of 1875 referred to the mean equator of 1875·0.

A similar set of observations round about the beginning of 1925 give the mean co-ordinates of the star at the beginning of 1925 referred to the mean equator of 1925·0. With our assumptions the interval between the two sets of observations is 50 years. Now if the star had no proper motion, the mean co-ordinates for 1925·0 ought to be the mean co-ordinates for 1875·0 *plus* the amount of precession in the interval. To find the actual amount of proper motion in right ascension and declination, we first derive the star's position at the beginning of 1875 referred to the mean equator of 1925·0. This involves the application of precession between 1875 and 1925—call this interval $t$. From the *A.G. Catalogue* we extract the following for Arcturus:

| | R.A. | Dec. |
|---|---|---|
| Annual Precession | $+\ 2^s{\cdot}8132$; | $-\ 16''{\cdot}915$ |
| [Chapter x: formulae (29), (30)] | | |
| Secular variation ($s$ and $s_1$) | $+\ 0^s{\cdot}0003$; | $+\ 0''{\cdot}228$ |

Using formulae (32) and (34) of Chapter X we have, for $t = 50$,

$$\frac{st}{200} = + 0^s{\cdot}0001; \quad \frac{s_1 t}{200} = + 0''{\cdot}057;$$

which, added to the annual precessions above, give respectively $+ 2^s{\cdot}8133$ and $- 16''{\cdot}858$. Hence the total changes in the co-ordinates due to precession are respectively

$$+ 140^s{\cdot}66; \quad - 842''{\cdot}9,$$

or

$$+ 2^m\, 20^s{\cdot}66; \quad - 14'\, 2''{\cdot}9.$$

Applying these to the mean co-ordinates for 1875·0, we obtain the mean co-ordinates of Arcturus, *as observed at the beginning of* 1875, referred to the mean equator of 1925·0; the results are

$$14^h\, 12^m\, 18^s{\cdot}29; \quad + 19°\, 35'\, 59''{\cdot}7 \quad \ldots\ldots(12).$$

We now consider the second set of observations made at the beginning of 1925. The mean co-ordinates of Arcturus, as observed in 1925 and referred to the mean equator of 1925·0, are given as follows:

$$14^h\, 12^m\, 14^s{\cdot}39; \quad + 19°\, 34'\, 19''{\cdot}9 \quad \ldots\ldots(13).$$

The co-ordinates given by (12) and (13) are all referred to the *same* co-ordinate system, namely, the mean equator and equinox of 1925·0. The differences between the two right ascensions and the two declinations in (12) and (13) are thus due to the proper motion of the star in 50 years. The R.A. has decreased by $3^s{\cdot}90$ in 50 years and the declination has decreased by $1'\, 39''{\cdot}8$. We then have

$$\mu_\alpha = - 0^s{\cdot}078; \quad \mu_\delta = - 2''{\cdot}00.$$

These are the components of the proper motion of Arcturus, with reference to the mean equator and equinox of 1925·0.

For methods of combining a large number of such determinations and for a discussion of the systematic errors of the various catalogues, the reader is referred to Newcomb's *Spherical Astronomy*.

For stars fainter than the ninth magnitude the determination of proper motions is undertaken with the photographic telescope; the method is described later in Chapter XII.

**148.** *The components of proper motion at different epochs referred to the same equatorial system.*

Let $S$ (Fig. 99) denote the mean position of a star in 1850, say, referred to the mean equator and equinox of 1850·0. To avoid

confusion, we shall at first express $\mu$ in circular measure. Owing to proper motion the star will move along the great circle $SAC$ at the rate of $\mu$ radians per annum. We shall assume that $\mu$ is constant. While changes in $\mu$ cannot be ignored, their omission here will greatly clarify the actual problem under consideration. After an interval of $t$ years let the star's position on the celestial sphere be $A$.

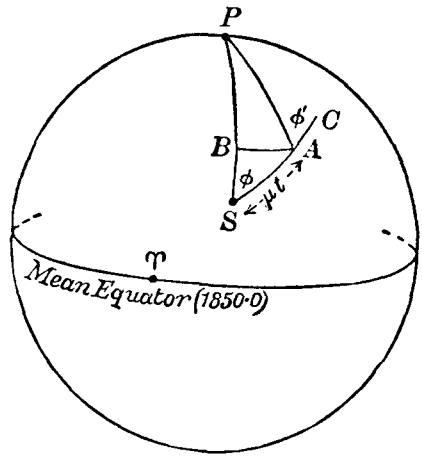


Fig. 99.

Let us write $\mu_\alpha$, $\mu_\delta$ for the components of proper motion when the star is at $S$, that is, in 1850. Then by (10) and (11), $\mu_\alpha$ and $\mu_\delta$ being expressed in radians per annum, we have

$$\mu_\alpha = \mu \sin \phi \sec \delta; \quad \mu_\delta = \mu \cos \phi \quad \ldots\ldots(14),$$

the co-ordinates of $S$ being $(\alpha, \delta)$ and $\phi$ the angle $PSA$.

Let us write $\mu_\alpha'$, $\mu_\delta'$ for the components of proper motion when the star is at $A$, that is, in $1850 + t$. Then if $(\alpha', \delta')$ are the co-ordinates of $A$ (referred to the mean equator of 1850·0) and $\phi'$ is the angle $PAC$, we have similarly

$$\mu_\alpha' = \mu \sin \phi' \sec \delta'; \quad \mu_\delta' = \mu \cos \phi' \quad \ldots\ldots(15).$$

In these formulae we neglect variation of proper motion, for the effect of which see Exercise 7 below. It is thus clear from (14) and (15) that the components of proper motion alter with regard to the epoch for which they are defined, even although the fundamental plane of reference (the mean equator of 1850·0) is the same and the total proper motion $\mu$ is constant.

Through $A$ draw the small circle arc $AB$ parallel to the equator. In the triangle $PSA$, $PS = 90° - \delta$, $PA = 90° - \delta'$, $S\hat{P}A = \alpha' - \alpha$, $P\hat{S}A = \phi$, $P\hat{A}S = 180° - \phi'$ and $SA = \mu t$. By the sine-formula **B** we have

$$\sin \phi' \cos \delta' = \sin \phi \cos \delta.$$

Writing $\phi + \Delta\phi$ for $\phi'$ and $\delta + \Delta\delta$ for $\delta'$ we obtain (regarding $\Delta\phi$, $\Delta\delta$ as small angles, for $SA$ ($\equiv \mu t$) is small even for considerable intervals of time)

$$(\sin\phi + \Delta\phi\cos\phi)(\cos\delta - \Delta\delta\sin\delta) = \sin\phi\cos\delta,$$

from which, neglecting infinitesimals of the second order, we have

$$\Delta\phi\cos\phi\cos\delta = \Delta\delta\sin\phi\sin\delta \qquad \text{.....(16).}$$

This equation gives the variation of $\phi$ with $\delta$.

In the infinitesimal triangle $SAB$, $SB = \delta' - \delta = \Delta\delta$, and therefore

$$\Delta\delta = \mu t\cos\phi \qquad \text{.....(17).}$$

Hence from (16) and (17),

$$\Delta\phi = \mu t\sin\phi\tan\delta \qquad \text{.....(18).}$$

From (15) we have

$$\mu_\alpha' = \frac{\mu(\sin\phi + \Delta\phi\cos\phi)}{\cos\delta - \Delta\delta\sin\delta}$$

$$= \mu\sec\delta(\sin\phi + \Delta\phi\cos\phi)(1 + \Delta\delta\tan\delta),$$

or

$$\mu_\alpha' = \mu\sin\phi\sec\delta + \mu\Delta\phi\cos\phi\sec\delta + \mu\Delta\delta\sin\phi\sec\delta\tan\delta,$$

again neglecting small quantities of the second order. Hence using (14), (17) and (18) we obtain

$$\mu_\alpha' - \mu_\alpha = 2\mu^2 t\sin\phi\cos\phi\sec\delta\tan\delta,$$

and by (14)

$$\mu_\alpha' - \mu_\alpha = 2t\mu_\alpha\mu_\delta\tan\delta \qquad \text{.....(19).}$$

This equation has been derived on the supposition that $\mu$, $\mu_\alpha$, $\mu_\alpha'$ and $\mu_\delta$ are all expressed in circular measure. Express $\mu_\alpha$, $\mu_\alpha'$ in seconds of time and $\mu_\delta$ in seconds of arc; then (19) is written

$$\mu_\alpha' - \mu_\alpha = 2t\mu_\alpha\mu_\delta\tan\delta\sin 1'' \qquad \text{.....(20).}$$

In a similar way, we have from (15), the proper motions being expressed in circular measure,

$$\mu_\delta' = \mu(\cos\phi - \Delta\phi\sin\phi),$$

or

$$\mu_\delta' - \mu_\delta = -\mu\Delta\phi\sin\phi$$

$$= -\mu^2 t\sin^2\phi\tan\delta \qquad \text{[by (18)]}$$

$$= -t\mu_\alpha^2\sin\delta\cos\delta \qquad \text{[by (14)].}$$

Expressing $\mu_\alpha$, $(\mu_\delta' - \mu_\delta)$ in seconds of time and arc respectively, we obtain

$$\mu_\delta' - \mu_\delta = -t(15\mu_\alpha)^2\sin\delta\cos\delta\sin 1'' \qquad \text{.....(21).}$$

Equations (20) and (21) thus give the components of proper motion at the epoch $(1850 + t)$ when the components for epoch 1850·0 are known; and *vice versa*. It is only, however, when the components of proper motion are large that the quantities on the right of (20) and (21) need be taken into account.

*Example.* The mean co-ordinates of the star Groombridge 1830 for 1900·0 are $11^h\ 47^m\ 13^s{\cdot}0$, $+\ 38°\ 26'\ 10''$, and the components of the proper motion for 1900 are

$$\mu_\alpha = +\ 0^s{\cdot}3405, \quad \mu_\delta = -\ 5''{\cdot}801.$$

To find the components of proper motion $\mu_\alpha'$, $\mu_\delta'$ for this star in the year 2000, referred to the mean equator of 1900·0.

Apply (20) in which $t = 100$, $\tan\delta = 0{\cdot}793$ and $\sin 1'' = 1/206265$. Then

$$\mu_\alpha' - \mu_\alpha = -\ 0^s{\cdot}0016,$$

so that $\mu_\alpha'$ is $+\ 0^s{\cdot}3389$.

Similarly from (21), $\mu_\delta' = -\ 5''{\cdot}801 - 0''{\cdot}006$ or $-\ 5''{\cdot}807$.

**149.** *The components of proper motion referred to the mean equators of two different epochs.*

In Fig. 100 let $S$ be the position of a star in 1900·0, referred to the mean equator and equinox of that year. Owing to luni-solar precession the pole $P$ will move to $Q$ after $t$ years, describing a

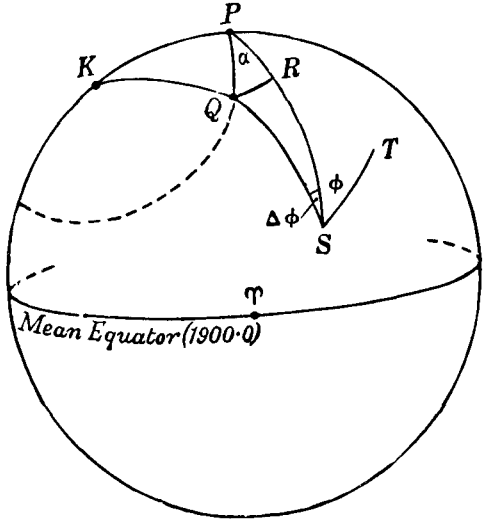



Fig. 100.

small circle arc around $K$, the pole of the ecliptic. The angle $PKQ$ is the luni-solar precession $\psi t$ in $t$ years, the value of $\psi$ as we have seen in Chapter X being about 50″. As in the previous section we shall at first express $\psi$ and the proper motions in circular measure. We regard $P\hat{K}Q$ as a small angle, so that the great circle arc joining $P$ and $Q$ will be indistinguishable from the small circle arc along which the precessional motion of the pole takes place. Let $(\alpha, \delta)$ be the co-ordinates of $S$ referred to the mean equator and equinox of 1900·0 and $(\alpha_1, \delta_1)$ the co-ordinates of $S$ referred to the mean equator and equinox of $1900 + t$. Then $PS = 90° - \delta$ and $QS = 90° - \delta_1$. Also $K\hat{P}S = 90° + \alpha$, so that $Q\hat{P}S = \alpha$.

Let the proper motion $\mu$ of the star be directed along the great circle $ST$. Then if $\mu_\alpha$, $\mu_\delta$ are the components of proper motion of *S in 1900* referred to the mean equator of 1900·0 we have, by (10) and (11),

$$\mu_\alpha = \mu \sin \phi \sec \delta; \quad \mu_\delta = \mu \cos \phi \quad \ldots\ldots(22),$$

where $\phi$ is the angle $PST$.

Let $\mu_\alpha''$, $\mu_\delta''$ denote the components of proper motion of the star, *also in 1900*, referred to the mean equator and pole $Q$ for $1900 + t$. Then

$$\mu_\alpha'' = \mu \sin \phi_1 \sec \delta_1; \quad \mu_\delta'' = \mu \cos \phi_1 \quad \ldots\ldots(23),$$

where $\phi_1$ is the angle $QST$. We shall write $\phi_1 = \phi + \Delta\phi$, so that $P\hat{S}Q = \Delta\phi$. Then by formula **B** we have, from the triangle $PQS$,

$$\sin \Delta\phi = \sin PQ \sin \alpha \sec \delta_1,$$

or, since $PQ$ and $\Delta\phi$ are supposed to be small and $PQ = \psi t \sin \epsilon$,

$$\Delta\phi = \psi t \sin \alpha \sec \delta \sin \epsilon \quad \ldots\ldots(24),$$

in which we have written $\delta$ for $\delta_1$ without sensible loss of accuracy.

Draw the small circle arc $QR$ which has $S$ as its pole. With this construction, $PR = \delta_1 - \delta$. Writing $\Delta\delta$ for $\delta_1 - \delta$, we obtain from the infinitesimal triangle $PQR$,

$$\delta_1 - \delta \equiv \Delta\delta = \psi t \cos \alpha \sin \epsilon \quad \ldots\ldots(25).$$

From the first of (23), we have

$$\begin{aligned} \mu_\alpha'' &= \frac{\mu \sin (\phi + \Delta\phi)}{\cos (\delta + \Delta\delta)} \\ &= \frac{\mu (\sin \phi + \Delta\phi \cos \phi)}{\cos \delta (1 - \Delta\delta \tan \delta)} \\ &= \mu \sec \delta (\sin \phi + \Delta\phi \cos \phi)(1 + \Delta\delta \tan \delta), \end{aligned}$$

or, neglecting small quantities of the second order,

$$\mu_\alpha'' = \mu \sec\delta \sin\phi + \mu\Delta\delta \sec\delta \sin\phi \tan\delta + \mu\Delta\phi \sec\delta \cos\phi,$$

or, using (22), (24) and (25),

$$\mu_\alpha'' - \mu_\alpha = \mu\psi t \sin\epsilon \sec\delta (\cos\alpha \tan\delta \sin\phi + \sin\alpha \sec\delta \cos\phi).$$

But $\mu_\alpha = \mu \sin\phi \sec\delta$ and $\mu_\delta = \mu\cos\phi$.

Hence $$\mu_\alpha'' - \mu_\alpha = \psi t \sin\epsilon (\mu_\alpha \cos\alpha \tan\delta + \mu_\delta \sin\alpha \sec^2\delta) \quad (26).$$

Similarly, $\mu_\delta'' = \mu(\cos\phi - \Delta\phi \sin\phi)$

$$= \mu\cos\phi - \mu\psi t \sin\epsilon \sin\alpha \sec\delta \sin\phi,$$

so that $$\mu_\delta'' - \mu_\delta = -\psi t \mu_\alpha \sin\alpha \sin\epsilon. \qquad \ldots\ldots(27).$$

In the derivation of the formulae (26) and (27), $\mu_\alpha$, $\mu_\delta$ and $\psi$ are all supposed to be expressed in circular measure. Express $\mu_\alpha$, $\mu_\alpha''$ in seconds of time and $\mu_\delta$, $\mu_\delta''$ and $\psi$ in seconds of arc, and these equations become

$$\mu_\alpha'' - \mu_\alpha = \psi t \sin\epsilon (\mu_\alpha \cos\alpha \tan\delta + \tfrac{1}{15}\mu_\delta \sin\alpha \sec^2\delta) \sin 1'' \qquad \ldots\ldots(28),$$

$$\mu_\delta'' - \mu_\delta = -15\psi t \mu_\alpha \sin\alpha \sin\epsilon \sin 1'' \qquad \ldots\ldots(29).$$

*Example.* To find the value of $\mu_\alpha''$ for the motion of the star Groombridge 1830 in 1900 referred to the mean equator of 1930·0, given that $\mu_\alpha = +\ 0^s{\cdot}3405$, $\mu_\delta = -\ 5''{\cdot}801$ (referred to the mean equator for 1900·0). Here $t = 30$, $\alpha = 11^h\ 47^m\ 13^s$, $\delta = 38°\ 26'\ 10''$, $\cos\alpha = -\ 0{\cdot}998$, $\sin\alpha = 0{\cdot}056$, $\tan\delta = 0{\cdot}793$, $\sec^2\delta = 1{\cdot}63$, $\psi = 50$, $\sin\epsilon = 0{\cdot}398$. Hence, from (28),

$$\mu_\alpha'' - \mu_\alpha = -\ 0^s{\cdot}0009,$$

so that $$\mu_\alpha'' = +\ 0^s{\cdot}3396.$$

The computation for the value of $\mu_\delta''$ can be performed in a similar manner by means of (29).

The principal formulae of this and the preceding section enable us to derive the components of proper motion of a star at any epoch referred to the fundamental co-ordinate system at that or a different epoch. These formulae need only be applied when the components of proper motion are very large.

### 150. *Mean and apparent co-ordinates.*

In the chapter on precession and nutation we omitted the consideration of proper motion, and we now revise the fundamental definitions, taking proper motion into account. Suppose

we start with the mean co-ordinates $(\alpha_0, \delta_0)$ of a star for 1900·0. The mean co-ordinates $(\alpha, \delta)$ for $1900 + t$ are then given by applying the precession for $t$ years with the secular variation, together with the proper motion for $t$ years. The annual precession plus the annual proper motion is called the *annual variation*, and this quantity is generally tabulated in the catalogues for each star. The mean R.A. and declination for $1900 + t$ are then obtained by means of formulae (32) and (34) of Chapter X, in which $\frac{d\alpha_0}{dt}$ and $\frac{d\delta_0}{dt}$ are now taken to mean the annual variation in right ascension and declination respectively.

The *apparent place* of a star, say, on 1931 March 9, is its position on this date on the celestial sphere (with the earth as centre) and referred to the actual equator and equinox of date. The apparent place on the date is obtained from the mean place at the beginning of the year by applying precession, nutation, aberration (by means of the Besselian or independent day numbers) and the proper motion for the fraction of the year in question.

**151.** *The solar motion and parallactic motion.*

We have seen that the proper motion of a star is its rate of change of direction as viewed from the sun. It has been convenient to describe the cause of the star's proper motion as the star's linear space-velocity relative to the sun, but the question might well be asked: "Is it not possible that all the stars are 'fixed in space' and that their proper motions are the results of the sun's motion in some definite direction?" In this section we examine the problem thus suggested.

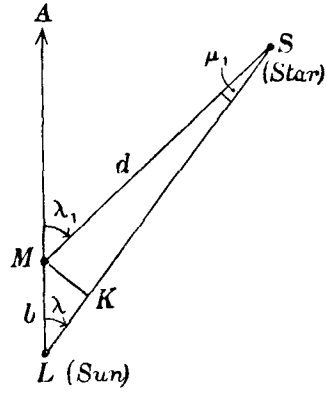


Fig. 101.

In Fig. 101 consider a star $S$ at rest with respect to certain hypothetical axes in space. Suppose the sun moves with respect to these axes along the straight line $LA$, traversing the distance $LM$ in one year. If $U$ is its velocity (in kilometres per second) and $n$ the number of seconds in one year,

then the distance $LM$—which we denote by $b$—is given in kilometres by

$$b = nU \qquad \text{.....(30).}$$

Let $\lambda$ denote the angle $ALS$—it is the angle between the direction of the solar motion and the direction in which the star is seen from $L$. Let $\lambda_1$ denote the angle $AMS$. Further let $\lambda_1 - \lambda = \mu_1$, so that $M\hat{S}L = \mu_1$. Then $\mu_1$ is the change in the direction of the star, as seen from the sun, in one year. This appears in the observations as proper motion.

If $d$ is the distance $MS$ in kilometres, we have

$$\frac{\sin \mu_1}{\sin \lambda} = \frac{b}{d},$$

or, since $\mu_1$ is a very small angle, we can write

$$\mu_1 = \frac{b}{d} \sin \lambda \operatorname{cosec} 1'' \qquad \text{.....(31),}$$

in which $\mu_1$ is expressed in seconds of arc.

We can transform (31) by introducing the annual parallax $\Pi$ of the star by means of

$$\Pi = \frac{a}{d} \operatorname{cosec} 1'' \qquad \text{.....(32),}$$

where $a$ is the radius of the earth's orbit around the sun in kilometres. From (30), (31) and (32) we derive

$$\mu_1 = \frac{n}{a} \Pi U \sin \lambda.$$

But $n = 31{\cdot}56 \times 10^6$ and $a = 149{\cdot}6 \times 10^6$ km, hence

$$\mu_1 = \frac{\Pi U \sin \lambda}{4{\cdot}74} \qquad \text{.....(33).}$$

It is to be noted that formula (33) provides a method of deriving the annual parallax $\Pi$ of a star, provided that the solar speed $U$ and its direction are known and provided also that the assumption regarding the "fixity" of the stars still holds.

The proper motion $\mu_1$ which arises from the solar motion is called the *parallactic motion* of the star. It varies directly as the parallax $\Pi$ and, consequently, by (32) it varies inversely as the distance of the star from the sun.

The point on the celestial sphere towards which the solar motion is directed is called the *solar apex*.

In Fig. 102 let $A$ be the solar apex and $S$ a star. We shall show later how the position of $A$ is determined; meanwhile we shall assume that its co-ordinates are known. The great circle arc $AS$ defines the angle between the direction in which the sun is moving and the direction of the star. Thus $AS$ is $\lambda$. Also, referring to Fig. 101, we see that the change in direction, or $\mu_1$, takes place in the plane $ALS$, so that the star appears to move on the celestial sphere along the great circle $AST$ (Fig. 102). As $\lambda_1$ is greater than $\lambda$, the star appears to move away from $A$, that is, in the direction $ST$.

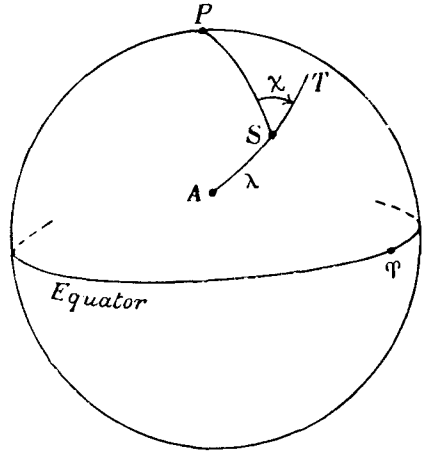


Fig. 102.

Denote by $\chi$ the position angle $PST$; $\chi$ is measured from 0° to 360° in the direction of the arrow. Then since the parallactic motion is $\mu_1$ along $ST$ we have, applying (10) and (11) and denoting the components of the parallactic motion in R.A. and declination by $P_\alpha$, $P_\delta$ respectively,

$$P_\alpha = \mu_1 \sin\chi \sec\delta; \quad P_\delta = \mu_1 \cos\chi \qquad \text{.....(34).}$$

Inserting the expression for $\mu_1$ given by (33), we obtain

$$P_\alpha = \frac{\Pi U}{4{\cdot}74} \sin\lambda \sin\chi \sec\delta \qquad \text{.....(35),}$$

and

$$P_\delta = \frac{\Pi U}{4{\cdot}74} \sin\lambda \cos\chi \qquad \text{.....(36).}$$

These formulae are of importance when it is desired to find the components of the parallactic motion of a star, or of a group of stars, when the quantities on the right of (35) and (36) are known.

### 152. *Secular parallax.*

It is evident that if the distance $LM$ (Fig. 101) through which the sun moves in one year is known, then this line forms a convenient base-line for the measurement of stellar distances.

By analogy with the annual parallax $\Pi$ the *secular parallax* $H$, as it is called, is defined by

$$\sin H = \frac{b}{d},$$

or, since $H$ is small, by

$$H = \frac{b}{d} \operatorname{cosec} 1''.$$

Also, from (30), $b = nU$, so that

$$H = \frac{nU}{d} \operatorname{cosec} 1'' \qquad \ldots\ldots(37),$$

in which $H$ is expressed in seconds of arc. Hence, from (32) and (37),

$$H = \frac{\Pi U}{4{\cdot}74} \qquad \ldots\ldots(38).$$

The measurement of $U$ will be discussed later; meanwhile we state its value, which is 19·5 kms. per second. Thus we obtain the numerical relation between the annual parallax $\Pi$ and the secular parallax $H$ in the form:

$$\Pi = 0{\cdot}243\, H \qquad \ldots\ldots(39).$$

By means of (38) the formulae (35) and (36) take on a simpler form: they become

$$P_\alpha = H \sin\lambda \sin\chi \sec\delta \qquad \ldots\ldots(40),$$

$$P_\delta = H \sin\lambda \cos\chi \qquad \ldots\ldots(41).$$

**153.** *The solar motion and radial velocity.*

In Fig. 101 let $MK$ be drawn perpendicularly to $LS$. Now $LM$ is very small in comparison with $LS$ and we may consider $MS$ and $SK$ equal. Thus in one year, as the sun moves from $L$ to $M$, the distance of the star decreases from $LS$ to $MS$, that is, by the distance $LK$. This is equivalent to the effect of a negative radial velocity $\rho_1$ (in kilometres per second). Hence $LK = n\rho_1$, where $n$ is the number of seconds in one year. Also $LK = b \cos\lambda$ and, since $b = nU$ by (30), we obtain

$$\rho_1 = -\, U \cos\lambda \qquad \ldots\ldots(42),$$

in which the minus sign has been inserted, since the radial velocity is negative. Formula (42) gives the radial velocity of the star arising from the solar motion.

**154.** *The solar motion in the general case.*

In section 151 we started from the hypothesis that the stars are "fixed in space" and that only the sun was in motion with respect to fixed axes. This assumption can be examined by means of the observed proper motions. Referring to (35) and (36) or (40) and (41), we see that there are two points on the celestial sphere—the solar apex and the diametrically opposite point, the *ant-apex*—where the parallactic motion vanishes, for then $\lambda$ is 0° or 180°, so that both $P_\alpha$ and $P_\delta$ vanish. Thus there ought to be two regions of the sky in which the observed proper motions of all the stars concerned should be zero. This is contrary to the facts. More generally, if the hypothesis is correct, the *total proper motion of any star ought to be in a great circle* passing through a definite point $A$ (Fig. 102)—the solar apex—so that by considering only two stars in different parts of the sky it should be possible to determine the apex as one of the two points of intersection of the respective great circles. Actually it is found from the proper motions that the points of intersection derived from one pair of stars differ widely from the points of intersection derived from any other pair of stars. The conclusion is that the stars have individual motions of their own in addition to the parallactic motion due to the sun's velocity.

Under these conditions the solar motion must be given a more precise interpretation. We have no longer fixed points in space to which the motion of the sun can be referred, for all the stars as well as the sun are in motion. If we are considering the proper motions of 10,000 stars scattered over the sky, the solar motion is defined with reference to all these stars regarded as a group or, more precisely, as the velocity relative to the mean centre of the group.

Consider Fig. 103, in which $S$ is a star and $A$ is the solar apex whose position we shall assume to be known. Let the proper motion $\mu$ of the star be along the great circle $SU$, and let $P\hat{S}U$ be denoted by $\phi$. Both $\mu$ and $\phi$ can be calculated from the observed values of $\mu_\alpha$ and $\mu_\delta$ by means of (10) and (11). The parallactic motion $\mu_1$ is along the great circle $AST$, from $S$ towards $T$; the angle $PST$ is $\chi$. Let $(\alpha, \delta)$ be the co-ordinates of $S$ and $(A, D)$ the co-ordinates of the apex $A$. Then, in the triangle $PAS$, $PS = 90° - \delta$, $PA = 90° - D$, $A\hat{P}S = \alpha - A$, $P\hat{S}A = 180° - \chi$,

$AS = \lambda$. The values of $\lambda$ and $\chi$ can be easily found as follows By the cosine-formula **A**,

$$\cos\lambda = \sin D \sin\delta + \cos D \cos\delta \cos(\alpha - A) \quad \ldots(43),$$

from which $\lambda$ can be computed, and by formula **C**,

$$-\sin\lambda\cos\chi = \sin D\cos\delta - \cos D\sin\delta\cos(\alpha - A) \ldots(44),$$

from which $\chi$ can be obtained.

$\lambda$ is the angular distance of the star from the apex. The direction $ST$ is the direction of the ant-apex from $S$ and the angle $\chi$ between the meridian $SP$ and the great circle $ST$ is called

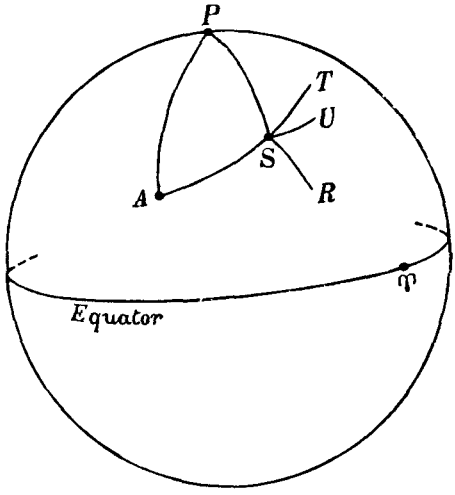



Fig. 103.

the *position angle of the ant-apex*. As the ant-apex is 180° from $A$, the angular distance of the star from the ant-apex is $180° - \lambda$. These angles, $(180° - \lambda)$ and $\chi$, can be found with sufficient accuracy by means of specially prepared charts,* thus saving the rather tedious computations by (43) and (44).

Now resolve the observed proper motion $\mu$, which is in the direction $SU$ (Fig. 103), along the great circle $ST$ and along the

* W. M. Smart: Charts published by the Royal Astronomical Society and described in *Monthly Notices*, vol. LXXXIII, p. 465; J. A. Pearce and S. N. Hill: *Publications of the Dominion Astrophysical Observatory, Victoria, B.C.*, vol. IV, No. 4; J. M. Baldwin: Chart published by the Royal Astronomical Society and described in *Monthly Notices*, vol. LXXXIX, p. 453.

great circle $RS$ drawn at right angles to $AST$. Let $v$ and $\tau$ denote these components. Then, since $T\hat{S}U = \phi - \chi$,

$$v = \mu \cos(\phi - \chi),$$

and

$$\tau = \mu \sin(\chi - \phi).$$

Since, by (10) and (11), $\mu_\alpha = \mu \sin\phi \sec\delta$ and $\mu_\delta = \mu \cos\phi$, we can write the formulae for $v$ and $\tau$ as follows:

$$v = \mu_\alpha \cos\delta \sin\chi + \mu_\delta \cos\chi \quad \ldots\ldots(45),$$

$$\tau = \mu_\delta \sin\chi - \mu_\alpha \cos\delta \cos\chi \quad \ldots\ldots(46).$$

Assuming the co-ordinates of the solar apex to be known, we can obtain the value of $\chi$ by formula (44) or from charts, and thus we can calculate the values of $v$ and $\tau$, the values of $\mu_\alpha$ and $\mu_\delta$ being supposed known.

Let us write

$$v = \mu_1 + v_1 \quad \ldots\ldots(47),$$

where $\mu_1$ is the parallactic motion along $ST$. Then $v_1$ is the portion of the observed proper motion in the direction $ST$ due to the star's individual motion, $\mu_1$ being due to the solar motion. The component $\tau$ is evidently due entirely to the star's individual motion.

**155.** *Statistical parallaxes.*

The determination of the annual parallax of even a single star involves a large amount of observational and computational work. In many investigations it is of great importance to know the mean parallax of a particular class of stars, say stars between the tenth and eleventh magnitudes in a particular region of the sky, and this mean parallax can be determined from the $v$ and $\tau$ components of proper motion. We shall consider the $v$ components of $N$ stars close together in the sky, the stars belonging, for example, to some definite magnitude class.

We shall define $v_1$ to be positive when it is in the direction $ST$, and negative when it is in the opposite direction $SA$ (Fig. 103). If the $N$ stars have haphazard individual motions, the expectation will be that there will be as many stars with positive values of $v_1$ as with negative values, so that the algebraic sum of all the $v_1$'s (denoted by $\Sigma v_1$) will either vanish or be very small compared with $\Sigma\mu_1$. Adding the $N$ equations of the type (47), we have

$$\Sigma v = \Sigma\mu_1 + \Sigma v_1,$$

and, neglecting $\Sigma v_1$ as the result of the previous remark, we are left with

$$\Sigma\mu_1 = \Sigma v \qquad \text{.....(48).}$$

For a given star, $v$ is supposed to be known—it is calculated by means of (45) from known data. Hence for all the stars the sum $\Sigma v$ will be easily found. Now for a star whose annual parallax is $\Pi$, $\mu_1$ is given by (33), namely,

$$\mu_1 = \frac{\Pi U \sin\lambda}{4{\cdot}74}.$$

Hence

$$\Sigma\mu_1 = \frac{U \sin\lambda}{4{\cdot}74}\Sigma\Pi.$$

If $\Pi_0$ denotes the mean parallax of the $N$ stars, $N\Pi_0 = \Sigma\Pi$, and therefore we obtain, by means of (48),

$$\Pi_0 = \frac{4{\cdot}74}{NU \sin\lambda}\Sigma v \qquad \text{.....(49).}$$

As all the quantities on the right of (49) are supposed known, the mean parallax $\Pi_0$ is readily derived.

Using equations (38) and (49) the mean secular parallax, $H_0$ say, may be similarly computed, and it is this quantity, rather than the mean parallax itself, that is normally derived. It is found that the mean secular parallax decreases with increasing magnitude class and increases with galactic latitude. The galactic latitude of a star is defined with reference to the plane of the Milky Way in the same way as its celestial latitude is referred to the plane of the ecliptic.

153. *Determination of the solar apex from proper motions.*

If we are given that the sun is moving in a certain direction with velocity $U$, then relatively to the sun each star will appear to have an additional velocity of $U$ in the opposite direction. In Fig. 104 let $O$ (the sun) be the centre of the celestial sphere and let $Ox$, $Oy$, $Oz$ be rectangular axes: $Ox$ is drawn through the vernal equinox $\Upsilon$, $Oy$ through the point $B$ on the equator with right ascension 90°, and $Oz$ through the pole $P$. Then if $-X$, $-Y$, $-Z$ (in kilometres per second) denote the components of the solar motion with respect to the axes, then relatively to the sun a star at $S$ will have a velocity whose components are

$+X, +Y, +Z$. We neglect for the moment the star's individual linear velocity, so that we are simply investigating the effect of parallactic motion. Let the star be at a distance of $d$ kms. from the sun, and for simplicity we shall take the radius of the sphere to be $d$.

The velocity of the star $S$ has components $+X$ parallel to $O\Upsilon$, $+Y$ parallel to $OB$ and $+Z$ parallel to $OP$. It is our first object to find the components of the star's linear velocity along

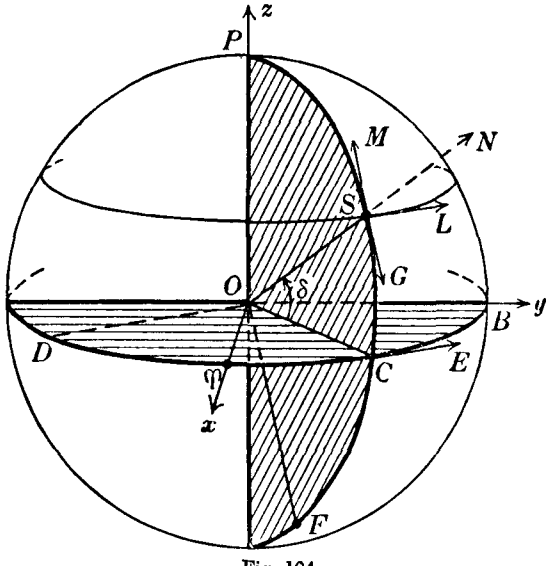



Fig. 104.

the rectangular axes $SL$, $SM$ and $SN$, where $SL$ is tangential at $S$ to the parallel of declination (in the sense of increasing R.A.), $SM$ is tangential to the meridian $PSC$ (in the sense of increasing declination) and $SN$ is the radius $OS$ produced.

Now the velocity $X$ parallel to $O\Upsilon$ is equivalent to a velocity $X\cos\alpha$ parallel to $OC$ (the angle $\Upsilon OC$ is $\alpha$) and a velocity $X\sin\alpha$ parallel to $OD$, where $DOC$ is 90°. Similarly the velocity $Y$ parallel to $OB$ is equivalent to a velocity $Y\sin\alpha$ parallel to $OC$ and a velocity $-Y\cos\alpha$ parallel to $OD$. We thus can replace $X$ and $Y$ by

$X\cos\alpha + Y\sin\alpha$ parallel to $OC$ .....(50),

and $X\sin\alpha - Y\cos\alpha$ parallel to $OD$ .....(51).

Let $CE$ be drawn tangential to the equator at $C$. Then $CE$ i perpendicular to $OC$ and therefore $CE$ is parallel to $OD$. Sinc $CE$ is drawn in the direction opposite to that of $OD$, we ca write for (51),

$$-X\sin\alpha + Y\cos\alpha \quad \text{parallel to } CE \quad \ldots\ldots(52).$$

Also $SL$ is parallel to $CE$, so that (52) can be replaced by

$$-X\sin\alpha + Y\cos\alpha \quad \text{parallel to } SL \quad \ldots\ldots(53).$$

Let $OF$ be drawn in the plane of the meridian $PSC$ perpen dicularly to $OS$. Then $OF$ is parallel to the tangent $MSG$ at $S$

Consider (50). We resolve this velocity parallel to $OS$ an $OF$. Now $C\hat{O}S = \delta$. Hence $(X\cos\alpha + Y\sin\alpha)$ parallel to $O$ is equivalent to

$$(X\cos\alpha + Y\sin\alpha)\sin\delta \quad \text{parallel to } OF,$$

or
$$-(X\cos\alpha + Y\sin\alpha)\sin\delta \quad \text{parallel to } SM \quad \ldots\ldots(54),$$

and
$$(X\cos\alpha + Y\sin\alpha)\cos\delta \quad \text{along } OS \text{ or } SN \quad \ldots(55).$$

Similarly the velocity $Z$ parallel to $OP$ is equivalent t $Z\cos P\hat{O}S$ along $OS$ and $Z\sin P\hat{O}S$ along $SM$. Bu $P\hat{O}S = 90^\circ - \delta$, so that $Z$ is equivalent to

$$Z\cos\delta \quad \text{along } SM \quad \ldots\ldots(56),$$

and
$$Z\sin\delta \quad \text{along } SN \quad \ldots\ldots(57).$$

Collecting the components of velocity along $SL$, $SM$ and $SN$ in turn and denoting the sums by $\xi$, $\eta$ and $\zeta$ respectively, w have:

from (53), $$\xi = -X\sin\alpha + Y\cos\alpha \quad \ldots\ldots(58);$$

from (54) and (56),

$$\eta = -X\cos\alpha\sin\delta - Y\sin\alpha\sin\delta + Z\cos\delta\ldots\ldots(59);$$

from (55) and (57),

$$\zeta = X\cos\alpha\cos\delta + Y\sin\alpha\cos\delta + Z\sin\delta \quad \ldots\ldots(60).$$

As $X$, $Y$, $Z$ are expressed in kilometres per second, so also ar $\xi$, $\eta$, $\zeta$. Now consider the velocity $\xi$ along $SL$. Due to thi velocity alone, the star will move along $SL$ through a distanc $n\xi$ kms. in one year, where $n$ is the number of seconds in on year, and therefore, since $OS$ is $d$ kms., the angle through whic $S$ moves along the parallel of declination is $n\xi/d$ in circula

measure or $\frac{n\xi}{d}$ cosec 1″ in seconds of arc. Hence the right ascension of the star increases by $\frac{n\xi}{d}$ sec δ cosec 1″ seconds of arc per annum. But this is the component of the parallactic motion in right ascension, $P_\alpha$, which we assume to be expressed in seconds of arc. Hence

$$\frac{n\xi}{d} = P_\alpha \cos\delta \sin 1''.$$

Introducing the annual parallax Π by (4) we obtain, after inserting the values of $n$ and $a$ and writing for $\xi$ the expression given by (58),

$$-X \sin\alpha + Y \cos\alpha = \frac{4{\cdot}74}{\Pi} P_\alpha \cos\delta \quad \ldots\ldots(61).$$

This is an accurate equation for the star $S$ and clearly it holds for any other star. The reader is reminded that the parallactic motion is alone considered at the moment.

In a similar manner, by considering the component $\eta$ along $SM$ which gives rise to the component of parallactic motion in declination, $P_\delta$, we derive from (59),

$$-X \cos\alpha \sin\delta - Y \sin\alpha \sin\delta + Z \cos\delta = \frac{4{\cdot}74}{\Pi} P_\delta \quad \ldots\ldots(62).$$

Now consider the observed components of proper motion, $\mu_\alpha$ and $\mu_\delta$. The parallactic motion contributes to $\mu_\alpha$ and so does the star's individual motion. Similarly for $\mu_\delta$. We can then write

$$\mu_\alpha = P_\alpha + \mu_\alpha'; \quad \mu_\delta = P_\delta + \mu_\delta' \quad \ldots\ldots(63),$$

where $\mu_\alpha'$, $\mu_\delta'$ are the contributions of the star's individual motion to the proper motion in R.A. and declination respectively. All the quantities in (63) are assumed to be expressed in seconds of arc.

By means of (63), equation (61) becomes

$$-X \sin\alpha + Y \cos\alpha = \frac{4{\cdot}74}{\Pi} \mu_\alpha \cos\delta - \frac{4{\cdot}74}{\Pi} \mu_\alpha' \cos\delta \quad \ldots\ldots(64).$$

In this equation, $X$, $Y$ are quantities to be found, $\alpha$ and $\delta$ are known and $\mu_\alpha$ is given from proper motion observations. In general Π is unknown and $\mu_\alpha'$ is also unknown.

Consider $N$ stars in a small area of the sky, so that we can regard each star as having the same values of $\alpha$ and $\delta$. Each star contributes an equation of the type (64), and adding the $N$ equations we derive

$$-X\sin\alpha + Y\cos\alpha = \frac{4{\cdot}74}{N}\cos\delta\,\Sigma\left(\frac{\mu_\alpha}{\Pi}\right) - \frac{4{\cdot}74}{N}\cos\delta\,\Sigma\left(\frac{\mu_\alpha'}{\Pi}\right) \quad \ldots\ldots(65).$$

If $\Pi_0$ corresponds to the mean distance $d_0$ of the stars, we can write (65) as

$$-X\sin\alpha + Y\cos\alpha = \frac{4{\cdot}74}{N\Pi_0}\cos\delta\,\Sigma\mu_\alpha - \frac{4{\cdot}74}{N\Pi_0}\cos\delta\,\Sigma\mu_\alpha' \quad \ldots\ldots(66),$$

in which the assumption is now made that the stars have each the same parallax $\Pi_0$.

Now consider the last term of (66). $\mu_\alpha'$ is the component of proper motion due to the star's individual velocity, and it is as likely to be positive as negative. We can either assume that $\Sigma\mu_\alpha'$ vanishes or that, being small, it is of the nature of an accidental error in the measurement of the proper motions. Neglecting the last term we have

$$-X\sin\alpha + Y\cos\alpha = \frac{4{\cdot}74}{N\Pi_0}\cos\delta\,\Sigma\mu_\alpha \quad \ldots\ldots(67),$$

in which $\Pi_0$ is unknown and $\Sigma\mu_\alpha$ is obtained from the observed proper motions. Write $\frac{1}{N}\Sigma\mu_\alpha = \bar{\mu}_\alpha$, so that $\bar{\mu}_\alpha$ is the average observed value of $\mu_\alpha$ for the $N$ stars.

Suppose that we deal with a large number of regions in the same way, and for simplicity assume that there are $N$ stars in each region and that the mean distance of the stars in each region is the same, so that $\Pi_0$ is constant. Then we have for each region an equation of the type

$$-X\sin\alpha + Y\cos\alpha = K\bar{\mu}_\alpha\cos\delta \quad \ldots\ldots(68),$$

where $K \equiv 4{\cdot}74/\Pi_0$ and is regarded as an unknown constant.

In a similar manner and with the same assumptions we derive from (62),

$$-X\cos\alpha\sin\delta - Y\sin\alpha\sin\delta + Z\cos\delta = K\bar{\mu}_\delta \ldots(69).$$

If there are $M$ regions, there are $M$ equations of the type (68) and $M$ of the type (69). The solution of these equations is effected

by the method of least squares. The procedure is as follows. Multiply (68) throughout by $-\sin\alpha$ (the coefficient of $X$) for each of the $M$ equations and add together all the resulting equations. Denoting summation by $\Sigma$ we obtain

$$X\Sigma\sin^2\alpha - Y\Sigma\sin\alpha\cos\alpha = -K\Sigma\bar{\mu}_\alpha\sin\alpha\cos\delta \quad ...(70).$$

Similarly by multiplying by $\cos\alpha$ (the coefficient of $Y$) and summing we derive

$$-X\Sigma\sin\alpha\cos\alpha + Y\Sigma\cos^2\alpha = K\Sigma\bar{\mu}_\alpha\cos\alpha\cos\delta \quad ...(71).$$

Treating (69) in a similar manner, we obtain the three equations·

$$X\Sigma\cos^2\alpha\sin^2\delta + Y\Sigma\sin\alpha\cos\alpha\sin^2\delta - Z\Sigma\cos\alpha\sin\delta\cos\delta = -K\Sigma\bar{\mu}_\delta\cos\alpha\sin\delta \quad .....(72),$$

$$X\Sigma\sin\alpha\cos\alpha\sin^2\delta + Y\Sigma\sin^2\alpha\sin^2\delta - Z\Sigma\sin\alpha\sin\delta\cos\delta = -K\Sigma\bar{\mu}_\delta\sin\alpha\sin\delta \quad .....(73),$$

$$-X\Sigma\cos\alpha\sin\delta\cos\delta - Y\Sigma\sin\alpha\sin\delta\cos\delta + Z\Sigma\cos^2\delta = K\Sigma\bar{\mu}_\delta\cos\delta \quad .....(74).$$

In the five equations, (70) to (74), the coefficients of $X$, $Y$ and $Z$ can all be calculated from the known values of $\alpha$ and $\delta$, and also the coefficients of $K$ on the right-hand sides of these equations, since the values of $\bar{\mu}_\alpha$, $\bar{\mu}_\delta$ are obtainable from the observed proper motions. Adding (70) and (72), adding (71) and (73) and rewriting (74), we have the three equations in symbolical form

$$\left.\begin{aligned} aX + bY + cZ &= KS_1 \\ bX + BY + dZ &= KS_2 \\ cX + dY + CZ &= KS_3 \end{aligned}\right\} \quad .....(75),$$

in which the coefficients of $X$, $Y$ and $Z$ are all calculable functions of $\alpha$ and $\delta$ and $S_1$, $S_2$, $S_3$ are also known. The solution of these equations (75) determines

$$\frac{X}{K}, \quad \frac{Y}{K}, \quad \frac{Z}{K},$$

but *not* $X$, $Y$ and $Z$, since $K$ is unknown.

We can now derive the co-ordinates of the solar ant-apex towards which the sun is moving with a velocity whose components are $+X$, $+Y$, $+Z$. (We assumed originally that the components of the solar motion are $-X$, $-Y$, $-Z$, and, as this motion is directed towards the apex, the velocity whose com-

ponents are $+X, +Y, +Z$ will be directed towards the opposite point on the celestial sphere, that is, the ant-apex.)

In Fig. 105 let $J$ be the ant-apex and take rectangular axes as shown. Let $PJC$ be the meridian through $J$ and let $A$, $D$ be the equatorial co-ordinates of $J$. Then $\Upsilon\hat{O}C = A$ and $J\hat{O}C = D$. The components $X$ and $Y$ combine to give a velocity $(X^2 + Y^2)^{\frac{1}{2}}$

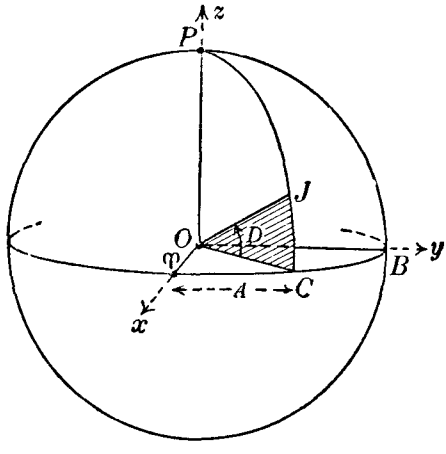



Fig. 105.

along $OC$ and $A$ is given by $\tan A = Y/X$. Also $(X^2 + Y^2)^{\frac{1}{2}}$ along $OC$ combines with $Z$ along $OP$ to give the reverse of the solar motion in the direction $OJ$, so that

$$\tan D = Z/(X^2 + Y^2)^{\frac{1}{2}}.$$

We rewrite these equations in $\tan A$ and $\tan D$ as follows:

$$\tan A = \frac{Y}{K}\Big/\frac{X}{K}; \quad \tan D = \frac{Z}{K}\Big/\left[\left(\frac{X}{K}\right)^2 + \left(\frac{Y}{K}\right)^2\right]^{\frac{1}{2}} \ldots(76).$$

The ratios $\frac{X}{K}, \frac{Y}{K}, \frac{Z}{K}$ are known from the solution of (75); hence $A$ and $D$ are determined. The co-ordinates of the solar apex are then $180° + A$ and $-D$.

The method described above, which makes use of proper motions only, cannot determine the magnitude of the sun's velocity but only its direction, that is the position of the solar apex on the celestial sphere. The co-ordinates derived, however, depend to some extent on the particular stellar class that is used

in the investigation. One further point should perhaps be recognised. The assumption that the stars' individual motions are entirely random is not justified on account of the systematic rotation of the galaxy, and allowance must be made for this in the investigation.

**157.** *Determination of the solar motion from radial velocities.*

We now consider formula (60), which gives the component velocity $\zeta$ in the line of sight $OS$ (Fig. 104) due to the solar velocity. Thus $\zeta$ is simply $-\rho_1$ of section 153, and by (42) we can write

$$\zeta = U \cos \lambda \qquad \text{.....(77),}$$

where $U$ is the solar velocity. Let $\rho$ now denote the observed radial velocity.* Then the solar motion contributes $U \cos \lambda$ to $\rho$ and the star's individual space-velocity contributes $\rho_2$, say. Hence

$$\rho = U \cos \lambda + \rho_2 \qquad \text{.....(78).}$$

Hence, from (60), (77) and (78), we clearly have

$$X \cos \alpha \cos \delta + Y \sin \alpha \cos \delta + Z \sin \delta = \rho - \rho_2 \qquad \text{.....(79),}$$

in which $\rho$ is given by spectroscopic observations and $\rho_2$ is unknown. If the radial velocities of $N$ stars, scattered over the sky, are observed, each star contributes an equation of the form (79) and the $N$ equations are solved by the method of least squares. Multiplying each equation in turn by the coefficient of $X$ and summing we obtain

$$X \Sigma \cos^2 \alpha \cos^2 \delta + Y \Sigma \sin \alpha \cos \alpha \cos^2 \delta + Z \Sigma \cos \alpha \sin \delta \cos \delta = \Sigma \rho \cos \alpha \cos \delta - \Sigma \rho_2 \cos \alpha \cos \delta \qquad \text{.....(80).}$$

Now the radial component $\rho_2$ of the star's individual linear velocity is as likely to be positive as negative, and the sum $\Sigma \rho_2 \cos \alpha \cos \delta$ will tend to vanish. Neglecting this term in (80) we write

$$X \Sigma \cos^2 \alpha \cos^2 \delta + Y \Sigma \sin \alpha \cos \alpha \cos^2 \delta + Z \Sigma \cos \alpha \sin \delta \cos \delta = \Sigma \rho \cos \alpha \cos \delta \qquad \text{.....(81).}$$

* The *observed radial velocity* with which we are concerned here is the radial velocity relative to the sun. But the observation of a star's radial velocity made on the earth includes an effect due to the earth's orbital motion round the sun and an effect due to the earth's rotation (pp. 213–217). We suppose that these effects are removed from the actual observations of radial velocity.

Similarly,

$$X \Sigma \sin\alpha \cos\alpha \cos^2\delta + Y \Sigma \sin^2\alpha \cos^2\delta + Z \Sigma \sin\alpha \sin\delta \cos\delta = \Sigma \rho \sin\alpha \cos\delta \quad \ldots\ldots(82),$$

$$X \Sigma \cos\alpha \sin\delta \cos\delta + Y \Sigma \sin\alpha \sin\delta \cos\delta + Z \Sigma \sin^2\delta = \Sigma \rho \sin\delta \quad \ldots\ldots(83).$$

The coefficients of $X$, $Y$, $Z$ in these equations can be computed and also the quantities on the right-hand sides. Thus we have three linear equations from which $X$, $Y$ and $Z$ are derived. It is to be remarked that $X$, $Y$ and $Z$ are given from this solution in kilometres per second.

The co-ordinates of the solar ant-apex $A$ and $D$ are then given by (76) as in the preceding section.

The solar velocity $U$ is given by

$$U^2 = X^2 + Y^2 + Z^2 \quad \ldots\ldots(84).$$

The treatment given above again neglects the effect of galactic rotation. On account of this, that part of the star's radial velocity that we have called $\rho_2$ will have a systematic component which will depend on the star's direction and on its distance from the sun.

Recent determinations* of the solar motion using both proper motions and radial velocities give the following results:

| | |
|---|---|
| Right Ascension of the Solar Apex | 271°, |
| Declination of the Solar Apex ... | + 30°, |
| The Solar Velocity ... ... ... | 19·7 km per second. |

## EXERCISES

**1.** The parallax of Lalande 21258 is 0″·177 and the total annual proper motion is 4″·52. Show that the star's velocity at right angles to the line of sight is 121 kms. per second. [*Lond.* 1928.]

**2.** If the parallax of a star is 0″·037 and its total annual proper motion is 0″·52, show that its tangential linear velocity is 2·24 times the orbital velocity of the earth, the latter's orbit being assumed circular. [*Lond.* 1925.]

**3.** The parallax of 61 Cygni is 0″·30, and its motion perpendicular to the line of sight is 5″·2 per year. Compare its tangential linear velocity with that of the earth in its motion round the sun.

* For a more detailed discussion of the solar motion and its determination, see D. Mihalis, *Galactic Astronomy*, chapter 5 (Freeman, 1968).

**4.** The total annual proper motion of a star (R.A. $3^h$, declination $+ 10°$) is $0''{\cdot}1$. Assuming that this proper motion is entirely parallactic, determine the star's parallax, given that the solar motion is 19·5 kms. per second towards the point ($18^h$, $+ 34°$). [*Lond.* 1927.]

**5.** A star $(\alpha, \delta)$ has a tangential linear velocity, relative to the sun, of $V$ kms. per second in the direction of a point on the celestial sphere whose right ascension is $A$ and declination is $D$. Prove that

$$\mu_\alpha = 0^s{\cdot}0141\, V\Pi \cos D \sin (A - \alpha) \sec \delta,$$

where $\Pi$ is the star's parallax. [*Lond.* 1926.]

**6.** The R.A. and declination of Sirius are $6^h\,41^m$ and $-16°\,35'$ respectively, and the proper motions in R.A. and declination are $- 0^s{\cdot}0374$ and $- 1''{\cdot}209$; the observed radial velocity is $- 7{\cdot}5$ kms. per second, and the parallax is $0''{\cdot}38$.

Show that the velocity of Sirius relative to the sun is inclined to the line of sight at an angle of about $114\frac{1}{2}°$, and calculate the magnitude of this velocity in kms. per second. [*Lond.* 1929.]

**7.** If $\mu_\alpha$, $\mu_\delta$ are the components of annual proper motion of a star $(\alpha, \delta)$ at time $t$ with reference to a particular equinox and equator, and if $\mu_\alpha'$, $\mu_\delta'$ are the components of proper motion at time $t'$ with reference to the *same* equinox and equator, prove that, $t' - t$ being expressed in years,

$$\mu_\alpha' - \mu_\alpha = \{2\mu_\alpha \mu_\delta \tan \delta - 2V\Pi \mu_\alpha/4{\cdot}74\}(t' - t) \sin 1'',$$
$$\mu_\delta' - \mu_\delta = -\{225\mu_\alpha^2 \sin \delta \cos \delta + 2V\Pi \mu_\delta/4{\cdot}74\}(t' - t) \sin 1'',$$

$\Pi$ being the star's parallax in seconds of arc and $V$ its line of sight velocity in kms. per second.

[$\mu_\alpha$ is expressed in seconds of time and $\mu_\delta$ in seconds of arc.] [*Lond.* 1930.]

**8.** $\mu_\alpha$, $\mu_\delta$ are the components of proper motion, in equatorial co-ordinates, of a star $S(\alpha, \delta)$. If $(l, b)$ are the galactic co-ordinates of $S$ and $\mu_l$, $\mu_b$ the corresponding components of proper motion, show that

$$\mu_l = \mu_\alpha \cos \eta \cos \delta \sec b + \mu_\delta \sin \eta \sec b,$$
$$\mu_b = -\mu_\alpha \sin \eta \cos \delta + \mu_\delta \cos \eta,$$

where $\eta$ is the angle $PSK$, $P$ and $K$ being the north poles of the equator and galactic plane respectively.

[N.B. $\eta$ is to be reckoned positive if, on going round the triangle $SPK$ in the order named, the area of the triangle is on the left.] [*Lond.* 1926.]

**9.** The R.A. of the node of the galactic plane on the equator is $\Omega$, and $i$ is its inclination to the equator.

$(\xi, \eta, \zeta)$ are the components of velocity (in kms. per sec.) of a star $(\alpha, \delta)$ referred to galactic rectangular axes, the direction of the $\xi$-axis being given by the node $N$.

If $\mu_\alpha$, $\mu_\delta$ are the components of annual proper motion in seconds of time and seconds of arc respectively, and $\Pi$ is the parallax of the star, show that

$$\xi \sin(\alpha - \Omega) - \eta \cos i \cos(\alpha - \Omega) + \zeta \sin i \cos(\alpha - \Omega) = -\frac{K}{\Pi} \mu_\alpha \cos \delta,$$

where $K$ is a certain numerical constant, and derive the corresponding equations in $\mu_\delta$ and in the radial velocity. What is the value of $K$ in the system of units employed? [*Lond.* 1926.]

**10.** If $(u, v, w)$ are the components of the linear velocity $V$, referred to equatorial axes for a fixed date, of an open cluster, prove for any cluster star $(\alpha, \delta)$ with components of annual proper motion $\mu_\alpha$, $\mu_\delta$ (both in seconds of arc) that

$$u(\mu_\delta \sin\alpha - \mu_\alpha \cos\alpha \sin\delta \cos\delta) - v(\mu_\delta \cos\alpha + \mu_\alpha \sin\alpha \sin\delta \cos\delta) + w\mu_\alpha \cos^2\delta = 0.$$

Hence show how the co-ordinates $(A, D)$ of the convergent point can be derived. [*Lond.* 1930.]

**11.** Show that if $\theta$ be the angle between the convergent point of a moving cluster and one of its component stars, whose radial velocity is $v$, the velocity of the cluster relative to the sun is $v \sec\theta$, and the distance of the star is $v \tan\theta \div$ observed proper motion. If $v = 45{\cdot}6$ km./sec., $\theta = 60°$, and the P.M. is $1''{\cdot}8$, find the parallax of the star. [*Lond.* 1922.]

**12.** A star has radial velocity $\rho$, parallax $\pi$, and proper motion $\mu$. These quantities are specified respectively in kilometres per second, arc seconds, and arc seconds per year. Show that their annual rates of change are given by

$$\frac{d\rho}{dt} = 4{\cdot}74\frac{\mu^2}{\pi}\sin 1'',$$

$$\frac{d\mu}{dt} = -\frac{\rho\mu\pi}{2{\cdot}37}\sin 1'',$$

$$\frac{d\pi}{dt} = -\frac{\rho\pi^2}{4{\cdot}74}\sin 1''.$$

[*Glas.* 1974.]

**13.** Prove that the space velocity of a star $(\alpha, \delta)$ is directed towards the point on the celestial sphere with equatorial co-ordinates $(A, D)$, where

$$\tan(A - \alpha) = \frac{71{\cdot}1\mu_\alpha}{(\rho\pi - 4{\cdot}74\mu_\delta \tan\delta)}.$$

(Here all symbols have their usual meanings and are measured in their usual units). Derive a formula to determine $D$. [*Glas.* 1973.]

CHAPTER XII

# ASTRONOMICAL PHOTOGRAPHY

**158.** *The photographic refractor.*

The two principal features of the photographic refractor with which we need concern ourselves in this chapter are, first, the object-glass and, second, the photographic plate (Fig. 106). We

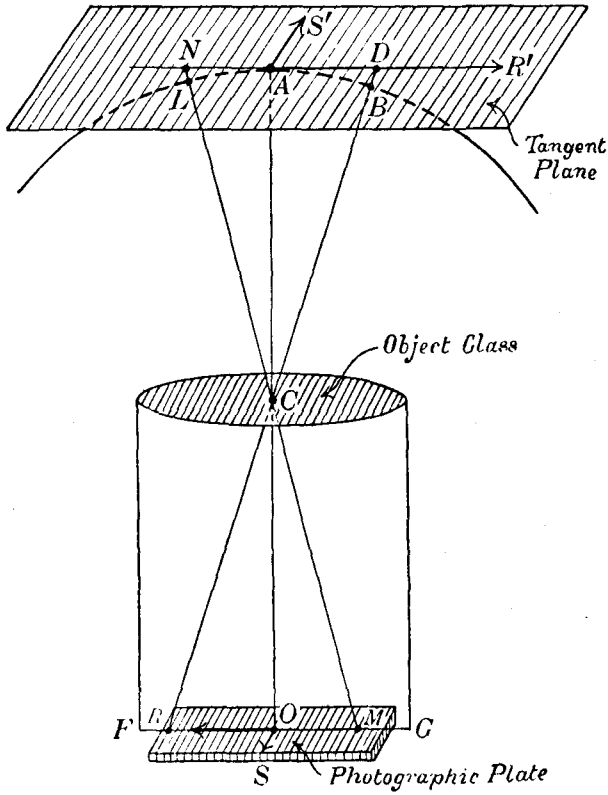



Fig. 106.

shall assume that the object-glass (which is specially designed for photographic purposes) is correctly adjusted in the telescope tube and that $CO$ is the optical axis, $C$ being the centre of the

object-glass. The focal plane of the object-glass is $FG$, at right angles to $CO$. The photographic plate (which we shall assume to be a square) is held in position by a carrier (not shown in the figure) in such a way that the sensitive side of the plate is exactly in the focal plane. We shall also suppose for simplicity that the optical axis $CO$ passes through the geometrical centre of the plate. Consider a star seen in the direction $CA$ ($OC$ produced). All the rays from the star, falling on the object-glass, will be brought to a focus at $O$ and a small circular image of the star will be formed on the photographic plate at this point. Consider another star seen in the direction $CB$. The rays falling on the object-glass from this star will be brought to a focus at $R$ (on $BC$ produced). Let the plane $ACB$ or $RCO$ be parallel to an edge of the plate. Suppose that the image of the star $B$ is just on the plate at $R$. Then if $l$ denotes the length of the side of the plate, $l = 2OR$ and, since $OR = OC \tan RCO = OC \tan ACB$,

$$l = 2OC \tan \beta \qquad \ldots\ldots(1),$$

where $\beta$ denotes the angular distance between the stars $A$ and $B$. $OC$ is of course the focal length of the object-glass and is presumed known. If the dimensions of the plate are known we can, by (1), calculate the area of the sky (in angular measure) which can be photographed. For example, suppose that we have an f19 refracting telescope with a focal length of 589 cm, and the plates used are squares of side 16 cm. (The diameter of the object-glass will be 31 cm.) Then, by (1),

$$\tan \beta = \frac{16}{2 \times 589},$$

so that $\beta = 47'$ approximately. Thus a square region of the sky, $1\frac{1}{2}°$ by $1\frac{1}{2}°$, can be photographed on the plates.

Of great importance is the scale of the plate or the relation between a linear distance on the plate and the corresponding angular displacement in the sky. Thus for the plates just referred to, 1 cm. on the plate is equivalent to $5\frac{5}{8}'$, or 1 mm. is equivalent to $35''$ approximately. We can express this somewhat differently; if two stars are separated by $5\frac{5}{8}'$ in the sky, the centres of their images on the photographic plate will be 1 cm. apart. Similar considerations apply equally to the reflecting telescope when used for photographic purposes.

**159.** *The tangent plane.*

In Fig. 106 the celestial sphere, with $C$ as centre, is drawn. The tangent plane at $A$ is drawn—this plane is at right angles to the radius $CA$ and is therefore parallel to the photographic plate. It is to be remembered that $A$ is the point on the celestial sphere towards which the optical axis of the telescope is directed. Produce $CB$ to meet the tangent plane in $D$; then $D$ will be called the projection of $B$ on the tangent plane. The projection of any other point on the celestial sphere can be constructed in a similar manner by joining the centre $C$ to the point under consideration and producing the radius to meet the tangent plane. Consider a star $L$ whose projection on the tangent plane is $N$ and whose image on the photographic plate is at $M$. If $\phi = O\hat{C}M = A\hat{C}L$, we have

$$\tan \phi = \frac{OM}{OC} = \frac{AN}{AC} \qquad \text{.....(2).}$$

It follows generally that the system of stellar images on the plate is *similar* to the system of the projections on the tangent plane, one system however being on a different linear scale from the other. Let $AR'$, $AS'$ be the positive directions of rectangular axes in the tangent plane; let $OR$, $OS$ be parallel to $AR'$, $AS'$ respectively, in the plane of the plate, their positive directions being opposite to those of $AR'$ and $AS'$. Let $\xi'$, $\eta'$ be the co-ordinates of the projection of a star on the tangent plane and $\xi$, $\eta$ the co-ordinates of the image on the plate; then by the principle of similarity, we have

$$\frac{\xi'}{AC} = \frac{\xi}{OC} \qquad \text{.....(3)}$$

and

$$\frac{\eta'}{AC} = \frac{\eta}{OC} \qquad \text{.....(4).}$$

**160.** *Standard co-ordinates.*

For the sake of geometrical simplicity we shall suppose the celestial sphere (centre $C$) and the tangent plane at $A$ to be drawn as in Fig. 107. It is to be understood that $A$ is the point on the celestial sphere towards which the telescope is pointed. If $S$ is a star near $A$, its projection $T$ on the tangent plane is obtained by joining $C$ to $S$ and producing $CS$ to meet the

tangent plane in $T$. Draw the great circle arc $AS$; then, since the plane of this great circle passes through $C$, it follows that all radii joining $C$ to points on $AS$ lie in one plane and this plane intersects the tangent plane in a straight line $AT$. More generally, we can say that any great circle projects into a straight line in the tangent plane. Let $P$ be the north pole of the celestial sphere. $AP$ is the meridian of $A$ and it projects into the straight line $AQ$. We shall take $AQ$ as the $\eta'$-axis of the tangent plane. The $\xi'$-axis is taken to be $AU$, which is drawn perpendicular to

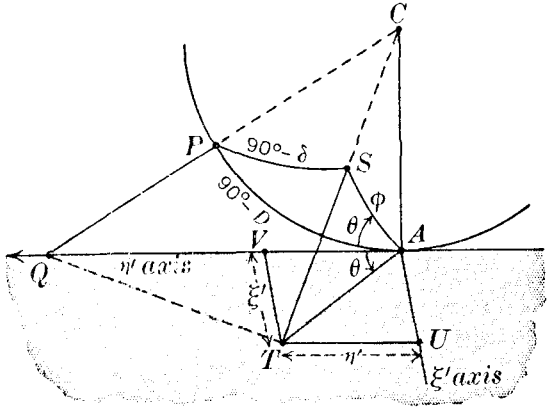



Fig. 107.

$AQ$, and its positive direction is taken to be eastwards of the meridian $AP$ so that increasing values of $\xi'$ correspond to increasing values of the right ascension.

Since $AT$ lies in the tangent plane, $AT$ is perpendicular to $AC$ and is therefore the tangent at $A$ to the great circle arc $AS$. Similarly, $AQ$ is the tangent to the great circle arc $AP$. Now $Q\hat{A}T$ defines the spherical angle $PAS$; hence $Q\hat{A}T = P\hat{A}S$. Thus we see that the angle between any two great circle arcs, intersecting at the tangential point $A$, is exactly reproduced on the tangent plane as the angle between the two straight lines into which the great circles project. This remark holds only for great circles passing through the tangential point; for example, the great circle $SP$ projects into the straight line $TQ$ and $AP$ projects into $AQ$; but $A\hat{Q}T$ (the angle between $AQ$ and $TQ$) is

*not* equal to $S\hat{P}A$ (the angle between the great circles $AP$ and $SP$).

Denote the arc $AS$ by $\phi$ and $S\hat{A}P$ by $\theta$; then $Q\hat{A}T = \theta$. Draw perpendiculars $TU$, $TV$ to $AU$, $AQ$ respectively. Then

$$VT = \xi' = AT \sin\theta \qquad \ldots\ldots(5)$$

and

$$UT = \eta' = AT \cos\theta \qquad \ldots\ldots(6).$$

Now

$$AT = AC \tan A\hat{C}T = AC \tan\phi.$$

Hence

$$\frac{\xi'}{AC} = \tan\phi \sin\theta \qquad \ldots\ldots(7),$$

$$\frac{\eta'}{AC} = \tan\phi \cos\theta \qquad \ldots\ldots(8).$$

Hence, by (3) and (4),

$$\frac{\xi}{OC} = \tan\phi \sin\theta \qquad \ldots\ldots(9)$$

and

$$\frac{\eta}{OC} = \tan\phi \cos\theta \qquad \ldots\ldots(10),$$

in which $\xi$ and $\eta$ are the co-ordinates of the image of $S$ on the photographic plate with reference to rectangular axes through the centre $O$ of the plate (Fig. 106), and drawn parallel, but oppositely directed, to the axes $AU$, $AQ$ on the tangent plane. $OC$ is the focal length of the telescope. Suppose that the focal length is known in millimetres and that the plate co-ordinates $\xi$ and $\eta$ are derived also in millimetres by processes which will be described later; then the values of $\phi$ and $\theta$ can be calculated from (9) and (10). As we shall see immediately, $\phi$ and $\theta$ are functions of the right ascensions and declinations of $A$ and $S$; if the right ascension and declination of $A$ are known, the right ascension and declination of $S$ can then be deduced from the values of the co-ordinates $\xi$ and $\eta$.

If we take the focal length $OC$ to be the unit of length and $\xi$, $\eta$ to be expressed in terms of this unit, we have from (9) and (10),

$$\xi = \tan\phi \sin\theta \qquad \ldots\ldots(11),$$

$$\eta = \tan\phi \cos\theta \qquad \ldots\ldots(12).$$

$\xi$ and $\eta$ are then called the *standard co-ordinates* of the star concerned. In this definition of standard co-ordinates—first

introduced by Professor H. H. Turner*—the following points have to be noted: (i) the origin of the co-ordinate axes corresponds to a definite position, whose right ascension and declination are specified with respect to a standard mean equinox; the epoch of this mean equinox may be chosen as 1900·0; (ii) the $\xi$- and $\eta$-axes are correctly oriented for the epoch 1900·0; (iii) the definition, being a purely geometrical one, excludes the effects of instrumental imperfections and of refraction and aberration (all of which will be considered later). The standard co-ordinates of a particular star thus specify the position of the star uniquely, and can therefore be used in place of right ascension and declination.

**161.** *Formulae for the standard co-ordinates.*

Let $A$, $D$ be the right ascension and declination (referred to 1900·0) of the point $A$ on the celestial sphere, and $\alpha$, $\delta$ the corresponding co-ordinates of the star $S$. We shall now show how the relations between $\xi$, $\eta$ and $A$, $D$, $\alpha$ and $\delta$ are obtained. In the spherical triangle $ASP$ (Fig. 107) we have: $AP = 90° - D$, $SP = 90° - \delta$, $A\hat{P}S = \alpha - A$ (in the figure, $S$ is eastwards of the meridian $AP$), $AS = \phi$, $S\hat{A}P = \theta$. Then by formulae **A**, **B** and **C**, we have

$$\cos\phi = \sin\delta\sin D + \cos\delta\cos D\cos(\alpha - A) \quad \text{.....(13),}$$

$$\sin\phi\sin\theta = \cos\delta\sin(\alpha - A) \quad \text{.....(14),}$$

$$\sin\phi\cos\theta = \sin\delta\cos D - \cos\delta\sin D\cos(\alpha - A) \quad \text{.....(15).}$$

Dividing (15) by (13) we obtain, using (12),

$$\eta = \frac{\cos D - \cot\delta\sin D\cos(\alpha - A)}{\sin D + \cot\delta\cos D\cos(\alpha - A)} \quad \text{.....(16).}$$

Define $q$ as follows:

$$\cot q = \cot\delta\cos(\alpha - A) \quad \text{.....(17).}$$

Then

$$\eta = \frac{\cos D - \sin D\cot q}{\sin D + \cos D\cot q},$$

from which

$$\eta = \tan(q - D) \quad \text{.....(18).}$$

* *Monthly Notices*, vol. LIV, p. 13 (1893).

Again, dividing (14) by (13) and using (11), we have

$$\xi = \frac{\cot\delta \sin(\alpha - A)}{\sin D + \cos D \cot\delta \cos(\alpha - A)} \qquad \ldots\ldots(19)$$

$$= \frac{\cot q \tan(\alpha - A)}{\sin D + \cos D \cot q}$$

by (17), so that

$$\xi = \frac{\cos q \tan(\alpha - A)}{\cos(q - D)} \qquad \ldots\ldots(20).$$

The auxiliary quantity $q$, which has been introduced into (18) and (20) in order to simplify the logarithmic computations of $\xi$ and $\eta$ in the case when $A$, $D$, $\alpha$ and $\delta$ are all known, is readily seen to have a simple geometrical interpretation. Draw a great

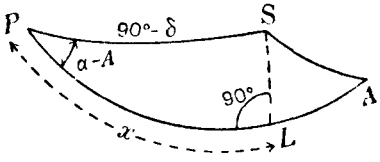



Fig. 108.

circle arc $SL$ to cut $AP$ at right angles in $L$ (Fig. 108). Denote $PL$ by $x$; then by formula **D**,

$$\cos x \cos(\alpha - A) = \sin x \tan\delta - \sin(\alpha - A)\cot 90°,$$

from which $\qquad \tan x = \cot\delta \cos(\alpha - A).$

Hence, by (17), $x = 90° - q$, so that $q$ is the declination of $L$.

The formulae (18) and (20) enable the calculation of $\xi$ and $\eta$ to be made when $A$, $D$, $\alpha$ and $\delta$ are known.

We now derive the formulae which will give $\alpha$ and $\delta$ in terms of $A$, $D$, $\xi$ and $\eta$. We have from (16),

$$\eta \sin D + \eta \cot\delta \cos D \cos(\alpha - A)$$
$$= \cos D - \cot\delta \sin D \cos(\alpha - A),$$

from which

$$\cot\delta \cos(\alpha - A)\{\eta \cos D + \sin D\} = \cos D - \eta \sin D,$$

and hence,

$$\cot\delta \cos(\alpha - A) = \frac{1 - \eta \tan D}{\eta + \tan D} \qquad \ldots\ldots(21).$$

Again, from (19),

$$\cot\delta\sin(\alpha-A)=\xi\sin D+\xi\cos D\cot\delta\cos(\alpha-A)$$
$$=\xi\left\{\sin D+\frac{\cos D(1-\eta\tan D)}{\eta+\tan D}\right\},$$

by means of (21), whence we derive

$$\cot\delta\sin(\alpha-A)=\frac{\xi\sec D}{\eta+\tan D} \qquad \ldots\ldots(22).$$

Divide (22) by (21); then

$$\tan(\alpha-A)=\frac{\xi\sec D}{1-\eta\tan D} \qquad \ldots\ldots(23),$$

from which $(\alpha-A)$ can be calculated and $\alpha$ obtained. When $(\alpha-A)$ has been found, $\delta$ can be obtained from (21) or (22).

In astronomical photography there are two fundamental processes employed directly or indirectly. The first is the calculation of the standard co-ordinates of one or more stars whose right ascensions and declinations are known; this process involves the use of equations (18) and (20). The second is the calculation of the right ascensions and declinations of stars from the values of their standard co-ordinates; this process is carried out by means of (23) and (21) or (22).

**162.** *The measurement and scale of photographic plates.*

For convenience we shall consider the plates taken with astrographic telescopes. These instruments, constructed according to a standard design, were in use in about a score of observatories scattered over the globe, the work to which they have been principally devoted being a complete photographic survey of the heavens. It must be understood, however, that the methods described below are now mainly of historical interest, since the photographic survey is now completed. Modern computer methods that are now being used in astrometric work are highly specialised and, as such, are beyond the scope of this book.

On each astrographic plate, a net-work system of parallel lines (Fig. 109) is photographed, either before or after the plate is exposed to the stars, so that on development the plate shows the stellar images and the réseau system of lines. The lines are equally spaced at intervals of five millimetres. We shall suppose that the central lines $XOY$ and $UOV$ correspond exactly to the

$\xi$- and $\eta$-axes already defined. Consider a star-image at $S$. The distance $AS$, parallel to $OX$, is measured by a machine and we shall suppose that $AS = 4{\cdot}14$ mm. The distance of $S$ from the axis $UOV$ (that is to say $OC$) is thus $10 + 4{\cdot}14$ or $14{\cdot}14$ mm. In a similar way, $BS$ is measured and the distance $OD$ obtained.

Now the standard co-ordinates are expressed in terms of the focal length regarded as the unit; hence the standard $\xi$ co-ordinate

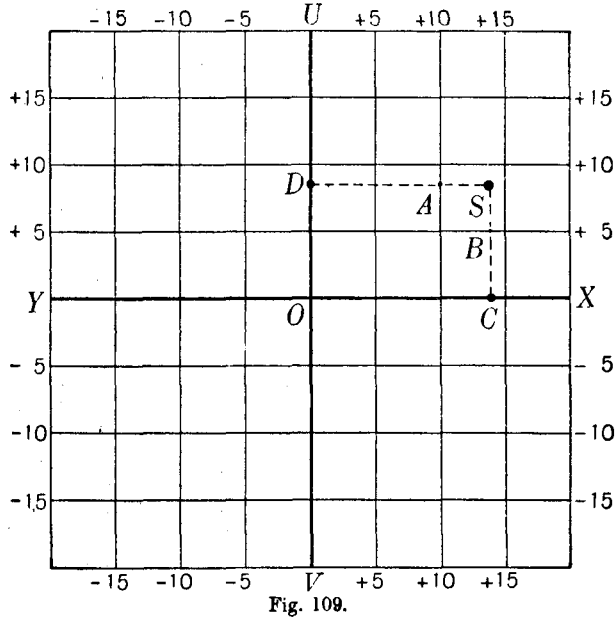


Fig. 109.

of $S$, for example, in Fig. 109 is $OC$ (measured in mm) divided by the focal length (measured in mm). For simplicity, consider two stars (of declination $\delta_1$ and $\delta_2$) on the central meridian $AP$ and let $Y_1$, $Y_2$ (along the $\eta$-axis) be their standard co-ordinates. Then, by (17) and (18),

$$Y_1 = \tan(\delta_1 - D) \quad \text{and} \quad Y_2 = \tan(\delta_2 - D).$$

Assuming that $(\delta_1 - D)$ and $(\delta_2 - D)$ are small angles and expressed in minutes of arc, we have

$$Y_1 = (\delta_1 - D)\sin 1' \quad \text{and} \quad Y_2 = (\delta_2 - D)\sin 1',$$

so that

$$Y_2 - Y_1 = (\delta_2 - \delta_1)\sin 1'.$$

If $d$ is the distance between the images measured in mm and $f$ is the focal length in the same unit, we have $Y_2 - Y_1 = d/f$, and consequently,

$$\delta_2 - \delta_1 = s.d \qquad \ldots\ldots(24),$$

where $s = (1/f)$ cosec 1′; the value of $s$ can thus be regarded as known. $s$ is the "scale" of the plate; it gives the number of minutes of arc corresponding to 1 mm on the plate. Hence, by measuring the distance (in mm) between the images of any two stars, we are enabled by (24) and the known value of $s$ to deduce the angular separation of the two stars in the sky.* The astrographic telescope was designed to give the scale of 1 mm to 1′.

In certain classes of work it is sometimes more convenient to measure small distances on the plate in terms of revolutions of the micrometer attached to the measuring machine. Proceeding as before, we measure the distance between two stellar images in terms of the micrometer scale (say it is 3·456 revolutions), and this number corresponds to the number of minutes (or seconds) of arc by which the stars are separated in the sky. For example, if one revolution of the micrometer head corresponds to 12″·34 on the photographic plate, then the angular separation of the two stars will be 42″·65.

**163.** *The measured co-ordinates.*

In defining the standard co-ordinates of a star we have assumed (*a*) that the optical axis of the telescope passes through the origin of co-ordinates on the plate, (*b*) that the plate is perpendicular to the optical axis, (*c*) that the $\eta$-axis corresponds precisely to the projection of the central meridian, for the epoch 1900·0, on the tangent plane, (*d*) that the $\xi$-axis is perpendicular to the $\eta$-axis. In practice it is impossible to attain the geometrical perfection just indicated, and consequently the co-ordinates of a star-image, measured with reference to the axes on the plate, must be expected to differ (generally slightly) from the theoretical standard co-ordinates. But this is not all. Hitherto, in referring to star-images, we have ignored the effects of refraction and aberration. We have seen in previous chapters that owing to refraction and aberration the apparent position of a star on the celestial sphere is displaced by measurable or

* The focal length, $f$, and the scale of the measuring machine change slightly with temperature; allowance for these changes is made automatically by the "plate constants" $a$ and $e$ (section 167).

calculable amounts from its true position; consequently the actual image of a star on the photographic plate will be somewhat displaced from the position it would occupy were these effects inoperative. It will thus be realised that standard co-ordinates are ideal co-ordinates, whereas the measured co-ordinates of a star-image include the effects of geometrical (or mechanical) imperfections and the effects of refraction and aberration. At first sight the problem of deriving the standard co-ordinates of a star from the measured co-ordinates of its image on the photographic plate seems one of great difficulty; actually, as we shall see, the solution in practice is extremely simple. We shall now examine in detail the differences between the standard and measured co-ordinates.

**164.** *Discussion of errors.*

We shall consider the errors individually. We shall denote by $\xi$ and $\eta$ the true standard co-ordinates of a star and by $x$ and $y$ the co-ordinates as influenced by the particular error concerned.

(*a*) *Error of orientation.*

In Fig. 110 let $XOY$ and $UOV$ be the axes of co-ordinates correctly centred and oriented for the epoch 1900·0; let $X'OY'$

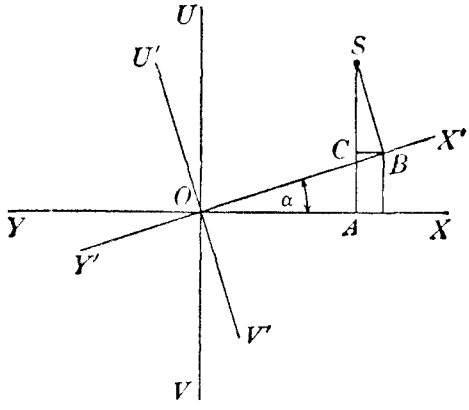



Fig. 110.

and $U'OV'$ be the axes on the plate (the central réseau lines) correctly centred but erroneously oriented. Let $X'\hat{O}X = \alpha$.

Let $S$ be a point whose standard co-ordinates are $\xi$ and $\eta$—referred to $OX$ and $OY$—and whose co-ordinates referred to $OX'$

and $OY'$ are $x$ and $y$. Draw perpendiculars $SA$, $SB$ to $OX$, $OX'$ respectively. Then $OA = \xi$, $AS = \eta$, $OB = x$ and $BS = y$. We have

$$OA = OB \cos \alpha + BS \cos (90° + \alpha),$$

or

$$\xi = x \cos \alpha - y \sin \alpha,$$

whence

$$\xi - x = -2x \sin^2 \frac{\alpha}{2} - y \sin \alpha.$$

Similarly,

$$\eta - y = x \sin \alpha - 2y \sin^2 \frac{\alpha}{2}.$$

We write these in the form:

$$\left.\begin{aligned} \xi - x &= a_1 x + b_1 y \\ \eta - y &= d_1 x + e_1 y \end{aligned}\right\} \quad \ldots\ldots(25),$$

where $a_1, \ldots e_1$ are simple functions of $\alpha$. In practice $\alpha$ is always a small angle and in consequence the coefficients $a_1, \ldots e_1$ are also small. The formulae (25) are essentially linear in $x$ and $y$.

(*b*) *Non-perpendicularity of axes.*

If the $x$-axis on the plate is not exactly perpendicular to the $y$-axis, it is clear that we shall again obtain linear formulae of the type (25) for the quantities $\xi - x$ and $\eta - y$.

A similar result is also obtained if the corresponding axes of the micrometer scale are not exactly perpendicular.

(*c*) *Centering error.*

Suppose firstly that during an exposure the direction of the optical axis corresponds to a given direction $(A, D)$ referred to the mean equinox of 1900·0, and that $\xi$ and $\eta$ are the standard co-ordinates of a star $(\alpha, \delta)$ with reference to the position $(A, D)$ as centre. It is hardly to be expected that the straight line joining the centre of the object-glass to the origin of the impressed réseau co-ordinate axes will coincide exactly with the optical axis, and consequently the origin will correspond to slightly different values of $A$ and $D$. Secondly, the optical axis may not be directed quite accurately to the position $(A, D)$ for 1900·0. As a result, we must therefore assume that the origin of co-ordinates on the plate corresponds to a position $(A + \Delta A, D + \Delta D)$, where $\Delta A$ and $\Delta D$ may be supposed to be small quantities.

All other errors and influences being assumed absent, let $x$ and $y$ be the co-ordinates of an image with respect to the réseau

axes. Then $x$ and $y$ may be taken to be the standard co-ordinates of the star concerned with reference to the position $(A + \Delta A,\ D + \Delta D)$ as centre. We shall denote $x - \xi$ and $y - \eta$ by $\Delta\xi$ and $\Delta\eta$ respectively.

Now, by (11) and (12),

$$\xi = \tan\phi \sin\theta, \quad \eta = \tan\phi \cos\theta,$$

where $\phi$, $\theta$ are functions of $A$ and $D$. Corresponding to increments $\Delta A$ and $\Delta D$, we shall have increments $\Delta\phi$, $\Delta\theta$. Hence we have

$$\Delta\xi = \Delta\phi\,(1 + \tan^2\phi)\sin\theta + \Delta\theta \tan\phi \cos\theta,$$
$$\Delta\eta = \Delta\phi\,(1 + \tan^2\phi)\cos\theta - \Delta\theta \tan\phi \sin\theta.$$

For a star at an angular distance of 1° from $(A,\ D)$, $\tan\phi = 1/57$, and in the above formulae we can neglect such terms as have factors $\Delta\phi \tan^2\phi$. We thus have

$$\left.\begin{aligned}\Delta\xi &= \Delta\phi \sin\theta + \eta\Delta\theta\\ \Delta\eta &= \Delta\phi \cos\theta - \xi\Delta\theta\end{aligned}\right\} \qquad \ldots\ldots(26).$$

We have now to express $\Delta\phi$ and $\Delta\theta$ in terms of $\Delta A$, $\Delta D$.

From (13) we have

$$-\sin\phi\,\Delta\phi = \Delta D\,\{\sin\delta\cos D - \cos\delta\sin D\cos(\alpha - A)\} + \Delta A\cos\delta\cos D\sin(\alpha - A),$$

and, using (14) and (15), this becomes

$$\Delta\phi = -\Delta D\cos\theta - \Delta A\cos D\sin\theta \qquad \ldots\ldots(27).$$

From (14) we obtain

$$\Delta\theta\sin\phi\cos\theta + \Delta\phi\cos\phi\sin\theta = -\Delta A\cos\delta\cos(\alpha - A),$$

or, using (27),

$$\Delta\theta\sin\phi\cos\theta = \cos\phi\sin\theta\,(\Delta D\cos\theta + \Delta A\cos D\sin\theta) - \Delta A\cos\delta\cos(\alpha - A).$$

Multiply (13) by $\cos D$ and (15) by $\sin D$ and subtract. Then

$$\cos\delta\cos(\alpha - A) = \cos\phi\cos D - \sin\phi\cos\theta\sin D.$$

Hence

$$\Delta\theta\sin\phi\cos\theta = \Delta D\cos\phi\sin\theta\cos\theta + \Delta A\,\{\cos\phi\sin^2\theta\cos D - \cos\phi\cos D + \sin\phi\cos\theta\sin D\},$$

from which

$$\Delta\theta\sin\phi = \Delta D\cos\phi\sin\theta + \Delta A\,(\sin\phi\sin D - \cos\phi\cos\theta\cos D).$$

Multiplying this last equation by $\sec\phi\cos\theta$, we have

$$\eta\Delta\theta = \Delta D\sin\theta\cos\theta + \Delta A\,(\eta\sin D - \cos^2\theta\cos D).$$

Inserting this expression for $\eta\Delta\theta$ and the expression for $\Delta\phi$ given by (27) in the first of (26), we obtain

$$\Delta\xi = -\Delta A\cos D + \eta\Delta A\sin D,$$

or
$$\xi - x = \Delta A\cos D - \eta\Delta A\sin D.$$

As $(\xi - x)$ is of order $\Delta A$, we can write this last equation with sufficient accuracy as

$$\left.\begin{aligned}\xi - x &= \Delta A\cos D - y\Delta A\sin D.\\ \text{Similarly,}\quad \eta - y &= \Delta D + x\Delta A\sin D\end{aligned}\right\} \quad \ldots\ldots(28).$$

These formulae have the linear forms:

$$\begin{aligned}\xi - x &= \quad b_2y + c_2,\\ \eta - y &= d_2x \quad + f_2.\end{aligned}$$

(*d*) *Error of tilt.*

This error is due to the non-perpendicularity of the optical axis to the plane of the plate. If $i$ is the angle between the optical axis and the normal to the plate, the expressions for $\xi - x$, $\eta - y$ are of the form

$$\begin{aligned}\xi - x &= \tan i\,(px^2 + qxy),\\ \eta - y &= \tan i\,(pxy + qy^2).\end{aligned}$$

As the angle $i$ is in practice only a few minutes of arc and as the squares and products of $x$ and $y$ only are involved, the correction for tilt can generally be neglected.

The total effect of the various errors considered here is to give the displacements $(\xi - x)$ and $(\eta - y)$ in terms of essentially linear expressions; we can thus write the general formulae

$$\left.\begin{aligned}\xi - x &= ax + by + c\\ \eta - y &= dx + ey + f\end{aligned}\right\} \quad \ldots\ldots(29),$$

in which $a$, $b$, ... $f$ are small quantities depending on the small errors involved.

**165.** *Refraction.*

We now investigate the displacements of the stellar images due only to refraction. In Fig. 111 the tangent plane to the celestial sphere at $A$ is drawn as in Fig. 107. $Z$ is the zenith and

$W$ is its projection on the tangent plane; let the co-ordinates of $W$ be $X$, $Y$. Owing to refraction, a star $S$ is seen at $S'$, the displacement $SS'$ being along the great circle arc joining $S$ to $Z$. We shall use the formula [no. (7) of Chapter III]

$$SS' = k \tan ZS \qquad \text{.....(30),}$$

in which $k$ is expressed in circular measure and $ZS$ is written, without sensible loss of accuracy, in place of the observed zenith distance $ZS'$. The great circle $ZS'S$ projects into the straight line $WT'T$ on the tangent plane, $T$ and $T'$ being the

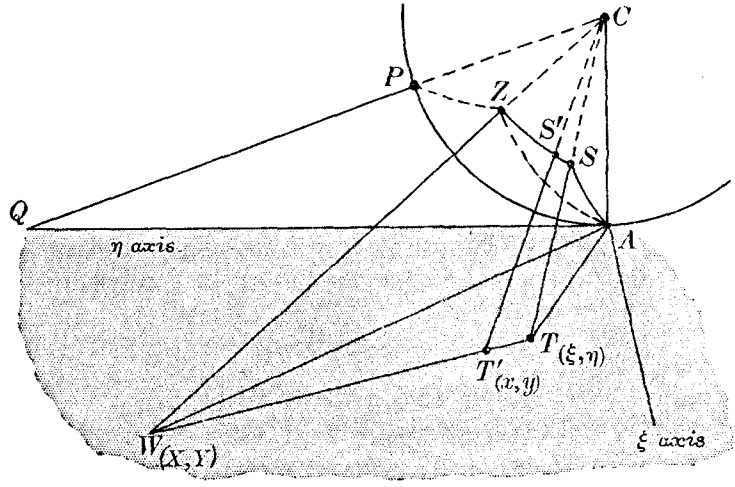



Fig. 111.

projections of $S$ and $S'$ respectively. Let $\xi, \eta$ be the co-ordinates of $T$ and $x, y$ the co-ordinates of $T'$. If $\xi, \eta$ and $x, y$ are expressed in terms of $AC$ as the unit of length, these quantities are the standard co-ordinates of the star and the measured co-ordinates of its image on the plate respectively.

Since the region to be photographed is generally no more than $2° \times 2°$, the different points on that part of the spherical surface concerned are actually very close to the corresponding projected points on the tangent plane; we accordingly assume that $SS' = TT'$. From (30) we have

$$TT' = k \tan ZS \qquad \text{.....(31).}$$

Since $T$, $T'$, $W$ are collinear, we have

$$\frac{x-\xi}{TT'} = \frac{X-\xi}{TW} \text{ and } \frac{y-\eta}{TT'} = \frac{Y-\eta}{TW},$$

and writing $\Delta\xi$ for $(\xi - x)$ and $\Delta\eta$ for $(\eta - y)$ we obtain, from (31),

$$\Delta\xi = -\frac{k(X-\xi)}{TW}\tan ZS \qquad \text{.....(32),}$$

$$\Delta\eta = -\frac{k(Y-\eta)}{TW}\tan ZS \qquad \text{.....(33).}$$

We remind the reader that the different co-ordinates $\xi$, $\eta$, $x$, $y$, $X$, $Y$ are all supposed to be expressed in terms of $AC$ as the unit of length. In particular, $\xi$ and $\eta$ will thus be small quantities and in the sequel we shall neglect the much smaller quantities $\xi^2$, $\xi\eta$, $\eta^2$ and higher powers and products of $\xi$ and $\eta$.

Now from the plane triangle $TAW$, we have

$$TW^2 = AT^2 + AW^2 - 2AT.AW\cos TAW,$$

so that

$$(X-\xi)^2 + (Y-\eta)^2 = (\xi^2+\eta^2) + (X^2+Y^2) - 2AT.AW\cos TAW,$$

which gives us

$$AT.AW\cos TAW = X\xi + Y\eta \qquad \text{.....(34).}$$

From the spherical triangle $ZAS$, we have by the cosine-formula **A**,

$$\cos ZS = \cos AS\cos AZ + \sin AS\sin AZ\cos ZAS$$
$$= \frac{AC}{CT}\cdot\frac{AC}{CW} + \frac{AT}{CT}\cdot\frac{AW}{CW}\cos ZAS.$$

Since $T\hat{A}W$ defines the spherical angle $ZAS$, we have from (34), putting $AC = 1$,

$$\cos ZS = \frac{1 + X\xi + Y\eta}{CT.CW}.$$

**But**

$$CT^2 = 1 + \xi^2 + \eta^2$$
$$= 1, \text{ when we neglect } \xi^2 \text{ and } \eta^2.$$

**Hence**

$$\cos ZS = \frac{1 + X\xi + Y\eta}{CW} \qquad \text{.....(35),}$$

**from which**

$$\sin^2 ZS = \frac{CW^2 - (1 + X\xi + Y\eta)^2}{CW^2}$$
$$= \frac{(CW^2 - 1) - 2(X\xi + Y\eta)}{CW^2}$$
$$= \frac{AW^2 - 2(X\xi + Y\eta)}{CW^2}.$$

Hence, using the binomial theorem, and neglecting $\xi^2$, etc., we obtain

$$\sin ZS = \frac{AW}{CW}\left(1 - \frac{X\xi + Y\eta}{AW^2}\right) \qquad \text{.....(36).}$$

From (35) and (36),

$$\tan ZS = \frac{AW\left(1 - \frac{X\xi + Y\eta}{AW^2}\right)}{1 + X\xi + Y\eta} \qquad \text{.....(37).}$$

Now $TW^2 = (X - \xi)^2 + (Y - \eta)^2$

$= X^2 + Y^2 - 2(X\xi + Y\eta)$, neglecting $\xi^2$, $\eta^2$,

$= AW^2 - 2(X\xi + Y\eta)$;

$$\therefore\ TW = AW\left(1 - \frac{X\xi + Y\eta}{AW^2}\right) \qquad \text{.....(38).}$$

Hence from (37) and (38),

$$\tan ZS = TW(1 + X\xi + Y\eta)^{-1}$$
$$= TW(1 - X\xi - Y\eta).$$

We thus obtain from (32),

$$\Delta\xi = -k(X - \xi)(1 - X\xi - Y\eta),$$

or
$$\Delta\xi = -kX + k\{(1 + X^2)\xi + XY\eta\} \qquad \text{.....(39).}$$

Since $(\xi - x)$ and $k$ (expressed in circular measure) are both small, we can write $x$ and $y$ for $\xi$ and $\eta$ respectively on the right-hand side of (39) without introducing any appreciable error. Write $(\xi - x)$ for $\Delta\xi$; then (39) gives us

$$\Delta\xi \equiv \xi - x = -kX + k\{(1 + X^2)x + XYy\} \text{.....(40).}$$

Similarly, we obtain

$$\Delta\eta \equiv \eta - y = -kY + k\{XYx + (1 + Y^2)y\} \text{.....(41).}$$

The displacements due to refraction for the centre of the plate are $-kX$ and $-kY$, and these quantities, since they appear in the values of $(\xi - x)$ and $(\eta - y)$ for all the images on the plate, may be supposed to be incorporated in the undetermined constants $c$ and $f$ of equations (29). When $-kX$ and $-kY$ are omitted from (40) and (41) the remaining terms express the values of $(\xi - x)$ and $(\eta - y)$ for the *differential refraction*; these equations are then of the linear form

$$\xi - x = ax + by \qquad \text{.....(42),}$$

$$\eta - y = dx + ey \qquad \text{.....(43),}$$

in which, for example, $a = k(1 + X^2)$.

The coefficients $a$, $b$, $d$ and $e$ can be calculated if necessary. It is sufficient to consider the values of $X$ and $Y$ at the middle of the exposure. Let the sidereal time then be $S$. Then as $PZ$ is the observer's meridian (Fig. 111), the hour angle of the equinox $\Upsilon$ is equal to the right ascension of any point that lies on the observer's meridian $PZ$; in particular, the right ascension of $Z$ is $S$. Also $PZ = 90° - \phi$, where $\phi$ is the latitude, so that regarding $Z$ as a particular point on the celestial sphere, its declination is $\phi$. We now apply the formulae (17), (18) and (20). Let $Q$ be defined by

$$\cot Q = \cot \phi \cos (S - A) \qquad \text{.....(44);}$$

then

$$Y = \tan (Q - D) \qquad \text{.....(45),}$$

$$X = \frac{\cos Q \tan (S - A)}{\cos (Q - D)} \qquad \text{.....(46).}$$

The value of $k$ being known—it is $58{\cdot}2 \sin 1''$ in circular measure—we can then deduce the values of $a$, $b$, $d$ and $e$ in (42) and (43). In practice this calculation is hardly ever necessary, for it is sufficient to know that the differences between the standard co-ordinates and the corresponding co-ordinates of the image (as affected by refraction) are given with sufficient accuracy, as a general rule, by linear expressions in $x$ and $y$ with small coefficients $a$, $b$, etc.

However, when the altitude of the celestial region to be photographed is less than 30° or so, the simple formula (30) for the refraction is not sufficiently accurate; also the omission of quadratic terms can no longer be justified. Assuming that the refraction $R$ is given by

$$R = A \tan ZS' + B \tan^3 ZS'$$

[see formulae (29) and (30) of Chapter III] and taking account of second order terms, we can write the formulae for $(\xi - x)$ and $(\eta - y)$ as follows:

$$\xi - x = \alpha x + \beta y + g x^2 + h xy + k y^2,$$
$$\eta - y = \gamma x + \delta y + l x^2 + m xy + n y^2.$$

These give the effect of differential refraction on the standard co-ordinates. The coefficients $\alpha$, $\beta$, ... $m$, $n$ are expressible in terms of $A$ and $B$ and the co-ordinates $X$ and $Y$ of the zenith. In practice, their values are derived without considering the theoretical formulae by which they can be expressed.

**166.** *Aberration.*

The investigation of the effect of aberration on the standard co-ordinates of a star is very similar to that in the previous section. From Chapter VIII, we know that if $F$ is the position on the celestial sphere towards which the earth is moving at the time of the observation, the position of a star is displaced from its true position $S$ to a position $S'$ on the great circle arc $SF$, $S'$ being nearer to $F$ than $S$. The displacement $SS'$ is given by

$$SS' = \kappa \sin FS \qquad \text{.....(47),}$$

where $\kappa$ is the aberration constant whose value in circular measure is 20·5 sin 1″. Confining ourselves to the effects of aberration only, we write, as before, $\xi$ and $\eta$ for the standard co-ordinates of a star and $x$ and $y$ for the co-ordinates of its image on the photographic plate. $F$ is, of course, a definite point on the celestial sphere; we shall suppose that its projection on the tangent plane is $W_1$, with co-ordinates $U$ and $V$. We shall also suppose that $F$, $U$ and $V$ correspond simply to the time of the middle of the exposure. Following the procedure of the previous section, we have the formulae corresponding to (32) and (33),

$$\Delta\xi = -\frac{\kappa(U-\xi)}{TW_1}\sin FS \qquad \text{.....(48),}$$

$$\Delta\eta = -\frac{\kappa(V-\eta)}{TW_1}\sin FS \qquad \text{.....(49),}$$

and from (36) and (38),

$$\sin FS = \frac{AW_1}{CW_1}\left(1 - \frac{U\xi + V\eta}{AW_1{}^2}\right) \qquad \text{.....(50),}$$

$$TW_1 = AW_1\left(1 - \frac{U\xi + V\eta}{AW_1{}^2}\right),$$

so that

$$\frac{\sin FS}{TW_1} = \frac{1}{CW_1}.$$

Now

$$CW_1{}^2 = 1 + U^2 + V^2.$$

We accordingly obtain

$$\Delta\xi = -\frac{\kappa U}{(1+U^2+V^2)^{\frac{1}{2}}} + \frac{\kappa\xi}{(1+U^2+V^2)^{\frac{1}{2}}} \qquad \text{.....(51),}$$

$$\Delta\eta = -\frac{\kappa V}{(1+U^2+V^2)^{\frac{1}{2}}} + \frac{\kappa\eta}{(1+U^2+V^2)^{\frac{1}{2}}} \qquad \text{.....(52).}$$

As in the previous section, we can write $x$ for $\xi$ and $y$ for $\eta$ without any sensible loss of accuracy on the right-hand sides of (51) and (52); also we extract the constant terms (independent of $\xi$ and $\eta$) on the right of these equations. We then obtain the expressions for *differential aberration* in the form

$$\xi - x = a_1 x \qquad \text{.....(53),}$$

$$\eta - y = d_1 y \qquad \text{.....(54),}$$

in which $a_1$ and $d_1$ are small (they have as a common factor 20·5 sin 1″, which is of the order $10^{-4}$). Again it is unnecessary, as a rule, to calculate the values of $a_1$ and $d_1$ from their theoretical expressions.

**167.** *The general relations between standard co-ordinates and measured co-ordinates. (Turner's method.)*

From the previous three sections we have seen that refraction, aberration and the instrumental errors taken separately produce in each co-ordinate a displacement of the image of a star on the plate from the position corresponding to the standard co-ordinates of the star, and that this displacement is given, generally with sufficient accuracy, as a linear expression in the co-ordinates. If we combine all the various effects we clearly obtain linear formulae for $(\xi - x)$ and $(\eta - y)$, which we can write in the general forms

$$\xi - x = ax + by + c \qquad \text{.....(55),}$$

$$\eta - y = dx + ey + f \qquad \text{.....(56),}$$

where $\xi$, $\eta$ are the standard co-ordinates of the star and $x$, $y$ are the measured co-ordinates of its image on the plate. In these equations $a$, $b$, etc. are small and dependent, in a composite way, on the instrumental errors, on refraction and on aberration. Thus in general $x$ differs from $\xi$ by a small quantity and we can write the equations, without loss of accuracy, in an alternative form, namely,

$$\xi - x = a\xi + b\eta + c \qquad \text{.....(57),}$$

$$\eta - y = d\xi + e\eta + f \qquad \text{.....(58),}$$

in which the quantities $a$, $b$, etc. are small. These quantities are called the *plate constants*. We shall now consider three practical applications: (i) the measurement of astrographic plates, (ii) the

measurement of proper motions, (iii) the measurement of stellar parallaxes.

**168.** *The measurement of astrographic plates.*

The precise determination of the right ascensions and declinations of stars brighter than about magnitude 9 can be made directly from meridian observations. Beyond this magnitude the stars are too faint to be observed, owing to the illumination necessary to show up the wires of the instrument. Actually, of course, the observation with this instrument (if it were practicable) of the multitudinous faint stars would be utterly beyond the astronomical resources of the world's observatories even if they were increased a thousandfold in number and employed solely for this purpose. Amongst the different investigations to which the photographic telescope is applied, the study of the positions of the stars is the one to which we first turn our attention.

We shall now suppose that a photograph has been taken of a region, the co-ordinates $A$, $D$ of whose centre are known, and that a system of réseau lines has been impressed on the plate. With the measuring machine the co-ordinates $x$ and $y$ (with reference to the two central lines regarded as the axes of co-ordinates on the plate) of every stellar image can be derived. As the standard co-ordinates $\xi$ and $\eta$ are defined in terms of the focal length as unit of length, we shall suppose that $x$ and $y$ are defined in terms of the same unit; thus if the $x$-measure of an image on the plate is found to be $p$ mm and if $f$ is the focal length in millimetres, we have $x = p/f$.

Having regard to the size of the astrographic field ($2° \times 2°$), we can take it as certain that any plate will contain the images of several stars whose right ascensions and declinations are known from meridian circle observations. Suppose for the moment that there are three such stars. Then by (18) and (20) the standard co-ordinates of these stars, which we shall designate *reference stars*, can be calculated. The measured co-ordinates of the images of the reference stars are also known. Then from (55) we have

$$\left.\begin{aligned} \xi_1 - x_1 &= ax_1 + by_1 + c \\ \xi_2 - x_2 &= ax_2 + by_2 + c \\ \xi_3 - x_3 &= ax_3 + by_3 + c \end{aligned}\right\} \quad \text{.....(59),}$$

in which the suffixes 1, 2, 3 refer to the three stars. These three equations are sufficient to determine the three plate constants $a$, $b$ and $c$. For any other star we have

$$\xi - x = ax + by + c,$$

and, as $x$ and $y$ are known from the measures of the image and the values of $a$, $b$ and $c$ have now been determined, we are therefore able to calculate the value of $\xi$—the standard $\xi$-co-ordinate of the star in question. In a similar way $\eta$ is obtained, the values of the plate constants $d$, $e$ and $f$ having been calculated by means of the three reference stars. If it is desired, the calculation of the right ascension $\alpha$ and of the declination $\delta$ of this star can be carried out by means of (23) and (21). Actually the standard co-ordinates $\xi$ and $\eta$ of the star define its position uniquely with reference to the centre of the region ($A$, $D$) and, except for special objects, the calculation of the equatorial co-ordinates $\alpha$ and $\delta$ need not be undertaken.

In practice, more than three stars are chosen as reference stars. If there are $N$ such stars, we shall have $N$ equations of the form shown in (59). The constants $a$, $b$ and $c$ are then derived from a solution of the $N$ equations according to the method of least squares (or an analogous process). This applies also to the determination of the plate constants $d$, $e$ and $f$.

Our treatment of standard co-ordinates, immediately above, is not complete without further reference to the epoch to which they refer. *The epoch chosen for the standard co-ordinates* of star-places in the astrographic catalogue is 1900·0. Let us suppose that a plate was taken on 1904 March 4. For the epoch 1900·0, the standard co-ordinates of the reference stars can be found from the catalogues. The reduction of the measured co-ordinates of any other star leads to the standard co-ordinates, referred to the mean equator of 1900·0, of the star's position in the sky on the date 1904 March 4. If the proper motion and parallax of the star are negligible, the standard co-ordinates so obtained define the position of the star for the epoch 1900·0. The measured co-ordinates $x$ and $y$ of any one of the reference stars will contain the components $\mu_x$ and $\mu_y$ of proper motion, corresponding to the interval 1900·0 to 1904 March 4. (We denote by $\mu_\alpha$ the component of proper motion along the $\xi$-axis

and by $\mu_y$ the component along the $\eta$-axis; thus $\mu_x = \mu_\alpha \cos\delta$ and $\mu_y = \mu_\delta$, all the quantities concerned being expressed in circular measure.) If $\mu_x$ and $\mu_y$ are known, the effects of $\mu_x$ and $\mu_y$ ought strictly to be removed from the values of $x$ and $y$ before the solution for the plate constants is made.

169. *Photographic observations of minor planets and comets.*

We shall suppose that a photograph of a minor planet (the general procedure is applicable also to comets) is taken at an instant $t$ on a given date—in practice $t$ is taken to be the time corresponding to the middle of the exposure. If we define the standard co-ordinates of selected reference stars with reference to the mean equator and equinox of a standard epoch, say 1950·0, the measurement of the planet's image on the photographic plate and the subsequent reduction, as indicated in section 168, will give the position of the *image* referred to the mean equator and equinox of the same chosen epoch. It is to be remembered that the reduction automatically removes, in particular, the effects of annual aberration at the date in question. The reduced co-ordinates include the effect of the planet's parallax, which depends on the observer's position on the earth. When this effect has been removed, the resulting co-ordinates are called *astrometric co-ordinates.*

The *Astronomical Ephemeris* gives the astrometric positions of the four brightest of the minor planets. The differences between the apparent and the astrometric co-ordinates are also given. We should observe that apparent co-ordinates are readily derived from meridian observations, while the astrometric position is obtained from photographic plate measurements.

The theory of planetary motions yields the geometric position of a body; this is derived from its astrometric position (i) by adding the effects of annual aberration,

$$Cc + Dd \quad \text{in R.A.},$$
$$Cc' + Dd' \quad \text{in declination}$$

[formulae (25) and (26), p. 184], and (ii) by ante-dating the observed time $t$ of observation by $\tau$, the interval required by light to travel from the planet to the earth (section 112).

**170.** *The measurement of proper motions.*

For the accurate determination of proper motions an adequate interval $T$ between the two epochs of observation is necessary. Let us suppose that a region of the sky is photographed on 1904 March 4 and again on 1926 April 3. We can then take the interval $T$ to be 22·1 years. After the measures have been reduced, according to the method of section 168, the first plate will give the standard co-ordinates $\xi_1$, $\eta_1$ (with reference to the mean equator for 1900·0 and to a definite central point with equatorial co-ordinates $A$ and $D$ for this epoch) of the position of a star $X$ in the sky on 1904 March 4. If $x_1$, $y_1$ are the measured co-ordinates of its image on the first plate, we have

$$\xi_1 - x_1 = a_1 x_1 + b_1 y_1 + c_1 \qquad \text{.....(60),}$$

in which $a_1$, $b_1$ and $c_1$ are the constants of this plate, determined from a suitable number of reference stars. Now suppose that the second plate has been measured in a similar way. (We are assuming that a réseau system of lines has been impressed on each plate.) Generally, the constants of the second plate will be different from the constants of the earlier plate; let their values, determined from the reference stars, be $a_2$, $b_2$ and $c_2$. Let $x_2$, $y_2$ be the measured co-ordinates of the image of the star $X$ on the second plate; let $\xi_2$, $\eta_2$ denote the standard co-ordinates (with reference to the mean equator for 1900·0 and to the same central point of the region) of the position of $X$ in the sky on 1926 April 3. Then

$$\xi_2 - x_2 = a_2 x_2 + b_2 y_2 + c_2 \qquad \text{.....(61).}$$

If the star has no proper motion, $\xi_2$ and $\xi_1$ will clearly be the same. If the star has a proper motion with components $\mu_x$ and $\mu_y$ per annum, then, since the difference between $\xi_2$ and $\xi_1$ is the displacement (parallel to the $\xi$-axis) due to proper motion in $T$ years, we have

$$\xi_2 - \xi_1 = T\mu_x \qquad \text{.....(62).}$$

Similarly,

$$\eta_2 - \eta_1 = T\mu_y \qquad \text{.....(63).}$$

The values of $\xi_1$, $\xi_2$, $\eta_1$ and $\eta_2$ are obtained from (60) and (61) and the corresponding equations in $\eta$. Thus the values of $\mu_x$ and $\mu_y$ are found from (62) and (63).

It is to be noticed that in these formulae the unit in which $\mu_x$ and $\mu_y$ are expressed is the same as that by which standard

co-ordinates are defined. In practice, it is much more convenient to express the standard co-ordinates either in millimetres or in terms of the micrometer scale unit, or in minutes or seconds of arc. If we suppose that $\xi$, $\eta$, $x$ and $y$ are all expressed in seconds of arc, we obtain the values of $\mu_x$ and $\mu_y$ expressed in the familiar way, that is, in seconds of arc per annum. Then $\mu_y$ is the same as $\mu_\delta$ (the proper motion in declination) and $\mu_x$ is $\mu_\alpha \cos \delta$, where $\delta$ is the star's declination and $\mu_\alpha$ is the proper motion (in seconds of arc) along the equator.

Instead of using the linear formulae (60) and (61) it may be necessary to include the quadratic terms mentioned at the end of section 165. From the equations of a sufficient number of reference stars the individual constants of the formulae for $(\xi - x)$ and $(\eta - y)$ are derived.

171. *The measurement of proper motions (Kapteyn's method).*

The method just described involves the use of réseau lines as intermediate axes of reference. For each plate, the measure of a co-ordinate $x$ or $y$ of an image requires a setting of the micrometer wire or scale on a neighbouring réseau line and then a setting on the image. For the second plate the procedure is the same. In the method often adopted in practice the work of measuring is reduced by half by eliminating the réseau system altogether. We shall first consider the plan suggested by Kapteyn, as the principle involved is most clearly seen in this application. A plate on which an exposure has been made is not developed, but is carefully stored. After a suitable interval of time, $T$, has elapsed the plate is again placed in the telescope and a new exposure made of the same region. To prevent the earlier and the later images of the same star from coinciding or over-

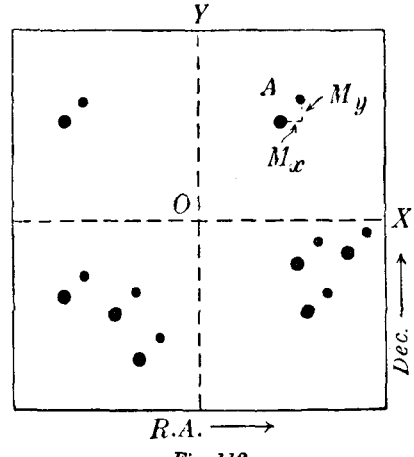


Fig. 112.

lapping, the plate-carrier is given a small displacement perpendicular to the optical axis. The plate is developed after the later exposure. Each star is represented by two images, one corresponding to the earlier epoch and the other to the later. Fig. 112 illustrates this result; we shall suppose that the larger dot at $A$ represents the image of a particular star at the earlier epoch, say, 1900 and the smaller dot the image at the second epoch, say, 1920. If, at the earlier exposure, a bright star is allowed to register a trail across the plate (this is effected by stopping the driving mechanism of the telescope) the trail will define with sufficient accuracy the $\xi$-axis; by means of the trail the plate can then be oriented correctly in the measuring machine.

The procedure consists in measuring the components, $M_x$ and $M_y$, of the displacement between the 1900 and 1920 images. Referred to hypothetical axes on the plate (represented by broken lines in Fig. 112) we shall have, using (57), for the image of $A$ corresponding to the position of the star in 1900,

$$\xi_1 - x_1 = a_1\xi_1 + b_1\eta_1 + c_1 \qquad \text{.....(64),}$$

and, for its image in 1920,

$$\xi_2 - x_2 = a_2\xi_2 + b_2\eta_2 + c_2 \qquad \text{.....(65).}$$

(We cannot assume that $a_1 = a_2$, etc., because the effects of refraction and aberration and the orientation of the plate, for example, are hardly likely to be identical in 1900 and in 1920.) The differences between the standard co-ordinates in 1900 and 1920 will be small—the differences are simply the effects of proper motion in the interval—and as $a_2$, $b_2$ are small, we can replace $\xi_2$ and $\eta_2$ by $\xi_1$ and $\eta_1$ on the right of (65) without sensible error and thus obtain

$$\xi_2 - x_2 = a_2\xi_1 + b_2\eta_1 + c_2 \qquad \text{.....(66).}$$

From (64) and (66) we have by subtraction

$$\xi_2 - \xi_1 = x_2 - x_1 + (a_2 - a_1)\,\xi_1 + (b_2 - b_1)\,\eta_1 + (c_2 - c_1) \text{...(67).}$$

In this equation, $\xi_2 - \xi_1 = T\mu_x$ and $(x_2 - x_1)$ is simply the measured displacement $M_x$. Also we can replace $\xi_1$ by $x_1$ and $\eta_1$ by $y_1$ on the right of (67) without altering the linear character of the equation. We thus have the general expression for $T\mu_x$ in the form

$$T\mu_x = M_x + ax_1 + by_1 + c \qquad \text{.....(68).}$$

There is a similar equation giving $T\mu_y$.

The plate constants $a$, $b$ and $c$ in (68) are determined by means of several stars—the *comparison stars*—scattered over the plate, whose proper motions are presumed to be small; faint stars are more likely to satisfy this condition than the brighter stars. If we assume that the proper motions of all the comparison stars are zero, then to each of these stars corresponds an equation of the type

$$ax + by + c + M_x = 0 \qquad \ldots\ldots(69),$$

where $x$, $y$ are the measured co-ordinates of the star. The values of $x$, $y$ need not be determined with great accuracy, as their factors $a$ and $b$ respectively are small quantities. If there are $N$ comparison stars the solution of the $N$ equations of the type (69) by the method of least squares yields the appropriate values of the plate constants $a$, $b$ and $c$. A similar procedure is adopted for the $N$ equations of the form

$$dx + ey + f + M_y = 0.$$

The substitution of the values of $a$, $b$ and $c$ in the general formula (68) gives for any other star the component $\mu_x$ of the proper motion. The value of $\mu_y$ is obtained similarly.

We have assumed that the proper motions of all the comparison stars are zero; by substituting the values of $a$, $b$ and $c$ appropriate to any comparison star in the equation (68), we can readily see how satisfactory (or otherwise) this assumption really is, for by this process we effectively determine the value of $\mu_x$ for each of the comparison stars. If it should prove that a comparison star has an appreciably large proper motion, it must be discarded and another star selected, if possible. This entails a new calculation of the plate constants. Actually, all the comparison stars cannot be expected to have zero proper motions; and the derived values of $\mu_x$ and $\mu_y$ for the stars in general are the values of their *relative proper motions*, that is, relative to the mean motion of the group of comparison stars.

**172.** *The reduction of relative to absolute proper motions.*

The proper motions of stars derived from meridian circle observations are *absolute proper motions*. Suppose that there is one star on the plate whose absolute proper motions $\mu_x'$ and $\mu_y'$ are known; the plate measures give the relative proper motions $\mu_x$ and $\mu_y$. Then $\mu_x' - \mu_x$ and $\mu_y' - \mu_y$ are the corrections to be

applied to all the relative proper motions to convert them into absolute proper motions. Meridian proper motions are generally not so accurate as one would wish, and therefore the above correction can only be obtained with sufficient accuracy if there are at least about ten stars with well-determined proper motions. This is a condition that is hardly likely to be fulfilled in general and consequently another method of deriving the correction has to be adopted.

We have seen on p. 263 that the components $P_\alpha$ and $P_\delta$ of parallactic motion, in seconds of arc per annum, are given by

$$P_\alpha = H \sin \lambda \sin \chi \sec \delta,$$

$$P_\delta = H \sin \lambda \cos \chi,$$

in which $H$ is the secular parallax of the star, $\lambda$ is its angular distance from the ant-apex of the solar motion and $\chi$ is the position angle of the ant-apex with reference to the star. Writing $P_x$ for $P_\alpha \cos \delta$ and $P_y$ for $P_\delta$ and denoting by $H_m$ the secular parallax of a star of magnitude $m$, we have

$$P_x = H_m \sin \lambda \sin \chi \qquad \ldots\ldots(70),$$

$$P_y = H_m \sin \lambda \cos \chi \qquad \ldots\ldots(71).$$

For the stars in a photographic region we can take the average values of $\lambda$ and $\chi$ to be the values at the centre of the plate. Consider a group of stars, $N$ in number, all of magnitude $m$. Each star will have its own random motion in space but, as it is observed relative to the sun, there will be superimposed the parallactic motion. The observed proper motion will thus consist of the effects of the random motion and of the parallactic motion. Adding up all these effects for the $N$ stars, the effects of the random motions on the proper motions will tend to cancel out, leaving $N$ times the parallactic motion of a star of magnitude $m$ (at the mean distance of the stars in the group) in the region of the sky concerned. Thus taking the average for the $N$ stars, the components of their mean absolute proper motion ought to be very nearly the values of $P_x$ and $P_y$ in (70) and (71). Let $\bar{\mu}_x$, $\bar{\mu}_y$ denote the average values of the relative proper motions of the $N$ stars as obtained from the measurement of the photographic plate. The corrections which we must apply to the components of the relative proper motions of all the stars photographed in

order to convert the relative proper motions into absolute proper motions is evidently given by the quantities

$$P_x - \bar{\mu}_x \text{ and } P_y - \bar{\mu}_y.$$

In practice several magnitude groups are formed and the weighted mean of the corrections derived from the several groups is taken as the final correction to be applied. The following table* illustrates the method. The centre of the photographic region is the star $\epsilon$ Cygni; the values of $\lambda$ and $\chi$ are 147° and 102° respectively; then by (70),

$$P_x = 0{\cdot}53 H_m.$$

The values of $H_m$ are Kapteyn's values corresponding to the galactic latitude of $\epsilon$ Cygni which is $-6°$.

| Group | Mag-nitude $m$ | No. of stars $N$ | $H_m$ | $P_x$ | $NP_x$ | $N\bar{\mu}_x$ | $N(P_x - \bar{\mu}_x)$ |
|---|---|---|---|---|---|---|---|
| 1 | 8·1 | 7 | 0″·0234 | + 0″·01242 | + 0″·087 | + 0″·049 | + 0″·038 |
| 2 | 9·2 | 11 | 0″·0163 | + 0″·00865 | + 0″·095 | + 0″·065 | + 0″·030 |
| 3 | 9·5 | 20 | 0″·0147 | + 0″·00780 | + 0″·156 | + 0″·035 | + 0″·121 |
| 4 | 9·7 | 11 | 0″·0138 | + 0″·00733 | + 0″·080 | + 0″·072 | + 0″·008 |
| | | 49 | | | | | + 0″·197 |

The correction to be applied to all the components $\mu_x$ of relative proper motion to convert them into absolute proper motion is thus + 0″·197 ÷ 49 or + 0″·004. The correction to be applied to the components $\mu_y$ is obtained in a similar way.

**173.** *The film-to-film method of measuring proper motions.*

Kapteyn's method of measuring proper motions, which we have described in section 171, suffers from several disadvantages, amongst which we mention the following: the photographic film is likely to deteriorate during the rather long interval between the early and late exposures, in which event the images of the later exposure will fall below the standard of quality expected in precise work of this kind; the images of the early exposure may be of poor quality owing to bad atmospherical conditions or indifferent "guiding" during the exposure, and this state of affairs is not brought to light until many years later. The method now commonly adopted is as follows. The photograph at the earlier epoch is taken in the usual way and, after a star has

* *Cambridge Astronomical Observations*, vol. xxvi, p. 18 (1928).

been allowed to register its trail, the plate is developed at once. If the plate is unsatisfactory another can be taken to replace it. After a suitable interval has elapsed, a photograph is again taken of the region, but in this case the plate is placed in its carrier with its film-side away from the object-glass. The stellar images are consequently made by the starlight after its passage through the glass of the plate. The plate—we shall call it the "reversed" plate—is developed in the usual way. The plate of the earlier epoch and the reversed plate are now bound together, film to film, by clips allowing for a small displacement of the pairs of images of the several stars. The appearance of the double images is then similar to that in Kapteyn's method as illustrated in Fig. 112. The superposition of the plates must be done with as little rotation of one plate relative to the other as possible. The measurement of the images and the derivation of the proper motions follow the same course as in Kapteyn's method. In practice, several plates are generally exposed on each region at each epoch and at least two separate exposures made on each plate. The material is then adequate to give precise values of the relative proper motions.

**174.** *The determination of the plate constants by the method of Christie and Dyson.*

We shall write equation (68) in the simple form

$$ax + by + c = M - p \qquad \text{.....(72),}$$

in which $p$ is the quantity $T\mu_x$ and the suffix in $M_x$ has been dropped. This equation holds strictly for every star in the region. In order to determine the plate constants by the simple method* proposed by Christie and Dyson, the comparison stars are selected in four equal groups, according to quadrants on the plate, so that the mean centres of position of the groups are as nearly symmetrical about the centre as possible. Assume that the co-ordinates of the group-centres are as in Fig. 113 and that there are $N$ stars in each group.

The values of $p$ in (72) are of course unknown. Let us assume, however, that the selected comparison stars have small relative proper motions, for example, less than 0″·02 per annum.

* *Monthly Notices, R.A.S.* vol. LV, p. 61.

A preliminary solution shows up stars which do not fulfil this condition; they can be removed from the appropriate groups and replaced by others.

Adding the $N$ equations (72) for the comparison stars in group 1, we have

$$N(-aX+bY+c)=\sum_1 p-\sum_1 M \quad \ldots\ldots(73),$$

in which $\sum_1 p$ denotes the sum of the proper motions (parallel to the $x$-axis) of the $N$ stars in group 1 and $\sum_1 M$ denotes the sum of the measured displacements between pairs of images of the $N$ stars. We have similarly, for the other groups,

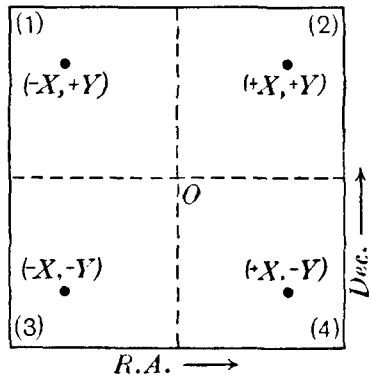


Fig. 113.

$$N(+aX+bY+c)=\sum_2 p-\sum_2 M \quad \ldots\ldots(74),$$

$$N(-aX-bY+c)=\sum_3 p-\sum_3 M \quad \ldots\ldots(75),$$

$$N(+aX-bY+c)=\sum_4 p-\sum_4 M \quad \ldots\ldots(76).$$

Add (73) and (75); then

$$-2NaX+2Nc=\sum_1 p+\sum_3 p-\sum_1 M-\sum_3 M \quad \ldots\ldots(77).$$

Add (74) and (76); then

$$+2NaX+2Nc=\sum_2 p+\sum_4 p-\sum_2 M-\sum_4 M \quad \ldots\ldots(78).$$

Subtracting (77) from (78) we obtain

$$4NaX=\{\sum_2 p-\sum_1 p+\sum_4 p-\sum_3 p\}+\{\sum_1 M-\sum_2 M+\sum_3 M-\sum_4 M\} \quad \ldots\ldots(79).$$

Similarly,

$$4NbY=\{\sum_1 p+\sum_2 p-\sum_3 p-\sum_4 p\}-\{\sum_1 M+\sum_2 M-\sum_3 M-\sum_4 M\} \quad \ldots\ldots(80),$$

$$4Nc=\{\sum_1 p+\sum_2 p+\sum_3 p+\sum_4 p\}-\{\sum_1 M+\sum_2 M+\sum_3 M+\sum_4 M\} \quad \ldots\ldots(81).$$

In these equations $N$, $X$, $Y$ and $\sum_1 M, \ldots \sum_4 M$ are known and

the plate constants $a$, $b$ and $c$ are now derived on the assumption that

$$\sum_2 p - \sum_1 p + \sum_4 p - \sum_3 p = 0,$$

$$\sum_1 p + \sum_2 p - \sum_3 p - \sum_4 p = 0,$$

$$\sum_1 p + \sum_2 p + \sum_3 p + \sum_4 p = 0,$$

from which $\sum_1 p = \sum_4 p$; $\sum_2 p = \sum_3 p$; $\sum_1 p = -\sum_2 p$.

In a group of $N$ stars, some values of $p$ may be expected to be positive and some negative; the value of $\sum_1 p$, for example, consequently may be expected to be small and it will be practically negligible in comparison with the other quantities on the right of (79), ... (81). The third assumption mentioned above—that $\sum_1 p + \sum_2 p + \sum_3 p + \sum_4 p = 0$—is simply another way of expressing the fact that the proper motions derived in this way are *relative proper motions*. They can be converted into absolute proper motions by the method already described in section 172. A similar procedure is adopted for the components parallel to the $y$-axis. The values of the plate constants derived by this method are little inferior in accuracy to the values obtained by the rigorous, but more lengthy, method of least squares.

**175.** *The measurement of stellar parallaxes.*

In Chapter IX we have derived the displacements, in right ascension and declination, of a star due to parallax. We consider here only the displacement in right ascension, measures of which alone suffice in practice. If $\alpha$, $\delta$ are the heliocentric right ascension and declination of a star whose parallax $\Pi$ is to be measured, and $\alpha_1$ is the right ascension as viewed from the earth on a particular date $T_1$, then by (67), p. 222, we have

$$\alpha_1 - \alpha = \Pi F_1 \sec \delta \qquad \ldots\ldots(82),$$

in which $F_1$ is the parallax factor, the value of which can be calculated as the quantities on which it depends are all known. At a later date $T_2$, the heliocentric right ascension of the star will have increased owing to proper motion to $\alpha + T\mu_\alpha$, $T$ being the interval (expressed in terms of the year as unit) between $T_1$ and $T_2$ and $\mu_\alpha$ being the proper motion in right ascension. If

$\alpha_2$ is the geocentric right ascension at the date $T_2$ and $F_2$ the corresponding parallax factor, we shall have

$$\alpha_2 - (\alpha + T\mu_\alpha) = \Pi F_2 \sec \delta \qquad \text{.....(83).}$$

From (82) and (83),

$$\alpha_2 - \alpha_1 = \Pi (F_2 - F_1) \sec \delta + T\mu_\alpha \qquad \text{.....(84).}$$

Let $\xi_1$, $\eta_1$ be the standard co-ordinates (with reference to the centre of the region whose equatorial co-ordinates are $A$, $D$) corresponding to the geocentric co-ordinates $\alpha_1$, $\delta_1$ of the parallax star, and $\xi_2$, $\eta_2$ the standard co-ordinates corresponding to $\alpha_2$, $\delta_2$. Then by (23),

$$\tan (\alpha_1 - A) = \frac{\xi_1 \sec D}{1 - \eta_1 \tan D}.$$

As the parallax star is always chosen to be near the centre of the region, so that $\xi_1$, $\eta_1$ are very small and $D$ differs hardly at all from $\delta_1$ or $\delta$, we can write, with all necessary accuracy,

$$\alpha_1 - A = \xi_1 \sec \delta.$$

Similarly,
$$\alpha_2 - A = \xi_2 \sec \delta.$$

Hence
$$\alpha_2 - \alpha_1 = (\xi_2 - \xi_1) \sec \delta \qquad \text{.....(85).}$$

Denote the proper motion along the parallel of declination $\delta$ by $\mu_x$; then $\mu_x = \mu_\alpha \cos \delta$ or $\mu_\alpha = \mu_x \sec \delta$. Inserting this value of $\mu_\alpha$ in (84) and using (85) we have

$$\xi_2 - \xi_1 = \Pi (F_2 - F_1) + T\mu_x \qquad \text{.....(86).}$$

Now let $x_1$, $y_1$ be the co-ordinates of the image with respect to rectangular axes on the plate taken on the date $T_1$; then by (57),

$$\xi_1 - x_1 = a_1\xi_1 + b_1\eta_1 + c_1.$$

Similarly,
$$\xi_2 - x_2 = a_2\xi_2 + b_2\eta_2 + c_2.$$

Since $\xi_2$ differs very little from $\xi_1$, the difference being due to parallax and proper motion, and as either differs little from $x_1$, $y_1$, we can write the difference between these two equations in the form

$$\xi_2 - \xi_1 = x_2 - x_1 + ax_1 + by_1 + c.$$

In practice the difference $(x_2 - x_1)$ between a pair of images of the parallax star is measured; denote it by $M_x$; then making use of (86) we obtain

$$\Pi (F_2 - F_1) + T\mu_x = M_x + ax + by + c \qquad \text{.....(87),}$$

on the right of which we have dropped the subscripts in $x_1$ and $y_1$.

The plate constants $a$, $b$ and $c$ are derived from the measures of several faint stars, whose parallaxes and proper motions may be regarded as negligible. Each comparison star thus contributes an equation of the type

$$ax + by + c + M_x = 0 \qquad \text{.....(88).}$$

The co-ordinates $x$, $y$ of the comparison stars are supposed known. The solution of the several equations (88) gives the values of $a$, $b$ and $c$.

Generally, the parallax star is chosen as the origin of the $x$ and $y$ co-ordinate axes, so that for this star $x = y = 0$ and (87) becomes

$$\Pi (F_2 - F_1) + T\mu_x = M_x + c \qquad \text{.....(89).}$$

This equation, derived from the measurement of a pair of plates taken on dates $T_1$ and $T_2$ such that the quantity $F_2 - F_1$ is as large as possible, contains two unknowns, the parallax $\Pi$ and the component $\mu_x$ of proper motion (strictly, relative to the mean motion of the group of comparison stars).

Several other pairs of plates (perhaps ten in number) provide a corresponding number of equations of the type (89); these equations, when solved by the method of least squares, give the values of $\Pi$ and $\mu_x$. In practice $M_x$ and $c$ are found in terms of the unit of the micrometer attached to the measuring machine; thus by (89), $\Pi$ and $\mu_x$ are found in terms of this unit. They are converted into seconds of arc by means of the known equivalent of the micrometer unit. [See also p. 411.]

Several methods are employed to obtain the measures $M_x$ between the pairs of images of the parallax and comparison stars. Kapteyn's method, similar in principle to that described for the measurement of proper motions (section 171), has been tried. A second method, adopted at the Royal Greenwich Observatory, depends on the use of reference lines and the principle is the same as if a system of réseau lines were marked on each plate. In some observatories a special kind of measuring machine—the stereo-comparator—enables the measurement of the quantities $M_x$ to be made without the use of any intermediary lines of reference. This instrument is also used for the measurement of proper motions.

As an example, the measures of two pairs of plates, by Dr van

Maanen at Mt Wilson Observatory, of the star "Boss 3650" and other relevant details are summarised in the table below:

| Plate no. | Date | Parallax factor $F$ | $F_2 - F_1$ | $T$ | $M_x + c$ |
|---|---|---|---|---|---|
| 136 | 1914 June 15 | − 0·73 | + 1·64 | 0·64 | − 31 |
| 470 | 1915 Feb. 6 | + 0·91 | | | |
| 486 | 1915 Feb. 7 | + 0·90 | − 1·50 | 0·32 | − 150 |
| 555 | 1915 June 4 | − 0·60 | | | |

The last column contains the values of $M_x + c$ in terms of the measuring-machine unit; one unit = 0″·001649. The first pair of plates gives the equation

$$+ 1{\cdot}64\,\Pi + 0{\cdot}64\,\mu_x = -\ 31,$$

and the second pair gives

$$-1{\cdot}50\,\Pi + 0{\cdot}32\,\mu_x = -\ 150.$$

Eliminating $\mu_x$, we obtain $\Pi = +\ 58$; we then obtain $\mu_x = -\ 197$. Multiplying these numbers by 0·001649, we find

$$\Pi = +\ 0''{\cdot}095,$$

$$\mu_x = -\ 0''{\cdot}315.$$

From the measures of six pairs of plates, van Maanen found

$$\Pi = +\,0''{\cdot}096 \pm 0''{\cdot}003,$$

$$\mu_x = -\,0''{\cdot}302 \pm 0''{\cdot}008.$$

Here 0″·096 is the *most probable value* of $\Pi$ derived from the measures; ± 0″·003 is the *probable error* which indicates the degree of precision with which the value of $\Pi$ has been obtained. Having regard to the inevitable errors of measurement and the like, we interpret the complete result for $\Pi$ by saying that the probability that the *true* value of $\Pi$ lies between the limits 0″·096 + 0″·003 and 0″·096 − 0″·003, that is between 0″·099 and 0″·093, is equal to the probability that it lies outside these limits. The probable error given above is, in fact, 0·6745 times the more usually quoted standard error.

All the methods considered above are based on the use of standard co-ordinates. While these co-ordinates illustrate well the geometrical nature of the problem of imaging part of the celestial sphere on to a flat photographic surface, they are

usually dispensed with in modern astrometric techniques which depend heavily on computer methods. As we have already mentioned, these techniques are too specialised to be considered here, but it should be recognised that some of the methods described in this chapter are now mainly of historical interest. These methods still illustrate, however, the basic principles involved, and they still have considerable relevance in particular problems, such as the immediate determination of the position of a newly discovered asteroid or comet so that a preliminary orbit can be quickly derived.

## EXERCISES

**1.** Two plates are centred at $P$ and $Q$ and $(x, 0)$ and $(\xi, 0)$ are the corresponding standard co-ordinates of a star whose image appears on each plate. Prove that

$$\xi = \frac{(x + c)}{1 - cx},$$

where $c$ is the $\xi$-co-ordinate of $P$ with respect to the plate centred at $Q$.

[The $x$ and $\xi$ axes are the projections on the respective plates of the great circle $PQ$.] [*Lond.* 1930.]

**2.** Show that, to a first approximation, the trail of a star of declination $\delta$ on a photographic plate is

$$y = \text{const.} + \tfrac{1}{2}x^2 \tan \delta,$$

and that the projection of a meridian distant $\Delta\alpha$ from the central meridian makes with the latter the angle $\tan^{-1}(\tan \Delta\alpha \sin D)$, where $D$ is the declination of the plate-centre.

**3.** A photographic plate has been taken of a region of the sky containing a comet and a number of field stars whose proper motion components and whose mean equatorial co-ordinates for 1950·0 are known. Describe in detail how the right ascension and declination of the comet may be determined from measurement of this photographic plate. Would these co-ordinates differ from those determined by transit instrument measurements, and if so what are the causes of these differences? [*Glas.* 1973.]

CHAPTER XIII

# DETERMINATION OF POSITION AT SEA

**176.** *The sextant.*

In this chapter we shall consider the problem of determining a ship's position at sea from observations of heavenly bodies. The instrument used is the sextant, by means of which the altitude of a heavenly body above the sea-horizon can be measured. Fig. 114 is a diagram embodying the more important features of the

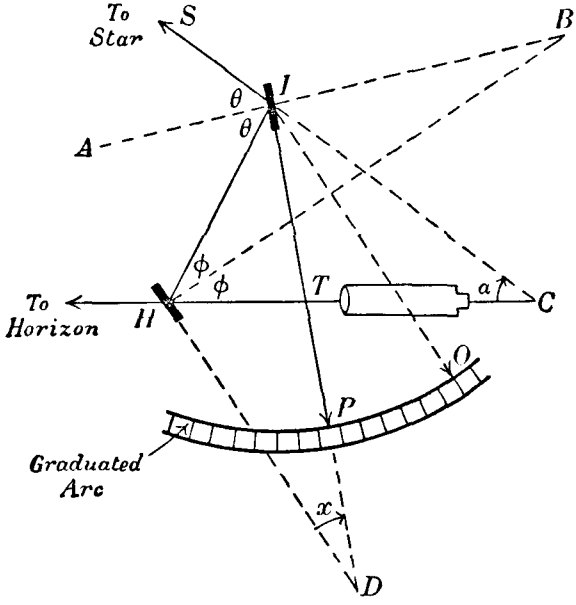



Fig. 114.

instrument. $I$ (called the *index-glass*) represents a mirror perpendicular to the plane of the paper. Rigidly attached to the index-glass is an arm $IP$. The index-glass and the arm $IP$ can rotate about an axis (perpendicular to the plane of the paper) at $I$. The centre of rotation at $I$ is also the centre of a graduated arc $OP$. To any given position of the index-glass and of the arm $IP$ corresponds a reading on the graduated arc. $H$ is a small

rectangular piece of glass, fixed to the framework of the instrument and perpendicular to the plane of the paper; the upper half of its surface is clear and the lower half is silvered, the latter acting as a mirror. $T$ is a small telescope attached to the framework. To find the altitude of a star (for example) above the sea-horizon, the observer holds the instrument in a vertical plane and points the telescope so as to see the horizon through the upper half of $H$ (called the *horizon-glass*). He then moves the index-glass $I$ by means of the arm $IP$ until the image of the star appears in the field of view. When the image appears on the line of the horizon, he notes the reading on the graduated arc.

Let $IS$ denote the direction of the star. A ray in the direction $SI$ is reflected by the mirror $I$ along $IH$; it is then reflected by the mirror portion of the horizon-glass $H$ along $HT$, and the star is thus observed in the telescope in the direction in which the sea-horizon is seen. The altitude of the star is simply related to the inclination of the index-glass $I$ to the horizon-glass $H$ —in Fig. 114 the inclination is $I\hat{D}H$, which we denote by $x$. Let $AIB$ and $HB$ be the normals to the mirrors $I$ and $H$; then $I\hat{B}H$ is evidently $x$. The laws of reflection give

$$S\hat{I}A = A\hat{I}H \equiv \theta,$$

and

$$I\hat{H}B = B\hat{H}C \equiv \phi.$$

If $a$ is the star's altitude above the sea-horizon, then $S\hat{C}H = a$. From the triangle $IHC$, the exterior angle $SIH = I\hat{C}H + I\hat{H}C$, so that

$$2\theta = 2\phi + a \qquad \ldots\ldots(1).$$

Similarly, from the triangle $IBH$,

$$\theta = \phi + x \qquad \ldots\ldots(2).$$

Hence from (1) and (2),

$$a = 2x \qquad \ldots\ldots(3),$$

or the star's altitude is twice the angle between the mirrors $I$ and $H$ (or between their normals). The altitude $a$ is zero when $x$ is zero, that is, when the mirrors $I$ and $H$ are parallel. In Fig. 114, $IO$ is parallel to the fixed direction $HD$; $O$ is the zero-point of the scale. The angle $OIP$ can thus be found from the reading on the graduated arc and, by (3), the star's altitude is twice this angle. The arc of the sextant is generally about one-sixth of the circumference of a circle (hence the name "sextant"), but instead

of having 60 divisions each representing one degree, the arc is divided into 120 divisions. In this way, the altitude is read directly from the scale without the necessity of applying the factor 2 of equation (3). With the aid of sub-divisions and a vernier, altitudes can be read with a first-class instrument to one-tenth of a minute of arc.

**177.** *The errors of a sextant.*

In this section we shall mention very briefly the errors* to which a sextant is liable.

(i) *Error of perpendicularity.* The index-glass $I$ ought to be perpendicular to the plane of the graduated arc. (ii) *Side error.* The horizon-glass $H$ ought to be perpendicular to the plane of the graduated arc. Errors (i) and (ii) can be removed by means of appropriate screws at $I$ and $H$. (iii) *Collimation error.* The line of collimation of the telescope ought to be parallel to the plane of the arc. (iv) *Index error.* When $I$ is parallel to $H$ the pointer $P$ ought to indicate $0°$ on the graduated arc. It is possible to remove the index error by means of a screw at $H$, but the usual custom is to determine the reading when $I$ and $H$ are parallel and to apply the appropriate correction to all altitude readings. The condition of parallelism of $I$ and $H$ is achieved most simply as follows (we assume that errors (i) and (ii) have first been eliminated). With the pointer set near $0°$, the telescope is pointed to a star. Two images of the star are then in the field of view; one is the direct image of the star as seen through the upper half of the horizon-glass $H$ and the other is the image formed as the result of the reflections at $I$ and $H$. With a slow-motion screw, the two images can be superimposed and the reading on the arc then corresponds to the position of $I$ when it is parallel to $H$. As the instrument is not likely to be indefinitely in proper adjustment, the index error should be determined at frequent intervals. (v) *Centering error.* The pivot about which the arm $IP$ revolves ought to be the centre of the graduated arc. In the best instruments this error is generally very small; it varies according to the particular position of the arm $IP$. In

* For the practical methods of eliminating or determining the errors, the reader is referred to more detailed treatises, e.g. *The Admiralty Manual of Navigation* and Doolittle's *Practical Astronomy as applied to Geodesy and Navigation.*

Great Britain, sextants are tested at the National Physical Laboratory before being put on the market, and the values of the centering errors determined there are noted on the sextant box.

**178.** *Corrections to the observed altitude.*

(i) *Dip of the horizon.* In making an observation, the observer's eye is at some distance above sea-level, and consequently the visible horizon will appear somewhat depressed below the horizontal plane, that is, the plane perpendicular to the direction of the observer's zenith. Thus the zenith distance of the visible horizon will be a little over 90°, and all altitudes measured from the visible or sea-horizon will require correction. In Fig. 115 let

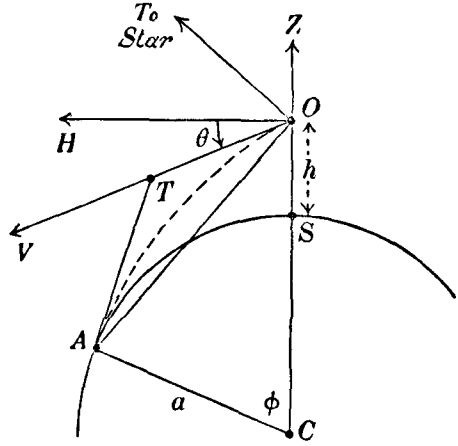



Fig. 115.

$COZ$ be the direction of the zenith of an observer at $O$ at a height $h$ feet above sea-level. Consider the vertical plane $ZOA$ containing the direction of the star observed. Let $OH$ be perpendicular to $OZ$ in this plane. Let $A$ be the point of the sea-surface corresponding to the visible horizon. Owing to atmospherical refraction the path of a ray from $A$ to $O$ will be slightly curved and the direction in which $A$ is actually observed will be along $OV$, the tangent at $O$ to the dotted curve representing the path of the ray. The angle $HOV$ (denoted by $\theta$) is called the *dip of the horizon*. As $A$ is the most distant point visible, the path of the

ray will be tangential to the earth's surface at $A$. Consequently if $AT$ is a tangent at $A$ to the curved path of the ray, $AT$ is perpendicular to $CA$. Since $h$ is small compared with the earth's radius $a$, we assume that the curved path is approximately a circular arc; hence, as $OV$ and $AT$ are tangents to this arc, then $T\hat{O}A = O\hat{A}T$. Denote $O\hat{C}A$ by $\phi$. It is usually assumed that $T\hat{O}A$ or $O\hat{A}T$ is a constant fraction of $\phi$. We can then write

$$O\hat{A}T = \beta\phi \qquad \text{.....(4)},$$

where $\beta$ is a constant numerical factor whose value is deduced to be 1/13 approximately. We now have: $O\hat{A}C = 90° - \beta\phi$; $A\hat{O}C = 90° - (\theta + \beta\phi)$, and hence

$$90° - \beta\phi + 90° - (\theta + \beta\phi) + \phi = 180°,$$

from which $$\phi\,(1 - 2\beta) = \theta \qquad \text{.....(5)}.$$

The earth's radius being denoted by $a$ (in feet) we have, from the triangle $AOC$,

$$\frac{\sin(90° - \beta\phi)}{a + h} = \frac{\sin(90° - \theta - \beta\phi)}{a},$$

or $$\frac{\cos\beta\phi}{\cos(\theta + \beta\phi)} = 1 + \frac{h}{a} \qquad \text{.....(6)},$$

whence $$\frac{2\sin(\theta/2)\sin\frac{1}{2}(\theta + 2\beta\phi)}{\cos(\theta + \beta\phi)} = \frac{h}{a}.$$

As $\theta$ and $\phi$ are small angles, we can write this last equation as

$$\theta(\theta + 2\beta\phi) = 2h/a,$$

or, using (5), $$\theta^2 = 2(1 - 2\beta)\,h/a \qquad \text{.....(7)}.$$

Inserting the value of $\beta$ and expressing $\theta$ in minutes of arc, we have

$$\theta = \sqrt{\frac{22h}{13a}}\ \text{cosec}\ 1'.$$

Now $a = 3960 \times 5280$ feet and cosec $1' = 3438$. Hence we obtain

$$\theta = 0{\cdot}98\,(h)^{\frac{1}{2}} \qquad \text{.....(8)},$$

or, as a sufficiently good working rule: "The dip, in minutes of arc, is equal to the square root of the height, above sea-level, in feet". For example, if the observer's eye is 36 feet above sea-level the angle of dip, $\theta$, is practically 6′·0.

As the observed altitude of the heavenly body is measured with respect to $OV$—the direction of the sea-horizon—the angle of dip is to be subtracted from the observed altitude; thus the altitude with respect to $OH$—the direction of the theoretical horizon—is obtained.

(ii) *Astronomical refraction.* We have seen in Chapter III that the effect of atmospherical refraction is to make the heavenly body appear nearer the zenith than it would be if the atmosphere were non-effective in deviating the rays of light in their passage through the air. The observed altitude is thus too great by the amount of refraction $R$ given by formula (7) of Chapter III,

$$R = k \tan \zeta.$$

(This formula is sufficiently accurate for navigational purposes.)

(iii) *Semi-diameter* (S.D.). In observations of the sun, moon and the nearer major planets, the altitude of the centre of the disc cannot be accurately measured directly; the observation consists in measuring the altitude of the upper or lower limb of the body concerned, and applying to the observed altitude the value of the semi-diameter given in the almanacs. In this way, the altitude of the centre of the sun, moon or planet is derived.

(iv) *Parallax.* As we have seen in Chapter IX, the effect of parallax is to make the observed altitude less than it would be if the dimensions of the earth were negligible in comparison with the distance of the heavenly body observed. If $P$ denotes the horizontal parallax of the body and $a$ its altitude (previously corrected for dip, refraction and semi-diameter), the correction due to parallax is given with sufficient accuracy for navigational purposes by $P \cos a$, which is to be added to the observed altitude. This correction is important only in the case of the moon.

When the index error and the corrections (i) to (iv) have been applied, in the order indicated, to the observed altitude we obtain, by subtracting this corrected altitude from 90°, the true zenith distance of the heavenly body.

As regards star observations, only the corrections (i) and (ii) are of practical moment. In nautical tables,* the sum of these

* For example, *Inman's Tables*; Bowditch's *Practical Navigator*.

corrections can be taken directly from a simple double-entry table.

As regards the sun, simple tables have been prepared to enable the total effect of corrections (i) to (iv) to be found by inspection.

Corrections (iii) and (iv) for the planets are small and in nautical observations are usually neglected.

*Example.* To find the true zenith distance of the moon's centre at 1931 March 24, $10^h$ U.T.; given $h = 25$ feet; observed altitude of the moon's lower limb = 32° 20′·0 (corrected for index error). From the *Nautical Almanac*, S.D. = 15′·2; $P$ = 55′·8.

| | | |
|---|---|---|
| Obs. altitude ($a$) | | 32° 20′·0 |
| Dip | − | 5′·0 |
| Refraction | − | 1′·5 |
| Semi-diameter | + | 15′·2 |
| Parallax ($P \cos a$) | + | 47′·0 |
| Hence the corrected altitude = | | 33° 15′·7 |

and thus the true zenith distance is 56° 44′·3.

**179.** *The position circle.*

An essential part of the navigator's observation concerns the exact U.T. at which the observation of altitude was made. For this purpose a reliable chronometer keeping U.T. as nearly as possible is necessary. The daily radio time-signals enable the navigator to determine the error of the chronometer at suitable intervals; consequently, when he measures the altitude of a heavenly body he knows the exact U.T. at which the observation was made. The various corrections in section 178 being applied, a complete observation yields two definite pieces of information:

(i) the true zenith distance of the heavenly body,
(ii) the U.T. of the observation.

Now, from the U.T. it is easy to find the position on the earth's surface at which the body concerned is exactly overhead at the moment of observation. Let $U$ be this point in Fig. 116, which represents the earth with its centre at $C$. At the moment of observation the body is thus in the direction $CUS$. $CP$ is the direction of the north pole and therefore $S\hat{C}P$ is the north polar distance of the heavenly body, that is, the great circle arc $PU$

is $(90° - \delta)$, in which $\delta$ is the declination of the body at the U.T. of the observation. But $PU$ is the colatitude of $U$ and thus the latitude of $U$ is simply the declination of the heavenly body. Again, if $PGQ$ is the Greenwich meridian, $G\hat{P}U$ is the longitude (measured west of Greenwich) of $U$. But $G\hat{P}U$ is also equal to the angle between the celestial meridian of the heavenly body and the celestial meridian corresponding to the Greenwich meridian $PGQ$. But this latter angle is simply the hour angle of the heavenly body, at the particular U.T., with respect to the Greenwich meridian. Thus the longitude of $U$ (measured westwards) is the hour angle of the heavenly body at Greenwich at the U.T. concerned. This hour angle can be calculated from the data in the almanacs. Accordingly, the position of $U$ on the earth's surface can be definitely specified. $U$ is known as the *geographical position* (or the sub-solar or sub-stellar point, in the case of the sun or a star respectively) of the body observed at the particular U.T.

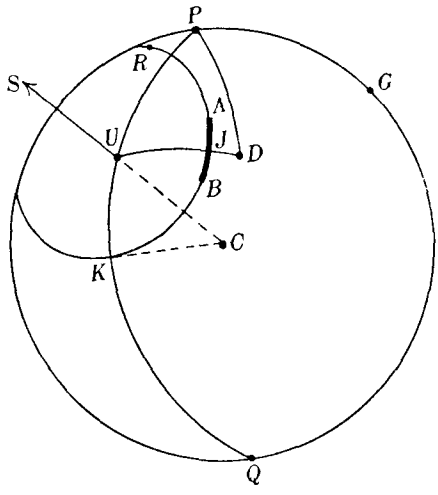


Fig. 116.

Consider now the observation of altitude from which the true zenith distance $z$ is derived. If the observer were actually at such a point as $K$ on the earth's surface, the direction of his zenith would be along the radius $CK$; also the direction of the heavenly body at the time of observation is along the radius $CU$; hence $K\hat{C}U$ or the great circle arc $UK$ is the true zenith distance $z$. Now since the U.T. of the observation is known, the geographical position $U$ is a definite point on the earth's surface; the true zenith distance $z$ is also known, and therefore the observer must be situated somewhere on a small circle $KJR$ of which $U$ is the pole, every point of this small circle being at

an angular distance $z$ from $U$. This small circle is called the *position circle*. It is to be noted that a single observation of a heavenly body leads only to a certain small circle, on which the observer is situated. If we suppose that a similar observation of another heavenly body is made at the same U.T., a second position circle will be derived on which the observer must be situated. Therefore his actual position must be at one or other of the two points of intersection of the two position circles. As the approximate position of the ship is always known, there is no difficulty in deciding which of the two points is the correct position.

The approximate position of the ship is obtained by the navigator by the process of *dead reckoning*. He knows with fair accuracy the ship's various speeds and courses since leaving harbour, or since the last determination of position. He can also estimate, perhaps with moderate success, the effects of winds and currents on his progress. Using these data, he plots on a chart the ship's position hour by hour, and when he makes an astronomical observation he obtains from his chart the position according to dead reckoning—the D.R. position, so called. We shall suppose that in Fig. 116 $D$ represents the D.R. position of the ship at the time of the altitude observation. It is to be noted that the D.R. position is, at best, only an approximate estimate; after 24 hours or more without astronomical observations it is not unlikely that the D.R. position may be in error by 10 miles or more.

**180.** *The position line (St Hilaire's method).*

As we have seen, the altitude observation of a heavenly body yields the information that at the time of observation the ship is situated on a certain small circle $KJR$ (Fig. 116); at this moment the estimated position of the ship is at $D$. It is clear that the only part of the position circle with which the navigator need concern himself is that part in the immediate neighbourhood of $D$. His object then is to represent on his chart this part $AJB$ of the position circle (shown with a heavy line in Fig. 116). Now the latitude and longitude of $U$ and of $D$ are known; hence the length of the great circle arc $UD$ can be calculated. But this arc is simply the angle between the radii $CU$ and $CD$, and as $CD$ produced gives the direction of the zenith at the particular

point $D$, the arc $UD$ is thus the zenith distance of the heavenly body at the U.T. of the observation for a hypothetical observer situated at $D$. We shall call this zenith distance $UD$ the *calculated zenith distance*. Now the length of the arc $UJ$ is known from the observation—it is the true zenith distance $z$. Hence by subtraction we obtain the length of the arc $DJ$. This arc $DJ$ is known as the *intercept*. Expressed in minutes of arc, it gives the distance in nautical miles of the D.R. position $D$ from the nearest point $J$ on the position circle. The arc $DJ$ is perpendicular to the position circle at $J$, for $U$ is the pole of $KJR$.

Also, the spherical angle $UDP$ is easily seen to be the azimuth of the heavenly body, at the U.T. concerned, for an observer at $D$; it can be calculated or found by inspection in such tables as Burdwood's *Azimuth Tables*. In Fig. 116 the azimuth of $J$ is the same as the azimuth of $U$, which we now suppose to be known. Thus, under the circumstances depicted in Fig. 116, where the calculated zenith distance $UD$ is greater than the true zenith distance $UJ$, the navigator can draw from the D.R. position on his chart a straight line in the direction given by the azimuth of the heavenly body; he then marks off along this line a distance equal to the intercept, and through the point so obtained he draws a straight line, called the *position line*,* perpendicular to the line of azimuth. The position line represents on his chart (which we will consider in greater detail in section 182) the portion $AJB$ of the position circle $KJR$ (Fig. 116). It is clear that if $D$ is within the position circle, that is, if the calculated zenith distance is less than the true zenith distance, the intercept will be marked off in the direction opposite to that given by the azimuth.

**181.** *Example of the calculation of the intercept.*†

The observed altitude of the sun's lower limb was 17° 27′·0 at $16^h\ 31^m\ 2^s$ U.T. on 1931 March 10, the height of the observer's eye above sea-level being 25 feet and the index error of the sextant being −2′·0. The ship's estimated position $D$ by dead reckoning was: Lat. 48° 15′ N, Long. 7° 28′ W. To calculate the intercept.

* Sometimes called the *Sumner line*.

† We retain U.T. in this chapter although G.M.T. is used in navigation circles, e.g. in the *Nautical Almanac* (for use in navigation).

The longitude 7° 28′ W is equivalent, in time-measure, to $29^m\ 52^s$ W.

For the U.T. and date, we have from the almanac:

| | |
|---|---|
| Sun's declination | 4° 18′·0 S |
| Sun's Greenwich H.A. | $4^h\ 20^m\ 29^s$ |
| Sun's semi-diameter | 16′·1 |

In Fig. 117, $Z$ is the zenith of the ship's estimated position. Thus $PZ = 41°\ 45'$ (the colatitude). Also $X$ is the position of

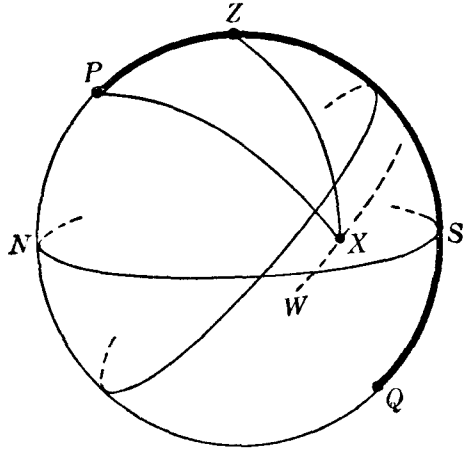



Fig. 117.

the sun's centre on the celestial sphere and $PX$ is 94° 18′·0. The *angle ZPX is the hour angle* of the sun (H.A.T.S.). We shall use the haversine formula (section 13) to compute $ZX$ (the calculated zenith distance). From equation (23) on page 19 we can write the haversine formula in this case as

$$\text{hav}\, ZX = \text{hav}\,(PX - PZ) + \text{hav}\,\theta,$$

where $\text{hav}\,\theta$ is a short-hand notation, defined by

$$\text{hav}\,\theta = \sin PX \sin PZ \,\text{hav}\, ZPX.$$

| | | | | |
|---|---|---|---|---|
| G.H.A.T.S. | $4^h\ 20^m\ 29^s$ | | | |
| Long. (W) | $-\ 29^m\ 52^s$ | | | |
| H.A.T.S. | $3^h\ 50^m\ 37^s$ | log hav | | $\bar{1}$·366 40 |
| $PX$ | 94° 18′·0 | log sin | | $\bar{1}$·998 78 |
| $PZ$ | 41 45 ·0 | log sin | | $\bar{1}$·823 40 |
| | 52° 33′·0 | log hav $\theta$ | | $\bar{1}$·188 58 |

| | |
|---|---|
| hav $\theta$ | 0·154 37 |
| hav 52° 33′·0 | 0·195 97 |
| hav $ZX$ | 0·350 34 |

$\therefore$ $ZX$ = 72° 35′·0 (the calculated zenith distance).

The reader may verify this calculation by means of the cosine-formula. We now find the true zenith distance.

| | |
|---|---|
| Observed alt. of sun's lower limb ... | 17° 27′·0 |
| Index error ... ... ... ... | − 2·0 |
| Dip (for height of eye, 25 feet) ... | − 5·0 |
| Refraction ... ... ... ... | − 3·1 |
| Semi-diameter ... ... ... | + 16·1 |
| $\therefore$ Corrected observed altitude ... | = 17° 33′·0 |
| Thus the true zenith distance is ... | 72° 27′·0 |
| But the calculated zenith distance is | 72° 35′·0 |

**Hence the intercept is 8′·0.**

The sun's azimuth is 118° W or S 62° W. (We leave this calculation as an exercise to the reader.)

Since the calculated zenith distance is greater than the true zenith distance (72° 27′·0), the position line on the chart (Fig. 118) is obtained by drawing from $D$ (the estimated position)

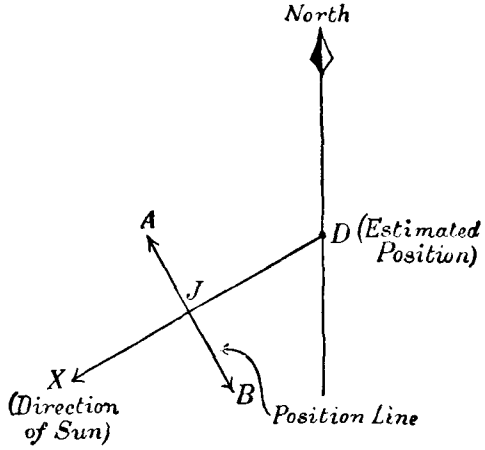



Fig. 118.

a line $DX$ in the direction S 62° W, and taking a point $J$ such that $DJ$ represents eight nautical miles; the straight line $AB$

drawn through $J$ at right angles to $DX$ is the position line on which the ship's position must be at the time of observation. This position line is also shown later in Fig. 122.

**182.** *Mercator's chart.*

Unless a ship is manœuvring, its course is generally constant for several hours at least; in these circumstances, its track on the earth's surface is called a *rhumb line* or a *loxodrome*, which can be more precisely defined as a curve on the earth's surface such that the tangent at any point of the curve cuts the meridian through that point at a constant angle. In Fig. 119, $AXZB$ is a rhumb line cutting all the meridians between $A$ and $B$ at a

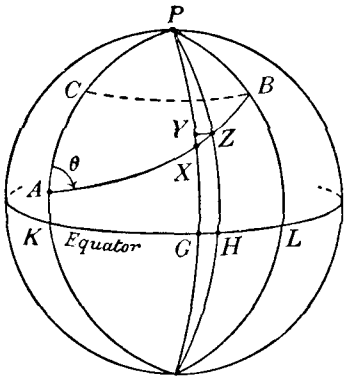



Fig. 119.

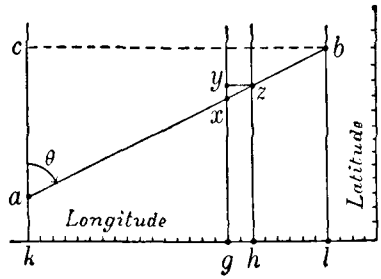



Fig. 120.

constant angle $\theta$. It is evident that a projection of the earth's surface on which rhumb lines are represented by straight lines would be of inestimable convenience to the navigator, for then the track of his ship would be *represented* on the chart by a straight line. Such a chart is Mercator's projection. We now state the two main principles of its construction. (i) All rhumb lines on the surface of the earth are represented by straight lines on the chart. (ii) The angle between any two intersecting rhumb lines is correctly represented on the chart; for example, if two rhumb lines intersect at an angle of 30°, then on the chart the angle between the two corresponding straight lines is also 30°. Now the equator is a rhumb line, and all meridians of longitude are rhumb lines intersecting the equator at right angles. Hence on the chart the equator will be represented by a straight line

(*kl* in Fig. 120) and the meridians *KAP* and *LBP*, for example, will be represented by parallel straight lines *ka*, *lb* perpendicular to *kl*. On the chart the meridians are equally spaced, so that the length of *kl* is proportional to the difference of longitude between *K* and *L* (or between *A* and *B*). Again, all parallels of latitude are rhumb lines intersecting the meridians at right angles; hence on the chart parallels of latitude such as *CB* will be represented by straight lines parallel to *kl* (for example, *cb* in Fig. 120) and perpendicular to *ka* and *lb*.

Consider two neighbouring points *X* and *Z* on the rhumb line *AB* (Fig. 119). Draw *YZ*, a parallel of latitude. We can regard *XYZ* as a small plane triangle in which $Y\hat{X}Z = \theta$ (the constant course along the rhumb line). If $\Delta L$ denotes the angle *XPZ* (the difference of longitude between *X* and *Z*) and $\phi$ the latitude of *Z*, then $YZ = \Delta L \cos \phi$. If the latitude of *X* is $\phi - \Delta\phi$, then $XY = \Delta\phi$. Now $YZ = XY \tan \theta$, so that

$$\Delta\phi \sec \phi = \Delta L \cot \theta \qquad \text{.....(9),}$$

in which we suppose that $\Delta\phi$ and $\Delta L$ are expressed in circular measure. Now on the chart the meridians of *X* and *Z* are represented by the straight lines *gx* and *hz*, and the rhumb line *AB* is represented by the straight line *ab* making the angle $\theta$ with the meridians. The distance *gh* represents the difference of longitude $\Delta L$ between the meridians *GXP* and *HZP*. Let us suppose, for example, that the scale of the chart is chosen so that one minute of arc of longitude is represented by one millimetre on the chart; then if $\Delta L$ is equivalent to *n* minutes of arc, the length of *gh* will be *n* mm. But since $\Delta L$ is expressed in circular measure, we have $\Delta L = n \sin 1'$. Hence (9) becomes

$$\Delta\phi \sec \phi = n \cot \theta \sin 1' \qquad \text{.....(10).}$$

In the small triangle *xyz* (Fig. 120), which represents on the chart the small triangle *XYZ* on the sphere, we have $\hat{yxz} = \theta$ and $yz \equiv gh = n$ mm. Denote the length of *xy* in millimetres by $\Delta y$. Then

$$\Delta y = n \cot \theta \qquad \text{.....(11).}$$

From (10) and (11), putting $\sin 1' = 1/3438$, we obtain

$$\Delta y = 3438 \sec \phi \Delta\phi \qquad \text{.....(12).}$$

If the difference of latitude between *X* and *Z* is 1′, so that $\Delta\phi$ (in circular measure) is 1/3438, then by (12) $\Delta y = \sec\phi$ mm.

It is thus seen that as $\phi$ increases the actual distance (measured in millimetres) between the parallels of latitude $\phi$ and $(\phi + 1')$ on the chart increases from the equator northwards or southwards, becoming infinite when the north and south poles are reached; for this reason, the north and south poles cannot be represented on the Mercator chart. On the chart, the latitude scale is marked on one or more lines parallel to *lb*.

From (12), by integration, we have for the length of *lb* (expressed in millimetres), which we denote by $y_2$,

$$y_2 = 3438 \log_e \tan\left(\frac{\pi}{4} + \frac{\phi''}{2}\right) \qquad \text{.....(13),}$$

in which $\phi''$ is the latitude of $B$. Similarly, if $y_1 = ka$ and $\phi'$ is the latitude of $A$,

$$y_1 = 3438 \log_e \tan\left(\frac{\pi}{4} + \frac{\phi'}{2}\right) \qquad \text{.....(14).}$$

These formulae give the distances on the chart between the equator and the parallels of latitude, on which $A$ and $B$ are situated, in millimetres, that is to say, in terms of the distance, as unit, on the longitude scale corresponding to one minute of arc of longitude; this latter scale is engraved on *kl* or on one or more lines parallel to *kl*.

We now consider the problem of deriving, from measures made on the chart, the distance in nautical miles between any two positions on the earth measured along the rhumb line joining them. Consider the neighbouring points $X$ and $Z$; on the chart they are represented by $x$ and $z$. The triangles $XYZ$ and $xyz$ are similar; hence

$$XZ : XY = xz : xy = \sec\theta : 1 \qquad \text{.....(15).}$$

If $XY = 1'$ (one nautical mile), then by (15) $XZ = \sec\theta$ nautical miles. Now in the triangle $xyz$ the distance $xy$ *represents* one nautical mile and since, by (15), $xz = xy \sec\theta$, the measurement of $xz$ will give the number of nautical miles between $X$ and $Z$, provided $xz$ is measured in terms of $xy$ as the unit. The latitude scale thus provides the unit by which a distance measured on the chart gives correctly in nautical miles the rhumb-line distance between the corresponding points on the earth. As this unit varies in length according to sec $\phi$, the process of finding the number of nautical miles represented by a line such as *ab* on

the chart is theoretically complicated, for it involves the division of *ab* into a large number of sections and deriving the number of nautical miles in each section by reference to the unit of the latitude scale immediately opposite. In practice, however, it is sufficiently accurate to divide *ab* into sections of 30 or 40 nautical miles in length and to measure each in terms of the latitude unit opposite the mid-point of the section concerned.

**183.** *Determination of the ship's position from two observations of altitude.*

We consider now the general problem of determining the ship's position from two observations, making due allowance for the "run" of the ship between the observations, which we shall suppose to be made at times $t_1$ and $t_2$. Let $D$ (Fig. 121) on the

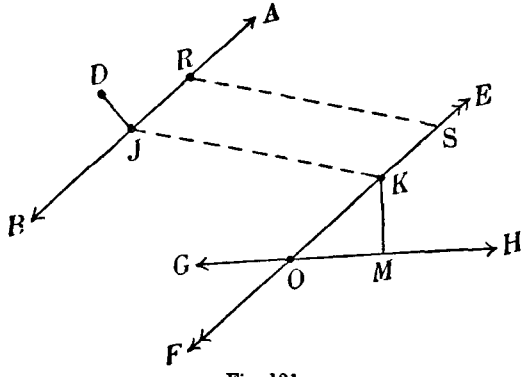



Fig. 121.

chart be the estimated (D.R.) position at $t_1$. Computing the zenith distance of the heavenly body observed at $t_1$ for the latitude and longitude of $D$, and using the corrected observed altitude, we obtain the intercept. Then having found the azimuth of the body we can draw on the chart the position line at $t_1$, according to the procedure of sections 180 and 181. Let $AB$ be the deduced position line at $t_1$; then the position of the ship a $t_1$ is on the line $AB$. Let $JK$ be parallel to the ship's course and equal in length to the distance travelled in the interval $(t_2 - t_1)$. Through $K$ draw $EF$ parallel to $AB$. Take any point $R$ on $AB$ and draw $RS$ parallel to $JK$ to meet $EF$ in $S$. If the ship were actually at $J$ at time $t_1$, then its position at $t_2$ would be at $K$.

Similarly, if its position at $t_1$ were at $R$, its position at $t_2$ would be at $S$. Hence it is evident that, as the ship's position at $t_1$ is somewhere on $AB$, its position at $t_2$ must lie on $EF$. This line $EF$ is called the *transferred position line.*

We now utilise the second observation made at $t_2$. We can use $K$ as the estimated position at this time and, calculating the intercept as before, we obtain the position line $GH$ on which the ship's position must lie at $t_2$. But the first observation and the application of the ship's run place the ship on $EF$ at $t_2$. Hence its position at $t_2$ is at $O$, the intersection of $EF$ and $GH$.

The ship's true position at $t_1$ (if it is required) is obtained by drawing through $O$ a line parallel to $JK$; the point of intersection of this line with $AB$ is the ship's position at $t_1$.

It is evident that we can utilise observations of the same body made at $t_1$ and $t_2$ (for example, the sun), provided the change of azimuth in the interval is such that the two derived position lines do not intersect at too small an angle.

We illustrate the principles of this section in the following example involving observations of the sun and a star.

**184.** *Example of finding the ship's position from two observations.*

The estimated position of a ship steaming N 84° E at 8 knots was Lat. 48° 15′ N, Long. 7° 28′ W at $16^h\ 31^m\ 2^s$ U.T. on 1931 March 10. The following observations were made:

At $16^h\ 31^m\ 2^s$ U.T., observed altitude of sun's lower limb was 17° 27′·0.

At $18^h\ 46^m\ 10^s$ U.T., observed altitude of Betelgeuse was 48° 55′·0.

To find the ship's position at $18^h\ 46^m$ U.T., given that the index error of the sextant is $-2'{\cdot}0$ and that the height of eye is 25 feet.

The first observation is that considered in detail in section 181. We now plot the results on the chart (Fig. 122). $D$ is the estimated position given; $DJ$ is the intercept 8′·0 drawn in the direction S 62° W and $AB$ is the position line at $16^h\ 31^m$ U.T.

The interval between the two observations is $2\frac{1}{4}$ hours; the run is therefore 18′·0. We draw $JK$ in the direction N 84° E (the ship's course) and make $JK$ equal to eighteen divisions of the latitude scale opposite. $EF$ is drawn through $K$ parallel to $AB$; $EF$ is the transferred position line on which the ship must

be situated at $18^h\ 46^m$ U.T. We take $K$ as the estimated position at this latter time, and use this position to derive the position line given by the observation of Betelgeuse. From the chart we find that $K$ is the point, Lat. 48° 13′ N, Long. 7° 12′·5 W.

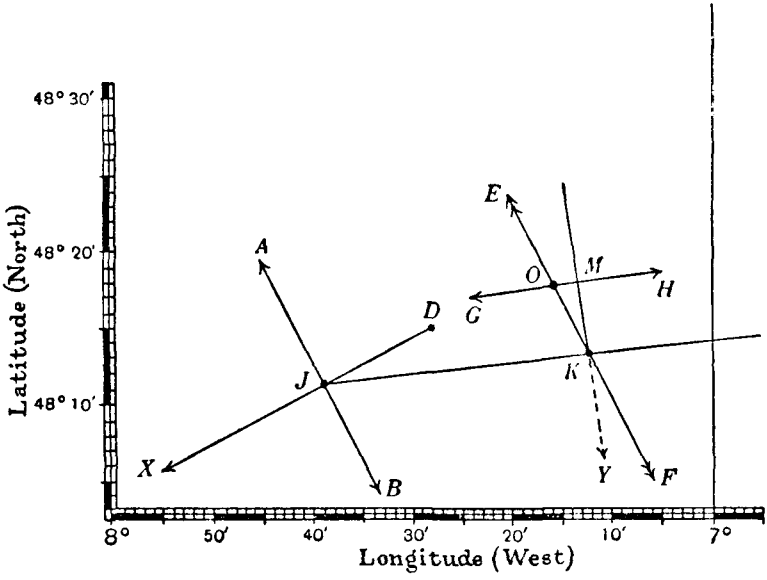



Fig. 122.

We first calculate the hour angle of the star as follows (its right ascension and declination are $5^h\ 51^m\ 27^s$ and + 7° 23′·8 respectively):

| | | | |
|---|---|---|---|
| G.S.T. at $0^h$ U.T. | $11^h$ | $06^m$ | $49^s$ |
| U.T. of observation | 18 | 46 | 10 |
| | 29 | 52 | 59 |
| U.T. to G.S.T. conversion | | +3 | 05 |
| G.S.T. of observation | 29 | 56 | 04 |
| Long. (W) | | −28 | 50 |
| L.S.T. | 29 | 27 | 14 |
| R.A. star | 5 | 51 | 27 |
| H.A.* | 23 | 35 | 47 |

* The value of the star's Greenwich hour angle can be obtained directly from the *Nautical Almanac* and the procedure is then slightly different. The working given here is required if the data is taken from the *Astronomical Ephemeris*.

If $X$ denotes the position of Betelgeuse on the celestial sphere and $Z$ the zenith of the point $K$, we have

$PX = 82° 36'·2$; $PZ = 41° 47'·0$; $Z\hat{P}X \equiv 24^h - \text{H.A.} = 0^h 24^m 13^s$.

We calculate $ZX$ by the cosine-formula:

| | | | |
|---|---|---|---|
| log cos $PX$ | $\bar{1}·109\ 71$ | log sin $PX$ | $\bar{1}·996\ 37$ |
| log cos $PZ$ | $\bar{1}·872\ 55$ | log sin $PZ$ | $\bar{1}·823\ 68$ |
| | | log cos $Z\hat{P}X$ | $\bar{1}·997\ 57$ |
| | $\bar{2}·982\ 26$ | | $\bar{1}·817\ 62$ |

Hence $\cos ZX = 0·096\ 00 + 0·657\ 08 = 0·753\ 08$.

$\therefore\ ZX = 41° 8'·5$.

The azimuth is found by calculation to be 171° E, so that the bearing of the star is S 9° E—in the direction $KY$ in Fig. 122.

We now correct the observed altitude:

| | |
|---|---|
| Observed alt. of star... ... ... | 48° 55'·0 |
| Index error ... ... ... ... | − 2·0 |
| Dip ... ... ... ... ... | − 5·0 |
| Refraction ... ... ... ... | − 0·9 |
| ∴ corrected observed alt. ... = | 48° 47'·1 |
| Thus the true zenith distance is ... | 41° 12'·9 |
| But the calculated zenith distance is | 41° 8'·5 |
| **Hence the intercept is 4'·4.** | |

As the calculated zenith distance is smaller than the true zenith distance, the point $K$ is evidently within the position circle given by the observation of Betelgeuse. Hence to obtain the position line on the chart we draw $KM$ equal to the intercept (4'·4), and in the direction *opposite* to that given by the star's azimuth. The line $GH$ drawn through $M$ perpendicular to $KM$ is the position line resulting from the observation of the star.

The ship's position at $18^h 46^m$ U.T. is given by $O$, the intersection of $GH$ with $EF$. From the chart it is found that $O$ is the position: Lat. 48° 17'·8 N, Long. 7° 16' W.

**185.** *Special methods.*

When the pole star is observed the position line can be obtained very easily by means of simple tables based on the formula in Exercise 20, p. 55. These tables are given in the *Nautical Almanac*, the *Astronomical Ephemeris* and also in

*Inman's Tables.* Since the bearing of the pole star is in general very nearly north, the resulting position line is practically a parallel of latitude; thus an observation of the pole star yields the ship's latitude.

When the hour angle of the heavenly body observed is within 30 or 40 minutes from the meridian, simple tables based on the formula of Exercise 21, p. 55, facilitate the calculation of the zenith distance. Such observations are known as *ex-meridian* observations.

In the general problem, the arithmetical computations can be greatly lightened by means of special tables, amongst which may be mentioned *Altitude Tables** by R. de Aquino and *Position Line Tables (Sine Method)** by W. M. Smart and F. N. Shearme, the general principles of which are used in other tables. The position circle on the earth is deducible from two items of observation: (i) the observed altitude of the heavenly body concerned, (ii) the U.T. at which the observation is made. As the position circle is independent of the estimated position of the ship, so also is the position line on the chart, and although we employ the estimated position ($D$) to derive the position line, any other point $C$ within 30 or 40 miles of $D$ would serve equally well. In the sine method, a particular point $C$ is selected in a way that can be best illustrated by means of an example. Suppose that the ship's estimated position $D$ is: Lat. 48° 39′ N, Long. 7° 18′ W, and that the hour angle of the heavenly body computed for $D$ is $2^h\ 18^m\ 32^s$. We choose the latitude of $C$ to be the integral number of degrees nearest to that of $D$ and its longitude such that the hour angle for $C$ is the nearest multiple of $4^m$. In this example, the latitude of $C$ is 49° N and its longitude such that the hour angle for $C$ is $2^h\ 20^m$; hence $C$ must be $1^m\ 28^s$ or 22′·0 *east* of $D$, so that its longitude is 6° 56′ W. The point $C$ is plotted on the chart. The choice of $C$ in the general case—its latitude a whole number of degrees and its longitude such that the hour angle of the heavenly body is a multiple of $4^m$—allows the tabulation in a compact form of certain quantities which are used in the subsequent calculations.

In Fig. 123 let $Z$ be the zenith of $C$, $X$ the heavenly body (declination $\delta$) and $H$ the hour angle for the longitude of $C$. We

* Published by J. D. Potter, 145 Minories, London, E. 1.

shall first suppose that this hour angle is within $6^h$ of the meridian. Draw the great circle arc $ZY$ to cut the great circle $PX$ at right angles. Let $PY$ and $ZY$ be denoted by $U$ and $p$ respectively, and let $\phi$ denote the latitude of $C$. From the right-angled triangle $PZY$, in which $PZ = 90° - \phi$, we have

$$\tan U = \cot \phi \cos H \qquad \text{.....(16).}$$

$$\cos p = \sin \phi \sec U \qquad \text{.....(17).}$$

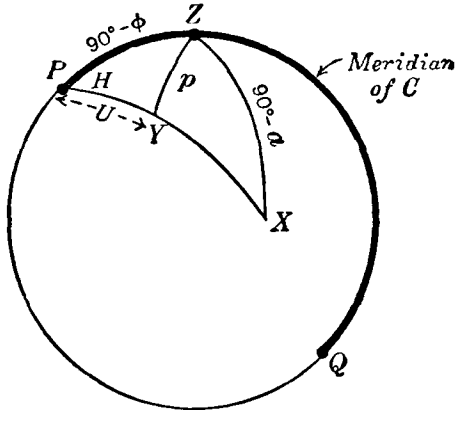



Fig. 123.

Let the zenith distance $ZX$ be denoted by $(90° - a)$; we call $a$ the calculated altitude. From the triangle $ZYX$, in which $YX = PX - U = 90° - \delta - U$, we have

$$\sin a = \sin (\delta + U) \cos p \qquad \text{.....(18).}$$

The values of $U$, calculated by means of (16), for each degree of latitude and at intervals of $4^m$ in the hour angle, from $0^h\ 0^m$ to $6^h\ 0^m$ or from $18^h\ 0^m$ to $24^h\ 0^m$, are given in the tables. Adjacent to each of these entries, the corresponding values of $\log \cos p$, given by (17), are tabulated. The computation of the calculated altitude $a$ follows simply from (18). The difference between $a$ so derived and the observed altitude (corrected) is the intercept, which, of course, has to be drawn from $C$ on the chart. In this way the position line can be drawn with a minimum of arithmetical computation.

When the hour angle ($H$) of the heavenly body is more than $6^h$ from the meridian, a slight modification is necessary. Let $H_1$

denote $(12^h - H)$ or $(H - 12^h)$. The quantities $\log \cos p$ and $U$ are taken from the tables for the appropriate value of $H_1$ and the formula corresponding to (18) is

$$\sin a = \sin (\delta - U) \cos p.$$

**186.** *Equation of a great circle on Mercator's chart.*

The great circle distance between two points on the earth's surface is shorter than the rhumb-line distance, and the difference becomes of economic importance in long oceanic voyages. In such instances it is the practice to follow the appropriate great circle as nearly as possible. Suppose that the great circle is represented on the chart. It is divided into a suitable number of sections, and between the beginning and end of a section the ship follows the appropriate rhumb-line course (represented by the straight line joining the ends of the section), which is easily obtained from the chart. When the ship reaches the end of a section, its course is altered to that corresponding to the next section; and so on.

To find the equation of the curve representing the great circle joining two points $U$ and $V$ (Fig. 124) we proceed as follows. Let the great circle $UV$ cut the equator in $O$ and let the longitude (measured east) of $O$ be $L_0$, $PG$ being the Greenwich meridian. Let the inclination of $UV$ to the equator (in the direction of increasing east longitudes) be denoted by $i$; $i$ can have all values between 0° and 180°. Let $L$ and $\phi$ denote the east longitude and north latitude respectively of any point $X$ on the great circle $UV$. Let $x$ and $y$ be the co-ordinates of $X$ on the chart; we shall suppose that $x$ and $y$ are expressed in terms of the radian as unit. Then we have

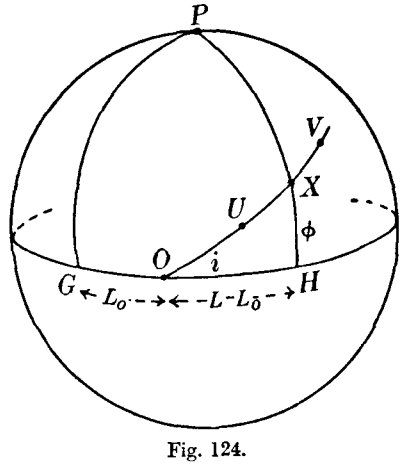


Fig. 124.

$$x = L \qquad \text{.....(19)},$$

where $L$ is expressed in circular measure, and, by (13),

$$y = \log_e \tan\left(\frac{\pi}{4} + \frac{\phi}{2}\right) \qquad \text{.....(20).}$$

From the spherical triangle $XOH$, in which $H$ is the intersection with the equator of the meridian through $X$, we have: $OH = L - L_0$, $HX = \phi$, $X\hat{O}H = i$ and $X\hat{H}O = 90°$. Formula **D** gives

$$\tan \phi = \sin (L - L_0) \tan i \qquad \text{.....(21).}$$

But (20) can be written

$$e^y = \tan\left(\frac{\pi}{4} + \frac{\phi}{2}\right) = \frac{1 + \sin \phi}{\cos \phi} = \frac{\cos \phi}{1 - \sin \phi},$$

Hence

$$\sec \phi + \tan \phi = e^y,$$

$$\sec \phi - \tan \phi = e^{-y},$$

from which

$$\tan \phi = \tfrac{1}{2}(e^y - e^{-y}) = \sinh y \qquad \text{.....(22).}$$

Hence, using (19) and (22) with (21), we obtain the equation of the great circle in the form

$$\sinh y = \sin (x - L_0) \tan i \qquad \text{.....(23),}$$

in which we regard $L_0$ and $i$ as two constants associated with the great circle. The values of $L_0$ and $i$ can now be expressed in terms of the longitudes $L_1$ and $L_2$ of $U$ and $V$ respectively, and the corresponding latitudes $\phi_1$ and $\phi_2$. From (21), we have

$$\left.\begin{aligned} \tan \phi_1 &= \sin (L_1 - L_0) \tan i \\ \tan \phi_2 &= \sin (L_2 - L_0) \tan i \end{aligned}\right\} \qquad \text{.....(24).}$$

By division,

$$\frac{\tan \phi_1}{\tan \phi_2} = \frac{\sin (L_1 - L_0)}{\sin (L_2 - L_0)},$$

from which we obtain

$$\frac{\tan \phi_2 - \tan \phi_1}{\tan \phi_2 + \tan \phi_1} = \frac{\sin (L_2 - L_0) - \sin (L_1 - L_0)}{\sin (L_2 - L_0) + \sin (L_1 - L_0)},$$

giving, on further simplification,

$$\tan\left(\frac{L_2 + L_1}{2} - L_0\right) = \tan\left(\frac{L_2 - L_1}{2}\right) \cdot \frac{\sin (\phi_2 + \phi_1)}{\sin (\phi_2 - \phi_1)} \qquad \text{.....(25).}$$

The value of $L_0$ is calculated from (25). $\tan i$ is then obtained from one of the formulae (24).

To find the position of a point on the curve with any assumed longitude $L$, we calculate the corresponding latitude $\phi$ by means of (21). The $x$ and $y$ co-ordinates are then given by (19) and (20).

The plotting of a great circle track on the Mercator chart can be much simplified by means of the Gnomonic Projection.

## EXERCISES

**1.** Using the effect of refraction as given in section 178, p. 318, prove that the distance, in nautical miles, of the horizon for an observer $h$ feet above sea-level is

$$\left(\frac{26}{11}\frac{h}{a}\right)^{\frac{1}{2}} \operatorname{cosec} 1',$$

where $a$ is the earth's radius in feet.

**2.** An observer, on the mast of a ship, 80 feet above sea-level, can just see a light which is 100 feet above sea-level. Show that his distance from the light is $21\frac{3}{4}$ nautical miles.

**3.** Show that in a place whose latitude is $\phi$ sunrise at the equinoxes will be visible at the top of a mountain $h$ feet high about $4\sqrt{h}\sec\phi$ seconds sooner than at its foot. [*Coll. Exam.*]

**4.** The path of the setting sun makes an angle $\theta$ with the horizon. Prove that if the sun's declination is $\delta$ a mountain in latitude $\phi$, of height $\frac{1}{n}$ of the earth's radius, will have its summit illuminated

$$12\sqrt{2}\operatorname{cosec}\theta\sec\delta/\pi\sqrt{n} \text{ hours}$$

after the sun has set on the plain at its base.

Determine to the nearest minute the value of this expression at the summer solstice for a mountain three miles high in latitude 45°. [*Coll. Exam.*]

**5.** The angle of depression of the sun's upper limb at setting is observed from an eminence to be $d$, and from a neighbouring eminence $h$ feet higher to be $d + \Delta d$, where $\Delta d$ is expressed in seconds of arc. Prove that the earth's radius in feet is approximately

$$\frac{h\cos d\cot d}{\Delta d\sin 1''}.$$

**6.** If $s$ is the length of the rhumb line joining two points in latitudes $\phi_1$ and $\phi_2$ and $\theta$ is the angle which the rhumb line makes with the meridians, prove that

$$s = a(\phi_2 - \phi_1)\sec\theta,$$

where $a$ is the earth's radius. [*Ball.*]

**7.** Show that on Mercator's projection the equation of a small circle on the earth (assumed spherical) is of the form

$$\cosh y/c\sec x/c = \text{const.},$$

provided that the small circle does not include a pole of the earth and $x$, $y$ are measured on the chart from a certain origin.

Obtain the radius of curvature of this curve, at a point where the tangent lies N and S, in the form

$$\rho = c \sin\theta (\cos^2\delta - \sin^2\theta)^{-\frac{1}{2}},$$

where $\delta$ is the latitude of the centre of the above small circle and $\theta$ is its angular radius.

Determine the error made in neglecting the curvature of the "position line" when determining longitude from a timed observation of altitude 30° in latitude 51° 30′, the error in the assumed latitude being 100′; the observation is made near the prime vertical and the assumed latitude corresponds to an ideal observation exactly on the prime vertical. [*Lond.* 1926.]

**8.** An observation of the altitude of a star at a place of known latitude $\phi$ is made when the star is on the prime vertical. Show that an error in $\phi$ will have no sensible effect on the calculated longitude.

**9.** The latitudes and west longitudes of two places $F$ and $G$ are $(\phi_1, \lambda_1)$ and $(\phi_2, \lambda_2)$ respectively. When simultaneous observations of the same heavenly body (declination $\delta$) are made at $F$ and $G$, the corrected altitude is $a$ in each case. Prove that

$$\sin^2(\lambda_2 - \lambda_1)\cos^2\delta = P_1^2 - 2P_1P_2\cos(\lambda_2 - \lambda_1) + P_2^2$$
$$= S_1^2 - 2S_1S_2\cos(\lambda_2 - \lambda_1) + S_2^2,$$

where $P_i = \sin a \sec\phi_i - \sin\delta\tan\phi_i$

and $S_i = \cos a \sin A_i$ $\Big\}$ $(i = 1, 2)$,

$A_1$ and $A_2$ being the azimuths at $F$ and $G$ respectively. [*Lond.* 1930.]

**10.** At $15^h\ 17^m\ 48^s$ U.T. on a certain date the sun's observed altitude (corrected) was 89° 34′. Assuming that the ship's latitude was known to be 23° 3′, that the sun's declination was 23° 13′ and the equation of time $+\ 0^m\ 12^s$, prove that the ship's longitude was either 49° 4′ W or 49° 56′ W.

**11.** When the latitude and longitude are found by simultaneous observations of the altitudes $a_1$, $a_2$ of two known stars, prove that the two possible places of observation will have the same longitude if

$$\sin a_1/\sin a_2 = \sin\delta_1/\sin\delta_2,$$

where $\delta_1$, $\delta_2$ are the declinations of the stars. [*Coll. Exam.*]

**12.** Assuming that an observation of the sun's altitude is made correctly but that the U.T. used in the subsequent calculations is in error by $\Delta t$ seconds, show that the resultant position line is displaced through $\frac{1}{4}\Delta t \sin A \cos\phi$ nautical miles, where $A$ is the sun's azimuth and $\phi$ is the latitude.

[*Lond.* 1930.]

**13.** Two altitudes $a_1$, $a_2$ of the sun are taken, at an interval of time $2h$, and the position lines cut orthogonally. Show that

$$\sin a_1 \sin a_2 = 1 - 2\sin^2 h \cos^2\delta.$$ [*Coll. Exam.*]

**14.** In determining the hour angle of a star a sailor makes errors $\Delta a$ in his altitude and $\Delta\phi$ in his assumed latitude. Show that the error in hour angle will be given by

$$\Delta H = \Delta\phi \cot A \sec \phi - \Delta a \sec \phi \operatorname{cosec} A,$$

where $\phi$ is the latitude and $A$ is the azimuth. [*M.T.* 1923.]

**15.** At 6 p.m. a star $X$ is observed (all corrections having been applied) as follows: altitude, 33° 20′; true bearing, 220°.

At 7 p.m. a star $Y$ is observed similarly, as follows: altitude, 63° 50′; true bearing, 290°.

The calculated altitudes are *both* computed for the position Lat. 60° N, Long. 30° W: they are 33° 15′ and 63° 56′ respectively. If the ship is steaming 070° at 10 knots find, by plotting on squared paper, the position of the ship at 6 p.m. and at 7 p.m. [*Lond.* 1928.]

**16.** An intercept $p$ is drawn in the direction $\theta$ east of north from the position: east longitude, $L$; north latitude, $\phi$. If the true coordinates of the ship are $L + \Delta L$, $\phi + \Delta\phi$ prove that

$$\Delta L \cos \phi \sin \theta + \Delta\phi \cos \theta - p = 0$$

where $\Delta L$, $\Delta\phi$ and $p$ are expressed in minutes of arc.

**17.** A ship's navigator observes that the altitude of a known star as it crosses the southern meridian is $a°$, and the time of the transit is $t$ hours after Greenwich mean noon. The star's declination is $\delta°$ and its right ascension is $\alpha$ hours. From the almanac he finds that the mean sun's right ascension was $\sigma$ hours at Greenwich mean noon on the day of the observation. Show that the ship is in latitude $90° + \delta - a$ and west longitude $15\left(\sigma - \alpha + t\,\frac{366\frac{1}{4}}{365\frac{1}{4}}\right)$, both measured in degrees. [*M.T.* 1921.]

CHAPTER XIV

# BINARY STAR ORBITS

**187.** *Visual binary stars.*

The name "double star" is ordinarily applied to a pair of stars seen very close together in the telescope. This apparent closeness is due to one of two causes: (i), the two stars may be at greatly differing distances from the earth, but nearly in the same direction as viewed from the earth; (ii), they may actually be close together in space, forming a system in which their mutual gravitational attraction might be expected to be exhibited by orbital motion. In 1803, Sir W. Herschel first demonstrated, from observations carried on over a quarter of a century, that certain double stars showed relative orbital motion. A large number of stars of this character have since been discovered, and to these the name *binary stars* is given. The class of double stars in (i) above are called *optical double stars*; we do not consider these further.

Generally, the two members of a binary star are of unequal brightness. The brighter star is called the *primary* and the fainter is called the *companion*. Fig. 125 represents part of the celestial sphere in which $A$ is the primary and $B$ the companion. Let $AN$ define the direction of the north celestial pole—$AN$ is then part of the meridian through $A$. The angle $NAB$, denoted by $\theta$, is the *position angle* of $B$ with respect to $A$. The position angle is measured from 0° to 360° eastwards, in the direction indicated by the arrow. The angular distance between $A$ and $B$ is called simply the distance $AB$ or, sometimes, the separation, and is generally denoted by $\rho$. Thus $\rho$ and $\theta$ define the position of $B$ with respect to $A$. If $A$ and $B$ form a binary system then,

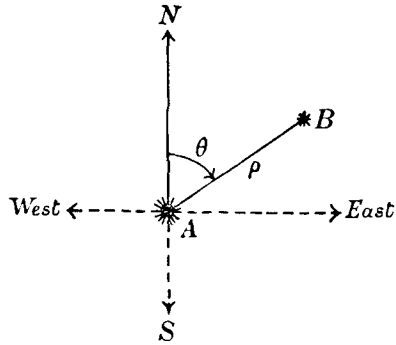


Fig. 125.

owing to the mutual gravitational attraction of the two stars—which we assume to be given by the same law as in the planetary system—the companion will describe an elliptic orbit relative to the primary. This is the *true orbit* and its plane is the true orbital plane. In general, this plane will be distinct from the plane perpendicular to the line of sight and consequently the observed orbit, called the *apparent orbit*, will be the projection of the true orbit on the plane perpendicular to the line of sight. This latter plane is the plane of the apparent orbit. The observations furnish the details concerning the apparent orbit, and it will be our purpose to show how the elements of the true orbit can be deduced. The importance of this subject in astronomy lies in the fact that valuable information concerning the masses of the stars can be obtained.

**188.** *The micrometer.*

The instrument used in practice for measuring distance and position angle is the micrometer. This is attached to the eye-end of the telescope. In the field of view of the micrometer eye-piece are generally three spider-wires, one of which, $XY$ (Fig. 126*),

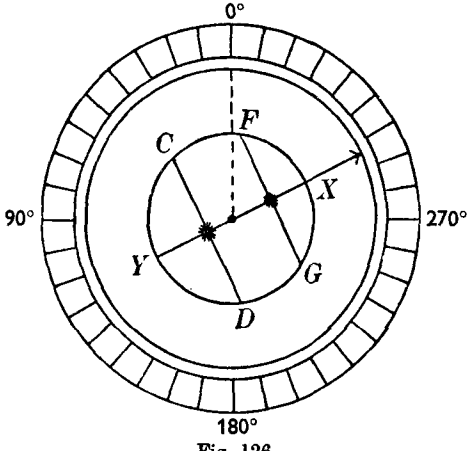


Fig. 126.

is fixed centrally and two, $CD$ and $FG$, are perpendicular to $XY$. The plate to which these wires are fitted can be rotated about the telescopic axis. By adjustment, the images of the two stars

* Unlike Fig. 125, Fig. 126 represents the observer's view.

$A$ and $B$ lie on $XY$. A graduated circle enables the position angle $\theta$ to be derived, provided the reading corresponding to the direction $ON$ (position angle 0°) can first be obtained. This is effected in practice as follows. The telescope is directed to any star and its image placed on $XY$. If the telescope is stopped, the diurnal motion will carry the star along a parallel of declination; if $XY$ is perpendicular to the meridian $ON$, the star will thus appear to travel along $XY$. The procedure then is to rotate the plate carrying the wires until this condition is fulfilled. In this position, the reading on the graduated scale corresponds to position angle 90° or 270°; hence the reading corresponding to position angle 0° is deduced. Alternatively, by rotating the graduated circle about the optical axis of the telescope, the scale can be made to correspond exactly with position angle; in this case, no correction is necessary to the readings.

The distance $\rho$ between $A$ and $B$ is derived by means of the two wires $CD$ and $FG$, each of which can be moved perpendicularly to $XY$ by fine screws to which micrometer heads are attached. When the instrument is adjusted, as shown in Fig. 126, the observation for distance consists in placing $CD$ over $A$ and $FG$ over $B$. The micrometer readings furnish the value of $\rho$, expressed in terms of the number of revolutions of the micrometer heads. From observations made on two stars whose angular distance is accurately known, the value of one revolution of a micrometer can be expressed in seconds of arc.

The complete observation of a binary star leads to the value of the distance $\rho$, expressed in seconds of arc, and the value of the position angle $\theta$.

**189.** *The elements of the true orbit of a visual binary.*

Let a sphere be drawn with the primary star $S$ as centre (Fig. 127). The straight line joining the earth to $S$ cuts the sphere at $K$. The plane of the great circle, of which $K$ is the pole, represents the plane of the apparent orbit. $KS$ is the line of sight. We suppose that the great circle $HLG$ defines the plane of the true orbit of the companion around $S$, the angle $GLD$ being the inclination $i$. The straight line $MSL$, which is the intersection of the two planes considered, is the line of nodes. Let the radius $SN$ define the direction of position angle 0°. The true orbit of

the companion relative to the primary $S$ is shown shaded in Fig. 127. When the companion is at $F$ in its true orbit, its position in the apparent orbit will be obtained by drawing a perpendicular from $F$ to the plane of the great circle $NLD$. This perpendicular will lie in the plane of the great circle $KGD$, where $G$ is the point on the sphere obtained by producing $SF$. Thus the observed position of the companion will lie on the radius $SD$ and therefore its position angle $\theta$ is the arc $ND$. (We require position angles to be measured in the sense $NLD$.)

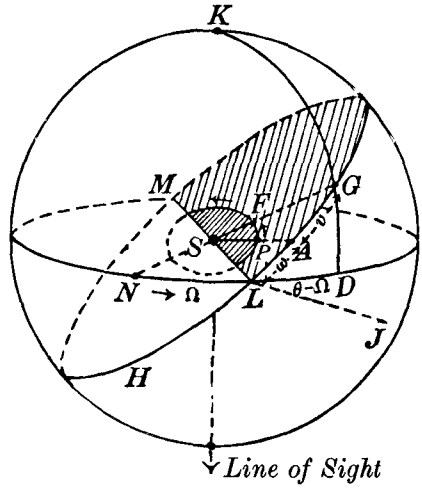


Fig. 127.

If $r$ is the radius vector $SF$, the distance $\rho$ is given by

$$\rho = r \cos GD \quad \ldots(1).$$

The position angle of that node which is less than 180° from $N$ is denoted by $\Omega$; in Fig. 127, $NL = \Omega$ and consequently $LD = \theta - \Omega$.

Let $P$ be the point on the true orbit at which the companion is nearest the primary; this point is called *periastron*; the other extremity of the major axis is called *apastron*. The semi-major axis (expressed in seconds of arc) of the true orbit will be denoted by $a$ and the eccentricity by $e$. Let $v$ be the true anomaly when the companion is at $F$. Then assuming that the companion moves in the direction of the arrow near $P$, the angle $FSP$ is $v$ and therefore $AG = v$. Let $\omega$ denote the arc $LA$. Then $LG = v + \omega$. Hence we obtain, from the triangle $LGD$ right-angled at $D$,

$$\cos LG = \cos LD \cos GD,$$

or

$$\cos GD = \cos (v + \omega) \sec (\theta - \Omega),$$

or, using (1),

$$\rho = r \cos (v + \omega) \sec (\theta - \Omega) \quad \ldots\ldots(2).$$

Also, from the triangle $LGD$, we obtain

$$\tan (\theta - \Omega) = \tan (v + \omega) \cos i \quad \ldots\ldots(3).$$

We denote the orbital period by $T$ (in years). The mean angular motion $n$ of the companion around $S$ in the true orbit is given by

$$n = 2\pi/T \qquad \ldots\ldots(4).$$

Let the time of periastron passage be denoted by $\tau$. Then the mean anomaly $M$ at time $t$ is given by

$$M = n\,(t - \tau) = E - e \sin E \qquad \ldots\ldots(5),$$

where $E$ is the eccentric anomaly. The true anomaly $v$ is related to $E$ by

$$\tan \frac{v}{2} = \left(\frac{1+e}{1-e}\right)^{\frac{1}{2}} \tan E/2 \qquad \ldots\ldots(6).$$

We have also $$n^2 a^3 = \frac{4\pi^2 a^3}{T^2} = G\,(m_1 + m_2) \qquad \ldots\ldots(7),$$

where $m_1$, $m_2$ are the masses of the two stars and $G$ is the constant of gravitation. We regard $T$ as an element, since the sum of the masses is defined in terms of $a$ and $T$.

The elements of the true orbit are: $a$, $e$, $i$, $\Omega$, $\omega$, $\tau$ and $T$. If all the elements are known, the value of $E$ at time $t$ can be derived from (4) and (5), and then the true anomaly $v$ is found by means of (6). Then from (3) and (2), the values of $\theta$ and $\rho$ can be finally obtained. In this way, the observed quantities $\rho$ and $\theta$ at any time can be compared with the calculated values based on the elements of the true orbit. The problem with which we shall be more intimately concerned is the derivation of the elements of the true orbit from the observations which enable the apparent orbit to be drawn.

It is to be remarked that although the value of the inclination may be known, the orbital plane is not uniquely determined. So far as observations with the micrometer are concerned, it is uncertain whether the true orbital plane is that given by the great circle $HLG$, or that given by the great circle of which an arc $LJ$ only is shown in Fig. 127, the angle $DLJ$ being also $i$. Now as regards the orbital plane $HLG$, in which the angular motion of the companion is in the direction from $L$ towards $A$, it is evident that as the companion passes through $L$ the component of its orbital motion in the line of sight (that is, perpendicular to the plane of the apparent orbit) is directed towards the earth. In this instance, the inclination is denoted by $-i$. If the companion's angular motion is in the direction $LJ$, the inclination is $+i$, although the distinction is artificial.

When the position angles increase with the time, as in the two orbits just considered whose planes are defined by $LG$ and $LJ$, the motion is *direct*. When the position angles decrease, as they would do if the angular motion were in the direction $GL$ or $JL$, the motion is *retrograde*.

**190.** *The apparent orbit of a visual binary.*

As we are considering the orbit of the companion relative to the primary, the latter will be situated at a focus of the true elliptic orbit. Now the apparent orbit is the projection of the true orbit on the plane perpendicular to the line of sight and is also an ellipse. But it does not necessarily follow that the focus of the true ellipse (that is, $S$) projects into a focus of the apparent ellipse. Let the ellipse in Fig. 128 represent the apparent orbit and $S$ the primary. From what has just been said, $S$ is generally not at a focus of this ellipse. If $SN$ denotes the direction defining position angle 0° and $SR$ that of $\theta = 90°$, the general equation of the ellipse referred to $SN$ and $SR$ as $x$ and $y$ axes respectively is

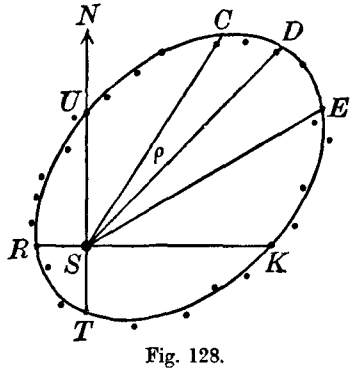


Fig. 128.

$$Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + 1 = 0 \quad \ldots\ldots(8),$$

in which there are five independent constants $A$, $B$, ... $F$, defining the particular ellipse concerned. If the companion is at $C$, an observation gives $\rho$ and $\theta$, from which the rectangular co-ordinates $x$ and $y$ of $C$ are given by

$$x = \rho \cos \theta, \quad y = \rho \sin \theta.$$

Five such observations spread over the orbit are in theory sufficient to determine the five constants $A$, $B$, ... $F$ of (8), but, owing to the unavoidable errors in measuring $\rho$ and $\theta$, the ellipse cannot be determined accurately in this way. A large number of observations, spread over many years, give a series of points such as $C$, $D$, $E$, ... from which a preliminary ellipse can be drawn. The test as to the accuracy of this ellipse is afforded by

the following consideration. In the true orbit the areas swept out by the moving radius vector are, by Kepler's second law, proportional to the relevant intervals of time, and the ratio of any two such areas is not altered by projection. Hence, in the apparent orbit, if $t_1$, $t_2$, $t_3$ are the times at which the companion is successively at $C$, $D$ and $E$, the ratio of area $CSD$ to area $DSE$ is equal to the ratio of $(t_2 - t_1)$ to $(t_3 - t_2)$. The latter ratio is, of course, accurately known. This test can be rapidly carried out with a planimeter, and the preliminary ellipse is modified until the requirements indicated are fulfilled. We assume now that the apparent ellipse has been satisfactorily drawn.

The constants $A$, $B$, ... $F$ of the general equation (8) can be conveniently derived as follows. Let the co-ordinate axes cut the apparent ellipse (drawn to a convenient scale) in $K$, $R$, $U$ and $T$. Let the co-ordinates of $U$ and $T$ be $(x_1, 0)$ and $(-x_2, 0)$. Since these co-ordinates satisfy (8), we have

$$Ax_1^2 + 2Gx_1 + 1 = 0,$$
$$Ax_2^2 - 2Gx_2 + 1 = 0,$$

from which $A$ and $G$ are determined. A similar procedure for the points $K$ and $R$ enables $B$ and $F$ to be found. If $(\xi, \eta)$ are the measured co-ordinates of a point $D$ on the apparent ellipse, $H$ is found from the relation

$$-2H\xi\eta = A\xi^2 + B\eta^2 + 2G\xi + 2F\eta + 1.$$

Amongst the many methods of deriving the elements of the true ellipse, there are the two well-known methods of Kowalsky and Zwiers, which we shall describe in detail. In each method the apparent orbit forms the basis of the subsequent procedure.

**191.** *Kowalsky's method of determining the elements of a visual binary orbit.*

The constants $A$, $B$, ... $F$ of the equation (8) for the apparent orbit are first derived according to the method given, or otherwise, the $x$-axis (Fig. 129) being $SN$ (in the direction of position angle 0°) and the $y$-axis being $SR$ (position angle 90°). $K$ is the pole of the great circle $NLR$. We take $SK$ as the $z$-axis.

Let $SA$ be the direction of periastron and $C$ the pole of the plane of the true orbit. Then since $S$ is the focus of the true orbit of the companion relative to the primary, the equation of the

orbital ellipse referred to rectangular axes $SA$ and $SB$ in the plane of the true orbit is

$$\frac{(\xi + ae)^2}{a^2} + \frac{\eta^2}{b^2} = 1 \qquad \ldots\ldots(9),$$

in which $b^2 = a^2(1 - e^2)$. The apparent orbit, whose equation is (8), is the projection of the ellipse, given by (9), on the plane $NLR$.

Let $(l_1, m_1, n_1)$ be the direction-cosines of $SA$ with respect to

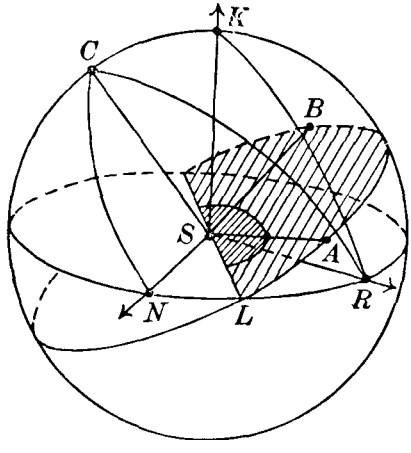



Fig. 129.

the axes $SN$, $SR$ and $SK$. Then if $A$ is joined by great circle arcs to $N$, $R$ and $K$, we have

$$l_1 = \cos AN, \quad m_1 = \cos AR, \quad n_1 = \cos AK.$$

Similarly, let $(l_2, m_2, n_2)$ and $(l_3, m_3, n_3)$ be the direction-cosines of $SB$ and $SC$ with respect to $SN$, $SR$ and $SK$. Then

$$l_2 = \cos BN, \quad m_2 = \cos BR, \quad n_2 = \cos BK,$$

and

$$l_3 = \cos CN, \quad m_3 = \cos CR, \quad n_3 = \cos CK.$$

From the spherical triangles $ANL$, $ARL$, $AKL$ we obtain

$$\left.\begin{aligned} l_1 &= \cos\Omega\cos\omega - \sin\Omega\sin\omega\cos i \\ m_1 &= \sin\Omega\cos\omega + \cos\Omega\sin\omega\cos i \\ n_1 &= \sin\omega\sin i \end{aligned}\right\} \qquad \ldots\ldots(10).$$

From the triangles $BLN$, $BLR$ and $BLK$—or simply writing $(90° + \omega)$ for $\omega$ in (10)—we obtain the direction-cosines of $SB$:

$$\left.\begin{aligned} l_2 &= -\cos\Omega\sin\omega - \sin\Omega\cos\omega\cos i \\ m_2 &= -\sin\Omega\sin\omega + \cos\Omega\cos\omega\cos i \\ n_2 &= \cos\omega\sin i \end{aligned}\right\} \quad \ldots\ldots(11).$$

From the triangles $CLN$, $CLR$, $CLK$ we obtain

$$\left.\begin{aligned} l_3 &= \sin\Omega\sin i \\ m_3 &= -\cos\Omega\sin i \\ n_3 &= \cos i \end{aligned}\right\} \quad \ldots\ldots(12).$$

Amongst the well-known relations connecting the various direction-cosines, we shall use the following:

$$l_1 m_2 - l_2 m_1 = n_3 \quad \ldots\ldots(13),$$

$$l_1^2 + l_2^2 + l_3^2 = 1 \quad \ldots\ldots(14),$$

$$m_1^2 + m_2^2 + m_3^2 = 1 \quad \ldots\ldots(15),$$

$$l_1 m_1 + l_2 m_2 + l_3 m_3 = 0 \quad \ldots\ldots(16).$$

Let $(\xi, \eta)$ denote the co-ordinates of any point on the true ellipse with reference to the axes $SA$, $SB$, and let $(x, y)$ be the co-ordinates of the projection of this point on the plane $NLR$ with reference to the axes $SN$, $SR$. Then from the circumstances of projection, we have

$$x = l_1\xi + l_2\eta,$$

$$y = m_1\xi + m_2\eta,$$

from which $$(l_1 m_2 - l_2 m_1)\,\xi = m_2 x - l_2 y,$$

or, using (13), $$\xi = \frac{m_2 x - l_2 y}{n_3} \quad \ldots\ldots(17).$$

Similarly, $$\eta = -\frac{m_1 x - l_1 y}{n_3} \quad \ldots\ldots(18).$$

Now these values of $\xi$ and $\eta$ satisfy (9); hence

$$\frac{(m_2 x - l_2 y + aen_3)^2}{a^2 n_3^2} + \frac{(m_1 x - l_1 y)^2}{b^2 n_3^2} = 1 \quad \ldots\ldots(19).$$

But $x$ and $y$ satisfy (8); hence (19) and (8) must represent the same ellipse on the plane of the apparent orbit. It follows that the coefficients of $x^2$, $xy$, $y^2$, ... in (8) must be severally propor-

tional to the coefficients of $x^2$, $xy$, $y^2$, ... in (19). Let $f$ denote the common ratio. Then we have

$$\left.\begin{aligned} A &= \frac{f}{n_3^2}\left(\frac{m_2^2}{a^2}+\frac{m_1^2}{b^2}\right) \\ B &= \frac{f}{n_3^2}\left(\frac{l_2^2}{a^2}+\frac{l_1^2}{b^2}\right) \\ H &= -\frac{f}{n_3^2}\left(\frac{l_2 m_2}{a^2}+\frac{l_1 m_1}{b^2}\right) \\ G &= \frac{fem_2}{an_3} \\ F &= -\frac{fel_2}{an_3} \end{aligned}\right\} \quad \text{.....(20).}$$

and $$1 = -f(1-e^2)$$

From these equations we obtain, after a simple reduction,

$$F^2 - G^2 + A - B = \frac{(l_1^2 - m_1^2 + l_2^2 - m_2^2)}{n_3^2 a^2 (1-e^2)^2}$$
$$= \frac{m_3^2 - l_3^2}{n_3^2 a^2 (1-e^2)^2},$$

by the aid of (14) and (15). Inserting the values of $l_3$, $m_3$ and $n_3$ given by (12), we have

$$F^2 - G^2 + A - B = \frac{\cos 2\Omega \tan^2 i}{p^2} \quad \text{.....(21),}$$

where $$p = a(1-e^2) \quad \text{.....(22).}$$

Again, we have $$FG - H = -\frac{(l_1 m_1 + l_2 m_2)}{n_3^2 a^2 (1-e^2)^2}$$
$$= \frac{l_3 m_3}{n_3^2 p^2},$$

by the aid of (16) and (22). Hence, from (12),

$$FG - H = -\frac{\sin 2\Omega \tan^2 i}{2p^2} \quad \text{.....(23).}$$

From (21) and (23), there results

$$(F^2 - G^2 + A - B)\sin 2\Omega + 2(FG - H)\cos 2\Omega = 0 \quad \text{...(24),}$$

from which $\Omega$ can be determined. It is to be remembered that the value of $\Omega$, according to our convention, lies between 0° and 180°.

Using the value of $\Omega$ just found, the value of $\frac{\tan^2 i}{p^2}$ is found by either (21) or (23).

Again, $$F^2 + G^2 - (A + B) = \frac{(l_1^2 + m_1^2 + l_2^2 + m_2^2)}{n_3^2 a^2 (1 - e^2)^2}$$
$$= \frac{(2 - n_1^2 - n_2^2)}{p^2 \cos^2 i},$$

using (14) and (15). But by (10) and (11),

$$2 - n_1^2 - n_2^2 = 2 - \sin^2 i$$
$$= 2\cos^2 i + \sin^2 i.$$

Hence $$F^2 + G^2 - (A + B) = \frac{2}{p^2} + \frac{\tan^2 i}{p^2} \quad \text{.....(25).}$$

But $\frac{\tan^2 i}{p^2}$ has already been determined; hence the value of $p^2$ can be determined from (25). When $p$ has been found, the value of $\tan^2 i$—and hence the inclination $i$—can be calculated (as mentioned on p. 344 the sign of $i$ is indeterminate).

Expressing $f$ as $-a/p$ by (22), we have from (20),

$$G = -\frac{e}{p}\frac{m_2}{\cos i},$$

$$F = \frac{e}{p}\frac{l_2}{\cos i}.$$

Inserting the values of $m_2$ and $l_2$ from (11), we have

$$\sin\Omega \sin\omega - \cos\Omega \cos\omega \cos i = \frac{Gp \cos i}{e} \quad \text{.....(26),}$$

$$\cos\Omega \sin\omega + \sin\Omega \cos\omega \cos i = -\frac{Fp \cos i}{e} \quad \text{...(27).}$$

Multiply (26) by $\sin\Omega$ and (27) by $\cos\Omega$ and add. Then

$$\sin\omega = \frac{p}{e}\cos i\,(G\sin\Omega - F\cos\Omega) \quad \text{.....(28).}$$

Multiply (26) by $\cos\Omega$ and (27) by $\sin\Omega$ and subtract. Then

$$\cos\omega \cos i = -\frac{p}{e}\cos i\,(G\cos\Omega + F\sin\Omega) \quad \text{.....(29).}$$

Hence, from (28) and (29),

$$\tan\omega = \frac{(F\cos\Omega - G\sin\Omega)\cos i}{F\sin\Omega + G\cos\Omega} \quad \text{.....(30).}$$

This formula allows $\omega$ to be calculated ($i$ and $\Omega$ being known). The eccentricity $e$ can then be found from (28) or (29), for all the quantities $\omega$, $p$, $i$, $\Omega$ have now been evaluated. The semi-major axis $a$ can now be derived from (22). We have thus shown how the elements $a$, $e$, $i$, $\Omega$ and $\omega$ can be obtained from the apparent ellipse.

There remains the problem of finding the period $T$ and $\tau$ (the time of periastron passage). For any point on the apparent ellipse corresponding to time $t$, we have the values of $\rho$ and $\theta$. Now by (3), $\tan(v + \omega) = \tan(\theta - \Omega)\sec i$,

from which the true anomaly $v$ can be found. The eccentric anomaly $E$ can then be calculated from (6). The mean anomaly $M$ is now found from $M = E - e\sin E$.

But $M = n(t - \tau)$, or

$$M = \frac{360^\circ}{T}(t - \tau) \qquad \text{.....(31).}$$

Hence, for any time $t$, we have an equation (31) involving two unknowns $T$ and $\tau$. Theoretically two equations of the type (31) are sufficient to determine $T$ and $\tau$. In practice, however, their values are generally determined by a least-square solution of several equations of type (31).

**192.** *Zwier's method of determining the elements of a visual binary orbit.*

The apparent ellipse is again supposed drawn—it is the ellipse whose area is shown shaded in Fig. 130. The determination of the elements can be carried out by (*a*) a graphical method, or (*b*) an analytical method based on the former.

(*a*) *Graphical method.* Let $C$ be the centre of the apparent ellipse and $S$ the primary star. The straight line joining $C$ and $S$ cuts the apparent ellipse in $A_1$, which is the projection of periastron in the true orbit, for $C$ is the projection of the centre of the true orbit. Since ratios are unaltered by projection, we have

$$CS : CA_1 = ae : a,$$

or

$$\frac{CS}{CA_1} = e,$$

where $e$ is the eccentricity of the true ellipse. Thus by determining $C$, the centre of the apparent ellipse, the eccentricity is

at once obtained. This procedure can be utilised as a check on the value of $e$ given by the formulae of Kowalsky's method.

Let $B_1CE_1$ be the diameter of the apparent ellipse conjugate to the diameter $A_1CD_1$. It can be obtained readily by drawing any chord $UW$ parallel to $A_1D_1$ bisecting it at $V$ and joining $C$ to $V$. Let $X$ be any point on the apparent ellipse. Draw $XR$ parallel to $CB_1$ meeting $CD_1$ in $R$. Produce $RX$ to $T$ so that

$$RT/RX = 1/(1-e^2)^{\frac{1}{2}} \equiv k \qquad \ldots\ldots(32).$$

If this construction is supposed made for every point on the apparent ellipse, the curve $D_1A_2B_3A_1D_2$ will be the result. This curve is an ellipse, as will be shown, known as the *auxiliary ellipse*. In particular, we have

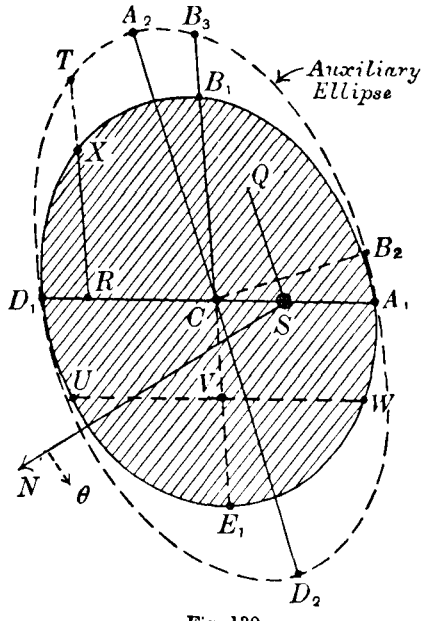


Fig. 130.

$$CB_3 = k.CB_1 \qquad \ldots\ldots(33),$$

where $k$ is defined by (32).

In the true ellipse, the minor axis is the conjugate of the major axis and this property holds for the corresponding projected lines. Hence $CB_1$ is the projection of the semi-minor axis of the true ellipse. Now consider the eccentric circle of the true ellipse. If a chord of the true ellipse, parallel to the minor axis, is produced both ways to meet the eccentric circle, the ratio of the chord of the eccentric circle to the chord of the true ellipse is $1/(1-e^2)^{\frac{1}{2}}$ or $k$. It follows from the construction of Fig. 130 that the curve $D_1A_2B_3A_1D_2$ is the projection of the eccentric circle on the plane of the apparent orbit and is consequently an ellipse; it touches the apparent ellipse at $A_1$ and $D_1$.

Let $CA_2$, $CB_2$ be the semi-major and semi-minor axes respectively of the auxiliary ellipse, and denote their measured

values by $a$ and $\beta$ respectively. We must then have

$$\frac{\beta}{a} = |\cos i| \qquad \ldots\ldots(34).$$

Now $\cos i$ is positive or negative according as the motion is direct or retrograde. So the inclination is determined by (34).

Also, the semi-major axis $a$ is equal to the radius of the eccentric circle, being the only radius not shortened by projection. But the radius of the eccentric circle is $a$, the semi-major axis of the true orbit; hence

$$a = a \qquad \ldots\ldots(35).$$

Thus $a$ is obtained from the auxiliary ellipse.

Now the diameter parallel to the line of nodes is not altered by projection; hence the major axis $A_2CD_2$ is parallel to the line of nodes. If $SN$ is the direction given by position angle $\theta = 0°$, then position angles are measured anticlockwise from the direction $SN$. Draw $SQ$ parallel to $CA_2$ as shown. Then the position angle of one node is $N\hat{S}Q$ measured anticlockwise. But this angle is greater than 180°, so, by the convention, $\Omega$ is $N\hat{S}Q - 180°$.

In Fig. 131 let $SA$ denote the direction of periastron, and let $KAG$ be part of a great circle through $A$ and the pole $K$ of the great circle $NLG$ which defines the plane of the apparent orbit. Let $\lambda$ denote $LG$. Then since $LA = \omega$, $A\hat{L}G = i$, $A\hat{G}L = 90°$ we have, by the four-parts formula **D**,

$$\tan \lambda = \tan \omega \cos i \qquad \ldots\ldots(36).$$

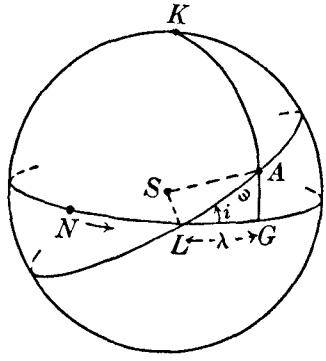


Fig. 131.

But $SL$ defines the line of nodes and $SG$ defines the projection of periastron. Hence, from Fig. 130, the angle $D_2CA_1$ must be $\lambda$. When this angle is measured, the element $\omega$ is obtained from (36).

We have thus found, from the auxiliary ellipse, the elements $a$, $e$, $i$, $\Omega$ and $\omega$. The remaining elements $T$ and $\tau$ are found as indicated in Kowalsky's method.

(*b*) *Analytical method.* In practice the auxiliary ellipse is not

actually drawn, but its properties are used. In Fig. 130 denote the various semi-diameters as follows:

$$CA_1 = a_1;\ CB_1 = b_1;\ CB_3 = b_2;\ CA_2 = \alpha;\ CB_2 = \beta.$$

Let $\theta_1$ be the position angle of $A_1$ and $\theta_2$ the position angle of $E_1$. Then $E_1\hat{C}A_1$ is $(\theta_1 - \theta_2)$.

Now $CB_1$ and $CA_1$ are conjugate semi-diameters of the apparent ellipse and, by the construction of the auxiliary ellipse, it is evident that $CA_1$ and $CB_3$ are conjugate semi-diameters of the auxiliary ellipse. We have the well-known relations connecting pairs of conjugate semi-diameters:

$$\alpha^2 + \beta^2 = a_1^2 + b_2^2 \qquad \text{.....(37),}$$

$$\alpha\beta = a_1 b_2 \sin(\theta_1 - \theta_2) \qquad \text{.....(38).}$$

But, by (33), $b_2 = kb_1$,

and we can write

$$(\alpha + \beta)^2 = a_1^2 + 2ka_1b_1 \sin(\theta_1 - \theta_2) + k^2 b_1^2 \equiv g^2 \text{ ...(39),}$$

and

$$(\alpha - \beta)^2 = a_1^2 - 2ka_1b_1 \sin(\theta_1 - \theta_2) + k^2 b_1^2 \equiv h^2 \text{ ...(40),}$$

whence $\alpha = \frac{1}{2}(g + h)$ and $\beta = \frac{1}{2}(g - h)$ .....(41).

The apparent ellipse is supposed drawn; the diameter conjugate to $CA_1$ is also actually drawn. We can thus obtain the values of $e$, $a_1$, $b_1$ and $(\theta_1 - \theta_2)$ by measurement; since $e$ is now known, $k$ can be calculated from (32). Thus the quantities $g$ and $h$ can be calculated and by (41) the values of $\alpha$ and $\beta$ are deduced.

Consider now the angle $D_2CA_1$ which we denoted previously by $\lambda$; it is the angle between the semi-major axis $CD_2$ of the auxiliary ellipse and the radius $CA_1$. Now the co-ordinates of $A_1$ referred to $CD_2$ and $CB_2$ as axes are $(a_1\cos\lambda,\ a_1\sin\lambda)$, and hence we have

$$\frac{a_1^2\cos^2\lambda}{\alpha^2} + \frac{a_1^2\sin^2\lambda}{\beta^2} = 1,$$

from which

$$\tan\lambda = \pm\frac{\beta}{\alpha}\left(\frac{\alpha^2 - a_1^2}{a_1^2 - \beta^2}\right)^{\frac{1}{2}},$$

or, using (34),

$$\tan\lambda = \pm\cos i\left(\frac{\alpha^2 - a_1^2}{a_1^2 - \beta^2}\right)^{\frac{1}{2}} \qquad \text{.....(42).}$$

As the sign of $\tan\lambda$ is the same as the sign of $\tan(\theta_1 - \theta_2)$, there are then two possible values of $\lambda$, differing by 180°, satis-

fying (42). Since the position angles of the directions $CA_1$ and $CD_2$ are respectively $\theta_1$ and $\Omega$ and since $D_2 \hat{C} A_1 = \lambda$, we have

$$\Omega = \theta_1 - \lambda \qquad \text{.....(43).}$$

Now $\Omega$ is defined to lie between $0°$ and $180°$ and since $\theta_1$ is known from measurement, the appropriate value of $\lambda$ satisfying (42) is indicated by (43); this last equation enables $\Omega$ to be found. The value of $\omega$ is then found from (36), which is

$$\tan \lambda = \tan \omega \cos i.$$

The remaining elements $T$ and $\tau$ are derived as before.

**193.** *The masses of the stars.*

The two elements of primary concern in this connection are the period $T$ (in years) and the semi-major axis $a$. From the period, we deduce the mean angular motion $n$ given by

$$n = 2\pi/T \qquad \text{.....(44).}$$

It is to be remembered that the micrometrical measures of the distances $\rho$ between the companion and the primary are expressed in seconds of arc, and that consequently we obtain $a$ expressed also in seconds of arc. Let $\Pi$ be the parallax of the binary. Then if $d$ is its distance from us measured in astronomical units, the value of $\Pi$ in seconds of arc is given by

$$\Pi = \frac{1}{d} \operatorname{cosec} 1'' \qquad \text{.....(45).}$$

Let $a_1$ denote the semi-major axis of the true orbit expressed in astronomical units. Then $a$ (in seconds of arc) is given by

$$a = \frac{a_1}{d} \operatorname{cosec} 1'' \qquad \text{.....(46).}$$

Hence, from (45) and (46),

$$a_1 = \frac{a}{\Pi} \qquad \text{.....(47).}$$

Now the relation between the mean angular velocity $n$ and the linear semi-major axis $a_1$ is

$$n^2 a_1^3 = G\,(m_1 + m_2),$$

where $G$ is the constant of gravitation and $m_1$, $m_2$ are the masses of the two stars. Hence, by (44),

$$\frac{a_1^3}{T^2} = \frac{G}{4\pi^2}(m_1 + m_2) \qquad \text{.....(48).}$$

Consider now the earth's orbit around the sun. We have a similar equation to (48), namely,

$$\frac{a_0^3}{T_0^2} = \frac{Gm}{4\pi^2} \qquad \text{.....(49),}$$

where $m$ is the sun's mass (we neglect the earth's mass) and $a_0$, $T_0$ refer to the earth's orbit. But $a_0 = 1$ astronomical unit and $T_0 = 1$ year. If the sun's mass $m$ is taken as the unit of mass, then $G$ expressed in terms of these units is given by

$$G = 4\pi^2 \qquad \text{.....(50).}$$

Hence, for the binary, equation (48) becomes

$$\frac{a_1^3}{T^2} = m_1 + m_2,$$

where the stellar masses are now expressed in terms of the sun's mass as unit. Then, using (47), we derive

$$m_1 + m_2 = \frac{a^3}{\Pi^3 T^2} \qquad \text{.....(51).}$$

This equation enables the sum of the masses of a binary star to be determined when the parallax and its orbit are known.

It is found that, for the majority of visual binaries, the mass of the system is about twice the mass of the sun. When other evidence is reviewed, it is exceptional to find the mass of a star outside the range $\frac{1}{10}m$ to $50m$, where $m$ denotes the sun's mass.

It is to be noted that only the sum of the masses can be calculated from (51). It is only in comparatively rare instances, when additional observational material of a different character is available, that the individual masses of primary and companion can be deduced.

**194.** *Dynamical parallaxes.*

The formula (51) can be used to determine the approximate value of $\Pi$ when this quantity cannot be measured by the usual method. We can write

$$\Pi = \frac{a}{T^{\frac{2}{3}}(m_1 + m_2)^{\frac{1}{3}}} \qquad \text{.....(52),}$$

in which $a$ and $T$ are supposed to be derived from the observations of the visual orbit. If we put $m_1 + m_2 = 2$, in accordance with the average result for *known* visual binaries, we obtain

what is called the dynamical parallax. In any given instance, of course, the value of $(m_1 + m_2)$ is actually unknown, but as the cube root of this quantity is involved in the formula (52), the resultant uncertainty in the calculated value of $\Pi$, due to the assumption concerning the mass of the system, is comparatively small. Dynamical parallaxes supply valuable information for certain investigations in which the distances of the stars are required.

**195.** *Spectroscopic binaries.*

A binary of this class consists of two members too close together to be resolved by the telescope into a double star; the duplicity of the system is inferred from spectroscopic observations of radial velocity. In a binary system which has its centre of gravity at $G$ (Fig. 132), each star describes an elliptic orbit around $G$. In general, the orbital plane will be inclined to the line of sight. The orbital elements are defined as for visual binaries ($\Omega$, the longitude of the node, is an exception, as it cannot be defined at all, owing to the absence of a definite reference point on the plane perpendicular to the line of sight).

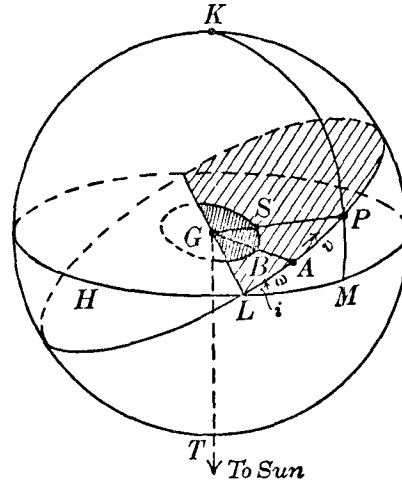


Fig. 132.

On page 213, reference has been made to the spectroscopic method of measuring the radial velocities of the stars and, in particular, the formula for the variable part of the radial velocity due to the earth's orbital motion round the sun was obtained. We shall suppose here that this variable part is removed from the observed radial velocity of a star; the value thus obtained represents the star's radial velocity relative to the sun.

In Fig. 132 let the straight line joining $G$ and the sun intersect the sphere centred at $G$ in $T$. The great circle of which $T$ is the

pole is $HLM$ and we can refer to this plane, without ambiguity, as the plane of the apparent orbit. Let $K$ be the other pole of $HLM$. But notice that Fig. 132 is inverted compared with Fig. 127.

Let $r$ (in kilometres) denote the radius vector $GS$, where $S$ is the position of one of the stars in its orbit with respect to $G$. Let $GB$ be the radius vector when the star is nearest $G$. Then $S\hat{G}B$ or $P\hat{G}A$ is the true anomaly $v$. (We suppose that the direction of motion is from $A$ towards $P$.) $L$ is the ascending node and we put $LA = \omega$ as before.

Let $z$ denote the distance of $S$ from the plane of $HLM$, reckoned positive when $S$ is on the same side of this plane as $K$. Then

$$z = r \sin PM,$$

where $KPM$ is the great circle arc through $P$. But from the triangle $PLM$, in which $LP = v + \omega$, $P\hat{L}M = i$ and $P\hat{M}L = 90°$, we have

$$\sin PM = \sin (v + \omega) \sin i.$$

Hence

$$z = r \sin (v + \omega) \sin i \qquad \text{.....(53),}$$

in which $z$ is expressed in kilometres. The rate at which $z$ varies with the time, that is, $\frac{dz}{dt}$, gives the radial velocity of the star $S$ relative to the centre of gravity $G$ of the system.

In general, the system itself will have a radial velocity with respect to the sun; denote it by $V$. Then $V$ is the radial velocity of $G$, regarded positive when the motion is one of recession from the sun. Hence the radial velocity $R$ of the star $S$ with reference to the sun is given by

$$R = V + \frac{dz}{dt} \qquad \text{.....(54).}$$

The value of $R$ is deduced from the spectroscopic observations. When both stars are sufficiently bright to register their spectra on the photographic plate, the radial velocities of both stars can be inferred. For the present, however, we shall consider the type of spectroscopic binary, only one of whose components enables radial velocity measures to be made.

Now we have the following formulae of elliptic motion applicable to the motion of the star $S$ around $G$:

$$r = \frac{a(1 - e^2)}{1 + e \cos v} \qquad \text{.....(55),}$$

$$r^2 \frac{dv}{dt} = h = \{\mu a (1-e^2)\}^{\frac{1}{2}} \quad \text{.....(56),}$$

$$\mu = n^2 a^3 \quad \text{.....(57),}$$

where $a$ is the semi-major axis in kilometres and $n$ is the mean angular motion. Now from (53),

$$\frac{dz}{dt} = \frac{dr}{dt} \sin(v+\omega) \sin i + r \cos(v+\omega) \sin i \frac{dv}{dt},$$

and we find easily from the preceding formulae that

$$\frac{dr}{dt} = \frac{nae \sin v}{(1-e^2)^{\frac{1}{2}}}; \quad r\frac{dv}{dt} = \frac{na(1+e\cos v)}{(1-e^2)^{\frac{1}{2}}}.$$

Hence we obtain

$$\frac{dz}{dt} = \frac{na \sin i}{(1-e^2)^{\frac{1}{2}}} [\cos(v+\omega) + e\cos\omega] \quad \text{.....(58).}$$

Thus from (54) we can express the radial velocity $R$ in terms of $V$, $v$ and the elements of the orbit.

**196.** *The velocity curve.*

The orbital periods of spectroscopic binaries are generally several days only, and as observations may be carried on over several months, or even years, the orbital periods can be found with great accuracy. From all the available observations a curve is drawn, giving the relation between the radial velocity $R$ and the time (through an interval equal to the period). Fig. 133

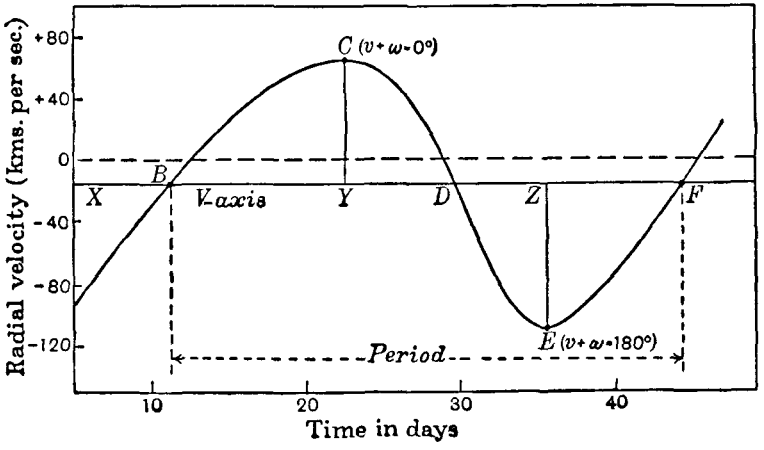



Fig. 133.

shows a typical velocity curve for which the period is $33\frac{1}{3}$ days; it refers to the star H.R. 8800, the spectroscopic observations of which were made at the Dominion Observatory, Victoria, B.C. (*Publications*, vol. I, p. 243). The curve between $B$ and $F$ represents the complete cycle of changes in the radial velocity $R$. At $C$ the radial velocity has its maximum positive value (about 60 kms. per second) and at $E$ its maximum negative value (about $-$ 115 kms. per second).

Actually, it is found (we shall explain this later) that the binary is approaching the sun with a speed of 15 kms. per second. Thus $V = -15$ kms. per second. The line $XYZ$ parallel to the time axis, in accordance with this value of $V$, is called the $V$-axis; ordinates measured from this line to the curve give the appropriate values of $\frac{dz}{dt}$. For the present, we shall suppose that the $V$-axis has been drawn.

There are many methods of deriving the orbital elements from the velocity curve; we shall consider only one of these, which is extensively used in practice.

**197.** *Method of Lehmann-Filhés.*

We shall write (58) as follows:

$$\frac{dz}{dt} = K\left[\cos(v+\omega) + e\cos\omega\right] \qquad \text{.....(59),}$$

where

$$K = \frac{na\sin i}{(1-e^2)^{\frac{1}{2}}} \qquad \text{.....(60).}$$

Now $dz/dt$ is a maximum when $(v+\omega)$ is $0°$ and, as the radial velocity is a maximum at $C$, the ordinate $YC$ (which we denote by $\alpha$), measured from the $V$-axis, is given by

$$\alpha = K(1 + e\cos\omega) \qquad \text{.....(61).}$$

Also, the radial velocity has its greatest negative value at $E$, and this occurs when $v+\omega = 180°$. If $\beta$ denotes the length of the ordinate $EZ$, we have, numerically,

$$\beta = K(1 - e\cos\omega) \qquad \text{.....(62).}$$

From (61) and (62),

$$K = \tfrac{1}{2}(\alpha+\beta) \qquad \text{.....(63),}$$

$$e\cos\omega = \frac{\alpha-\beta}{\alpha+\beta} \qquad \text{.....(64).}$$

The measurement of $\alpha$ and $\beta$ enables the values of $K$ and $(e \cos \omega)$ to be derived.

The $V$-axis is determined from the consideration that the area $BCD$ above this axis is equal to the area $DEF$ below. This is easily seen as follows. Any ordinate is $dz/dt$ and the abscissa is the time $t$. The area $BCD$ is thus given by the integral

$$q \int \frac{dz}{dt} dt,$$

where the limits are the values of $t$ corresponding to $B$ and $D$, and $q$ is a constant depending on the units in terms of which the area is measured. Hence this area is simply

$$q\,(z_D - z_B),$$

where $z_D$ denotes the value of $z$ corresponding to the point $D$ on the velocity curve. Similarly the area $DEF$ is

$$q\,(z_F - z_D).$$

But $z_F = z_B$, since $BF$ corresponds to a complete cycle, and therefore the numerical expressions for the two areas are equal. The drawing of the $V$-axis is then largely a matter of trial and error; a first attempt is made and the areas above and below this line are measured by means of a planimeter. After one or more trials the position of the $V$-axis can be finally obtained to satisfy the conditions of area. In Fig. 133 the position of the $V$-axis is found according to the principles described, and the value of $V$ obtained is $-15$ kms. per second. This is the radial velocity of the binary (H.R. 8800) relative to the sun.

Again, the area $CYD$, which we denote by $\Delta_1$, is given by

$$\Delta_1 = z_D - z_C \qquad \text{.....(65).}$$

Now at $C$, the value of $(v + \omega)$ is $0°$ and, from (53), $z_C = 0$. Hence

$$\Delta_1 = z_D \qquad \text{.....(66).}$$

In the same way, the area $DZE$ is equal to $(z_E - z_D)$, and as $z_E$ is zero, the area $DZE$ is equal to the area $CYD$, that is, $\Delta_1$. We can prove in a similar manner that the areas $BCY$ and $ZEF$ are equal.

If $v_1$ and $r_1$ are the values of the true anomaly and radius vector in the true orbit at the point corresponding to $D$ in Fig. 133, then by (53) and (66),

$$\Delta_1 = r_1 \sin (v_1 + \omega) \sin i \qquad \text{.....(67).}$$

But, at $D$, the value of $dz/dt$ is zero; hence by (59),

$$\cos(v_1 + \omega) = -e\cos\omega$$
$$= -\frac{\alpha - \beta}{\alpha + \beta} \quad \text{by (64).}$$

Hence $$\sin(v_1 + \omega) = \pm\frac{2\sqrt{\alpha\beta}}{\alpha + \beta}.$$

Now in passing along the velocity curve from $C$ towards $E$, the radial velocity changes from being positive to negative just before and after the point $D$. Hence, referring to Fig. 132, we see that the point $D$ on the velocity curve (Fig. 133) must correspond to the point on the true orbit for which $z$ has a maximum positive value. Hence by (53), $\sin(v_1 + \omega)$ is positive and therefore

$$\sin(v_1 + \omega) = +\frac{2\sqrt{\alpha\beta}}{\alpha + \beta} \qquad \text{.....(68).}$$

Consider now the area $ZEF$, which we denote by $\Delta_2$. Then

$$\Delta_2 = z_F - z_E,$$

or, since $z_E = 0$, we obtain $\Delta_2 = z_F$.

If $v_2$ and $r_2$ refer to the point on the true orbit corresponding to $F$ on the velocity curve, we have

$$\Delta_2 = r_2 \sin(v_2 + \omega)\sin i \qquad \text{.....(69).}$$

But, at $F$, $dz/dt$ is zero; thus

$$\cos(v_2 + \omega) = -e\cos\omega = -\frac{\alpha - \beta}{\alpha + \beta},$$

and now $$\sin(v_2 + \omega) = -\frac{2\sqrt{\alpha\beta}}{\alpha + \beta} \qquad \text{.....(70),}$$

the negative sign being taken in (70), since $F$ corresponds to the point on the true orbit for which $z$ has the maximum negative value. Hence, using (67), (68), (69) and (70), we find

$$\frac{\Delta_1}{\Delta_2} = -\frac{r_1}{r_2}.$$

The procedure has been such that $\Delta_2$ is expressed mathematically as a negative quantity. If $\Delta_1$ and $\Delta_2$ are both regarded as positive quantities, we can write

$$\frac{\Delta_1}{\Delta_2} = \frac{r_1}{r_2} \qquad \text{.....(71).}$$

But $r_1 = \dfrac{a(1-e^2)}{1+e\cos v_1}$, with a similar expression for $r_2$; hence

$$\frac{\Delta_1}{\Delta_2} = \frac{1+e\cos v_2}{1+e\cos v_1}.$$

We can write this last equation as follows:

$$\frac{\Delta_1}{\Delta_2} = \frac{1+e\cos(v_2+\omega-\omega)}{1+e\cos(v_1+\omega-\omega)}$$
$$= \frac{1+e\cos\omega\cos(v_2+\omega)+e\sin\omega\sin(v_2+\omega)}{1+e\cos\omega\cos(v_1+\omega)+e\sin\omega\sin(v_1+\omega)},$$

which, by the use of (64), (68) and (70), reduces to

$$\frac{\Delta_1}{\Delta_2} = \frac{2\alpha\beta - \sqrt{\alpha\beta}\,(\alpha+\beta)\,e\sin\omega}{2\alpha\beta + \sqrt{\alpha\beta}\,(\alpha+\beta)\,e\sin\omega},$$

from which it is easily found that

$$e\sin\omega = \frac{2\sqrt{\alpha\beta}}{\alpha+\beta}\cdot\frac{\Delta_2-\Delta_1}{\Delta_2+\Delta_1} \qquad \text{.....(72).}$$

By measuring the areas $\Delta_1$ and $\Delta_2$ (which are both positive quantities) the value of $e\sin\omega$ can be deduced from (72). But we had also

$$e\cos\omega = \frac{\alpha-\beta}{\alpha+\beta};$$

hence this equation and (72) determine $e$ and $\omega$.

To determine the time $\tau$ of periastron passage we note that, then, $v = 0$, and if $\dot{z}_1$ is the corresponding value of $dz/dt$ the value of $\dot{z}_1$ is given from (59) by

$$\dot{z}_1 = K(1+e)\cos\omega \qquad \text{.....(73).}$$

The value of $\dot{z}_1$ can now be found since $K$, $e$ and $\omega$ are all supposed known. There are two ordinates of the velocity curve given by (73). However, the ambiguity is resolved by noting that, at $C$, $v+\omega = 0$, and at $E$, $v+\omega = 180°$. Thus if $\omega$ is found to be, say, 60°, the value of $v$ at $C$ is 300° and at $E$ is 120°; in this instance periastron will correspond to the ordinate between $C$ and $D$ on the velocity curve and so the time $\tau$ (abscissa) of periastron passage, measured from the epoch corresponding, say, to $B$, is found.

Again, we have from (60) and (63),

$$K = \frac{na\sin i}{(1-e^2)^{\frac{1}{2}}} = \tfrac{1}{2}(\alpha+\beta) \qquad \text{.....(74).}$$

Also if the period $T$ is supposed known, $n = 2\pi/T$, and from (74),

$$a \sin i = \frac{T(\alpha + \beta)(1 - e^2)^{\frac{1}{2}}}{4\pi}.$$

The period $T$ is usually expressed in days and $\alpha$ and $\beta$ are measures of velocities expressed in kilometres per mean solar second. With these units, the last equation becomes (since 1 day = 86,400 seconds)

$$a \sin i = \frac{21600T}{\pi}(\alpha + \beta)(1 - e^2)^{\frac{1}{2}} \qquad \ldots\ldots(75).$$

This formula gives the value of the quantity $(a \sin i)$ expressed in kilometres, $T$ being expressed in days. Unless the inclination $i$ can be derived by other means, the semi-major axis $a$, with respect to the centre of gravity of the system, cannot be determined.

198. *Two spectra visible.*

When both members of the binary are sufficiently bright, the velocity curve associated with each star can be analysed by the previous, or any other, method. Let now $a_1$ denote the semi-major axis of one orbit, with respect to the centre of gravity of the system, and $a_2$ the semi-major axis of the other orbit. Then the semi-major axis of the orbit of one star relative to the other star is $(a_1 + a_2)$, which we denote by $a_0$. If $a_0$ is expressed in astronomical units and the period $T$ in days, the sum of the masses of the two stars (expressed in terms of the sun's mass as unit) is given by

$$m_1 + m_2 = a_0^3 \left(\frac{365\frac{1}{4}}{T}\right)^2 \qquad \ldots\ldots(76).$$

But $(a_1 \sin i)$ can be found by means of (75) in terms of kilometres, and from the analysis of the second velocity curve the value of $(a_2 \sin i)$ can be similarly found. Since $a_0 = a_1 + a_2$ and one astronomical unit = 149,600,000 km, formula (76) becomes, on multiplying both sides by $\sin^3 i$,

$$(m_1 + m_2)\sin^3 i = \left(\frac{a_1 \sin i + a_2 \sin i}{149{,}600{,}000}\right)^3 \left(\frac{365\frac{1}{4}}{T}\right)^2,$$

in which $a_1 \sin i$, $a_2 \sin i$ are expressed in km and $T$ in days, or, using (75),

$$(m_1 + m_2)\sin^3 i = 1{\cdot}295 \times 10^{-8} T(1 - e^2)^{\frac{3}{2}} \{(\alpha_1 + \beta_1) + (\alpha_2 + \beta_2)\}^3,$$

where $\alpha_1$, $\beta_1$ and $\alpha_2$, $\beta_2$ refer to the two velocity curves. On writing $K_1 = \frac{1}{2}(\alpha_1 + \beta_1)$ and $K_2 = \frac{1}{2}(\alpha_2 + \beta_2)$, this last equation is:

$$(m_1 + m_2)\sin^3 i = 10{\cdot}36 \times 10^{-8}\, T\,(1 - e^2)^{\frac{3}{2}}\,(K_1 + K_2)^3 \quad \ldots(77).$$

The value of $(m_1 + m_2)\sin^3 i$ can thus be obtained.

Now, since $G$ is the centre of gravity of the system,

$$m_1 a_1 = m_2 a_2 \quad \ldots\ldots(78),$$

or
$$m_1 (a_1 \sin i) = m_2 (a_2 \sin i).$$

Hence, by (75),
$$\frac{m_1}{m_2} = \frac{\alpha_2 + \beta_2}{\alpha_1 + \beta_1} = \frac{K_2}{K_1} \quad \ldots\ldots(79).$$

Thus the ratio of the masses can be derived. The formula (77) for the sum of the masses involves the unknown value of the inclination, and therefore the individual masses cannot be found unless $i$ can be derived by other means.

**199.** *The mass-function.*

We return now to the case when only one spectrum is visible. From (78), we have

$$\frac{a_1}{a_1 + a_2} = \frac{a_1}{a_0} = \frac{m_2}{m_1 + m_2};$$

$$\therefore\ (a_1 \sin i)^3 = \frac{m_2^3 a_0^3 \sin^3 i}{(m_1 + m_2)^3} \quad \ldots\ldots(80).$$

But from (76), expressing $a_0$ in kilometres,

$$m_1 + m_2 = \left(\frac{a_0}{149{,}600{,}000}\right)^3 \left(\frac{365\frac{1}{4}}{T}\right)^2,$$

or
$$m_1 + m_2 = 3{\cdot}985 \times 10^{-20} \frac{a_0^3}{T^2} \quad \ldots\ldots(81).$$

Hence from (80) and (81), eliminating $a_0$, we derive

$$\frac{m_2^3 \sin^3 i}{(m_1 + m_2)^2} = \frac{3{\cdot}985 \times 10^{-20} (a_1 \sin i)^3}{T^2} \quad \ldots\ldots(82).$$

The quantity on the right can be evaluated. The function on the left is the *mass-function*. It is generally given by computers of spectroscopic orbits.

**200.** *The masses of spectroscopic binaries.*

When one spectrum only is photographed, the information concerning the masses of the stars is contained in the value of the mass-function. When both spectra are visible, we have seen that the velocity curves enable the ratio of $m_1$ to $m_2$ to be determined, and also the quantity $(m_1 + m_2) \sin^3 i$. Thus for any individual binary, the actual values of the masses cannot be found owing to the presence of the unknown inclination $i$ in the formulae. Nevertheless, valuable information regarding the masses is derived from statistical considerations. Consider a large number of spectroscopic binaries of the same spectral type distributed over the sky. We assume that the masses of such binaries are much alike. If we take the average mass of a system to be $M$ and the number to be $N$, we can write, forming the sum $\Sigma (m_1 + m_2) \sin^3 i$, from the known values of this quantity,

$$\Sigma (m_1 + m_2) \sin^3 i = NMS,$$

where $S$ denotes the mean value of $\sin^3 i$, with reference to the random inclinations over the sky. The value of $S$ is generally taken to be 2/3. In this way we can derive an estimate of the average mass $M$ of a spectroscopic binary of given spectral type.

The most massive spectroscopic binary known is probably the star HD47129, for which a mass function of 12·88 solar masses is derived. Assuming that the two components have equal masses, each star's mass must exceed 50 times that of the sun.

**201.** *Eclipsing binary stars.*

If the line of sight is in or near the plane of the orbit of a binary star, it is evident that during each orbital period the component $A$ will pass wholly or partially in front of the component $B$, thus eclipsing the latter. The effect will be noted in the diminution of the light of the binary. In the same way the component $B$ will cause a total or partial eclipse of $A$. The analysis of the light curve of such a star cannot be given here, and the reader is referred to a more specialist text, such as A. H. Batten, *Binary and Multiple Systems of Stars* (Pergamon, 1973) for more detailed information. It may be added, however, that the light curve yields the inclination, $i$, and if both components can be observed spectroscopically, the mass of each is obtained.

## EXERCISES

**1.** The greatest apparent diameter of a visual binary orbit is 1″·8; the parallax is 0″·072 and the period 50 years. If the orbit is circular, show that the mass of the binary is 25/32 times the sun's mass. [*Lond.* 1925.]

**2.** The semi-major axis of the orbit of Krüger 60 is 2″·46; the period is 44·3 years; the parallax of the system is 0″·257. Calculate the mass of the binary in terms of the sun's mass. [*Lond.* 1928.]

**3.** The greatest angular separation of the components of $\tau$ Cygni is 0″·91, the period is 47·0 years and the parallax is 0″·05. Assuming that the orbit is *circular, calculate the mass of the binary.*

**4.** Calculate the dynamical parallax of the binary $\beta$ 7642, for which $a = 2''{\cdot}87$ and $T = 317{\cdot}5$ years.

The trigonometrical parallax is 0″·088; comment on the difference between the two values of the parallax. [*Lond.* 1926.]

**5.** Explain how the shape and position of the true orbit of a visual binary can be deduced from observations of position angle and distance extended over a whole period.

If $a$, $b$ be the semi-axes of the apparent orbit, $h$, $k$ the co-ordinates of the primary referred to these semi-axes, show that the inclination $i$ of the orbit to the plane at right angles to the line of sight is given by

$$\sin^2 i = \frac{2\lambda}{a^2 + b^2 - h^2 - k^2 + \lambda},$$

where $$\lambda^2 = (h^2 + k^2)^2 + (a^2 - b^2)^2 + 2(a^2 - b^2)(k^2 - h^2).$$

[*Lond.* 1922.]

**6.** Prove that if the complete apparent orbit of a visual binary has been obtained, and on any chord $POQ$ through the primary $O$ we take a point $R$ such that $OR$ is a harmonic mean between $PO$ and $OQ$, the locus of $R$ is an ellipse, of which the length of the major axis gives the latus rectum of the true orbit, the direction of this major axis gives the direction of the line of nodes, and the ratio of the minor to the major axis is the cosine of the inclination. [*Lond.* 1923.]

**7.** The true period of an eclipsing binary is 3 days, and its velocity in the line of sight (away from the sun) is 30 kms. per second. Show that its apparent period is greater than the true one by 26 seconds. [*Lond.* 1923.]

**8.** A variable star has ecliptic co-ordinates $\lambda$, $\beta$. If $T$ is the time of a maximum for an observer on the earth and $T_0$ the corresponding time with respect to the sun, show that, $c$ being the velocity of light in kilometres per second and $a$ the radius of the earth's orbit in kilometres,

$$T_0 = T - \frac{a}{c}\cos\beta\cos(\lambda - \odot),$$

where $\odot$ is the sun's longitude. [*Lond.* 1927.]

CHAPTER XV

# OCCULTATIONS AND ECLIPSES

**202.** *Occultations of stars by the moon.*

As the moon's sidereal period of orbital revolution around the earth is about $27\frac{1}{3}$ days, it moves eastwards with reference to the stars at an average rate of rather more than half a degree per hour. In its passage over the stellar background it is continually interposing its disc between us and the stars, and the sudden disappearance of a star in this way is called the *occultation* of the star by the moon. After an interval, which depends on a variety of factors, the star reappears. The disappearance and reappearance of the star are generally referred to as *immersion* and *emersion* respectively. The disappearance of the star and its reappearance are instantaneous phenomena and, if the time of one or the other is noted accurately, there is obtained at that instant a definite relation between the moon's position in the sky and the position of the observer, it being assumed that the star's position is known accurately. Formerly, occultations were utilised for the determination of longitude, but the introduction of radio time-signals has rendered the occultation method obsolete.

If the moon's position is known accurately, the particulars of the occultation of a star at any place can be predicted and, under these circumstances, it is to be expected that prediction and observation would agree. Now the moon's position is predicted in the almanacs for any instant of Ephemeris Time, while the recorded time of the observation of an occultation will be in Universal Time. The study of such occultations, therefore, provides a ready means of determining the relationship between Universal and Ephemeris Time, and, in particular, of deriving the correction $\Delta T$. The occultations of radio sources are also important, as precise radio positions are difficult to measure. The first positive optical identification of a quasar was made by timing the cessation of its radio signals in the course of a lunar occultation.

**203.** *The geometrical conditions for an occultation.*

Consider Fig. 134, in which the earth (regarded as a spheroid) and the moon (regarded as a sphere) are shown with their centres at $E$ and $M$ respectively. Let $MS$ be the straight line joining the moon's centre and the star concerned at a particular instant. Since the star can be regarded as at an infinite distance, the rays from the star lying within a circular cylinder, whose axis is $MS$ and whose cross-sectional radius is equal to the moon's radius, will be stopped by the moon. Suppose that this cylinder intersects the earth's surface in the curve $FGH$ (only part of the complete curve is shown in Fig. 134); then, at the particular

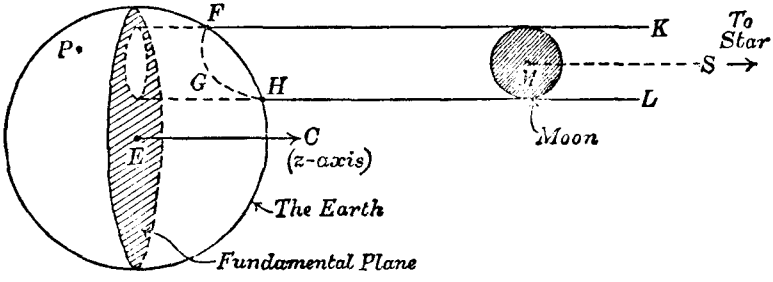



Fig. 134.

instant concerned, the star will be about to disappear or reappear behind the lunar disc, as viewed from points on the curve $FGH$, and at all places on the earth's surface within this curve the star will be invisible. For an occultation of a particular star to be visible at some point on the earth's surface, the cylinder must evidently intersect the earth's surface.

The plane passing through the earth's centre and perpendicular to the line $MS$ is called the *fundamental plane*; in the subsequent discussion the normal $EC$ to this plane is taken to be the $z$-axis. It is to be noted that, owing to the Earth's rotation, the fundamental plane and the $z$-axis are continually altering relative to fixed axes in the Earth.

As viewed from any point on the earth, the moon will be displaced, owing to parallax, away from the zenith by an angle which may be just over 1° (corresponding to the maximum or horizontal parallax); also its semi-diameter is 16′. Hence, for a star with the same right ascension as the moon at any instant,

an occultation may be possible at some place if the star's declination does not differ by more than about $1\frac{1}{3}°$ from the moon's tabulated declination. By comparing the moon's tabulated positions in the almanac with the positions of stars, the selection of stars, according to the criterion just mentioned, is made for which occultations are possible.

**204.** *Bessel's method of investigating an occultation.*

We now describe the method that is used for predicting the circumstances of an occultation. We shall use the following symbols:

$\alpha$, $\delta$—the apparent R.A. and Dec. of the star.
$\alpha_1$, $\delta_1$—the apparent R.A. and Dec. of the moon's centre.
$P_1$—the equatorial horizontal parallax of the moon.
$r_1$—the moon's geocentric distance.
$k$—the moon's radius.
$\rho$—the observer's geocentric distance.
} All expressed in terms of the earth's equatorial radius as the unit of length.
$\phi'$—the observer's geocentric latitude.

In Fig. 135 let $E$ (the earth's centre) be the centre of the celestial sphere shown. The radius $EC$ is drawn parallel to the straight line joining the moon's centre to the star; $EC$ is the $z$-axis and the plane $DBA$, of which $EC$ is the normal and which passes through the centre $E$, is the fundamental plane. Let $EP$ be the earth's axis of rotation, $P$ being the north pole. Since $C$ is the pole of the fundamental plane, the plane of any great circle drawn through $C$ is perpendicular to the fundamental plane; in particular, the great circle joining $C$ and $P$ defines a plane perpendicular to the fundamental plane. Let it intersect the latter in $EB$ and the equator in $F$; $BPCF$ is defined to be the $x$-plane. Since the star is regarded as at an infinite distance, $EC$ is the direction of the star, as viewed from the earth, and $PCF$ is the meridian through the star. The earth's rotation carries this meridian westwards, that is, in the direction $FD$. The positive direction of the $x$-axis is defined to be $EA$—the eastward drawn normal of the plane $BPCF$. It is evident that $A$ is on the celestial equator and the R.A. of this point is, accordingly, $90° + \alpha$ (since

$FA = 90°$ and $\Upsilon F = \alpha$, where $\Upsilon$ is the vernal equinox). Also $FC$ is $\delta$ (the star's declination). $EB$ is the $y$-axis.

Let $EX$ be the geocentric direction of the moon's centre. Let $(x, y, z)$ denote the rectangular co-ordinates of the moon's centre,

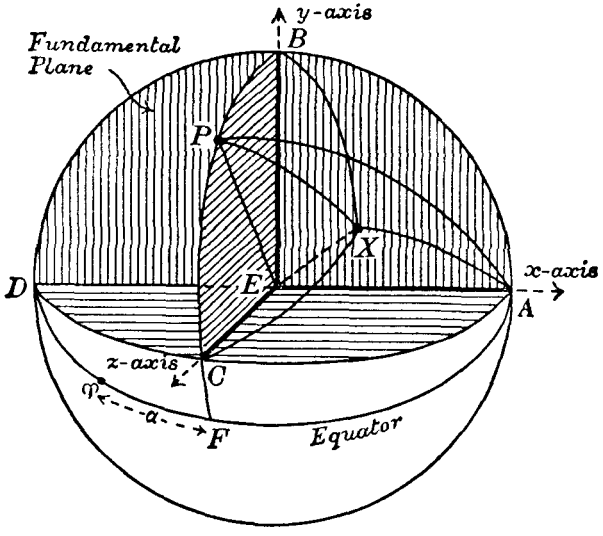



Fig. 135.

in terms of the earth's equatorial radius as unit, with respect to the co-ordinate system described. Joining $X$ to $A$, $B$ and $C$ by great circle arcs, we have

$$\frac{x}{r_1} = \cos AX, \quad \frac{y}{r_1} = \cos BX, \quad \frac{z}{r_1} = \cos CX.$$

In the spherical triangle $APX$, we have:

$$PX = 90° - \delta_1, \quad PA = 90°, \quad X\hat{P}A = (90° + \alpha) - \alpha_1.$$

We thus obtain, using the cosine-formula,

$$x = r_1 \cos \delta_1 \sin (\alpha_1 - \alpha) \qquad \ldots\ldots(1).$$

In the spherical triangle $BPX$, we have:

$$BP \equiv FC = \delta, \quad PX = 90° - \delta_1,$$

$$B\hat{P}X \equiv B\hat{P}A + A\hat{P}X = 180° + \alpha - \alpha_1.$$

Hence, by the cosine-formula,

$$y = r_1 [\sin \delta_1 \cos \delta - \cos \delta_1 \sin \delta \cos (\alpha_1 - \alpha)] \ldots\ldots(2).$$

We do not require the corresponding expression for $z$.

Since the fundamental plane is perpendicular to the straight line joining the moon's centre to the star, $x$ and $y$ are the co-ordinates, at any instant, on the fundamental plane of the centre of the shadow cast by the moon with reference to the light from the star or, in other words, $x$ and $y$ are the co-ordinates of the point of intersection of the axis of the cylinder with the fundamental plane.

Also $r_1$ and $P_1$ are related by the formula

$$\frac{1}{r_1} = \sin P_1 \qquad \text{.....(3).}$$

For an occultation, $(\alpha_1 - \alpha)$ and $(\delta_1 - \delta)$ may be regarded as small angles and, if $(\alpha_1 - \alpha)$ is expressed in seconds of time and $(\delta_1 - \delta)$ and $P_1$ in seconds of arc, the formulae (1) and (2) can be written, using (3), with sufficient accuracy for purposes of prediction, as

$$x = \frac{15}{P_1}(\alpha_1 - \alpha)\cos\delta_1 \qquad \text{.....(4),}$$

$$y = \frac{(\delta_1 - \delta)}{P_1} \qquad \text{.....(5).}$$

We now consider the co-ordinates $(\xi, \eta)$ of the observer with reference to the fundamental plane. To avoid complications in Fig. 135, we shall now suppose that $EX$ defines the geocentric direction of the observer, so that $PX$ is the observer's meridian. As before,

$$\frac{\xi}{\rho} = \cos AX, \quad \frac{\eta}{\rho} = \cos BX.$$

Now in the triangle $APX$, $PX = 90° - \phi'$, $PA = 90°$ and $X\hat{P}A = C\hat{P}A - C\hat{P}X$. If $h$ denotes the hour angle of the star at the instant concerned, $C\hat{P}X = h$, and therefore $X\hat{P}A = 90° - h$. Hence

$$\xi = \rho\cos\phi'\sin h \qquad \text{.....(6).}$$

Similarly, in the triangle $BPX$, we have $BP = \delta$, $PX = 90° - \phi'$, and $B\hat{P}X = 90° + (90° - h) = 180° - h$. Hence

$$\eta = \rho\,[\cos\delta\sin\phi' - \sin\delta\cos\phi'\cos h] \qquad \text{.....(7).}$$

Referring to Fig. 134 we see that, for immersion or emersion, the point of projection of the observer on the fundamental plane must lie on the circle of radius $k$ in which the cylinder cuts the fundamental plane. This condition is thus expressed by

$$(x - \xi)^2 + (y - \eta)^2 = k^2 \qquad \text{.....(8).}$$

For the calculation of occultations, the value of $k$ (the moon's radius expressed in terms of the earth's equatorial radius as unit) is taken to be 0·2725.

**205.** *The Besselian elements of an occultation.*

The quantities, relating to the occultation of any given star, are defined as follows:

$T_0$—the U.T. of conjunction in R.A. of the moon and the star.

$H$—the hour angle of the star at Greenwich at the instant $T_0$.

$Y$—the value of the $y$-co-ordinate [see formula (5)] at the instant $T_0$. At this instant, the value of the $x$-co-ordinate is evidently zero.

$x'$, $y'$—the rate of change of $x$ and $y$ *per hour* of mean solar time.

These quantities are called the Besselian elements of the occultation. They are the basis for the prediction of the time of the occurrence of the occultation at any particular observing station. Occultation data is not included in the *Astronomical Ephemeris* because it is inappropriate to an international almanac, since particular occultations are only observable from a limited region of the earth. Predictions of those occultations which are visible in the British Isles, New Zealand and parts of Australia are, however, given in the *Handbook of the British Astronomical Association.*

**206.** *The prediction of an occultation at any place.*

We shall assume that an approximate estimate of the U.T. of an occultation, visible at a particular point on the earth's surface, has first been obtained. We shall refer later to the methods of deriving this estimated U.T., which we shall suppose to be $(T_0+t)$.

If $\lambda$ is the longitude of the observer west of Greenwich, the hour angle $h$ of the star is given by

$$h = H - \lambda + t' \qquad \ldots\ldots(9),$$

where $t'$ is the number of sidereal hours equivalent to $t$ hours of mean solar time. Hence the values of $\xi$ and $\eta$ at U.T. $(T_0+t)$

can be calculated from (6) and (7) and the known terrestrial co-ordinates of the observer, $\rho$ and $\phi'$. The star's declination is given in the almanacs.

Also, since $x = 0$ at time $T_0$ (the moon and the star are then in conjunction as regards right ascension) and $y = Y$ at time $T_0$, then at time $(T_0 + t)$ we have

$$x = x't, \quad y = Y + y't \qquad \text{.....(10)},$$

$t$ being expressed in hours.

If $(T_0 + t)$ is actually the true time of the occultation, the equation

$$(x - \xi)^2 + (y - \eta)^2 = k^2$$

must be satisfied by the values of $\xi$, $\eta$, $x$, $y$ just calculated. In general, however, the estimated time is likely to differ by a minute or two from the true time, and we proceed as follows. We shall suppose that $(T_0 + t)$ is the estimated time and $(T_0 + t + \Delta t)$ is the accurate time concerned, where $\Delta t$ is expressed in hours. We calculate $x$, $y$ and $\xi$, $\eta$ for the U.T. $(T_0 + t)$, as already indicated; we denote these values by $x_1$, $y_1$, $\xi_1$ and $\eta_1$ respectively. The values of $x$, $y$, $\xi$ and $\eta$ corresponding to the true time $(T_0 + t + \Delta t)$ of the occultation are then derived from

$$\left.\begin{aligned} x &= x_1 + x' . \Delta t, \quad y = y_1 + y' . \Delta t \\ \xi &= \xi_1 + \xi' . \Delta t, \quad \eta = \eta_1 + \eta' . \Delta t \end{aligned}\right\} \qquad \text{.....(11)},$$

where $\xi'$, $\eta'$ denote the rate of change in $\xi$ and $\eta$ per mean solar hour. Since (8) must be satisfied for the occultation, we have

$$[x_1 - \xi_1 + \Delta t\,(x' - \xi')]^2 + [y_1 - \eta_1 + \Delta t\,(y' - \eta')]^2 = k^2,$$

or, writing

$$x_1 - \xi_1 = f, \quad y_1 - \eta_1 = g \qquad \text{.....(12)},$$

and neglecting squares of $\Delta t$, we deduce

$$\Delta t = \frac{k^2 - f^2 - g^2}{2\,(fx' + gy' - f\xi' - g\eta')} \qquad \text{.....(13)}.$$

In this formula, $k = 0{\cdot}2725$; $f$ and $g$ are derived from the quantities $x_1$, $\xi_1$, $y_1$ and $\eta_1$ computed for U.T. $(T_0 + t)$; also $x'$ and $y'$ are known Besselian elements. To calculate $\Delta t$, it is necessary to know the values of $\xi'$ and $\eta'$. Now by (6) and (7),

$$\xi' \equiv \frac{d\xi}{dt} = \rho \cos\phi' \cos h \frac{dh}{dt} \qquad \text{.....(14)},$$

$$\eta' \equiv \frac{d\eta}{dt} = \rho \cos\phi' \sin\delta \sin h \frac{dh}{dt} \qquad \text{.....(15)},$$

in which $dh/dt$ means the change in the star's hour angle (expressed in circular measure) in one hour of mean solar time. Now $h$ increases from 0 to $2\pi$ in $23^h\ 56^m$ of mean solar time,

$$\therefore \quad \frac{dh}{dt} = \frac{2\pi}{23{\cdot}93} = 0{\cdot}2625.$$

Write $$Q = \rho \cos \phi' \cos h \qquad \text{.....(16);}$$

then, from (6), (14), (15) and (16),

$$f\xi' + g\eta' = 0{\cdot}2625\,(fQ + g\xi \sin \delta) \qquad \text{.....(17)}$$

$$= 0{\cdot}2625 a_0, \quad \text{say.}$$

Hence $$\Delta t = \frac{k^2 - f^2 - g^2}{2\,[fx' + gy' - 0{\cdot}2625 a_0]} \qquad \text{.....(18),}$$

from which $\Delta t$ can be derived.

If necessary, greater accuracy can be attained by regarding $(T_0 + t + \Delta t)$ as the approximate time of the phenomenon, and repeating the calculations in the way just described.

There are several graphical methods* which enable an approximate estimate of the time of the phenomenon to be made. We can only refer very briefly to a method suggested by L. J. Comrie. Referred to the projection, on the fundamental plane, of the observer's position as origin, the co-ordinates of the centre of the shadow are $(f, g)$. Draw a circle of radius $k$ and compute the values of $f$ and $g$ for three times $t_1$, $t_2$ and $t_3$ in the neighbourhood of $T_0$. In actual practice $t_1$, $t_2$ and $t_3$ are always taken to be three consecutive values of the series $-2^h$, $-1^h$, $0^h$, $+1^h$, $+2^h$, in which $0^h$ corresponds to $T_0$. Plotting the positions defined by these values of the co-ordinates $(f, g)$, we obtain the track of the centre of the shadow with reference to the observer's position. This track or curve is approximately a straight line. If an occultation is possible, the curve will intersect the circle of radius $k$ in two points in general, and the corresponding times can be deduced from the data with which the curve was drawn. These times are the approximate times of immersion and emersion. Formula (18) can then be used to derive the time of occurrence of the occultation.

The position angle $\chi$ (measured eastwards from the north

* For example, see W. F. Rigge, *The Graphic Construction of Eclipses and Occultations* (Loyola Univ. Press, Chicago, 1924).

point of the moon's disc) corresponding to immersion or emersion is given from the figure by

$$\sin\chi = -\frac{f}{k}, \quad \cos\chi = -\frac{g}{k} \qquad \ldots\ldots(19),$$

where the values of $f$ and $g$ now refer to one or other of the points of intersection of the track and the circle.

In the *Handbook B.A.A.* predictions of occultations visible at Greenwich and Edinburgh are given. The approximate time of the phenomenon at a place $\Delta\lambda$ degrees west and $\Delta\phi$ degrees north of Edinburgh, say, can be derived from the formula

$$\text{Approx. time} = (\text{Predicted U.T. for Edinburgh}) + a\Delta\lambda + b\Delta\phi,$$

where $a$ and $b$ are quantities tabulated for each occultation. This approximate time serves as the basis for the accurate prediction according to the method already described.

**207.** *The reduction of occultations.*

Let $T$ denote the U.T. at the observed instant of an occultation Since the moon's ephemeris is given in E.T., the latest approximation to $\Delta T$ is first applied to the time of observation. The values of the moon's right ascension and declination are then derived from the almanac. From (4) and (5) we have

$$x = \frac{15}{P_1}(\alpha_1 - \alpha)\cos\delta_1 \qquad \ldots\ldots(20),$$

$$y = \frac{\delta_1 - \delta}{P_1} \qquad \ldots\ldots(21),$$

in which $\alpha_1$, $\delta_1$ and $P_1$ are the moon's right ascension, declination and horizontal parallax corresponding to time $T$; $(\alpha_1 - \alpha)$ is expressed in seconds of time and $(\delta_1 - \delta)$ in seconds of arc. Instead of (21), a more accurate formula is usually employed; it is derived as follows. From (2) and (3) we have

$$y = \frac{\sin(\delta_1 - \delta)}{\sin P_1} + \frac{2}{\sin P_1}\cos\delta_1 \sin\delta \sin^2\left(\frac{\alpha_1 - \alpha}{2}\right),$$

or—$(\delta_1 - \delta)$, $(\alpha_1 - \alpha)$ and $P_1$ being expressed as indicated above—

$$y = \frac{\delta_1 - \delta}{P_1} + \frac{(\alpha_1 - \alpha)\cos\delta_1 \sin\delta}{2P_1} \cdot 15^2 (\alpha_1 - \alpha)\sin 1'',$$

or, using (20), $$y = \frac{\delta_1 - \delta}{P_1} + \frac{x(\alpha_1 - \alpha)\sin\delta}{27500} \qquad \ldots\ldots(22).$$

The formulae (20) and (22) enable us to calculate the quantities $x$ and $y$ corresponding to time $T$.

If $h$ is the hour angle of the star at $T$, we calculate the values of $\xi$ and $\eta$ by means of (6) and (7).

We then derive $f$ and $g$ for the time $T$, where

$$f = x - \xi, \quad g = y - \eta \qquad \text{.....(23).}$$

In Fig. 136 let $X$ be the star's position at time $T$. We assume that, owing to an error in mean longitude, the moon's tabulated right ascension and declination are in error. Let $M$ be the centre of the moon, according to the tabulated values of $\alpha_1$ and $\delta_1$ at time $T$. The angular distance $XM$, calculated from the erroneous values of $\alpha_1$ and $\delta_1$, will thus be somewhat different from the observed angular distance, which is the moon's semi-diameter $S$. Denote the arc $XM$ by $S'$.

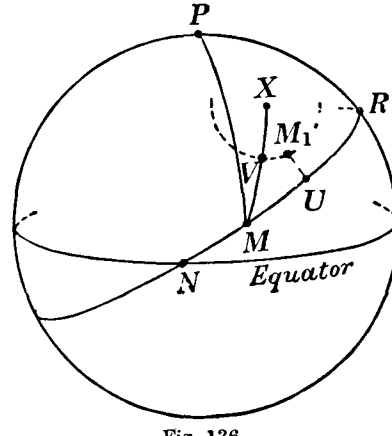


Fig. 136.

Denoting the angle $PMX$ —the position angle of the star with respect to $M$—by $\chi$, we have from (19) and (23)

$$\tan \chi = f/g = (x - \xi)/(y - \eta) \qquad \text{.....(24)}$$

from which $\chi$ is calculated.

Also, $\quad -r_1 S' = (x - \xi) \operatorname{cosec} \chi = (y - \eta) \sec \chi,$

where $S'$ is expressed in circular measure. We then calculate $S'$.

In Fig. 136 let $NMR$ be the great circle determined by the moon's orbital plane. Let $\rho$ denote the position angle $PMR$. Hence the angle $XMR$ is $\rho - \chi$.

$\rho$ is calculated as follows. Let $\Delta\alpha_1$ and $\Delta\delta_1$ denote the variation of $\alpha_1$ (in seconds of time) and of $\delta_1$ (in seconds of arc) in 1 minute. These quantities are obtained from the almanac. Then the components of the moon's motion along its orbit are, in circular measure, $15\Delta\alpha_1 \sin 1'' \cos\delta_1$ (along the parallel of declination through $M$) and $\Delta\delta_1 \sin 1''$ (along the meridian $MP$). Hence

$$\tan\rho = \frac{15\Delta\alpha_1 \cos\delta_1}{\Delta\delta_1} \qquad \text{.....(25).}$$

To summarise up to this point: In Fig. 136 we know for the observed instant $T$ of the actual occultation (i) the position of the star $X$, (ii) the angular distance $XM$ $(= S')$ and (iii) the angle $XMR$ $(= \rho - \chi)$. Now at the time, $T$, of occultation the angular distance of the star $X$ from the moon's centre is $S$ (the moon's semi-diameter). Since we know the position of $X$ definitely, the moon's centre must lie somewhere on the small circle (cutting $XM$ at $V$) whose centre is $X$ and whose angular radius is $S$. Let $M_1$ be the actual position of the moon's centre. Draw $M_1U$ perpendicular to $NMR$. Then $MU$ represents the correction to the moon's tabulated longitude (measured along $NMR$) and $UM_1$ the corresponding correction in latitude. Denote $MU$ by $\Delta\lambda$ and $UM_1$ by $\Delta\beta$. These quantities are small compared with $S$ (or $XV$) and so the arc $VM$ (or $S' - S$) is small. The moon's true centre $M_1$ is, accordingly, close to $V$ so that $M_1V$ is perpendicular to $XM$. Projecting $MU$ and $UM_1$ on $MV$ we obtain

$$S' - S = \Delta\lambda \cos(\rho - \chi) + \Delta\beta \sin(\rho - \chi) \quad \ldots\ldots(26).$$

Several observations of the same occultation made at different places provide the corresponding number of equations of the form of (26). The solution, by least squares, leads to the values of the corrections $\Delta\lambda$ and $\Delta\beta$.

### 208. *Eclipses of the moon.*

A lunar eclipse occurs when the moon passes into the shadow (with reference to the light from the sun) cast by the earth. The phenomenon can only take place when the earth is directly between the sun and the moon, that is to say, when the moon is in opposition; this corresponds to "full moon".

If the moon's orbital plane coincided with the plane of the ecliptic, lunar eclipses would occur at every full moon. As, however, the orbital plane is inclined at an angle of about 5° to the plane of the ecliptic, the conditions for the occurrence of a lunar eclipse require that the moon must be on or near the ecliptic when full, that is, the moon must be at or near one of the nodes.

When the whole of the moon's disc is obscured, the eclipse is said to be a *total eclipse*; when only a part at most is obscured, the eclipse is a *partial eclipse*.

In Fig. 137 let $S$ and $E$ be the centres of the sun and earth respectively. A cone can be drawn whose generators are the external tangents to the solar and terrestrial globes; the vertex of this cone is at $V$, and that part of the cone between $XV$ and $YV$ (dark shading) is called the *umbra*. In the figure the moon's orbit is shown and, if the moon passes completely within the umbra, a total eclipse occurs.

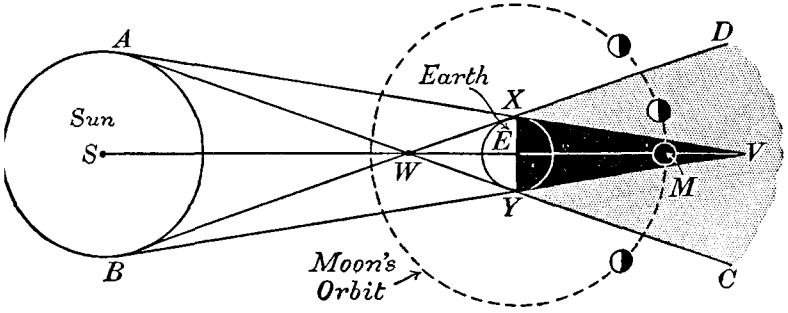



Fig. 137.

Another cone can be drawn whose generators are the internal tangents to the solar and terrestrial globes; its vertex is $W$. The parts of this cone represented by $DXV$ and $CYV$ (lightly shaded) receive a varying amount of illumination from the sun; along $XD$ and $YC$ the illumination is complete and along $XV$ and $YV$ it is zero. The lightly shaded portions of the internal cone form the *penumbra*, and when the moon passes through the penumbra towards the umbra its brightness gradually decreases until it is completely obscured when it is wholly immersed in the umbra.

**209.** *The angular radius of the shadow cone at the moon's geocentric distance.*

In the remainder of this chapter, we shall use the following notation:

$P$—equatorial horizontal parallax of sun.
$P_1$—equatorial horizontal parallax of moon.
$S$—sun's semi-diameter.
$S_1$—moon's semi-diameter.
$r$—geocentric distance of sun.
$r_1$—geocentric distance of moon.
$s$—angular radius (with respect to the earth's centre) of the umbral cone at the moon's distance.

We assume that $r$ and $r_1$ are expressed in terms of the earth's equatorial radius as the unit of distance.

Consider Fig. 138. The radius of the umbral cone at the moon's distance when a total eclipse is occurring is $MN$. The angular

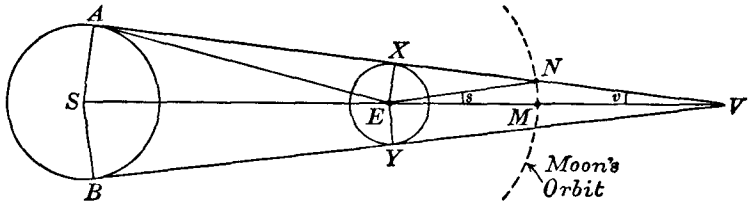



Fig. 138.

radius, $s$, with respect to $E$ is $N\hat{E}M$. Denote $X\hat{V}E$ by $v$. Then we have

$$X\hat{N}E = s + v.$$

But $X\hat{N}E$ is the angle subtended at the moon by the earth's radius and, regarding the earth as a sphere, we see that $X\hat{N}E = P_1$. Hence

$$P_1 = s + v \qquad \text{.....(27).}$$

Again, $S\hat{E}A = X\hat{A}E + v$. But $SEA$ is the angle subtended at the earth by the sun's radius and is therefore the sun's semi-diameter $S$. Also, $X\hat{A}E$ is evidently the sun's parallax $P$. Hence

$$S = P + v \qquad \text{.....(28).}$$

From (27) and (28) we have

$$s = P + P_1 - S \qquad \text{.....(29).}$$

In a similar manner it is easily shown that the angular radius $s'$ of the penumbral cone is given by

$$s' = P + P_1 + S \qquad \text{.....(30).}$$

The formulae for $s$ and $s'$ in (29) and (30) give the angular radii of the geometrical shadows concerned. It is found, however, that the earth's atmosphere increases, owing to absorption, the theoretical radii by about 2 per cent. For the prediction of lunar eclipses we use the expressions

$$s = \tfrac{51}{50}(P + P_1 - S) \qquad \text{.....(31),}$$

$$s' = \tfrac{51}{50}(P + P_1 + S) \qquad \text{.....(32).}$$

As an example, we find from (31), using the following values $P = 9''$, $P_1 = 57'$, $S = 16'$, that, very nearly,

$$s = 42'.$$

The angular diameter of the umbral cone is thus 84′. If the eclipse is *central*, that is, if the axis $EV$ of the shadow cone passes through the moon's centre, totality will last while the moon's centre moves through the angle $84' - 2S_1$ or 52′ if we take $S_1 = 16'$. Now the interval between two consecutive full moons is about 30 days; hence the moon's angular motion relative to the shadow is about 12° per day or 30′ per hour. Hence the central eclipse which we have been considering will last for about $\frac{52}{30}$ hour or about $1\frac{3}{4}$ hours.

The value of $s$ varies from a maximum of 46′·7 to a minimum of 38′·5; the former value occurs when the moon is nearest the earth (that is, in perigee) at the same time as the earth is furthest from the sun (that is, in aphelion); the minimum value occurs when the moon is in apogee and the earth in perihelion.

The length $EV$ of the earth's shadow is easily obtained. From (28), $v = S - P$, so that

$$EV = EX \operatorname{cosec} (S - P).$$

Putting $S = 16'$, $P = 9''$ and $EX = 3960$ miles, we find that

$$EV = 859{,}000 \text{ miles}.$$

**210.** *The ecliptic limits.*

We consider now the effect of the inclination to the ecliptic of the moon's orbital plane in restricting the number of eclipses. In Fig. 139 let $NM$ be the great circle, on the celestial sphere centred at $E$, which defines the plane of the moon's orbit. Let $M$ and $C$ be the centres of the moon and the shadow respectively, when an eclipse is about to take place. Let $C_1$ be the position of the shadow's centre when the moon is at the node $N$. Denote $NC_1$ by $\xi$ and let $t$ be the time required by the moon to go from $N$ to $M$, and for the shadow's centre to go from $C_1$ to $C$. Now the geocentric

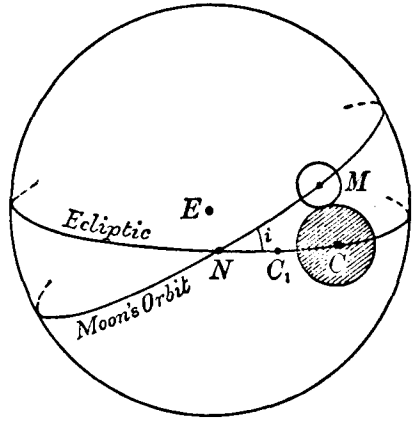



Fig. 139.

longitude of $C$ is the sun's longitude plus 180°. Hence, in finding the maximum value of $\xi$ for which an eclipse of some kind is possible we are, in effect, finding the maximum angular distance of the sun from the other node.

Let $\dot{\theta}$ be the rate at which the sun's longitude increases, and $\dot{\phi}$ the angular velocity of the moon in its orbit; for simplicity, we shall assume $\dot{\theta}$ and $\dot{\phi}$ to be constant. Then

$$NM = \dot{\phi}t, \quad NC = \xi + \dot{\theta}t.$$

If $\eta$ denotes the angular distance $CM$ and $i$ the inclination $M\hat{N}C$, we have, regarding the triangle $MNC$ as plane,

$$\eta^2 = (\xi + \dot{\theta}t)^2 + \dot{\phi}^2t^2 - 2\dot{\phi}t\,(\xi + \dot{\theta}t)\cos i,$$

or $$\eta^2 = \xi^2 - 2\xi t\,(\dot{\phi}\cos i - \dot{\theta}) + t^2\,(\dot{\theta}^2 + \dot{\phi}^2 - 2\dot{\theta}\dot{\phi}\cos i).$$

It is evident that $\eta$ is a minimum when $t$ is given by

$$\xi\,(\dot{\phi}\cos i - \dot{\theta}) - t\,(\dot{\theta}^2 + \dot{\phi}^2 - 2\dot{\theta}\dot{\phi}\cos i) = 0,$$

and, calling this minimum $\eta_0$, we have

$$\eta_0 = \frac{\xi\dot{\phi}\sin i}{(\dot{\theta}^2 + \dot{\phi}^2 - 2\dot{\theta}\dot{\phi}\cos i)^{\frac{1}{2}}} \qquad \text{.....(33),}$$

or, if $q$ is the ratio $\dot{\theta}/\dot{\phi}$, we can write (33) in the form

$$\xi = \eta_0\,(1 - 2q\cos i + q^2)^{\frac{1}{2}}\,\mathrm{cosec}\,i \qquad \text{.....(34).}$$

Now $q$ is the ratio of the earth's orbital angular velocity to the moon's orbital angular velocity, or the ratio of the moon's sidereal period to the year; taking mean values, $q$ is about 3/40. Also $i = 5°{\cdot}15$. Hence

$$\xi = 10{\cdot}3\eta_0 \qquad \text{.....(35).}$$

If the moon is about to enter the umbral cone,

$$\eta_0 = s + S_1,$$

the quantity $s$ being given by (31). For a partial eclipse to be possible it is evident that $\xi$ must not exceed $10{\cdot}3\,(s + S_1)$. For a total eclipse, $\xi$ must not exceed $10{\cdot}3\,(s - S_1)$. Taking the following numerical values,

$$S = 960'', \; S_1 = 933'', \; P = 9'', \; P_1 = 3423'',$$

we find that, for a partial eclipse to be possible, we must have

$$\xi < 10{\cdot}3\,[\tfrac{51}{50}\,(P + P_1 - S) + S_1],$$

or $$\xi < 9°{\cdot}9,$$

and, for a total eclipse to be possible,

$$\xi < 10{\cdot}3\,[\tfrac{51}{50}\,(P + P_1 - S) - S_1],$$

or $$\xi < 4^\circ{\cdot}6.$$

The value of $\xi$ required for a penumbral eclipse of the moon to take place may be derived in a similar manner.

An alternative way of examining the conditions for a lunar eclipse is to calculate the ecliptic limit. This is defined as the maximum possible distance of the sun from the node at the time of opposition in longitude (*full moon*). The method of calculation is almost identical to that for solar eclipses given in section 214. Since all the quantities on which it depends are not constants, the ecliptic limit is a variable quantity. Its maximum value is $12^\circ{\cdot}3$ (called the *superior ecliptic limit*) and its least value is $9^\circ{\cdot}6$ (called the *inferior ecliptic limit*).

**211.** *Calculation of a lunar eclipse.*

We shall suppose that the conditions for a partial or total eclipse are satisfied. In Fig. 140 let $M$ be the moon's position on the celestial sphere centred at $E$, and $C$ the centre of the earth's

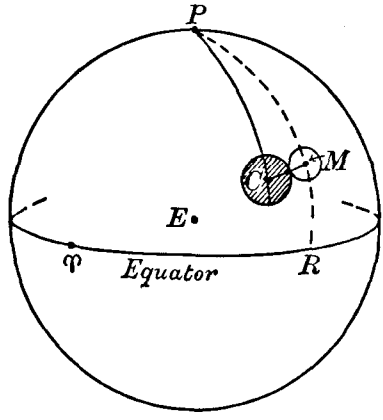



Fig. 140.

shadow. Let $(\alpha_1, \delta_1)$ and $(\alpha_0, \delta_0)$ be the equatorial co-ordinates of the moon's centre and of $C$ respectively. Denote $CM$ by $\eta$ and $P\hat{C}M$ by $Q$. Then we have

$$\sin\eta \sin Q = \cos\delta_1 \sin(\alpha_1 - \alpha_0),$$

$$\sin\eta \cos Q = \sin\delta_1 \cos\delta_0 - \cos\delta_1 \sin\delta_0 \cos(\alpha_1 - \alpha_0),$$

or, with sufficient accuracy,

$$\eta \sin Q = (\alpha_1 - \alpha_0) \cos \delta_1 \quad \text{.....(36),}$$

$$\eta \cos Q = \delta_1 - \delta_0 \quad \text{.....(37).}$$

As $C$ is diametrically opposite to the sun's centre, the R.A. $\alpha_0$ of $C$ is equal to the sun's R.A. + $12^h$, and the declination $\delta_0$ of $C$ is equal to $-\delta$ ($\delta$ being the sun's declination).

Write

$$x \equiv (\alpha_1 - \alpha_0) \cos \delta_1 = \eta \sin Q \quad \text{.....(38),}$$

$$y \equiv \delta_1 - \delta_0 \quad = \eta \cos Q \quad \text{.....(39).}$$

Let $x'$, $y'$ denote the hourly changes in $x$ and $y$. These are found by computing $x$ and $y$ at intervals of an hour, round about the time of the eclipse. Let $T_0$ be a convenient moment near opposition (say, the E.T.—to the nearest hour—of opposition in right ascension) and let $x_0$, $y_0$ be the computed values of $x$ and $y$ at this moment. At E.T. $(T_0 + t)$, where $t$ is in hours, we can write

$$x = x_0 + x't, \quad y = y_0 + y't.$$

Hence, by (38) and (39),

$$\eta \sin Q = x_0 + x't,$$

$$\eta \cos Q = y_0 + y't.$$

Write

$$x_0 = m \sin M, \quad y_0 = m \cos M \quad \text{.....(40),}$$

$$x' = n \sin N, \quad y' = n \cos N \quad \text{.....(41).}$$

From the known numerical values of $x_0$, $y_0$, $x'$, $y'$ the values of $m$, $M$, $n$, $N$ can be derived. We then have

$$\eta \sin Q = m \sin M + nt \sin N \quad \text{.....(42),}$$

$$\eta \cos Q = m \cos M + nt \cos N \quad \text{.....(43).}$$

Squaring these equations and adding, we obtain

$$\eta^2 = m^2 + 2mnt \cos (M - N) + n^2 t^2,$$

from which

$$t = -\frac{m}{n} \cos (M - N) \pm \left[\frac{\eta^2 - m^2 \sin^2 (M - N)}{n^2}\right]^{\frac{1}{2}} \quad \text{.....(44).}$$

Writing

$$m \sin (M - N) = \eta \sin \psi \quad \text{.....(45),}$$

formula (44) becomes

$$t = -\frac{m}{n} \cos (M - N) \pm \frac{\eta}{n} \cos \psi \quad \text{.....(46).}$$

The conditions for an eclipse can now be inserted. For the first and fourth contacts (when the moon is just entering or just leaving the umbra),

$$\eta = \tfrac{51}{50}(P + P_1 - S) + S_1.$$

Inserting this value of $\eta$ in (46), we obtain the times of first and fourth contacts.

For second and third contacts (the beginning and end of totality), the value of $\eta$ is given by

$$\eta = \tfrac{51}{50}(P + P_1 - S) - S_1,$$

and when this value of $\eta$ is inserted in (46), the times of the beginning and the end of the total phase are derived.

The time of the middle of the eclipse, whether partial or total, is given by $T_0 + t'$, where

$$t' = -\frac{m}{n}\cos(M - N).$$

The *magnitude* of a partial eclipse is the fraction of the moon's diameter obscured and is given by

$$\frac{\eta - m\sin(M - N)}{2S_1}.$$

where $m\sin(M-N)$ is taken positive for the time of mid-eclipse and $\eta$ is the mean of the values used for first and last umbral contacts.

There still remains the prediction of the position angle of the point on the moon's disc, where the eclipse is just beginning or ending. Referring to Fig. 140 we see that the figure has been drawn to represent fourth contact, when the moon is just about to pass out of the umbra. The great circle joining the point of contact of the moon's disc with the umbra and the centre of the umbra makes an angle $PMC$ with the meridian $PM$ through the moon's centre. As position angles are measured in the sense N E S W, the position angle $\theta$ of the point of contact is given by

$$\theta = 360° - P\hat{M}C,$$

or

$$\theta = 180° + C\hat{M}R.$$

But since $CM$ is small, $P\hat{C}M$ is approximately equal to $C\hat{M}R$. Hence

$$\theta = 180° + Q.$$

$Q$ is determined from (42) and (43), the value of $t$ for fourth contact being inserted in these equations. In a similar way, the

position angle of the point on the moon's disc where the eclipse begins can be derived.

**212.** *Eclipses of the sun.*

An eclipse of the sun is due to the interposition of the moon between the sun and the earth. The moon must then be at or near conjunction with the sun, that is, it must be new moon. Also, owing to the inclination of the moon's orbit to the ecliptic, it is evident that an eclipse cannot take place unless the moon is on or near the ecliptic, that is to say, the moon must be at or near one of the nodes.

As the moon's radius is much smaller than that of the earth, the earth cannot lie wholly within the shadow cone formed by the external tangents to the solar and lunar globes; consequently, an eclipse of the sun is visible only from a limited area

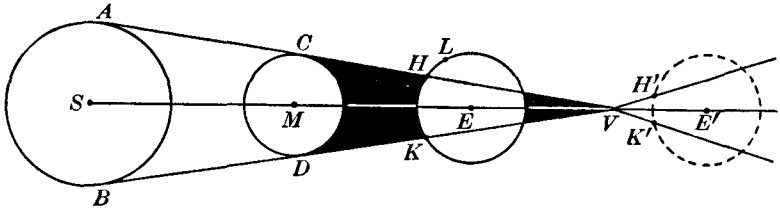



Fig. 141.

of the earth's surface. This is illustrated in Fig. 141, in which the shadow cone formed by the external tangents is shown shaded with its vertex at $V$; this shadow cone is called the *umbra*. As in the case of lunar eclipses, the *penumbra* is obtained by drawing the internal tangents. For points on the earth's surface between $H$ and $K$, within the umbra, the moon presents a complete barrier to the light from the sun and the eclipse is then said to be a *total solar eclipse*. At a point such as $L$, the moon evidently conceals only part of the solar disc; the eclipse is then said to be a *partial eclipse*. The condition for a partial eclipse at $L$ is that $L$ must be in the penumbra.

That total eclipses of the sun are at all possible is due to the fortunate fact that the moon's angular diameter is at times greater than that of the sun; the moon then covers a greater area of the sky than the sun. As the moon's orbit has an eccentricity of about 0·055, and the mean angular diameters of the

moon and sun are nearly equal, the value of the moon's diameter can fall below that of the sun (this occurs when the moon is in the neighbourhood of apogee) and consequently the moon cannot completely cover the solar disc; under these circumstances the eclipse, instead of being total, is said to be *annular*. This is illustrated in Fig. 141, where the centres of the sun, moon and earth are now supposed to be at $S$, $M$ and $E'$ respectively; for any place between $H'$ and $K'$, the eclipse is annular.

**213.** *The angle subtended at the earth's centre by the centres of the sun and moon at the beginning or end of a partial solar eclipse.*

In Fig. 142 the internal tangents between the sun and the moon are shown, forming the penumbra. Suppose that the tangent $AB$ is also tangential to the earth's surface at $C$. Then

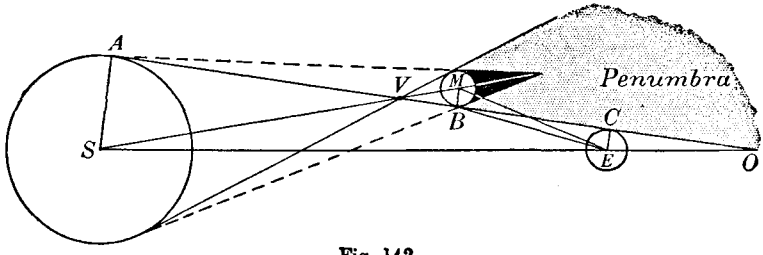



Fig. 142.

to an observer at $C$ the sun's surface is just wholly visible, and the geometrical circumstances illustrated in the figure evidently refer to the beginning or end of the partial phase, when $C$ is just about to enter or leave the penumbra. Let $D$ denote the angle $MES$. We have

$$D = B\hat{E}S + M\hat{E}B \qquad \text{.....(47).}$$

But under the circumstances depicted, $MB$ is very nearly perpendicular to $EB$, so that $M\hat{E}B = S_1$; hence

$$D = B\hat{E}S + S_1.$$

Also

$$B\hat{E}S = O\hat{B}E + E\hat{O}B \qquad \text{.....(48).}$$

But $O\hat{B}E \equiv C\hat{B}E =$ horizontal parallax of $B$, or very nearly the horizontal parallax of $M$; thus $O\hat{B}E = P_1$.

Again,

$$E\hat{O}B \equiv E\hat{O}A = A\hat{E}S - E\hat{A}C,$$

or

$$E\hat{O}B = S - P.$$

Hence, from (47) and (48),

$$D = S + S_1 + P_1 - P \qquad \ldots\ldots(49).$$

Thus $D$ can be evaluated at the time of any eclipse.

**214.** *Ecliptic limits.*

Consider Fig. 143, in which the moon and sun have the same geocentric longitude at $M$ and $S$—corresponding to new moon—near the descending node $N_1$. Let $\beta$ denote the latitude $SM$ and $i$ the inclination $M\hat{N}_1S$ of the moon's orbit to the ecliptic. Let $M_1$ and $S_1$ be the positions of the moon and sun a little later. Denote $MM_1$ by $y$. Regarding the triangle $MN_1S$ as plane, we see that the moon's longitude has increased by $MM_1 \cos i$ or $y \cos i$, and the sun's longitude by $SS_1$. If $m$ is the ratio of $SS_1$ to $y \cos i$, we have

$$SS_1 = my \cos i \qquad \ldots\ldots(50).$$

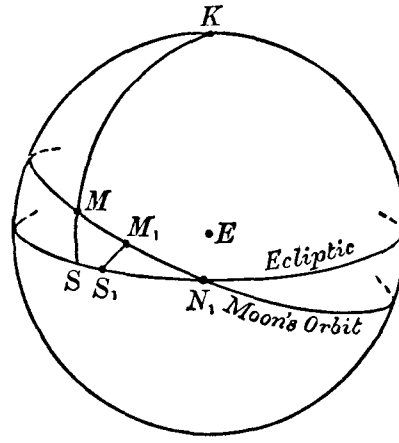



Fig. 143.

Also, $SN_1 = \beta \cot i$ and $MN_1 = \beta \operatorname{cosec} i$, so that

$$S_1N_1 = \beta \cot i - my \cos i,$$

and

$$M_1N_1 = \beta \operatorname{cosec} i - y.$$

Let $D$ denote the angular distance $M_1S_1$. Then from the triangle $M_1N_1S_1$ (regarded as plane), we have

$$D^2 = (\beta \cot i - my \cos i)^2 + (\beta \operatorname{cosec} i - y)^2 - 2(\beta \cot i - my \cos i)(\beta \operatorname{cosec} i - y) \cos i,$$

which may be written

$$D^2 = (1 - 2m \cos^2 i + m^2 \cos^2 i) \left\{ y - \frac{\beta \sin i}{1 - 2m \cos^2 i + m^2 \cos^2 i} \right\}^2 + \frac{\beta^2 (1 - m)^2 \cos^2 i}{1 - 2m \cos^2 i + m^2 \cos^2 i}.$$

Evidently the minimum value of $D$ (denoted by $D_0$) is given by

$$D_0 = \frac{\beta(1-m)\cos i}{(1-2m\cos^2 i + m^2\cos^2 i)^{\frac{1}{2}}},$$

or

$$D_0 = \frac{\beta(1-m)\cos i}{\{\sin^2 i + (1-m)^2\cos^2 i\}^{\frac{1}{2}}} \qquad \ldots\ldots(51).$$

Define an angle $j$ by

$$\tan j = \frac{\tan i}{1-m} \qquad \ldots\ldots(52).$$

Then we have

$$D_0 = \beta\cos j \qquad \ldots\ldots(53).$$

In (52), $\tan i$ is a small quantity (about 1/11), since $i = 5°{\cdot}15$; also $m = 3/40$ approximately. Using these values in (52), we deduce that

$$\cos j = \cos i - 0{\cdot}0006\cos i,$$

so that, with sufficient accuracy, we can write simply

$$D_0 = \beta\cos i = \beta\left(1 - 2\sin^2\frac{i}{2}\right).$$

For an eclipse to be possible, $D_0$ must not exceed the value of $D$ given by (49); hence

$$\beta\left(1 - 2\sin^2\frac{i}{2}\right) < S + S_1 + P_1 - P,$$

or, with sufficient accuracy, since $\sin^2\frac{i}{2}$ is a small quantity,

$$\beta < S + S_1 + P_1 - P + 2(S + S_1 + P_1 - P)\sin^2\frac{i}{2} \quad \ldots(54).$$

To evaluate the last term of (54), it is sufficient to use the mean values of the quantities concerned; the last term is thus $0'{\cdot}3$. We can also, with sufficient accuracy, put $P = 0'{\cdot}1$. Hence

$$\beta < S + S_1 + P_1 + 0'{\cdot}2 \qquad \ldots\ldots(55).$$

The maximum values of $S$, $S_1$ and $P_1$ are $16'{\cdot}3$, $16'{\cdot}8$ and $61'{\cdot}5$ respectively. Hence, from (55), if the moon's north or south latitude $\beta$ at the time of new moon is greater than $1°\,34'{\cdot}8$, a solar eclipse is not possible near this particular conjunction.

The *superior ecliptic limit* is defined to be the maximum distance of the sun from a node at the time of new moon if an eclipse is just possible. In Fig. 143 let $x$ denote $SN_1$. Then we have

$$\sin x = \tan\beta\cot i \qquad \ldots\ldots(56).$$

$x$ is thus a maximum when $\beta$ is greatest (1° 34′·8) and $i$ is least (4° 58′·8). With these values, $x = 18°{\cdot}4$. Hence, if the distance of the sun from a node at new moon exceeds 18°·4, an eclipse is impossible.

To calculate the *inferior ecliptic limit*, we take the minimum value of $\beta$ and the maximum value of $i$. The minimum values of $S$, $S_1$ and $P_1$ are 15′·8, 14′·7 and 53′·9 respectively; hence, from (55), the minimum value of $\beta$ is 1° 24′·6. The maximum value of $i$ is 5° 18′·6. Hence, from (54), the inferior ecliptic limit is 15°·4. We thus see that if the sun is within 15°·4 of a node at new moon an eclipse must take place.

**215.** *The Besselian elements for a solar eclipse.*

The method used in the prediction of eclipses is analogous to that already described in connection with occultations.

Through the earth's centre $E$, let a line $EC$ (Fig. 144) be drawn parallel, at a given moment, to the straight line joining

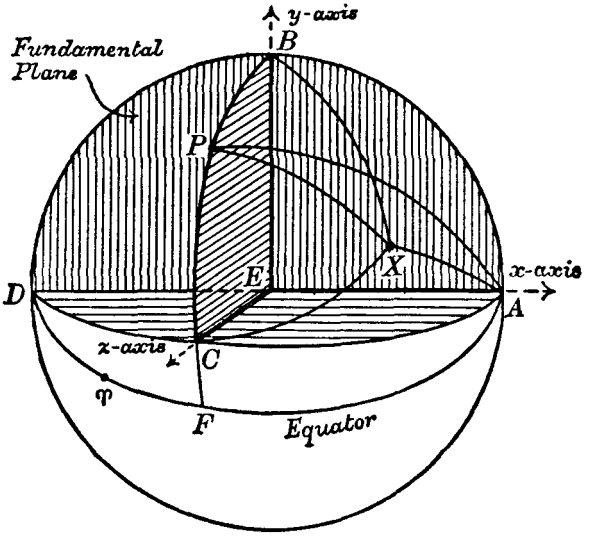



Fig. 144.

the centres of the moon and sun and meeting a sphere, centred at $E$, in $C$. $EC$ is the $z$-axis and the plane $DBA$ (shaded), to which $EC$ is normal at $E$, is the *fundamental plane*. If $P$ is the north celestial pole, the plane of the great circle through $C$ and

$P$ intersects the fundamental plane in $EB$. As in the corresponding figure for occultations, $EA$ and $EB$ are the $x$ and $y$ axes respectively.

(i) *The elements* x, y *and* d. We shall take $(\alpha, \delta)$ to be the apparent right ascension and declination of the sun and $(\alpha_1, \delta_1)$ the corresponding co-ordinates of the moon.

Let $(a, d)$ be the right ascension and declination of the point $C$ on the sphere.

Let $(x, y, z)$ be the rectangular co-ordinates of the sun, with reference to the axes described, in terms of the earth's equatorial radius as the unit. Then if $X$ is the sun's position on the sphere,

$$x = r\cos AX,\quad y = r\cos BX,\quad z = r\cos CX \quad ...(57),$$

$r$ being the sun's geocentric distance.

Now $A$ *is the pole of* $CPB$ *and must therefore be on the* equator; we have, then, $PA = 90°$, $FA = 90°$. Hence the right ascension of $A$ is $90° + a$ and therefore $X\hat{P}A = 90° + a - \alpha$. As $PX = 90° - \delta$, we have, from the first of (57), using the cosine-formula,

$$x = r\cos\delta\sin(\alpha - a) \quad .....(58).$$

In the triangle $PBX$, $BP = d$. Also, since $A\hat{P}B = 90°$, $X\hat{P}B = 180° + a - \alpha$. Hence

$$y = r[\sin\delta\cos d - \cos\delta\sin d\cos(\alpha - a)].....(59).$$

In the triangle $PCX$, $PC = 90° - d$, $PX = 90° - \delta$ and $X\hat{P}C = \alpha - a$; hence

$$z = r[\sin\delta\sin d + \cos\delta\cos d\cos(\alpha - a)].....(60).$$

In the same way we derive the corresponding equations for the moon:

$$x_1 = r_1\cos\delta_1\sin(\alpha_1 - a),$$
$$y_1 = r_1[\sin\delta_1\cos d - \cos\delta_1\sin d\cos(\alpha_1 - a)],$$
$$z_1 = r_1[\sin\delta_1\sin d + \cos\delta_1\cos d\cos(\alpha_1 - a)].$$

But since the $z$-axis is parallel to the line joining the centres of the moon and sun, we must have

$$x = x_1,\quad y = y_1.$$

The co-ordinates $(x, y)$ or $(x_1, y_1)$ are the *co-ordinates of the centre of the shadow* on the fundamental plane. Hence

$$r\cos\delta\sin(\alpha - a) = r_1\cos\delta_1\sin(\alpha_1 - a) \quad .....(61),$$
$$r[\sin\delta\cos d - \cos\delta\sin d\cos(\alpha - a)]$$
$$= r_1[\sin\delta_1\cos d - \cos\delta_1\sin d\cos(\alpha_1 - a)] ...(62).$$

At any instant, the values of $r$, $r_1$, $\alpha$, $\delta$, $\alpha_1$ and $\delta_1$ may be presumed known; the formulae (61) and (62) thus enable us to calculate $a$ and $d$. These formulae, however, can be put in a simpler form, since, at or near the time of eclipse, $\alpha$ and $\delta$ are little different from $\alpha_1$ and $\delta_1$ respectively. Let us write

$$\frac{r_1}{r} = b \qquad \text{.....(63),}$$

which can also be expressed as

$$b = \frac{\sin P}{\sin P_1} \qquad \text{.....(64),}$$

since, by definition, $1/r = \sin P$ and $1/r_1 = \sin P_1$. Thus $b$ can be calculated at any time; it is a small quantity of the order of 1/400. Writing $[\alpha_1 - \alpha + (\alpha - a)]$ for $(\alpha_1 - a)$ on the right of (61) and expanding, we find

$$\sin(\alpha - a)\{1 - b \sec\delta \cos\delta_1 \cos(\alpha_1 - a)\} = b \sec\delta \cos\delta_1 \cos(\alpha - a) \sin(\alpha_1 - \alpha),$$

or, with sufficient accuracy,

$$a = \alpha - \frac{b \sec\delta \cos\delta_1}{1 - b}(\alpha_1 - \alpha) \qquad \text{.....(65).}$$

In a similar way, from (62), we find that

$$d = \delta - \frac{b}{1 - b}(\delta_1 - \delta) \qquad \text{.....(66).}$$

The calculation of the quantities $a$ and $d$ is made at intervals of one hour.

In addition, the variations $x'$, $y'$ (per hour) of the co-ordinates $(x, y)$ of the centre of the shadow are required but these are easily derived from the differences of the values of $x$ and $y$ tabulated in the *Astronomical Ephemeris*.

(ii) *The element* $\mu$. For any meridian the hour angle of $C$ (Fig. 144) is the hour angle of the vernal equinox $\Upsilon$ less the right ascension of $C$. In particular, if $\mu$ denotes the hour angle of $C$ for the Ephemeris meridian when the Ephemeris sidereal time is $G$, then $\mu = G - a$. Hence, $a$ being found from (65), the value of $\mu$ can be calculated at any instant. Also the values of $\mu'$ (the variation of $\mu$ per hour) can evidently be found by simple processes.

(iii) *The elements* $f_1$, $f_2$. In Fig. 145 let $CD$ be the section of the fundamental plane with the plane of the paper. Consider first the penumbral cone with its vertex at $V_1$. Let $f_1$ denote the

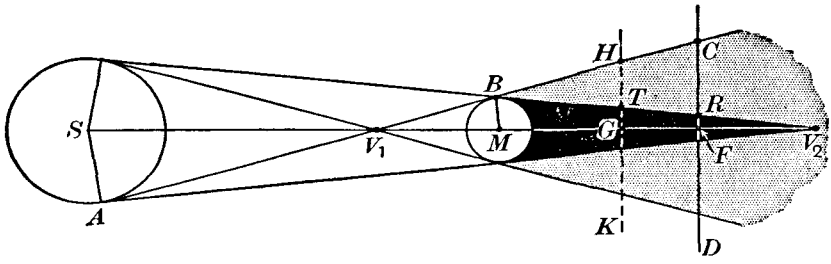



Fig. 145.

angle $AV_1S$ or $FV_1C$. Let $R$ be the linear radius of the sun and $k$ that of the moon. Then

$$\sin f_1 = \frac{R}{SV_1} = \frac{k}{V_1M} = \frac{R+k}{SM}.$$

But during any phase of the eclipse, $SM = r - r_1$, with sufficient accuracy. Hence, using (63),

$$\sin f_1 = \frac{R+k}{r(1-b)} \qquad \text{.....(67).}$$

In (67), the numerator $(R + k)$ is a constant, being the sum of the linear radii of the sun and moon.

Denoting the semi-vertical angle, $BV_2M$, of the umbral cone by $f_2$, we derive, in the same way,

$$\sin f_2 = \frac{R-k}{r(1-b)} \qquad \text{.....(68).}$$

The angles $f_1$ and $f_2$ are easily calculated from (67) and (68).

(iv) *The elements* $l_1$, $l_2$. Referring to Fig. 145, we see that $MF$ is the $z$-co-ordinate of the moon's centre, with reference to the axes already described. Hence $MF = z_1$. Also $V_1M = k \operatorname{cosec} f_1$. Denoting $V_1F$—the $z$-co-ordinate of the vertex of the penumbral cone—by $c_1$, we have

$$c_1 = z_1 + k \operatorname{cosec} f_1 \qquad \text{.....(69),}$$

and similarly, if $c_2$ denotes the $z$-co-ordinate of the vertex $V_2$ of the umbral cone,

$$c_2 = z_1 - k \operatorname{cosec} f_2 \qquad \text{.....(70).}$$

In each instance, $c_1$ and $c_2$ are algebraic quantities measured positively in the sense $\vec{FM}$ (the positive direction of the $z$-axis).

Let $l_1$ and $l_2$ denote the radii of the circles in which the penumbral and umbral cones intersect the fundamental plane. Then

$$l_1 \equiv FC = c_1 \tan f_1 \quad \text{and} \quad l_2 \equiv FR = c_2 \tan f_2 \quad \ldots(71),$$

or, using (69) and (70),

$$l_1 = z_1 \tan f_1 + k \sec f_1 \qquad \ldots\ldots(72),$$

$$l_2 = z_1 \tan f_2 - k \sec f_2 \qquad \ldots\ldots(73).$$

These formulae—(72) and (73)—enable the numerical values of $l_1$ and $l_2$ to be calculated.

The quantities $x$, $y$, $\sin d$, $\cos d$, $\mu$, $l_1$, and $l_2$ are called the Besselian elements of the eclipse. They are first calculated at hourly intervals and then subtabulated at intervals of 10 minutes in the *Astronomical Ephemeris*. Single values are also given of the quantities $\tan f_1$, $\tan f_2$, $\mu'$, and $d'$ as they are constant to the precision required. We now show how these elements are used to predict the circumstances of the eclipse.

**216.** *Eclipse calculations for any station.*

In Fig. 145 let $KH$ be the intersection, with the plane of the paper, of the plane through the observer parallel to the fundamental plane. Let $(\xi, \eta, \zeta)$ be the co-ordinates of the observer at any instant with reference to the fundamental axes. Then the plane $KH$ is given by $z = \zeta$. Consider now the radii of the circles on the plane $z = \zeta$, given by the intersection of this plane with the penumbral and umbral cones, and let $L_1$ and $L_2$ be these radii respectively. Then in the figure $L_1 = GH$ and $L_2 = GT$. We then have, since $FG = \zeta$,

$$L_1 = (c_1 - \zeta) \tan f_1,$$

$$L_2 = (c_2 - \zeta) \tan f_2,$$

or, using (71),

$$L_1 = l_1 - \zeta \tan f_1 \qquad \ldots\ldots(74),$$

$$L_2 = l_2 - \zeta \tan f_2 \qquad \ldots\ldots(75).$$

$L_1$ is always positive. Since $c_2$ has been used in its algebraic sense, $L_2$ is negative when the vertex $V_2$ of the umbral cone is situated (as in Fig. 145) to the right of $G$. This is the geometrical condition that certain areas on the earth's surface can lie within the umbral cone. Hence for any particular observer, at a distance $\zeta$

from the fundamental plane, a condition for a total eclipse is that $L_2$, computed by means of (75), shall be a negative quantity.

Let $\phi'$ and $\rho$ be respectively the geocentric latitude and distance of the observer, and $\lambda$ his longitude west of Greenwich. In Fig. 144 let $X$ now represent the observer's geocentric zenith on the celestial sphere. We then have

$$\xi = \rho \cos AX, \quad \eta = \rho \cos BX, \quad \zeta = \rho \cos CX \quad \ldots(76).$$

Now $PX$ is the meridian of the observer and, since $\mu$ is the Ephemeris hour angle of $C$, the hour angle of $C$ for the observer, $h$ say, is $X\hat{P}C = \mu - \lambda - 1{\cdot}0027\Delta T$. Then $X\hat{P}A = 90° - h$. Also $PX = 90° - \phi'$. The first of (76) then becomes, applying the cosine-formula to the triangle $APX$,

$$\xi = \rho \cos\phi' \sin h \qquad \ldots\ldots(77).$$

Similarly,

$$\eta = \rho[\sin\phi' \cos d - \cos\phi' \sin d \cos h] \qquad \ldots\ldots(78),$$
$$\zeta = \rho[\sin\phi' \sin d + \cos\phi' \cos d \cos h] \qquad \ldots\ldots(79).$$

The variations $\xi'$, $\eta'$ and $\zeta'$ per hour in the values of $\xi$, $\eta$ and $\zeta$ are calculated. For example,

$$\xi' = \mu' \rho \cos\phi' \cos h,$$

where $\mu'$ is the variation of $\mu$ per hour.

The values of $\xi$, $\eta$ and $\zeta$ are usually computed for the assumed time of contact; then by means of the calculated quantities $\xi'$, $\eta'$ and $\zeta'$, the values of $\xi$, $\eta$ and $\zeta$ are obtained at 10-minute intervals. The values of $L_1$ and $L_2$ can now be derived from (74) and (75) at the appropriate instants.

We now consider the conditions for the beginning and end of a partial or total eclipse at any station. When the partial phase is just beginning or ending, the observer is situated on the boundary of the penumbral cone, and his distance from the axis of the shadow is $L_1$—the radius of the circle in which the plane $z = \zeta$ cuts the penumbral cone. But the centre of this circle has co-ordinates $(x, y)$ and the observer's co-ordinates are $(\xi, \eta)$—both referred to axes in the plane $z = \zeta$ drawn parallel to the fundamental axes. Hence the condition required is

$$(x - \xi)^2 + (y - \eta)^2 = L_1^2 \qquad \ldots\ldots(80).$$

Similarly, the condition for the beginning or end of the total eclipse for the station concerned is

$$(x - \xi)^2 + (y - \eta)^2 = L_2^2 \qquad \ldots\ldots(81).$$

We shall now show how this last equation is to be used in calculating the beginning and end of the total phase. Let $T$ denote a suitably chosen E.T. near the time of totality, and let $T + t$ be the true E.T. of the beginning (or end) of totality Let $x_0$, $y_0$ be the values of the co-ordinates $x$, $y$ at time $T$, and $\xi_0$, $\eta_0$ the corresponding co-ordinates of the observer. Then we shall have, at time $T + t$, $t$ being expressed in hours,

$$x = x_0 + x't, \quad y = y_0 + y't; \quad \xi = \xi_0 + \xi't, \quad \eta = \eta_0 + \eta't.$$

Now, by (75), $$L_2 = l_2 - \zeta \tan f_2 \qquad \ldots\ldots(82),$$

and, since $f_2$ is a small angle, the value of $\zeta \tan f_2$ at time $T + t$ will not differ appreciably from its value at time $T$; also $l_2$ varies very slowly. Hence it will be sufficient to use, in (82), the value of $L_2$ computed for time $T$. We then have, for the beginning or end of the total phase (or of the annular phase),

$$[x_0 - \xi_0 + t(x' - \xi')]^2 + [y_0 - \eta_0 + t(y' - \eta')]^2 = L_2^2 \ldots(83).$$

All the quantities $L_2$, $x_0$, ... $\eta'$ being known, (83) is a quadratic equation in $t$, which will give the instants at which totality begins and ends.

Let auxiliary quantities $m$, $M$, $n$, $N$ be determined by means of

$$m \sin M = x_0 - \xi_0, \quad m \cos M = y_0 - \eta_0 \quad \ldots\ldots(84),$$

$$n \sin N = x' - \xi', \quad n \cos N = y' - \eta' \quad \ldots\ldots(85).$$

As $\tan M = (x_0 - \xi_0)/(y_0 - \eta_0)$, there are two values of $M$, differing by 180°, which can satisfy (84). Taking $m$ as the positive square root of $[(x_0 - \xi_0)^2 + (y_0 - \eta_0)^2]$, we shall choose that value of $M$ which will give to sin $M$ the same sign as $(x_0 - \xi_0)$. The procedure for $n$ and $N$ is similar. Geometrically, $m$ and $M$ are evidently the distance and position angle of the shadow axis relative to the observer; $n$ and $N$, in the same way, give respectively the magnitude and direction of the motion of the centre of the shadow relative to the observer. Inserting (84), (85) in (83), we obtain

$$n^2t^2 + 2mnt \cos(M - N) + m^2 - L_2^2 = 0 \ldots\ldots(86).$$

The two roots of this equation give the beginning and end of

totality. The solution is usually performed as follows. Let an angle $\psi$ be defined as follows:

$$L_2 \sin \psi = m \sin (M - N) \qquad \text{.....(87).}$$

The formula (87) gives two values of $\psi$—we shall refer to these immediately. We have, from (86),

$$\begin{aligned} n^2t^2 + 2mnt \cos (M - N) + m^2 \cos^2 (M - N) \\ = L_2^2 - m^2 + m^2 \cos^2 (M - N) \\ = L_2^2 \cos^2 \psi \quad \text{by (87).} \end{aligned}$$

Hence $$t = -\frac{m}{n} \cos (M - N) \pm \frac{L_2 \cos \psi}{n} \qquad \text{.....(88).}$$

If $\tau$ is the numerical value of $\frac{L_2 \cos \psi}{n}$, the beginning of totality occurs at E.T. $\left(T - \frac{m}{n} \cos \overline{M - N} - \tau\right)$ and the end of totality at E.T. $\left(T - \frac{m}{n} \cos \overline{M - N} + \tau\right)$. The duration of totality is $2\tau$. For some purposes, this result is not sufficiently accurate and we proceed as follows. Let $T_1$ be the approximate time of the beginning of totality. Calculate the quantities in (88) for $T_1$. As before, there will be two roots of (88), one referring to the beginning and the other to the end of the total phase. Let $\tau_1$ correspond to the former, so that the beginning of totality is at E.T.

$$T_1 - \frac{m_1}{n_1} \cos (M_1 - N_1) - \tau_1,$$

where $\tau_1$ is the numerical value of $\frac{L_2 \cos \psi}{n}$ computed for $T_1$, and the subscripts refer to the values of $m$, $n$, $M$, $N$ at $T_1$. Let $T_2$ be the approximate time of the ending of totality. Then, similarly, the end of the total phase is at E.T.

$$T_2 - \frac{m_2}{n_2} \cos (M_2 - N_2) + \tau_2,$$

where the subscripts refer to the values of $m$, $n$, $M$, $N$ at $T_2$. The difference between the computed times gives the duration of totality.

If $\psi_1$, $\psi_2$ are the values of $\psi$ at the beginning and end of totality, the quadrants of these angles are defined as follows, taking into account the two possible values of $\psi$ as given by (87). For the beginning of the total phase, the quadrant is such

that cos $\psi_1$ is positive, and, for the end of the total phase, the quadrant is such that cos $\psi_2$ is negative.

If it is considered necessary, the computations can be repeated, choosing a more accurate value of $T$.

For the calculations of the beginning and end of the partial phase at any station, a similar procedure is adopted, the value of $L_1$ being, of course, used in the calculations.

As in the case of occultations, the position angle $\theta$ for which the partial phase is beginning or ending is given by

$$L_1 \sin \theta = x_0 - \xi_0 + t\,(x' - \xi'),$$
$$L_1 \cos \theta = y_0 - \eta_0 + t\,(y' - \eta'),$$

the appropriate value of $t$ being that corresponding to the beginning or to the end of the partial phase, or, using the appropriate values of $N$ and $\psi$, by $\theta = N + \psi$.

The corresponding position angles for the total phase are found in the same way.

In the almanacs, the eclipse phenomena, as calculated by the preceding methods, are shown on charts.

**217.** *The frequency of eclipses.*

The interval between new moon and new moon—the synodical month or lunation—is 29·53 mean solar days. Also we have seen that the nodes of the moon's orbit (section 83) make a complete backward revolution of the ecliptic in 6798·3 days (about 18·6 years). The interval between two consecutive passages of the sun through a node is then 346·62 days. The sun thus separates from a node at the rate of 30·67 degrees per synodical month.

In considering the frequency of eclipses, we first summarise the relevant data concerning ecliptic limits:

| | Superior | Inferior |
|---|---|---|
| Ecliptic limits (lunar eclipse) | 12°·3 | 9°·6 |
| „ „ (solar eclipse) | 18°·4 | 15°·4 |

In each instance the ecliptic limit refers to the distance of the sun from a node when a partial eclipse (lunar or solar) is just possible.

Consider now Fig. 146, in which $N$ and $N'$ are the nodes. Let $NM_1 = NM_2 = N'M_1' = N'M_2'$ represent the ecliptic limits in

the case of a lunar eclipse, and $NS_1 = NS_2 = N'S_1' = N'S_2'$ the ecliptic limits in the case of a solar eclipse.

Now the arc $S_1S_2$ is at least 30°·8 (corresponding to the inferior limit for solar eclipses) and therefore this arc is always greater than the angle through which the sun separates from a node in a synodical month. Hence one new moon at least and, in consequence, one solar eclipse must occur when the sun is within the arc $S_1S_2$. Similarly one solar eclipse must occur when the sun is within the arc $S_1'S_2'$.

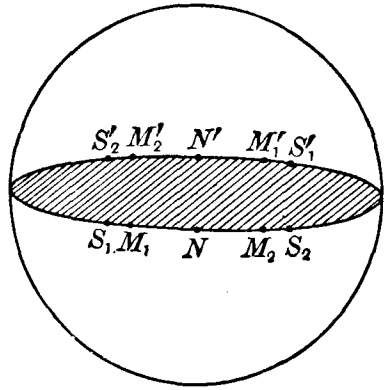


Fig. 146.

Again, the least value of $M_1M_2$ is twice 9°·6, or 19°·2, which is approximately two-thirds of the motion of the sun with reference to a node in a synodical month. It is thus possible for a full moon to occur, following a solar eclipse, near a node but outside the appropriate inferior limits; it follows that under these circumstances there will not be a lunar eclipse during the year.

We conclude that the *least* possible number of eclipses during a year is 2, and these are both solar eclipses.

Again, 6 synodical months = 177·2 days and the time required by the sun to pass from $N$ to $N'$ is 173·3 days. Therefore if it is full moon 2 days before the sun passes through $N$, it will also be full moon 2 days after the sun passes through $N'$. The angular distance of the sun from a node is then well within the limits for a lunar eclipse and hence, under these circumstances, there will be 2 lunar eclipses.

After $14\frac{3}{4}$ days from full moon, it will be new moon, and during this interval the sun will move through $15\frac{1}{3}°$ with reference to a node, and in 2 days will move 2° approximately. Hence if it is full moon 2 days before the sun reaches $N$, the sun will be $13\frac{1}{3}°$ beyond $N$ at the next new moon; at the preceding new moon it is $17\frac{1}{3}°$ from $N$. Also when it is full moon 2 days after the sun passes through $N'$, the sun will be $17\frac{1}{3}°$ beyond $N'$ at the next

new moon; at the preceding new moon it is $13\frac{1}{3}°$ from $N'$. Both $13\frac{1}{3}°$ and $17\frac{1}{3}°$ are within the superior limits for solar eclipses. Hence 2 solar eclipses are possible near each node, that is to say, 4 solar eclipses are possible within the interval of 346·6 days.

If the eclipses—when the sun is between $S$ and $S_1$—occur in January, the sun will be back again between $S$ and $S_1$ in the following December, that is, after 12 synodical months, and in the next half-lunation a solar and a lunar eclipse will be possible. Thus, in $12\frac{1}{2}$ synodical months, it is possible to have in all 5 solar and 3 lunar eclipses. But $12\frac{1}{2}$ lunations just exceed 365 days. Hence, in counting the possible eclipses in any year, we must omit either 1 solar eclipse or 1 lunar eclipse.

We conclude that the greatest number of eclipses possible in a year is 7, of which 4 are solar and 3 lunar or 5 solar and 2 lunar.

218. *The repetition of eclipses.*

The interval between two successive passages of the sun through a node is 346·62 days, and 19 such intervals are equivalent to 6585·8 days. The average value of the synodic month or lunation is 29·5306 days, and 223 lunations are equivalent to 6585·3 days. We thus have the approximate relation:

223 lunations = 19 revolutions of the sun with respect to a node
= 18 years 11 days, approximately.

This period of 18 years 11 days is called the *Saros*. Its significance with regard to eclipses is as follows. Suppose at new moon that the sun is within the ecliptic limits, say at $S$, so that a solar eclipse takes place. After 19 revolutions with respect to the node $N$, the sun will again be at the same distance $SN$ from $N$ and it will also be again new moon, for a period equal to 223 lunations has elapsed. Hence a solar eclipse will, in general, take place again.

The same argument applies to lunar eclipses. We thus see that eclipses are generally repeated at intervals of 18 years 11 days. As 19 revolutions of the sun are not, however, exactly equal to 223 lunations, it follows that the circumstances pertaining to one eclipse are not reproduced at the end of the Saros; in particular, the area on the earth's surface from which a total eclipse is visible will, in general, not be the same as that from which the next one (with reference to the Saros) is visible.

We add another interesting relation:

$$235 \text{ lunations} = 19 \text{ years of } 365\tfrac{1}{4} \text{ days.}$$

(235 lunations = 6939·69 days and 19 years = 6939·75 days.) This is the *Metonic cycle*. Thus if a full moon occurs on a certain date, there will be a full moon 19 years later. Clearly, the Metonic cycle applies in the same way to the repetition of new moons.

## EXERCISES

**1.** If the inclination of the moon's orbit to the ecliptic be 5° 20′ 6″, show that the moon will at some time occult every star whose latitude is numerically less than 6° 38′ 24″. [*Coll. Exam.*]

**2.** At midnight, on a certain date, the declination of the moon was 4° 36′ 46″·7 and its right ascension and declination were increasing at the rates of $23^s{\cdot}0$ and 164″ per 10 minutes, respectively. Show that a star in conjunction in right ascension with the moon at midnight could not be occulted at or near the time of this conjunction if the star's declination were less than 3° 10′·4, the sum of the moon's semi-diameter and horizontal parallax being 78′·0. [*Ball.*]

**3.** On the assumption of a spherical earth, show that, at a time when the moon's horizontal parallax and semi-diameter are respectively $P$ and $S$, any star of which the geocentric angular distance from the centre of the moon is less than $\sin^{-1}(\sin P + \sin S)$ is occulted by the moon at some part of the earth.

Show also that the greatest possible difference in declination of a star and the centre of the moon at the instant of conjunction in R.A., which will admit of an occultation occurring somewhere on the earth, is approximately

$$\sin^{-1}\{(\sin P + \sin S)\,(m^2\cos^2\delta + n^2)^{\frac{1}{2}}/(m\cos\delta)\},$$

where $\delta$ is the declination of the moon at conjunction, and $m$ and $n$ its rates of variation in R.A. and declination respectively. [*M.T.* 1919.]

**4.** Show that at the time of a partial or total lunar eclipse the geocentric angular distance of the moon's centre from the axis of the earth's shadow must be less than

$$\sin^{-1}(\sin P_1 + \sin S_1) - \sin^{-1}(\sin S - \sin P),$$

where $P$, $P_1$, $S$, $S_1$ are the horizontal parallaxes and semi-diameters of the sun and moon, respectively. The earth, sun and moon are considered to be spherical. [*M.T.* 1900.]

**5.** Show that the maximum duration of totality of a lunar eclipse is

$$\frac{2\,(P + P_1 - S - S_1)}{m}\left(1 + \frac{s\cos^2 i}{m}\right)\cos i \text{ hours},$$

approximately, if the atmospheric influence be neglected, where $P$, $P_1$, $S$, $S_1$, $s$, $m$ are the horizontal parallaxes, semi-diameters and hourly motions in longitude of the sun and moon, respectively, and $i$ is the inclination of the moon's orbit to the ecliptic.

**6.** Show that the interval between the middle of a lunar eclipse and the time of opposition is approximately

$$\frac{m\Delta}{m^2 + n^2 \cos\delta \cos\delta_1} \text{ hours,}$$

where $m$ and $n$ are the differences of hourly motion of the moon and the centre of the earth's shadow in declination and right ascension respectively, $\Delta$ is the difference in declination of the moon and the centre of the earth's shadow at the time of opposition, and $\delta$, $\delta_1$ are the average declinations of the shadow and the moon during the eclipse. [*Coll. Exam.*]

**7.** Show that the duration of a lunar eclipse does not necessarily contain the instant of opposition in longitude if the eclipse be partial, but must do so if the eclipse be total. [*Ball.*]

**8.** The horizontal parallaxes of the sun and moon being known, find the maximum inclination of the moon's orbit to the ecliptic which would ensure a solar eclipse every month.

**9.** Explain how a solar eclipse can be total at one place and annular at another. If at the place where the eclipse is just total at apparent noon the sun is in the zenith, what is the greatest possible breadth of the annulus at any other place? [*Lond.* 1925.]

**10.** Prove that at the instant of conjunction in right ascension the ratio of the distances of the sun from the moon and from the earth is

$$\{\sin P_1 - \sin P \cos(\delta - \delta_1)\}/\sin P_1,$$

$\delta$ and $\delta_1$ being the declinations of the sun and moon, $P$ and $P_1$ their horizontal parallaxes, and the square of sin $P$ being neglected.

Also prove that if $\dot{a}$ and $\dot{a}_1$ be the hourly changes in right ascension of the sun and moon, respectively, at the same instant, and $\dot{A}$ the hourly change in the right ascension of the line from the centre of the earth parallel to the line joining the centres of the sun and moon, then

$$\dot{A} = \dot{a} + \frac{\sin P \cos\delta_1}{\sin P_1 \cos\delta}(\dot{a} - \dot{a}_1). \qquad [\textit{Coll. Exam.}]$$

**11.** In 1917, within a few days of Dec. 21, an annular eclipse of the sun took place, visible near the South Pole. According to the *Nautical Almanac* the eclipse was exactly central at *midnight* in Latitude 89° 57′ S, Longitude 142° W. According to the *Connaissance des Temps* the eclipse was central at *noon* also in Latitude 89° 57′ S, but in Longitude 38° E. Show that the difference between the two statements would be accounted for if one almanac had made its calculations for the sea-level, and the other for a plateau about 15,000 feet above sea-level.

Show also that the difference would be accounted for if the positions of the moon adopted by the two almanacs differed by about $2\frac{1}{2}''$ in declination, the moon's parallax being 57′. [*M.T.* 1917.]

**12.** Obtain the following construction for the approximate determination of the point $Q$ at which a solar eclipse is central at any particular moment.

$M$ is the point of the earth's surface at which the moon is in the zenith, and $S$ the point at which the sun is in the zenith. Draw the great circle through $M$ and $S$, and on $SM$ produced take $Q$ such that

$$\sin SQ = (\alpha_{\mathbb{C}} - \alpha_{\odot}) \cos \delta / (P_{\mathbb{C}} - P_{\odot}) \sin \eta,$$

where

$$\tan \eta = (\alpha_{\mathbb{C}} - \alpha_{\odot}) \cos \delta / (\delta_{\mathbb{C}} - \delta_{\odot}).$$

The following data refer to the solar eclipse of 1922 September 20:

| | |
|---|---|
| G.M.A.T. of conjunction ... | $16^h\ 47^m\ 18^s$ |
| Equation of time ... ... | $+6^m\ 38^s$ |
| Sun's declination ... ... | $+1°\ 1'\ 42''{\cdot}7$ |
| Moon's declination ... | $+0°\ 48'\ 0''{\cdot}3$ |
| Sun's horizontal parallax | $8''{\cdot}8$ |
| Moon's horizontal parallax | $61'\ 24''{\cdot}1$ |

Show that the point at which the central eclipse takes place at noon is approximately in Longitude 106° 30′ E and Latitude 12° S. [*M.T.* 1922.]

# APPENDIX I

## *The Method of Dependences*

**219.** *Introduction.*

The method of dependences was first developed by F. Schlesinger* in 1911 in connection with the determination of stellar parallaxes and was subsequently applied to the measurement of the positions of asteroids and comets from photographs.†

In section 167 we obtained linear formulae connecting the standard co-ordinates of a star, or other object, in terms of the measured co-ordinates and the "plate constants" $a$, $b$, etc. To evaluate the plate constants, we make use of at least three comparison stars. In the astrographic problem, for example, if we have several plates giving the position of an asteroid, the plate constants have to be determined from each plate and this involves a large amount of numerical work. Consider a series of plates with the same plate-centre and in which the positions of the asteroid differ little from one plate to another. The same comparison stars can be used for each plate and instead of computing the plate constants for each plate we calculate certain quantities which depend on the comparison stars selected and on one position of the asteroid; these quantities, called *dependences*, are thus independent of the particular plate under investigation. The position of the asteroid is then expressed in terms of the dependences and certain measured quantities. A similar procedure is adopted for the measurement of stellar parallaxes.

**220.** *The astrographic problem* (3 *comparison stars*).

In this section we shall suppose that we employ three comparison stars. For a given plate, the measured and standard co-ordinates of the comparison stars are given by (57) and (58) on p. 297. We consider the measures in $x$ only—the procedure

* *Astrophysical Journal*, vol. XXXIII, p. 161 (1911).

† *Astronomical Journal*, vol. XXXVII, p. 77 (1926). See also H. C. Plummer, *Monthly Notices*, vol. XCII, p. 892 (1932).

for the measures in $y$ is similar. We then have for the three comparison stars

$$\xi_1 - x_1 = a\xi_1 + b\eta_1 + c \quad \ldots\ldots(1),$$
$$\xi_2 - x_2 = a\xi_2 + b\eta_2 + c \quad \ldots\ldots(2),$$
$$\xi_3 - x_3 = a\xi_3 + b\eta_3 + c \quad \ldots\ldots(3),$$

and for the asteroid

$$\xi - x = a\xi + b\eta + c \quad \ldots\ldots(4).$$

If $(\xi_0, \eta_0)$ are the standard co-ordinates of the asteroid for one of the plates—we shall refer to it as the "selected plate"—we can write (4) as

$$\xi - x = a\xi_0 + b\eta_0 + c + a(\xi - \xi_0) + b(\eta - \eta_0) \quad \ldots\ldots(5).$$

We are assuming for all the plates concerned that $(\xi - \xi_0)$ and $(\eta - \eta_0)$ are small quantities. Also, the constants $a$ and $b$, which involve principally the scale-correction and the orientation of the plate, are to be regarded as small quantities; accordingly, we neglect $a(\xi - \xi_0)$ and $b(\eta - \eta_0)$ in (5) which then becomes

$$\xi - x = a\xi_0 + b\eta_0 + c \quad \ldots\ldots(6).$$

It is to be remembered that, in (1), (2), (3) and (6), $x_1$, $x_2$, $x_3$ and $x$ are measured quantities obtained with all necessary accuracy. The standard co-ordinates $(\xi_1, \eta_1)$ etc. of the comparison stars are supposed to be known. We can then, if we please, solve (1), (2) and (3) in order to obtain $a$, $b$ and $c$ and then substitute their values in (6). Suppose for the moment that we know the values of $\xi_0$ and $\eta_0$. We then derive from (6) the value of $\xi$ for the plate concerned.

But this procedure is equivalent to the elimination of $a$, $b$ and $c$ between the four equations (1), (2), (3) and (6), and we can effect this elimination as follows.

Multiply (1), (2), (3) and (6) by $D_1$, $D_2$, $D_3$ and $-1$ respectively and add. We obtain

$$\begin{aligned} D_1(\xi_1 - x_1) + D_2(\xi_2 - x_2) + D_3(\xi_3 - x_3) - (\xi - x) &= a\{D_1\xi_1 + D_2\xi_2 + D_3\xi_3 - \xi_0\} \\ &+ b\{D_1\eta_1 + D_2\eta_2 + D_3\eta_3 - \eta_0\} \\ &+ c\{D_1 + D_2 + D_3 - 1\} \quad \ldots\ldots(7). \end{aligned}$$

The elimination of $a$, $b$ and $c$ is effected if

$$D_1\xi_1 + D_2\xi_2 + D_3\xi_3 = \xi_0 \quad \text{.....(8)},$$
$$D_1\eta_1 + D_2\eta_2 + D_3\eta_3 = \eta_0 \quad \text{.....(9)},$$
$$D_1 + D_2 + D_3 = 1 \quad \text{.....(10)},$$

and these are three equations from which $D_1$, $D_2$ and $D_3$ can be obtained.

The factors $D_1$, $D_2$ and $D_3$ are called the *dependences*. We then have from (7)

$$\xi - x = D_1(\xi_1 - x_1) + D_2(\xi_2 - x_2) + D_3(\xi_3 - x_3) \quad \text{.....(11)},$$

from which $\xi$ can be determined, all the other quantities being now supposed known.

Now $(\xi_1 - x_1)$, $(\xi_2 - x_2)$ and $(\xi_3 - x_3)$ are all small quantities and it will thus be sufficiently accurate to determine $D_1$, $D_2$ and $D_3$ if we substitute in (8) and (9) the *measured* co-ordinates of the comparison stars and the asteroid for the selected plate. Denoting these by $(X_1, Y_1)$, $(X_2, Y_2)$, $(X_3, Y_3)$ and $(X_0, Y_0)$ respectively, the equations to determine $D_1$, $D_2$ and $D_3$ become

$$D_1X_1 + D_2X_2 + D_3X_3 = X_0 \quad \text{.....(12)},$$
$$D_1Y_1 + D_2Y_2 + D_3Y_3 = Y_0 \quad \text{.....(13)},$$
$$D_1 + D_2 + D_3 = 1 \quad \text{.....(14)},$$

from which, solving in determinant form, we have for $D_1$

$$\frac{D_1}{\begin{vmatrix} X_0, & X_2, & X_3 \\ Y_0, & Y_2, & Y_3 \\ 1, & 1, & 1 \end{vmatrix}} = \frac{1}{\begin{vmatrix} X_1, & X_2, & X_3 \\ Y_1, & Y_2, & Y_3 \\ 1, & 1, & 1 \end{vmatrix}} \quad \text{.....(15)}.$$

In Fig. 147, let $S_1$, $S_2$ and $S_3$ be the positions on the selected plate of the images of the comparison stars, and $A_0$ the position of the image of the asteroid. Then the determinant under $D_1$ in (15) is simply twice the algebraic measure of the area of the triangle $A_0S_2S_3$ and the second determinant in (15) is twice the area of the triangle $S_1S_2S_3$. We obtain similar results for $D_2$ and for $D_3$ and we then have, as the solutions of (12), (13) and (14),

$$\frac{D_1}{A_0S_2S_3} = \frac{D_2}{A_0S_3S_1} = \frac{D_3}{A_0S_1S_2} = \frac{1}{S_1S_2S_3} \quad \text{.....(16)}.$$

Let the straight lines through $A_0$ and the vertices of the triangle cut the sides in $P_1$, $P_2$ and $P_3$.

Then $$\frac{A_0S_2S_3}{S_1S_2S_3} = \frac{A_0P_1}{S_1P_1}.$$

Hence

$$D_1 = \frac{A_0P_1}{S_1P_1},\ D_2 = \frac{A_0P_2}{S_2P_2} \text{ and } D_3 = 1 - D_1 - D_2 \quad ...(17).$$

The values of $D_1$, $D_2$ and $D_3$ can be readily obtained, with sufficient accuracy, by plotting the positions of the comparison stars and asteroid on squared paper, from the measured co-ordinates of the selected plate. It can be shown that the results of applying the method are most accurate when $A_0$ in Fig. 147 coincides with the centroid of the triangle $S_1S_2S_3$. In most instances it is possible to choose the comparison stars to satisfy this condition approximately. In any event, we choose the comparison stars so that the asteroid is within the triangle which they form.

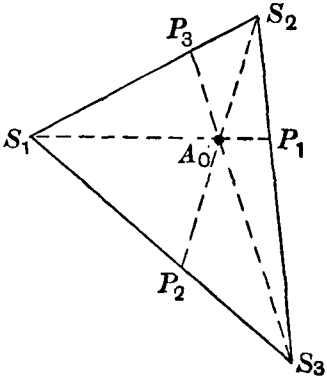


Fig. 147.

We rewrite (11) as

$$\xi = \sum_{i=1}^{3} D_i\xi_i + x - \sum_{i=1}^{3} D_i x_i \quad .....(18).$$

Similarly $$\eta = \sum_{i=1}^{3} D_i\eta_i + y - \sum_{i=1}^{3} D_i y_i \quad .....(19).$$

In (18) the expression $(x - \sum_{1}^{3} D_i x_i)$ is obtained, let us say, in millimetres and its actual value will depend on the scale of the plate taken with a particular telescope whose focal length we denote by $f_1$. The standard co-ordinates, on the other hand, are generally expressed in terms of the scale of the "astrographic telescope" the focal length of which we denote by $f_2$ ($f_2 = 3{\cdot}44$ metres so that 1 mm. on the plate corresponds to 60″). Consequently, we require to multiply $(x - \Sigma D_i x_i)$ by $f_2/f_1$ so

that $\xi$ may be given by (18) in astrographic units. A similar action is to be taken with respect to (19).

It is to be remembered that $(\xi_i, \eta_i)$ in (18) and (19) denote the standard co-ordinates of a comparison star with respect to the centre of the plate measured. If we make use of an Astrographic Catalogue we then obtain the standard co-ordinates of the comparison stars with respect to the "astrographic centre", that is, the centre of the astrographic plate from which the published standard co-ordinates are derived. If $\xi_1'$, $\xi_2'$, $\xi_3'$ and $\xi'$ denote the standard co-ordinates of the three comparison stars and the asteroid relative to the astrographic centre, we have, with sufficient accuracy,

$$\xi_1' = \xi_1 + k, \ldots, \xi' = \xi + k,$$

where $k$ is the standard co-ordinate of the centre of the measured plate with respect to the astrographic centre. Thus we obtain from (18), since $D_1 + D_2 + D_3 = 1$,

$$\xi' = \sum_{i=1}^{3} D_i \xi_i' + x - \sum_{i=1}^{3} D_i x_i \qquad \ldots\ldots(20),$$

with a similar equation for $\eta'$.

*Example.*

The following example has been given by H. E. Wood,* relating to Comet 1911*b*. The values of $\xi_1'$, $\xi_2'$ and $\xi_3'$ are taken from the *Oxford Astrographic Catalogue*. We give the data in tabular form for deriving the standard co-ordinate $\xi'$ of the comet.

| Star | $\xi'$ | $x$ | $y$ | $D$ | $D\xi'$ | $Dx$ |
|---|---|---|---|---|---|---|
| 1 | 7·692 | 5·6067 | 11·6605 | 0·233 | 1·7922 | 1·3064 |
| 2 | 17·650 | 15·8160 | 22·0935 | 0·273 | 4·8184 | 4·3178 |
| 3 | 23·547 | 21·6011 | 9·3951 | 0·494 | 11·6322 | 10·6709 |
| | | | | 1·000 | 18·2428 | 16·2951 |

For the comet,

$$x = 16{\cdot}3437, \quad y = 13{\cdot}3462 \qquad \ldots\ldots(21).$$

The dependences, $D$, in column 5 are found by means of a figure using the measured co-ordinates in columns 3 and 4 and the values of $x$ and $y$ for the comet as given in (21).

From (21) and the last column, we find that

$$x - \Sigma D_i x_i \equiv 16{\cdot}3437 - 16{\cdot}2951 = + 0{\cdot}0486.$$

* *Journal of the Brit. Ast. Association*, vol. XXXIX, p. 201 (1929).

We can obtain the value of the factor $f_2/f_1$ from the above data with sufficient accuracy, as follows. We have

$$\xi_2' - \xi_1' = 9{\cdot}958 \text{ and } x_2 - x_1 = 10{\cdot}2093.$$

Hence in astrographic co-ordinates

$$x - \Sigma D_i x_i = + 0{\cdot}0486 \times \frac{9{\cdot}958}{10{\cdot}209} = + 0{\cdot}0478.$$

Accordingly, from (20),

$$\begin{aligned} \xi' &= 18{\cdot}2428 + 0{\cdot}0478 \\ &= 18{\cdot}2906. \end{aligned}$$

This is the $\xi$ standard co-ordinate of the comet referred to the astrographic centre.

The $\eta$ co-ordinate is obtained in a similar way. If desired, the right ascension and declination of the comet can be computed by the formulae (22) and (23) of p. 285—or by equivalent formulae.

**221.** *The astrographic problem* (n *comparison stars*).

We suppose now that the number, $n$, of comparison stars is greater than 3. We now have $n$ equations of the form (1) to give the values of the plate constants $a$, $b$ and $c$. They are

$$\left.\begin{aligned} a\xi_1 + b\eta_1 + c &= \xi_1 - x_1 \\ a\xi_2 + b\eta_2 + c &= \xi_2 - x_2 \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ a\xi_n + b\eta_n + c &= \xi_n - x_n \end{aligned}\right\} \quad \ldots\ldots(22).$$

The values of $a$, $b$ and $c$ are to be obtained by the method of least squares. Multiply each by the corresponding coefficient of $a$ and, adding, we obtain

$$\begin{aligned} a\{\xi_1^2 + \xi_2^2 + \ldots + \xi_n^2\} + b\{\xi_1\eta_1 + \xi_2\eta_2 + \ldots + \xi_n\eta_n\} \\ + c\{\xi_1 + \ldots + \xi_n\} = \xi_1(\xi_1 - x_1) + \xi_2(\xi_2 - x_2). \end{aligned}$$

This is usually written in the notation

$$a[\xi_i^2] + b[\xi_i\eta_i] + c[\xi_i] = [\xi_i(\xi_i - x_i)] \quad \ldots\ldots(23),$$

the square brackets denoting summations for $i = 1, 2, \ldots, n$.

Similarly, by multiplying each equation of (22) by the corresponding coefficient of $b$, and adding, we obtain

$$a[\xi_i\eta_i] + b[\eta_i^2] + c[\eta_i] = [\eta_i(\xi_i - x_i)] \quad \ldots\ldots(24).$$

Also, adding the $n$ equations of (22), we have

$$a[\xi_i] + b[\eta_i] + nc = [\xi_i - x_i] \qquad \ldots\ldots(25).$$

As in (6), the equation for the asteroid is

$$a\xi_0 + b\eta_0 + c = \xi - x \qquad \ldots\ldots(26),$$

in which, as before, $\xi_0$ and $\eta_0$ denote the standard co-ordinates of the asteroid for a selected plate.

To eliminate $a$, $b$ and $c$ from the four equations (23)–(26) multiply these in order by $P$, $Q$, $R$ and $-1$ and add. Then if $P$, $Q$ and $R$ are given by

$$P[\xi_i^2] + Q[\xi_i\eta_i] + R[\xi_i] = \xi_0 \qquad \ldots\ldots(27),$$
$$P[\xi_i\eta_i] + Q[\eta_i^2] + R[\eta_i] = \eta_0 \qquad \ldots\ldots(28),$$
$$P[\xi_i] + Q[\eta_i] + Rn = 1 \qquad \ldots\ldots(29),$$

we have

$$\xi - x = P[\xi_i(\xi_i - x_i)] + Q[\eta_i(\xi_i - x_i)] + R[\xi_i - x_i]$$

which can be written

$$\xi - x = \sum_{i=1}^{n} (P\xi_i + Q\eta_i + R)(\xi_i - x_i).$$

Set $$D_i \equiv P\xi_i + Q\eta_i + R \qquad \ldots\ldots(30).$$

Then $$\xi - x = \sum_{i=1}^{n} D_i(\xi_i - x_i) \qquad \ldots\ldots(31).$$

In this last equation, $D_i$ is a function of the standard co-ordinates of the $n$ comparison stars and of $\xi_0$ and $\eta_0$. Since $(\xi_i - x_i)$ is a small quantity for any one of the stars it will be sufficient to regard $D_i$ as a function of the measured co-ordinates $x_1, \ldots, y_n, x_0, y_0$ for the selected plate; as before, we denote these co-ordinates by $X_i$, $Y_i$, $X_0$ and $Y_0$. Thus $P$, $Q$ and $R$ are now to be determined from (27), (28) and (29) in which we replace $\xi_i$ etc. by $X_i$ etc. Hence

$$P[X_i^2] + Q[X_iY_i] + R[X_i] = X_0 \qquad \ldots\ldots(32),$$
$$P[X_iY_i] + Q[Y_i^2] + R[Y_i] = Y_0 \qquad \ldots\ldots(33),$$
$$P[X_i] + Q[Y_i] + Rn = 1 \qquad \ldots\ldots(34).$$

We can simplify the calculation of $P$, $Q$ and $R$ by supposing that for the selected plate the values of $X_i$, $Y_i$, $X_0$ and $Y_0$ are measured from the centroid of the $n$ comparison stars. We then have

$$\sum_{i=1}^{n} X_i = \sum_{i=1}^{n} Y_i = 0$$

and $P$, $Q$ and $R$ are now to be determined from

$$P[X_i^2] + Q[X_iY_i] = X_0 \quad \text{.....(35)},$$

$$P[X_iY_i] + Q[Y_i^2] = Y_0 \quad \text{.....(36)},$$

$$Rn = 1 \quad \text{.....(37)}.$$

Then $$D_i = PX_i + QY_i + R \quad \text{.....(38)}.$$

The quantities $D_i$ are the *dependences*.

The practical procedure is first to calculate $P$, $Q$ and $R$ by means of (35), (36) and (37) and then to form the dependence $D_i$ for each star by means of (38).

As in the previous section,

$$\sum_{i=1}^{n} D_i = 1 \quad \text{.....(39)},$$

as we can see from (38) and (37), remembering that

$$\Sigma X_i = \Sigma Y_i = 0.$$

When the dependences have been calculated the value of $\xi$ for the asteroid can be then found from (31) for any number of plates.

It is to be remarked that the application of (31) requires that the values of $\xi_i$ for the $n$ comparison stars should be accurately known and that the values of $x$ and of $x_i$ for the $n$ comparison stars should be accurately measured.

**222.** *Application of the method of dependences to the determination of stellar parallax.*

The method of dependences can be used advantageously in the reduction of parallax measures.

Let $\xi_0$ denote the heliocentric standard co-ordinate of the parallax star at time $t_0$. As in section 175 we denote the parallax factor for an observation made at time $t$ by $F$, and the component of the annual proper motion by $\mu_x$ ($\equiv \mu_\alpha \cos\delta$). The geocentric standard co-ordinate $\xi$ at time $t$ is then given by

$$\xi = \xi_0 + F\Pi + T\mu_x \quad \text{.....(40)},$$

where $\Pi$ is the parallax of the star and $T \equiv t - t_0$ in years. But from the measurement of a plate obtained at time $t$, we have the value of $\xi$ given by (31) as

$$\xi = x + \sum_{i=1}^{n} D_i(\xi_i - x_i) \quad \text{.....(41)}.$$

Hence from (40) and (41)

$$F\Pi + T\mu_x + \xi_0 - \sum_{i=1}^{n} D_i\xi_i = x - \sum_{i=1}^{n} D_i x_i \quad \ldots\ldots(42).$$

Let

$$m = x - \sum_{i=1}^{n} D_i x_i \quad \ldots\ldots(43),$$

$$C = \xi_0 - \sum_{i=1}^{n} D_i\xi_i \quad \ldots\ldots(44).$$

Then

$$F\Pi + T\mu_x + C = m \quad \ldots\ldots(45).$$

In (43), $m$ depends on the measured quantities $x_i$ for the comparison stars and on the measured quantity $x$ for the parallax star. In (44), $C$ is a constant for each plate if we suppose the comparison stars to have negligible parallaxes and proper motions (actually, $\Pi$ and $\mu_x$ in (45) are the relative parallax and proper motion, that is, relative to the mean of the comparison stars). Thus each exposure provides an equation of the form of (45) and the values of $\Pi$, $\mu_x$ and $C$ (this is of no further interest) are determined from a least squares solution of all such equations. As the number of plates measured is usually at least a score, the convenience of the method and its economy in calculation can be readily appreciated.

*Example.*

The data below (in terms of a particular scale unit) are taken from the measures of a star, as given by Schlesinger.* Four comparison stars were used.

| Star | $X$ | $Y$ | $x$ | $D$ | $Dx$ |
|---|---|---|---|---|---|
| 1 | − 378 | + 62 | 198·766 | 0·25 | 49·6915 |
| 2 | − 8 | + 200 | 569·351 | 0·48 | 273·2885 |
| 3 | + 84 | − 130 | 660·934 | 0·11 | 72·7027 |
| 4 | + 302 | − 132 | 879·485 | 0·16 | 140·7176 |
| Parallax star | − 40·3 | + 75·5 | 535·133 | — | $\Sigma D_i x_i$ = 536·400 |

In columns 2 and 3 the co-ordinates of the four comparison stars and of the parallax star are with reference to the centroid

* *Astrophysical Journal*, vol. XXXIII, p. 163 (1911).

of the four comparison stars. From the values for the comparison stars we find

$$\sum_{i=1}^{4} X_i^2 = 241{,}208; \quad \sum_{i=1}^{4} Y_i^2 = 78{,}168; \quad \sum_{i=1}^{4} X_i Y_i = -75{,}820.$$

Inserting these values in (35), (36) and (37) and putting $X_0 = -40{\cdot}3$, $Y_0 = +75{\cdot}5$ (from the last line of the table), we find that

$$P = 0{\cdot}000196, \quad Q = 0{\cdot}00116, \quad R = 0{\cdot}250.$$

The values of $D_i$ are then calculated from (38)—they are shown in the fifth column. The products $D_i x_i$ are then formed for each star and the sum $\Sigma D_i x_i$ is shown at the bottom of the last column. Thus for this particular exposure the value of $m$, given by (43), is

$$535{\cdot}133 - 536{\cdot}400 \equiv -1{\cdot}267.$$

Also, for this particular exposure $F = +0{\cdot}880$ and $T = +0{\cdot}723$, so that we have from (45),

$$0{\cdot}880\Pi + 0{\cdot}723\mu_x + C = -1{\cdot}267 \qquad \ldots\ldots(46).$$

From the least square solution of all the equations of condition similar to (46), the values of $\Pi$ and $\mu_x$ in terms of the scale unit were found to be $0{\cdot}1055$ and $-0{\cdot}1307$ respectively, yielding the final results

$$\Pi = 0''{\cdot}281 \pm 0''{\cdot}004; \quad \mu_x = -0''{\cdot}348.$$

# APPENDIX II

## *Stellar Magnitudes*

**223.** *Apparent magnitude.*

The first classification of the stars, visible to the naked eye, according to their apparent brightness was made by Hipparchus about two thousand years ago. The fifteen brightest stars were designated "stars of the first magnitude" and stars just visible to the naked eye "stars of the sixth magnitude", stars of intermediate brightness being assigned to intermediate magnitude classes. With accurate instruments at present available for measuring the relative brightness of the stars, it is essential to have an accurate classification according to brightness and "magnitude" has now come to mean a number, on a certain scale, associated with the brightness of a star. If $m_1$ and $m_2$ denote the magnitudes of two stars on this scale, and $l_1$ and $l_2$ their apparent brightness or luminosity, the difference of magnitude, $m_2 - m_1$, is defined by the formula

$$\frac{l_1}{l_2} = 10^{-0\cdot4(m_1-m_2)} \qquad \ldots\ldots(1).$$

A difference of 5 magnitudes thus corresponds to a ratio of 100 : 1 in brightness and a difference of one magnitude to a ratio of 2·512 : 1. The zero of the magnitude scale is chosen arbitrarily. On the visual magnitude scale as adopted in practice magnitude zero corresponds approximately to the brightness of the star $\alpha$ Lyrae (Vega) whose magnitude is in fact 0·04. The visual magnitude scale ($V$) corresponds to that part of the spectrum to which the eye is most sensitive.

Other magnitude scales are introduced in the same way, corresponding to different parts of the spectrum. The two other systems in most frequent use are the blue ($B$) magnitude scale and the ultra-violet ($U$) scale. Again the zero-points of the scales are chosen arbitrarily. The choice is such that for a star like Vega the $V$, $B$ and $U$ magnitudes are all equal. In general the differences between these magnitudes for a particular star depend mainly on the star's temperature.

**224.** *Absolute magnitude.*

Although one star may appear to be brighter than another, it does not follow that the first star is *intrinsically* more luminous than the second. The brightness of a star varies inversely as the square of its distance from us; if $l$ denotes its apparent brightness corresponding to its distance $d$ and $L$ denotes what its apparent brightness would be if it were at a standard distance $D$ from us, we have

$$\frac{l}{L} = \frac{D^2}{d^2} \qquad \text{.....(2).}$$

Let $M$ denote what the star's apparent magnitude would be if it were at the standard distance $D$. Then, by (1) and (2),

$$10^{-0\cdot4(m-M)} = \frac{D^2}{d^2}$$

from which $$M - m = 5 \log_{10} (D/d) \qquad \text{.....(3).}$$

But the star's parallax $\Pi$ (in seconds of arc) is defined by

$$\Pi = \frac{a}{d} \operatorname{cosec} 1'' \qquad \text{.....(4),}$$

where $a$ is the radius of the earth's orbit. Similarly, if $\Pi_0$ is the parallax corresponding to the distance $D$,

$$\Pi_0 = \frac{a}{D} \operatorname{cosec} 1'' \qquad \text{.....(5).}$$

Hence, from (4) and (5),

$$\frac{D}{d} = \frac{\Pi}{\Pi_0}$$

and, from (3),

$$M = m + 5 \log_{10} \Pi - 5 \log_{10} \Pi_0 \qquad \text{.....(6).}$$

The standard distance, $D$, is by convention ten parsecs, so that $\Pi_0 = 0''\cdot1$. Hence we have

$$M = m + 5 + 5 \log_{10} \Pi \qquad \text{.....(7),}$$

in which $M$ is called the *absolute magnitude.*

It is found that the absolute magnitudes of the stars range from about $-5$ to $+15$, corresponding to a range in intrinsic luminosity of $100^4 : 1$ or 100,000,000 : 1.

# APPENDIX III

## *The Coelostat*

**225.** *General principles.*

For eclipse and certain astrophysical observations it is convenient to use a telescope which is fixed in position. Consequently, it is necessary to have subsidiary apparatus whose function is to reflect the light from the heavenly body under observation into the fixed telescope. Such an auxiliary instrument is called a *coelostat*. Its principal feature is a plane mirror

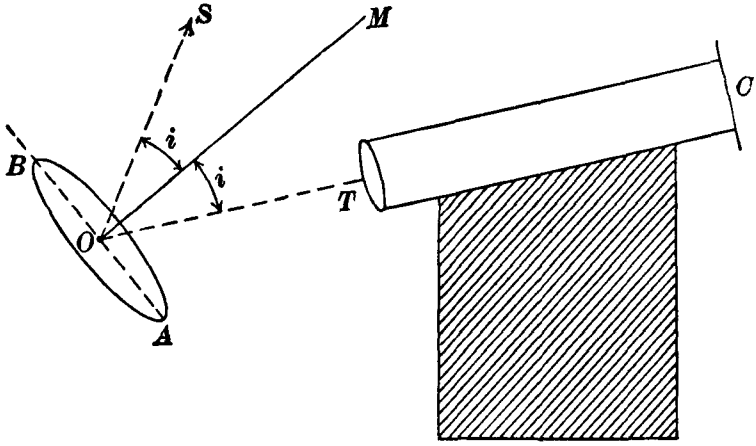



Fig. 148.

which can be made to rotate about a diameter at a uniform rate, this diameter being parallel to the earth's axis.

In Fig. 148, the mirror (which is generally circular) is shown with its axial diameter $AB$—the remainder of the instrument is not shown—and $OM$ is the normal to the mirror. The telescope is mounted as illustrated in the figure, $T$ being the object glass and $C$ the focal plane in which a photographic plate can be placed. The mirror is initially adjusted with its normal $OM$ in such a position that the beam of rays from a heavenly body $S$ is reflected along $OT$, which is collinear with the optical axis

of the telescope. As we shall see later, the mirror is made to rotate at half the angular rate of the diurnal motion of $S$ and this is a necessary condition that the rays from $S$ shall continue to fall normally on the object glass at $T$ after reflection at the mirror. We denote by $i$ the angle between the normal of the mirror and the direction $OS$ or $OT$.

Let $O$, the centre of the mirror, be the centre of the celestial sphere in Fig. 149. Then $S$ gives the direction of the heavenly

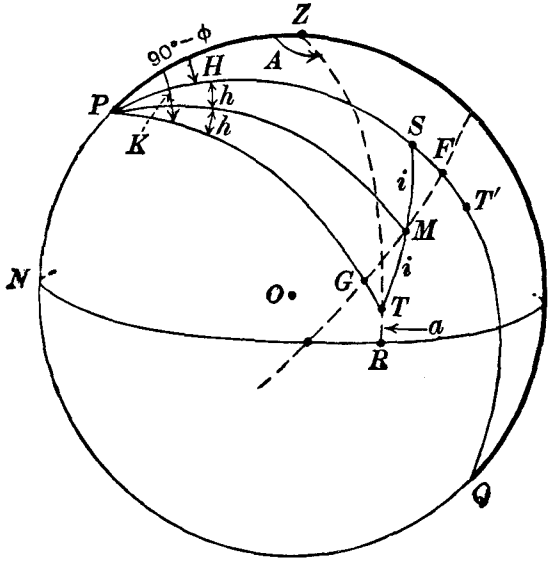



Fig. 149.

body, $M$ the direction of the normal of the mirror and $T$ the direction of the rays after reflection. Since the axis $AB$ of the mirror is parallel to the earth's axis of rotation, $M$ in Fig. 149 is on the celestial equator $FG$. If $\delta$ is the declination of $S$, the arc $FS$, measured along the meridian $PS$, is $\delta$. Let the meridian $PT$ intersect the equator in $G$. Then by the laws of reflection $OS$, $OM$ and $OT$ are coplanar and therefore $S$, $M$ and $T$ lie on a great circle; also the arcs $SM$ and $MT$ are equal and each is equal to $i$. It follows that $GT$ is equal to $SF$, so that $PT = 90° + \delta$. Again, $GM = MF$ and consequently $T\hat{P}M = S\hat{P}M$; we denote these angles by $h$.

Since the telescope is fixed in position, $T$ is a fixed point on the celestial sphere. Let the altitude and azimuth of $T$ be $a$ and $A$ respectively; then $a$ and $A$ are constants for a given position of the telescope.

Let $H$ be the hour angle of $S$, $H'$ the hour angle of $M$ and $K$ (a constant) the hour angle of $T$. We have

$$H' = H + h \text{ and } H + 2h = K$$

from which

$$2H' = H + K.$$

Hence

$$\frac{dH'}{dt} = \frac{1}{2}\frac{dH}{dt} \qquad \text{.....(1).}$$

This formula gives the rate at which the mirror must be rotated about its axis $AB$. For the sun, the rate may be expressed as 360° in 48 hours or $7\frac{1}{2}°$ per mean solar hour, very nearly. For a star, the rate is $7\frac{1}{2}°$ per sidereal hour.

226. *Formulae for the position of the telescope.*

We investigate now the various ways in which the telescope may be set up in a given latitude $\phi$.

(i) From the triangle $PZT$, in which

$PZ = 90° - \phi$, $ZT = 90° - a$, $PT = 90° + \delta$ and $P\hat{Z}T = A$,

we have by formula **A**,

$$-\sin\delta = \sin\phi \sin a + \cos\phi \cos a \cos A \qquad \text{.....(2),}$$

which gives the necessary relation between $a$ and $A$ for rays to be reflected into the telescope. Thus, if it is decided to have the telescope inclined to the horizontal at a certain angle, the azimuth $A$ is easily calculated from (2).

(ii) In particular, it may be found convenient to mount the telescope horizontally; in this case $a = 0°$ and $A$ is given by

$$\cos A = -\sin\delta \sec\phi \qquad \text{.....(3),}$$

with the condition for the numerical values of $\delta$ and $\phi$ that

$$\delta + \phi < 90°.$$

(iii) Again, it may be found convenient to mount the telescope in the meridian, in which event, $A = 180°$ and (2) becomes

$$\sin\delta = \cos(\phi + a)$$

whence

$$a = 90° - \phi - \delta \qquad \text{.....(4).}$$

(iv) A further practical consideration concerns the magnitude of the angle $i$ which, as a general rule, ought to be kept fairly small.

The cross section of the reflected beam is an ellipse with principal axes in the ratio $\cos i : 1$. It is always desirable to "fill" the object glass of the telescope and the condition for this is that $\cos i > \frac{d}{D}$, where $d$ and $D$ are respectively the diameters of the object glass and the mirror.

From the triangle $PSM$, in which $PM = 90°$, we have, by formula **A**,

$$\cos i = \cos \delta \cos h \qquad \ldots\ldots(5).$$

It is seen from (5) that $i$ is a minimum when $h = 0$. Thus the minimum value of $i$ is $\delta$. It follows that, in this case, $M$ and $T$ lie on the meridian $PS$, with $M$ at $F$ and $T$ at $T'$ where $FT' = FS = \delta$. The altitude, $a'$, of $T'$ is then easily found from the triangle $PZT'$ in which $PT' = 90° + \delta$: we have

$$\sin a' = -\sin \delta \sin \phi + \cos \delta \cos \phi \cos H \quad \ldots\ldots(6).$$

For example, with regard to a total eclipse of the sun, the hour angle $H$ for mid-totality can be found with sufficient accuracy and hence $a'$ can be calculated from (6). The azimuth, $A'$, of $T'$ can then be found from the formula

$$-\sin \delta = \sin \phi \sin a' + \cos \phi \cos a' \cos A' \quad \ldots\ldots(7).$$

# APPENDIX A

## ASTRONOMICAL CONSTANTS

Dimensions of the earth:

Equatorial radius, $a = 6378{\cdot}160$ km

Polar radius, $b = 6356{\cdot}775$ km

Flattening, $c \equiv \dfrac{a-b}{a} = \dfrac{1}{298{\cdot}25}$

Length of the day:

Mean sidereal day $= 23^{h}\ 56^{m}\ 4^{s}{\cdot}091$ mean solar time

Mean solar day $= 24\ \ 3\ \ 56\ {\cdot}555$ mean sidereal time

Length of the month (1900):

| | | | | | |
|---|---|---|---|---|---|
| Synodical | $= 29^{d}{\cdot}530589$ | $= 29^{d}$ | $12^{h}$ | $44^{m}$ | $2^{s}{\cdot}9$ |
| Tropical | $= 27\ {\cdot}321582$ | $= 27$ | 7 | 43 | 4 ·7 |
| Sidereal | $= 27\ {\cdot}321661$ | $= 27$ | 7 | 43 | 11 ·5 |
| Anomalistic | $= 27\ {\cdot}554551$ | $= 27$ | 13 | 18 | 33 ·2 |
| Nodical | $= 27\ {\cdot}212220$ | $= 27$ | 5 | 5 | 35 ·8 |

Length of the year (1900):

| | |
|---|---|
| Tropical | $= 365^{d}{\cdot}2422$ |
| Sidereal | $= 365\ {\cdot}2564$ |
| Anomalistic | $= 365\ {\cdot}2596$ |
| Eclipse | $= 346\ {\cdot}6200$ |

| | |
|---|---|
| Solar parallax | $= 8''{\cdot}794$ |
| Moon's equatorial horizontal parallax | $= 57'\ 2''{\cdot}608$ |
| Constant of aberration | $= 20''{\cdot}496$ |
| Constant of nutation | $= 9''{\cdot}210$ |
| General precession | $= 50''{\cdot}2564 + 0''{\cdot}0222T$* |
| Precession in R.A., | $m = 46''{\cdot}0850 + 0''{\cdot}0279T$ |
| | $= 3^{s}{\cdot}07234 + 0^{s}{\cdot}00186T$ |
| Precession in Dec., | $n = 20''{\cdot}0468 - 0''{\cdot}0085T$ |
| Obliquity of ecliptic, | $\epsilon = 23°\ 27'\ 8''{\cdot}26 - 46''{\cdot}85T$ |
| Mean geocentric distance of moon | $= 384{\cdot}401$ km |
| Mean geocentric distance of sun (1 astronomical unit) | $= 149{,}597{,}900$ km |
| 1 Parsec | $= 30{\cdot}86 \times 10^{12}$ km |
| 1 Light-year | $= 9{\cdot}46 \times 10^{12}$ km |
| Velocity of light | $= 299{,}792{\cdot}5$ km per second |

Light travels 1 astronomical unit in 499·012 seconds

Constant of gravitation, $G = 6{\cdot}670 \times 10^{-8}$ C.G.S. units or $6{\cdot}670 \times 10^{-11}$ S.I. units

Solar apex R.A. $= 18^{h}\ 04^{m}$ or 271°, Dec. $= +30°$

Pole of galactic plane (1900), R.A. $= 12^{h}\ 46^{m}{\cdot}6$, Dec. $= +27°\ 40'$

* $T$ is measured in Julian centuries from 1900·0.

# APPENDIX B

## DIMENSIONS OF THE SUN, MOON AND PLANETS

| | Symbol | Semi-diameter in km | Period of axial rotation | Reciprocal mass* (Sun = 1) | Density (water = 1) | Mean orbital velocity (km/sec) |
|---|---|---|---|---|---|---|
| Sun | ☉ | 696,000 | $25^{d}\cdot4$ | 1 | 1·41 | — |
| Moon | ☾ | 1,738 | $27^{d}\cdot3$ | 27,136,000 | 3·34 | 1·02 |
| Mercury | ☿ | 2,430 | $58^{d}\cdot6$ | 6,023,600 | 5·44 | 47·9 |
| Venus | ♀ | 6,050 | $243^{d}\cdot0$ | 408,590 | 5·23 | 35·0 |
| Earth | ⊕ | 6,378 (Eql.)<br>6,357 (Polar) | $23^{h}\ 56^{m}$ | 328,912 | 5·52 | 29·8 |
| Mars | ♂ | 3,395 (Eql.)<br>3,375 (Polar) | $24^{h}\ 37^{m}$ | 3,098,500 | 3·94 | 24·1 |
| Jupiter | ♃ | 71,600 (Eql.)<br>67,350 (Polar) | $9^{h}\ 50^{m}$ | 1,047·39 | 1·33 | 13·1 |
| Saturn | ♄ | 59,650 (Eql.)<br>53,850 (Polar) | $10^{h}\ 14^{m}$ | 3,498·5 | 0·70 | 9·7 |
| Uranus | ♅ | 23,550 (Eql.)<br>21,900 (Polar) | $10^{h}\ 49^{m}$ | 22,900 | 1·71 | 6·8 |
| Neptune | ♆ | 24,200 (Eql.)<br>23,700 (Polar) | $15^{h}\ 48^{m}$ | 19,300 | 1·77 | 5·4 |
| Pluto | PL | 5,900? | $6^{d}\cdot4$ | 4,000,000? | 4·6? | 4·7 |

* These masses include the mass of the satellite system, if any.

# APPENDIX C

## MEAN ELEMENTS OF THE PLANETARY ORBITS FOR THE EPOCH 1975 JANUARY 0·5 E.T.

| Name | Semi-major axis $a$ (in astronomical units) | Eccentricity $e$ | Sidereal period in tropical years | Sidereal mean daily motion $n$ | Synodic period (in days) |
|---|---|---|---|---|---|
| Mercury | 0·387099 | 0·205629 | 0·24085 | 4°·092339 | 115·88 |
| Venus | 0·723332 | 0·006785 | 0·61521 | 1°·602130 | 583·92 |
| Earth | 1·000000 | 0·016720 | 1·00004 | 0°·985609 | — |
| Mars | 1·523691 | 0·093382 | 1·88089 | 0°·524033 | 779·94 |
| Jupiter | 5·202804 | 0·048460 | 11·86224 | 0°·083091 | 398·88 |
| Saturn | 9·538844 | 0·055630 | 29·45771 | 0°·033460 | 378·09 |
| Uranus | 19·181854 | 0·047250 | 84·01247 | 0°·011732 | 369·66 |
| Neptune | 30·057960 | 0·008586 | 164·79558 | 0°·005981 | 367·49 |
| Pluto* | 39·29976 | 0·246115 | 246·378 | 0°·004001 | 366·75 |

| | | Mean longitude | | |
|---|---|---|---|---|
| Name | Inclination $i$ | Of the ascending node $\theta$ | Of perihelion $\varpi$ | At the epoch $\epsilon$ |
| Mercury | 7°·00427 | 48°·03493 | 77°·06645 | 320°·66305 |
| Venus | 3°·39438 | 76°·45475 | 131°.21928 | 310°·97453 |
| Earth | 0°·0 | 0°·0 | 102°·51044 | 99°·53431 |
| Mars | 1°·84983 | 49°·36466 | 335°·59881 | 249°·62919 |
| Jupiter | 1°·30450 | 100°·19608 | 13°·91992 | 355°·21414 |
| Saturn | 2°·48933 | 113°·43842 | 92°·55833 | 104°·17278 |
| Uranus | 0°·77316 | 73°·87283 | 170°·25472 | 205°·78286 |
| Neptune | 1°·77236 | 131°·50506 | 44°·40592 | 249°·91462 |
| Pluto* | 17°·14451 | 109°·9965 | 224°·2580 | 202°·3345 |

* The elements for Pluto are osculating elements, which define the instantaneous orbit of the planet round the sun.

# APPENDIX D

## ELEMENTS AND DIMENSIONS OF THE SATELLITES

| Planet | Satellite | | Mean distance from planet (unit = $10^3$ km) | Sidereal period (in days) | Eccentricity of mean orbit | Diameter (in km) |
|---|---|---|---|---|---|---|
| Earth | | Moon | 384·4 | 27·32166 | 0·05490 | 3476 |
| Mars | I | Phobos | *9·4* | *0·31891* | *0·0210* | 23 |
| | II | Deimos | 23·5 | 1·26244 | 0·0028 | 12 |
| Jupiter | I | Io | 422 | 1·76914 | 0·000 | 3660 |
| | II | Europa | 671 | 3·55118 | 0·000 | 2960 |
| | III | Ganymede | 1070 | 7·15455 | 0·001 | 5200 |
| | IV | Callisto | 1883 | 16·68902 | 0·007 | 4720 |
| | V | Almathea | 181 | 0·49818 | 0·003 | 160 |
| | VI | Himalia | 11478 | 250·57 | 0·158 | 100 |
| | VII | Elara | 11737 | 259·65 | 0·207 | 30 |
| | VIII* | Pasiphae | 23500 | 739 | 0·4 | 20 |
| | IX* | Sinope | 23600 | 758 | 0·275 | 20 |
| | X | Lysithea | 11720 | 259·22 | 0·12 | 20 |
| | XI* | Carme | 22600 | 692 | 0·207 | 20 |
| | XII* | Ananke | 21200 | 631 | 0·169 | 20 |
| | XIII | Leda | 11100 | 240 | ? | ? |
| Saturn | I | Mimas | 186 | 0·94242 | 0·0202 | 500 |
| | II | Enceladus | 238 | 1·37022 | 0·0045 | 600 |
| | III | Tethys | 295 | 1·88780 | 0·0000 | 1000 |
| | IV | Dione | 378 | 2·73692 | 0·0022 | 950 |
| | V | Rhea | 527 | 4·51750 | 0·0010 | 1300 |
| | VI | Titan | 1222 | 15·94545 | 0·0292 | 4800 |
| | VII | Hyperion | 1483 | 21·27666 | 0·1042 | 450 |
| | VIII | Iapetus | 3560 | 79·33085 | 0·0283 | 1100 |
| | IX* | *Phoebe* | 12950 | 550·34 | 0·1633 | 200 |
| | X | Janus | 159 | 0·7490 | 0·0 | 300 |
| Uranus | I | Ariel | 192 | 2·52038 | 0·0028 | 700 |
| | II | Umbriel | 267 | 4·14418 | 0·0035 | 500 |
| | III | Titania | 438 | 8·70588 | 0·0024 | 1000 |
| | IV | Oberon | 586 | 13·46326 | 0·0007 | 900 |
| | V | Miranda | 130 | 1·414 | $<0·01$ | 250 |
| Neptune | I* | Triton | 355 | 5·87684 | 0·00 | 3800 |
| | II | Nereid | 5562 | 359·88 | 0·75 | 300 |

* *Motion retrograde.*

## EPHEMERIS AND UNIVERSAL TIME

The rotation of the earth is a basis for time-measurement and as regards *Universal Time* (U.T.) this rate of rotation is assumed to be uniform. Recently, first crystal and then atomic clocks – now accurate to 1 part in $10^{13}$ – have shown that the earth's rotation is at times irregular, the deviations from uniformity being minute—of the order of 1 or 2 milliseconds per day—and unpredictable. In the gravitational theories of the bodies of the solar system, the passage of time is postulated to be *uniform*; this time is defined as *Ephemeris Time* (E.T.) and it is in terms of E.T. that astronomical quantities are now tabulated in the almanacs. The epoch from which E.T. is measured is

1900 January 0·5 [E.T.],

more elaborately defined in 1958 as "the instant near the beginning of the calendar year A.D. 1900 when the mean longitude of the sun was 279° 41′ 48″·04, at which instant the measure of E.T. was 1900 January 0, 12 h. *precisely*". The epoch for U.T. is 1900 January 0, 12 h. [U.T.]. Although the two epochs are apparently denoted by the same expression, they do not correspond to the same *instant* of time, the epoch of E.T. being 4 s. *later* than that of U.T.

The E.T. for any instant is then defined by the following formula for the geometric mean longitude of the sun:

$$L = 279^\circ\ 41'\ 48''{\cdot}04 + 129602768''{\cdot}13\,T + 1''{\cdot}089\,T^2.$$

Here $T$ is the ephemeris time measured in Julian centuries of 36525 ephemeris days from the fundamental epoch. The R.A. of the fictitious mean sun is given by the same expression with the effect of aberration added. The R.A. of the fiducial point for U.T., which we are calling simply the mean sun, has the same expression as that of the fictitious mean sun with universal time replacing ephemeris time as the argument.

It may be added that the fundamental unit of time is 1 second (E.T.) derived as 1/31556925·9747 of the length of the tropical year for 1900·0.

# INDEX

*The numbers refer to the pages*


**Aberration,** annual, 179; constant of, 185; constant of (in relation to the solar parallax), 213; differential, 297; diurnal, 80, 191; ellipse, 181; law of, 178; in astronomical photography, 296; planetary, 193
**Absolute** proper motions, 304
*Adams, C. E.*, 125
**Altitude,** 26; parallel of, 26
**Altitude Tables,** 333
**Analogies,** Delambre's, 22; Napier's, 23
**Analogue-formula,** 10
**Angle,** spherical, 2; hour, 29; of refraction, 61; of the vertical, 196; parallactic, 49; trigonometrical ratios for small, 21
**Annual aberration,** 179
**Annual parallax,** 218
**Annual variation,** 241, 260
**Annular eclipse,** 387
**Anomalistic month,** 133
**Anomalistic year,** 132
**Anomaly,** eccentric, 111; mean, 113; true, 111
**Ant-apex,** *264*; position angle of, *265*
**Antarctic Circle,** 48
**Apastron,** 343
**Apex,** solar, 261
**Aphelion,** 99
**Apogee,** 132
**Apparent co-ordinates,** 259
**Apparent noon,** 41
**Apparent orbit** of a visual binary, 345
**Apparent place** of a star, 245
**Apparent sidereal time,** 138
**Apparent solar day,** 41
*Aquino, R. de,* 333
**Arctic Circle,** 48
**Aries,** first point of, 37
**Artificial satellites,** 204
*Astrand, J. J.*, 115
**Astrographic plates,** measurement of, 298
**Astrometric co-ordinates,** 300
**Astronomical constants,** 420
**Astronomical latitude,** 196
**Astronomical unit,** of distance, 101
**Astronomical zenith,** 196
**Auxiliary ellipse,** 352
**Axis,** collimation, 75
**Azimuth,** 27, 30; rate of change of, 48
**Azimuth error,** 76; determination of, 89
**Azimuth Tables,** 323

*Baldwin, J. M.*, 265
*Ball, Sir R. S.*, 212
*Batten, A. H.*, 366
*Bauschinger, J.*, 115
**Besselian Day Numbers,** 184, 244, 246
**Besselian elements,** of an occultation, 373; of a solar eclipse, 390
**Besselian star constants,** 246
**Besselian year,** 145
**Bessel's formula,** 80
**Bessel's method** of investigating an occultation, *370*
**Binary,** eclipsing, 366; spectroscopic, 357
**Binary star,** 340; apparent orbit of, 345; elements of orbit of, 342
**Bohnenberger** eye-piece, 87
*Boss, Lewis*, 241
*Bradley*, 178; formula for refraction, 72
**Brightness,** of planets, 169
**Burdwood's Azimuth Tables,** 323

**Calendar,** 144
*Campbell, W. W.*, 83
**Cancer,** Tropic of, 151
**Capricorn,** Tropic of, 151
**Cardinal points,** 27
**Cassini's hypothesis** (refraction), 72
**Cataloguing** the stars, 246

**Celestial equator**, 28
**Celestial horizon**, 26
**Celestial latitude and longitude**, 40
**Celestial sphere**, 1, 33
**Centering error**, 289, 316
**Centre, equation of the**, 119, 148
*Christie and Dyson.* Method of determining plate constants, 307
**Chronograph**, 90
**Circle**, Antarctic, 48; Arctic, 48; great, 1; position, 320; small, 1; vertical, 26
**Circumpolar stars**, 32
**Clock stars**, 95
**Coelostat**, 416
**Colatitude**, 5
**Collimation axis**, 75
**Collimation error**, 76, 78; determination of, 83; in sextant, 316
**Collimation plane**, 75
**Collimator**, 84
**Companion** (*of a binary star*), 340
**Comparison stars**, 304
*Comrie, L. J.*, 375
**Conjunction**, 129
**Constant of aberration**, 185; measurement of, 187; related to solar parallax, 213; theoretical value of, 190
**Constant of gravitation**, 101
**Constant of nutation**, 247
**Co-ordinates**, galactic, 276; heliocentric ecliptic, 122; heliocentric equatorial, 124; heliographic, 169; mean and apparent, 259; standard, 280
**Correction for light-time**, 193
**Cosine-formula**, 6
**Culminate** (to), 29
**Culmination**, lower, 33; upper, 33
**Curve**, velocity, 359

**Day**, mean solar, 42; sidereal, 37, 138
**Day-Numbers**, Besselian, 184, 244, 246; independent, 184, 245
**Dead reckoning**, 322
**Declination**, 29, 35; measurement of, 91; parallel of, 29
**Deimos**, 103
*Delambre*, analogies, 22
**Dependences**, 404
**Dimensions** of the sun, moon and planets, 421
**Dip of the horizon**, 317
**Direct motion**, 162
**Disc**, 167; heliographic co-ordinates of centre of, 171
**Distance**, astronomical unit of, 101; north polar, 29; zenith, 26
**Diurnal aberration**, 191
**Diurnal** method (solar parallax), 210
*Doolittle*, 316
**Dynamical parallaxes**, 356
**Dynamical** principles of orbital motion, 104
*Dyson, Christie and*, 307

**E-terms** (aberration), 186
**Earth**, heliocentric rectangular co-ordinates of, 126; orbit of, 37, 131
**Eccentric anomaly**, 111; as a series, 117
**Eccentricity**, 99
**Eclipse**, of the moon, 378; of the sun, 386
**Eclipses**, frequency of, 398; repetition of, 400
**Eclipsing binary stars**, 366
**Ecliptic**, 38; mean, 238; obliquity of, 39
**Ecliptic limits**, 381, 388
**Elements**, determination of the, 129; of orbit of binary star, 342; of orbit of a planet, 116, 122; of the planetary orbits, 422; of the satellites, 423
**Ellipse**, aberrational, 181; auxiliary, 352; parallactic, 219
**Elliptic orbits**, elements of, 116, 122, 422
**Elongation**, 163
**Ephemeris Time**, 139, 424
**Epoch**, 122; mean longitude at the, 122
**Equation** of a planetary orbit, 106
**Equation of the centre**, 119, 148
**Equation of the equinoxes**, 139
**Equation of time**, 42, 146
**Equator**, celestial, 28; mean, 137, 238; reduction to the, 148; solar, 170; terrestrial, 4; true, 242
**Equatorial parallax**, mean horizontal, 200

**Equinox**, autumnal, 40; mean, 137, 238; precession of the, 226; true, 242; vernal, 37, 39
**Eros**, opposition of, 208, 212

**Factor**, parallax-, 222
**Fictitious mean sun**, 140, 424
**Film-to-film method** of measuring proper motions, 306
**First point** of Aries, 37
**Flattening**, 198
**Formula**, four-parts, 12; fundamental, 7; haversine, 18; polar, 15; sine, 9
**Four-parts formula**, 12
**Frequency** of eclipses, 398
**Fundamental plane**, 369, 390
**Fundamental stars**, 95

**Galactic co-ordinates**, 276
**Galactic latitude**, 267
**Galactic rotation**, 273, 275
**General precession**, 237
**Geocentric celestial sphere**, 33
**Geocentric distance**, 128
**Geocentric equatorial rectangular co-ordinates of sun**, 127
**Geocentric motion** of a planet, 160
**Geocentric parallax**, 199
**Geocentric right ascension and declination** of a planet, 127
**Geographical** position, 321
**Geoid**, 195; dimensions of, 198
**Gibbous**, 168
**Gladstone and Dale's Law**, 66
**G.M.T.**, note on, 44
**Gnomonic** Projection, 337
**Gravitation**, constant of, 101; law of, 101
**Great circle**, 1; equation of, on Mercator's chart, 335; pole of, 1
**Greenwich mean astronomical time**, 43
**Greenwich mean noon**, 43
**Greenwich tables** of refraction, 68
**Gregorian calendar**, 145

*Halley*, 249
**Haversine**, definition of, 18; formula, 19
**Heliocentric distance**, 99
**Heliocentric ecliptic rectangular co-ordinates**, 122
**Heliocentric equatorial rectangular co-ordinates**, 124, 125
**Heliocentric rectangular co-ordinates** of earth, 126
**Heliographic co-ordinates**, 169; of a sun-spot, 174
**Heliographic** latitude and longitude, 171
*Hill, S. N.*, 265
*Hipparchus*, 226, 249
**Horizon**, celestial, 26; dip of the, 317
**Horizon-glass**, 315
**Horizontal parallax**, 200
**Horizontal refraction**, 69
**Hour angle**, 29; of the mean sun, 42

*Immersion*, 368
**Independent** day numbers, 184, 246
**Index error**, 316
**Index-glass**, 315
**Index of refraction**, 59
**Inman's Nautical Tables**, 19
**Intercept**, 323

**Julian calendar**, 144
**Julian date**, 146
**Jupiter**, geocentric motion of, 163

*Kapteyn, J. C.*, 302, 306
**Kepler's equation**, 113; solution of, 114
**Kepler's Laws**, 98
**Knot**, 23
**Kowalsky's method** (visual binary), 346
*Küstner*, 190

**Lambert's theorem**, 135
**Laser**, distance of the moon, 203
**Latitude**, astronomical, 196; celestial, 40; galactic, 267; geocentric, 196; parallel of, 5; variation of, 190
**Leap-year**, 144
*Lehmann-Filhés*, 360
**Level error**, 76; determination of, 87
**Light**, velocity of, 420
**Light-year**, 224

**Line**, position, 323; rhumb, 326; Sumner, 323
**Local** sidereal time, 41
**Longitude**, celestial, 40; mean, 122; nutation in, 233, 246; of ascending node, 121; of perihelion, 121; terrestrial, 5; true, of a planet, 121
**Loxodrome**, 326
**Lunation**, 398
**Luni-solar precession**, 228

**Magnitude** (of a partial eclipse), 385
**Magnitude, Stellar**, 414; absolute, 415; apparent, 414
**Major axis**, 99
**Major Planets**, 98
**Masses**, of the planets, 102; of spectroscopic binaries, 366; of the stars, 355
**Mass-function**, 365
**Mean angular motion**, 100
**Mean anomaly**, 113
**Mean co-ordinates**, 259
**Mean ecliptic**, 238
**Mean equator**, 137, 238
**Mean equinox**, 137, 238
**Mean longitude**, 122
**Mean noon**, 43
**Mean place**, reduction from, to apparent place, 245
**Mean refraction**, 62
**Mean sidereal time**, 138
**Mean solar day**, 42, 140
**Mean solar time**, 41; conversion of, into sidreal time (and *vice versa*), 143
**Mean sun**, 42; hour angle of, 42; right ascension of, 42, 140
**Mercator's chart**, 326
**Meridian**, observer's, 29; terrestrial, 5
**Meridian circle**, description of, 76; errors of, 74
**Metonic cycle**, 401
**Micrometer**, 341
*Mihalis, D.*, 275
**Mile**, nautical, 6
**Milky Way**, pole of, 57
**Month**, anomalistic, 133; nodical, 133; sidereal, 133; synodic, 133
**Moon**, eclipses of, 378; orbit of, 132; parallax of, 200; phases of, 166; time of transit of, over any meridian, 155
**Motion**, mean angular, 100; parallactic, 260; solar, 260

*Napier*, analogies, 23; rules for right-angled and quadrantal triangles, 15
**Nautical mile**, 6
*Newcomb, S.*, 239
**New moon**, 133
*Newton*, law of gravitation, 101
**Node**, 120; longitude of ascending, 121
**Nodical month**, 133
**Noon**, apparent, 41; mean, 43
**North polar distance**, 29
**Nutation**, 136, 231; constant of, 247; in longitude, 233, 246; in the obliquity, 233, 246

**Obliquity** of the ecliptic, 39; nutation in, 233, 246
**Observed altitude**, corrections to the, 317
**Observer's meridian**, 29
**Occultations**, 368; Besselian elements of, 373; Bessel's method of investigating, 370; geometrical conditions for, 369; graphical method for, 375; prediction of, 373; reduction of, 376
**Opposition**, 130
**Orbit**, determination of elements of a planetary, 129; elements of elliptic, 116, 122; equation of, 106; in space, 120; of the earth, 37, 131; of the moon, 132; period in, 99; sun's apparent, 131
**Orbital motion**, dynamical principles of, 104
**Orientation**, error of, 288

**Parallactic** angle, 49; ellipse, 219; motion, 260
**Parallax**, annual, 218; geocentric, 199; horizontal, 200; in right ascension, and declination, 204; in zenith distance and azimuth, 206; mean equatorial horizontal, 200; of the moon, 200; solar, 207; stellar, 217, 309, 411
**Parallaxes**, dynamical, 356; secular, 262, 306; statistical, 266

**Parallax factor**, 222
**Parallel**, of altitude, 26; of declination, 29; of latitude, 5
**Parsec**, 223
**Partial eclipse**, of the moon, 378; of the sun, 386
*Pearce, J. A.*, 265
**Penumbra**, 379, 386
**Periastron**, 343
**Perigee**, 132
**Perihelion**, 99; longitude of, 121
**Periodic terms**, 231
**Periods**, orbital and synodic, 129
**Perpendicularity**, error of, 316
**Perturbations**, 103
**Phase**, 166
**Photographic plate**, measurement of, 285
**Photographic refractor**, 278
**place**, apparent, 245; mean, 300; true, 242
**Plane**, collimation, 75; fundamental, 369, 390; tangent, 280
**Planet**, brightness of, 169; geocentric motion of, 160; heliocentric coordinates of, 122; phase of, 166; velocity of, 108
**Planetary precession**, 235
**Planets**, major, 98; masses of the, 102; dimensions and elements of, 422
**Plate constants**, 297
*Plummer, H. C.*, 129, 404
**Pluto**, discovery of, 98
**Polar formulae**, 15
**Pole**, of a great circle, 1; of a small circle, 1; of Milky Way, 57; terrestrial, 4
**Position angle**, 173; of a binary star, 340; of the ant-apex, 265
**Position circle**, 320
**Position line**, 322; tables, 333
**Precession**, 136, 226; effect of, on right ascension and declination, 229; general, 237; luni-solar, 228; planetary, 235
**Primary**, 340
**Prime vertical**, 27
**Probable error**, 312
**Proper motion**, absolute, 304; components of, 252; definition of, 249; measurement of, 251, 301; relative, 304; the film-to-film method of measuring, 306; total, 252
**Pulkova Tables** (refraction), 68

**Quadrantal triangles**, 14
**Quadrature**, 168
**Quasar**, 368

**Radar**, distance of the moon, 203; distance of Venus, 212
**Radial velocity**, 213, 251; determination of solar motion from, 274; solar motion and, 263
**Radio-occultation**, 368
**Radius vector**, 99
**Reduction to the equator**, 148
**Reference stars**, 298
**Refraction**, angle of, 61; constant of mean, 62; differential, 294; effect of, on right ascension and declination, 70; effect of, on time of sunset, 69; general formula for, 62; horizontal, 69; in astronomical photography, 291; index of, 59; laws of, 58; mean, 62.
**Refractor**, photographic, 278
**Relative proper motions**, 304
**Retrograde motion**, 162
**Rhumb line**, 326
*Rigge, W. F.*, 375
**Right-angled triangles**, 14
**Right ascension**, 37; measurement of, 93; of the mean sun, 42
**Rising**, 46
*Roemer*, 178

**Saros**, 400
**Satellites**, elements and dimensions of, 423; artificial, 204
*Schlesinger, F.*, 404, 412
**Seasons**, 150; lengths of, 153
**Secular parallax**, 262, 306
**Secular terms**, 231
**Secular variation**, 240
**Semi-diameter**, 93, 203
**Semi-major axis**, 99
**Setting**, 46
**Sextant**, 314; errors of, 316
*Shearme, F. N.*, 333
**Side error**, 316
**Sidereal day**, 37, 138
**Sidereal month**, 133

**Sidereal time**, 37, 136; apparent, 138; conversion of, into mean time, 143; ephemeris, 159; local, 41; mean, 138
**Sidereal year**, 131, 141
**Simpson's hypothesis** (refraction), 72
**Sine-formula**, 9
**Small circle**, 1; length of arc of, 3
**Solar equator**, 170
**Solar motion**, 260; apex of, 261; determination of, from proper motions, 267; determination of, from radial velocities, 274
**Solar parallax**, 207; diurnal method, 210; related to constant of aberration, 213
**Solstice**, summer, 40, 150; winter, 40, 150
**Southern hemisphere**, diagram for, 31
**Spectroscopic binaries**, 357; masses of, 366
**Spectroscopic method** applied to solar parallax, 213
**Spherical angle**, 2
**Spherical triangle**, 1; solution of, 35
**Standard co-ordinates**, 280; formulae for, 283
**Standard error**, 312
**Stars**, circumpolar, 32; clock, 95; comparison, 304; fundamental, 95; masses of, 355; reference, 95
**Stationary point**, 162, 164
**Statistical parallaxes**, 266
**Stellar parallax**, 217; effect of, on right ascension and declination, 220; measurement of, 222, 309
**St Hilaire's method** (position line), 322
*Stracke, G.*, 129
**Sub-solar point**, 321
**Sub-stellar point**, 321
**Summer solstice**, 40, 150
**Sumner line**, 323
**Sun**, apparent orbit of, 131; axis of rotation of, 172; eclipses of, 386; mean, 42, 139; time of transit of, over any meridian, 155
**Sunset**, effect of refraction on time of, 69
**Sun-spot**, heliographic co-ordinates of, 174
**Synodic month**, 133
**Synodic period**, 129

**Tangential velocity**, 250
**Tangent plane**, 280
**Telescope**, photographic zenith, 190; photographic, 278; refracting, 279, 287; transit, 74
**Terrestrial** latitude and longitude, 5
*Thackery, A. D.*, 213
**Tilt**, error of, 291
**Time**, equation of, 42, 146; Greenwich civil, 44; Greenwich mean astronomical, 43; local civil, 44; local sidereal, 41; mean solar, 41; measurement of, 95; sidereal, 37, 136; universal, 44; zone, 44; ephemeris, 139, 424
**Torrid zone**, 152
**Total eclipse**, of the moon, 378; of the sun, 386
**Total proper motion**, 252
**Transferred position line**, 330
**Transit**, 29, 33; of Venus, 212
**Triangle**, quadrantal, 14; right-angled, 14; spherical, 1; solution of, 35
**Trigonometrical** ratios for small angles, 21
**Tropic** of Cancer, 151; of Capricorn, 151
**Tropical year**, 132, 141
**True anomaly** 111; as a series, 118
**True equator**, 242
**True equinox**, 242
**True place of a star**, 242
*Turner, H. H.*, 283, 297
**Twilight**, 51

**Umbra**, 379, 386
**Universal time**, 44, 139, 424

*Van Maanen, A.*, 312
**Variation**, annual, 241, 260; of latitude, 190; secular, 240
**Velocity**, components of, 110; radial, 213, 251; tangential, 250
**Velocity curve**, 359
**Venus**, transit of, 212
**Vernal equinox**, 37

**Vertical**, angle of the, 196; circle, 26; prime, 27
**Visual binary**, orbit of, 342

**Winter solstice**, 40
**Wire intervals**, 82
*Wood, H. E.*, 408

**Year**, anomalistic, 132; Besselian, 145; sidereal, 131, 141; tropical, 132, 141

**Zenith**, 25; astronomical, 196; geocentric, 196
**Zenith distance**, 26; rate of change of, 48; refraction for a small, 60; the calculated, 323
**Zone**, temperate, 152; torrid, 152
**Zone time**, 44
**Zwier's method** (visual binary star), 351

# Textbook
on
# Spherical Astronomy

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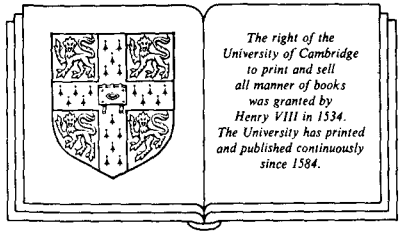
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Sixth edition
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CAMBRIDGE UNIVERSITY PRESS

CAMBRIDGE

LONDON · NEW YORK · NEW ROCHELLE

MELBOURNE · SYDNEY

Published by the Press Syndicate of the University of Cambridge
The Pitt Building, Trumpington Street, Cambridge CB2 1RP
32 East 57th Street, New York, NY 10022, USA
10 Stamford Road, Oakleigh, Melbourne 3166, Australia

This edition © Cambridge University Press 1977

First edition 1931
Second edition 1936
Third edition 1940
Fourth edition 1944
Reprinted 1947 1949 1956 1960
Fifth edition 1962
Reprinted 1965 1971
Sixth edition 1977
Reprinted 1986

Printed in Great Britain at the
University Press, Cambridge

*Library of Congress Cataloguing in Publication Data*
Smart, William Marshall, 1889–
Text-book on spherical astronomy.
Includes index.
1. Astronomy, Spherical and practical. I. Green,
Robin Michael, 1937– II. Title.
QB145.S6 1977 522'.7 76–50643

ISBN 0 521 21516 1 hard covers
ISBN 0 521 29180 1 paperback
(*Fifth edition* ISBN 0 521 06491 0 hard covers
ISBN 0 521 09159 4 paperback)

# PREFACE

THIS BOOK is based on lectures given annually in the University of Cambridge and on a parallel course of instruction in Practical Astronomy at the Observatory. The recent changes in the almanacs have, in many respects, affected the position of the older textbooks as channels of information on current practice, and the present work is intended to fill the gap caused by modern developments. In addition to the time-honoured problems of Spherical Astronomy, the book contains the essential discussion of such important subjects as heliographic co-ordinates, proper motions, determination of position at sea, the use of photography in precise astronomical measurements and the orbits of binary stars, all or most of which have received little attention in works of this kind. In order to make certain subjects as complete as possible, I have not hesitated to cross the traditional frontiers of Spherical Astronomy. This is specially the case as regards the spectroscopic determination of radial velocity which is considered, the physical principles being assumed, in relation to such problems as solar parallax, the solar motion and the orbits of spectroscopic binary stars.

Throughout, only the simplest mathematical tools have been used and considerable attention has been paid to the diagrams illustrating the text. I have devoted the first chapter to the proofs and numerical applications of the formulae of spherical trigonometry which form the mathematical foundation of the subsequent chapters. Although other formulae have been given for reference, I have limited myself to the use of the basic formulae only.

A writer of a textbook on Spherical Astronomy cannot avoid a certain measure of detailed reference to the principal astronomical instruments and, accordingly, general descriptions of instruments have been given in the appropriate places, usually with a simple discussion of the chief errors which must be taken into account in actual observational work.

In numerical applications, the almanac for 1931 has been used.

As regards notation, I have usually followed the recommendations of the International Astronomical Union, but I have made

several modifications when the avoidance of confusion or of misconception seemed to me of greater importance than the rigid adherence to the definite system proposed; for example, the angle of stellar parallax has been denoted by $\Pi$ instead of the usual symbol $\pi$ which the student is apt to associate on most mathematical occasions with the properties of the circle.

I have also followed the advice of the International Astronomical Union [*Transactions*, vol. III, p. 300 (1928)] as regards the nomenclature of standard mean time, and throughout the book G.C.T. (Greenwich Civil Time) denotes the mean time, for the meridian of Greenwich, reckoned from midnight. (See also footnote to p. 44 *infra*.) This is contrary to the present usage of the British *Nautical Almanac* which, since 1925, has employed the symbol G.M.T. in the sense in which G.C.T. has just been described, notwithstanding the fact that previous to 1925 G.M.T. universally signified the mean time reckoned from mean noon at Greenwich. The latter I have denoted (also in accordance with the recommendations of the I.A.U.) by the letters G.M.A.T. (Greenwich Mean Astronomical Time).

At the end of each chapter there is a collection of exercises many of which have been taken, by permission, from the papers set in the Mathematical Tripos at Cambridge and in the examinations of London University and Cambridge Colleges; several have also been taken from Ball's *Spherical Astronomy*.

I have the pleasant duty of recording my indebtedness to Professor W. E. Anderson, of Miami University, Oxford, Ohio, U.S.A., and to Mr M. J. Dean, B.A., Scholar of Trinity College, Cambridge, who have read the whole of the manuscript with the greatest care and removed many blemishes which might have escaped less vigilant eyes. I am also grateful to Dr F. S. Hogg, formerly of Amherst Observatory, Mass., U.S.A., who has read critically about half of the book, and to Dr L. J. Comrie, Superintendent of the *Nautical Almanac* Office, for helpful suggestions in the chapter on Occultations and Eclipses.

It is also a pleasure to express my thanks to the officials and staff of the University Press for their courtesy and care during the printing of the book.

W. M. S.

OBSERVATORY, CAMBRIDGE
1931 *January* 1

# PREFACE TO THE SIXTH EDITION

SINCE this book was first published there have been considerable changes in the terminology and the quantities tabulated in the *Astronomical Ephemeris* and other almanacs. In making this revision I have felt that it is important to recognise these changes and to ensure compatibility of the book with the *Astronomical Ephemeris*. While I hope that this has been generally achieved, slight differences do remain in the treatment of solar eclipses and in the definition of the *Besselian Day Numbers* for annual aberration.

Without doubt the most important change in the almanacs has been the introduction of Ephemeris Time. As it is this time that is used as the argument in almost all tabulation in the almanacs, it clearly requires an important place and adequate description in an introductory text such as this. Accordingly I have made substantial revision to the chapter on Time in stressing the distinction between Ephemeris and Universal Time. A difficulty arose, however, in connection with the exposition of this distinction. Professor Smart had used the term, mean sun, to define Universal Time. The mean sun is a wholly fictitious body that was introduced to define solar time long before the distinction, that we are concerned with, was recognised. Newcomb called it the fictitious mean sun and gave it a very precise and formal definition. Newcomb's work naturally related to the subsequent definition of Ephemeris Time and so I have retained, his term, the *fictitious mean sun*, as a reference point for *Ephemeris Time*. For continuity, I have also retained Smart's use of the term, mean sun, as a reference point for *Universal Time*. I hope that this dichotomy, which is not standard usage, will not lead to confusion in practice. It is not intended to imply that only one of the reference points is fictitious; both are.

I have taken the opportunity of adding a number of exercises at the end of several chapters. Some of these are taken, by permission from recent examination papers of Glasgow University. It is hoped that some of these examples will be helpful in illustrating new material that has been added to the text.

As this is intended to be a revision of an existing text and not a major reorganisation of it, the detailed subject matter of the book remains the same as in previous editions, with the exception of parts of Chapter VI on Time. I have, however, added fairly frequent comments in the text to indicate the sometimes modified relevance of the subject matter under discussion to modern astronomy.

A guiding principle of the revision has been to re-establish compatibility with the *Astronomical Ephemeris* and, to a lesser extent with the fuller explanations of the *Explanatory Supplement to the Astronomical Ephemeris and the American Ephemeris and Nautical Almanac*. In the original edition of this book Professor Smart took his numerical examples from the almanac (then the *Nautical Almanac*) of 1931. Where the quantities extracted from that almanac correspond either exactly or very nearly with those tabulated in the current *Astronomical Ephemeris*, I have retained that example with only minimal modification. Where, however, such detailed correspondence does not exist, I have substituted a new example based on the *Astronomical Ephemeris* of 1975. In no case is a quantity used that is not tabulated in the current *Astronomical Ephemeris*, whatever the apparent date of the example.

It is a pleasure to acknowledge my considerable debt to Dr G. A. Wilkins, Superintendent of the *Nautical Almanac* Office, who has drawn my attention to many points requiring revision in this book. Further I am indebted to Dr A. E. Roy for many informative discussions and guidance and to Mrs L. Williamson for her considerate typing of the often intricate material.

R. M. G.

DEPARTMENT OF ASTRONOMY,
UNIVERSITY OF GLASGOW
1976 *July* 7

# CONTENTS


CHAPTER I

PAGE

Spherical Trigonometry . . . . . 1

**1.** Introduction. **2.** The spherical triangle. **3.** Length of a small circle arc. **4.** Terrestrial latitude and longitude. **5.** The cosine-formula. **6.** The sine-formula. **7.** The analogue formula. **8.** The four-parts formula. **9.** Alternative proofs of the formulae **A**, **B** and **C**. **10.** Right-angled and quadrantal triangles. **11.** Polar formulae. **12.** Numerical example. **13.** The haversine formula. **14.** Another method. **15.** The trigonometrical ratios for small angles. **16.** Delambre's and Napier's analogies. Exercises on Chapter I

CHAPTER II

The Celestial Sphere . . . . . 25

**17.** Introduction. **18.** Altitude and azimuth. **19.** Declination and hour angle. **20.** Diagram for the southern hemisphere. **21.** Circumpolar stars. **22.** The standard or geocentric celestial sphere. **23.** Solution of the spherical triangle *PZX*. **24.** Right ascension and declination. **25.** The earth's orbit. **26.** Celestial latitude and longitude. **27.** Sidereal time. **28.** Mean solar time. **29.** Example. **30.** Hour angle of a heavenly body. **31.** Rising and setting. **32.** Rate of change of zenith distance and azimuth. **33.** Twilight. Exercises on Chapter II

CHAPTER III

Refraction . . . . . 58

**34.** The laws of refraction. **35.** Refraction for small zenith distances. **36.** General formula for refraction. **37.** Development of the general formula for refraction. **38.** The determination of the constants $A$ and $B$ in the formula (29). **39.** The effect of refraction on the time of sunset. **40.** Effect of refraction on the right ascension and declination of a star. Exercises on Chapter III

CHAPTER IV

The Meridian Circle . . . . . 74

**41.** General description. **42.** Instrumental errors. **43.** Azimuth error. **44.** Level error. **45.** Collimation error. **46.** The total correction to the observed time of transit. **47.** Bessel's formula. **48.** Wire intervals. **49.** Determination of the collimation error. **50.** Determination of the level error. **51.** Determination of the azimuth error. **52.** The chronograph (mainly of historical interest). **53.** The measurement of declination. **54** The measurement of right ascension. **55.** The measurement of time. Exercises on Chapter IV

CHAPTER V

PAGE

Planetary Motions . . . . . 98

**56.** Introduction. **57.** Kepler's first law. **58.** Kepler's second law. **59.** Kepler's third law. **60.** Newton's law of gravitation. **61.** The masses of the planets. **62.** Perturbations of the elements. **63.** The dynamical principles of orbital motion. **64.** The equation of the orbit. **65.** Velocity of a planet in its orbit. **66.** Components of the linear velocity perpendicular to the radius vector and to the major axis. **67.** The true and eccentric anomalies. **68.** Kepler's equation. **69.** Solution of Kepler's equation. **70.** Summary of the formulae of elliptic motion. **71.** The eccentric anomaly expressed as a series in terms of $e$ and the mean anomaly. **72.** The true anomaly expressed as a series in terms of $e$ and the eccentric anomaly. **73.** The equation of the centre. **74.** The orbit in space. **75.** The heliocentric ecliptic rectangular co-ordinates of a planet. **76.** The heliocentric equatorial co-ordinates of a planet. **77.** The heliocentric equatorial co-ordinates of a planet (alternative method). **78.** The heliocentric rectangular co-ordinates of the earth. **79.** The planet's geocentric right ascension and declination. **80.** The orbital and synodic periods of a planet. **81.** The earth's orbit. **82.** The sun's apparent orbit. **83.** The moon's orbit. Exercises on Chapter v

CHAPTER VI

Time . . . . . 136

**84.** Sidereal time. **85.** Ephemeris and Universal Time. **86.** The sidereal year and the tropical year. **87.** Relation between universal and mean sidereal time. **88.** The calendar. **89.** The Besselian year. **90.** The Julian Date. **91.** The equation of time. **92.** The seasons. **93.** Time of the sun's transit over any meridian. **94.** Time of the moon's transit over any meridian. Exercises on Chapter vi

CHAPTER VII

Planetary Phenomena and Heliographic Co-ordinates . 160

**95.** The geocentric motion of a planet. **96.** The heliocentric distance of a planet, when stationary, in terms of its elongation. **97.** The stationary points, the inclination being taken into account. **98.** The phases of the planets and the moon. **99.** The brightness of the planets. **100.** Heliographic co-ordinates. **101.** The heliographic co-ordinates of the centre of the disc. **102.** Position angle of the sun's axis of rotation. **103.** The heliographic co-ordinates of a sun-spot. Exercises on Chapter vii

CHAPTER VIII

Aberration . . . . . 178

**104.** The law of aberration. **105.** Annual aberration in ecliptic longitude and latitude. **106.** The aberrational ellipse. **107.** Aberration in right ascension and declination. **108.** The elliptic motion of the earth and aberration. **109.** The measurement of the constant of aberration. **110.** The theoretical value of the aberration constant. **111.** Diurnal aberration. **112.** The correction for light-time. Exercises on Chapter viii

CHAPTER IX

PAGE

Parallax . . . . . 195

113. Introductory. 114. The geoid. 115. Astronomical and geocentric latitude. 116. The figure of the earth. 117. Geocentric parallax. 118. The parallax of the moon. 119. Semi-diameter. 120. Parallax in right ascension and declination. 121. Parallax in zenith distance and azimuth. 122. The solar parallax. 123. The solar parallax (diurnal method). 124. The solar parallax (other methods). 125. Stellar parallax. 126. Effect of parallax on the star's longitude and latitude. 127. The parallactic ellipse. 128. The effect of stellar parallax on right ascension and declination. 129. The measurement of stellar parallax. 130. The parsec and light-year. Exercises on Chapter IX

CHAPTER X

Precession and Nutation . . . . . 226

131. Introductory. 132. The effect of precession on the right ascension and declination of a star. 133. The effect of precession on right ascension and declination (alternative method). 134. Nutation. 135. Nutation in the obliquity. 136. Planetary precession. 137. The mean equator and the mean co-ordinates of a star. 138. The secular variation. 139. The true equator and the true co-ordinates of a star. 140. The apparent place of a star. 141. Reduction from mean place to apparent place (or *vice versa*). 142. Cataloguing the stars. 143. The numerical values of the nutation in longitude and in the obliquity of the ecliptic. Exercises on Chapter X

CHAPTER XI

The Proper Motions of the Stars . . . . . 249

144. Definition of proper motion. 145. Relation between proper motion, tangential velocity and parallax. 146. Radial velocity. 147. The measurement of proper motion. 148. The components of proper motion at different epochs referred to the same equatorial system. 149. The components of proper motion referred to the mean equators of two different epochs. 150. Mean and apparent co-ordinates. 151. The solar motion and parallactic motion. 152. Secular parallax. 153. The solar motion and radial velocity. 154. The solar motion in the general case. 155. Statistical parallaxes. 156. Determination of the solar apex from proper motions. 157. Determination of the solar motion from radial velocities. Exercises on Chapter XI

CHAPTER XII

Astronomical Photography . . . . . 278

158. The photographic refractor. 159. The tangent plane. 160. Standard co-ordinates. 161. Formulae for the standard co-ordinates. 162. The measurement and scale of photographic plates. 163. The measured co-ordinates. 164. Discussion of errors. 165. Refraction. 166. Aberration. 167. The general relations between standard co-ordinates and measured co-ordinates (Turner's Method). 168. The measurement of astrographic

PAGE

plates. 169. Photographic observations of minor planets and comets. 170. The measurement of proper motions. 171. The measurement of proper motions (Kapteyn's method). 172. The reduction of relative to absolute proper motions. 173. The film-to-film method of measuring proper motions. 174. The determination of the plate constants by the method of Christie and Dyson. 175. The measurement of stellar parallaxes. Exercises on Chapter XII

CHAPTER XIII

Determination of Position at Sea . . . . . 314

176. The sextant. 177. The errors of a sextant. 178. Corrections to the observed altitude. 179. The position circle. 180. The position line (St Hilaire's method). 181. Example of the calculation of the intercept. 182. Mercator's chart. 183. Determination of the ship's position from two observations of altitude. 184. Example of finding the ship's position from two observations. 185. Special methods. 186. Equation of a great circle on Mercator's chart. Exercises on Chapter XIII

CHAPTER XIV

Binary Star Orbits . . . . . 340

187. Visual binary stars. 188. The micrometer. 189. The elements of the true orbit of a visual binary. 190. The apparent orbit of a visual binary. 191. Kowalsky's method of determining the elements of a visual binary orbit. 192. Zwier's method of determining the elements of a visual binary orbit. 193. The masses of the stars. 194. Dynamical parallaxes. 195. Spectroscopic binaries. 196. The velocity curve. 197. Method of Lehmann-Filhés. 198. Two spectra visible. 199. The mass-function. 200. The masses of spectroscopic binaries. 201. Eclipsing binary stars. Exercises on Chapter XIV

CHAPTER XV

Occultations and Eclipses . . . . . 368

202. Occultations of stars by the moon. 203. The geometrical conditions for an occultation. 204. Bessel's method of investigating an occultation. 205. The Besselian elements of an occultation. 206. The prediction of an occultation at any place. 207. The reduction of occultations. 208. Eclipses of the moon. 209. The angular radius of the shadow cone at the moon's geocentric distance. 210. The ecliptic limits. 211. Calculation of a lunar eclipse. 212. Eclipses of the sun. 213. The angle subtended at the earth's centre by the centres of the sun and moon at the beginning or end of a partial solar eclipse. 214. Ecliptic limits. 215. The Besselian elements for a solar eclipse. 216. Eclipse calculations for any station. 217. The frequency of eclipses. 218. The repetition of eclipses. Exercises on Chapter XV

PAGE

APPENDIX I. The Method of Dependences . . . 404

APPENDIX II. Stellar Magnitudes . . . . . 414

APPENDIX III. The Coelostat . . . . . 416

APPENDIX A. Astronomical Constants . . . . 420

APPENDIX B. Dimensions of the Sun, Moon and Planets 421

APPENDIX C. Mean elements of the planetary orbits for the epoch 1975 January 0·5 E.T. . 422

APPENDIX D. Elements and dimensions of the Satellites 423

APPENDIX E. Ephemeris and Universal Time . . 424

INDEX . . . . . 425